Lesson 01

INTRODUCTION TO FINANCIAL ECONOMICS

TOPIC 001: INTRODUCTION

Financial economics is a branch of economics that analyzes the use and distribution of resources in markets. Financial decisions must often take into account future events, whether those be related to individual stocks, portfolios, or the market as a whole. Making financial decisions is not always a straightforward process. Time, risk (uncertainty), opportunity costs, and information can create incentives or disincentives. Financial economics employs economic theory to evaluate how certain things impact decision making, providing investors with the instruments to make the right calls.

Financial economics usually involves the creation of sophisticated models to test the variables affecting a particular decision. Often, these models assume that individuals or institutions making decisions act rationally, though this is not necessarily the case. The irrational behavior of parties has to be taken into account in financial economics as a potential risk factor. This branch of economics builds heavily on microeconomics and basic accounting concepts. It is a quantitative discipline that uses econometrics as well as other mathematical tools.

TOPIC 002: FINANCE AND FINANCIAL SYSTEM

Finance is a term for matters regarding the management, creation, and study of money and investments. It involves the use of credit and debt, securities, and investment to finance current projects using future income flows. Because of this temporal aspect, finance is closely linked to the time value of money, interest rates, and other related topics.

- It is the study of how people allocate scarce resources over time.
- Allocation of financial decisions are different from other resource allocation decisions

Financial decisions:

1. Spread out over time and
2. Usually not known with certainty in advance by either the decision makers or anybody else.

A financial system is a set of institutions, such as banks, insurance companies, and stock exchanges that permit the exchange of funds. Financial systems exist on firm, regional, and global

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levels. Borrowers, lenders, and investors exchange current funds to finance projects, either for consumption or productive investments, and to pursue a return on their financial assets. The financial system also includes sets of rules and practices that borrowers and lenders use to decide which projects get financed, who finances projects, and terms of financial deals.

- The set of markets and other institutions used for financial contracting and the exchange of assets and risks.
- It includes stocks, bonds and other financial instruments; financial intermediaries such as banks and insurance companies; and regulatory bodies

Multiple components make up the financial system at different levels. The firm's financial system is the set of implemented procedures that track the financial activities of the company. Within a firm, the financial system encompasses all aspects of finances, including accounting measures, revenue and expense schedules, wages, and balance sheet verification.

On a regional scale, the financial system is the system that enables lenders and borrowers to exchange funds. Regional financial systems include banks and other institutions, such as securities exchanges and financial clearinghouses.

The global financial system is basically a broader regional system that encompasses all financial institutions, borrowers, and lenders within the global economy. In a global view, financial systems include the International Monetary Fund, central banks, government treasuries and monetary authorities, the World Bank, and major private international banks.

TOPIC 003: WHY STUDY FINANCIAL ECONOMICS?

The study of financial economics helps students understand these forces and provides the tools to assess their impacts. It provides a framework for analyzing how individuals or families, businesses, and governments make decisions as they face trade-offs.

- To manage your personal resources.
- To deal with the world of business.
- To pursue interesting and rewarding career opportunities.
- To make informed public choices as a citizen.
- To expand your mind.

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TOPIC 004: FINANCIAL DECISIONS

Households and firms have to make financial decisions. The financial decisions of households include:

Consumption and saving decisions: How much of the current wealth should they spend on consumption and how much of their current income should they save for the future?

Investment decision: How should they invest the money they have saved?

Financing decision: When and how should households use other people's money to implement their consumption and investment plans?

Risk-management decision: How and on what terms should households seek to reduce the financial uncertainties they face or when should they increase their risks?

There are four main financial decisions- Capital Budgeting or long-term Investment decision (Application of funds), Capital Structure or Financing decision (Procurement of funds), Dividend decision (Distribution of funds) and Working Capital Management Decision in order to accomplish goal of the firm viz., to maximize shareholder's (owner's) wealth.

TOPIC 005: MANAGEMENT AND OWNERSHIP OF A BUSINESS

A sole proprietorship is a business owned by only one person.

Advantages include: complete control for the owner, easy and inexpensive to form, and owner gets to keep all of the profits.

Disadvantages include: unlimited liability for the owner, complete responsibility for talent and financing, and business dissolves if the owner dies.

A general partnership is a business owned jointly by two or more people.

Advantages include: more resources and talents come with an increase in partners, and the business can continue even after the death of a partner.

Disadvantages include: partnership disputes, unlimited liability, and shared profits.

A limited partnership has a single general partner who runs the business and is responsible for its liabilities, plus any number of limited partners who have limited involvement in the business and whose losses are limited to the amount of their investment.

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A corporation is a legal entity that's separate from the parties who own it, the shareholders who invest by buying shares of stock. Corporations are governed by a Board of Directors, elected by the shareholders.

Advantages include: limited liability, easier access to financing, and unlimited life for the corporation.

Disadvantages include: the agency problem, double taxation, and incorporation expenses and regulations.

A limited liability company (LLC) is similar to a C-corporation, but it has fewer rules and restrictions than a C-corporation. For example, an LLC can have any number of members.

A cooperative is a business owned and controlled by those who use its services. Individuals and firms who belong to the cooperative join together to market products, purchase supplies, and provide services for its members.

A not-for-profit corporation is an organization formed to serve some public purpose rather than for financial gain. It enjoys favorable tax treatment.

A merger occurs when two companies combine to form a new company.

An acquisition is the purchase of one company by another with no new company being formed.

A hostile takeover occurs when a company is purchased even though the company's management and Board of Directors do not want to be acquired.

- Ability to run the business
- For the efficient scale of a business resources are to be pooled
- Diversification of risk is needed to overcome uncertainty
- Saving in the cost of gathering information
- The "learning curve" or "going concern" effect
- Primary goal of a corporate management
    - Maximize shareholder wealth
    - Maximize shareholder wealth

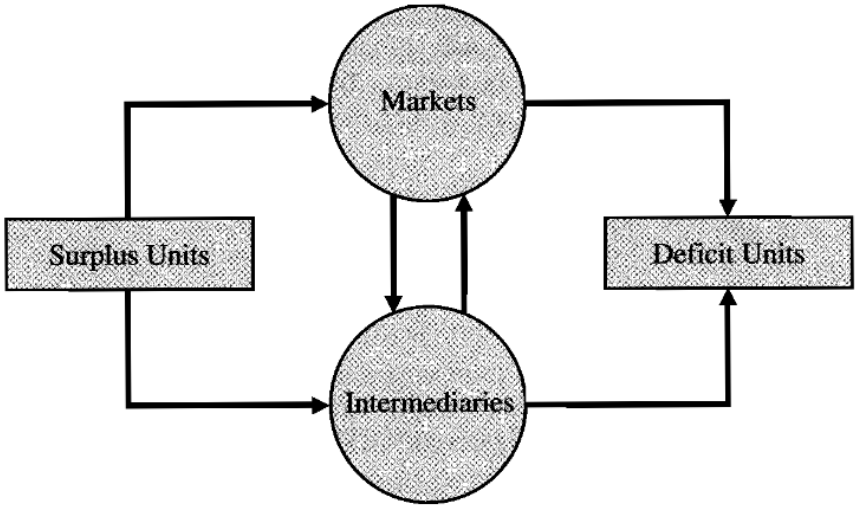
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Lesson 02

FINANCIAL MARKETS AND INSTITUTIONS

TOPIC 006: FINANCIAL MARKETS AND INSTITUTIONS

- Financial system encompasses the markets, intermediaries, service firms and other institutions used to carry out the financial decisions of households, business firms, and governments.



Basic Functions – Financial System

- Transferring resources across time and space
- Managing risk
- Clearing and settling payments to facilitate trade
- Pooling resources and subdividing shares
- Providing information
- Dealing with incentive problems

Financial markets refer broadly to any marketplace where the trading of securities occurs, including the stock market, bond market, forex market, and derivatives market, among others. Financial markets are vital to the smooth operation of capitalist economies.

Financial markets play a vital role in facilitating the smooth operation of capitalist economies by allocating resources and creating liquidity for businesses and entrepreneurs. The markets make it easy for buyers and sellers to trade their financial holdings. Financial markets create securities

products that provide a return for those who have excess funds (Investors/lenders) and make these funds available to those who need additional money (borrowers).

The stock market is just one type of financial market. Financial markets are made by buying and selling numerous types of financial instruments including equities, bonds, currencies, and derivatives. Financial markets rely heavily on informational transparency to ensure that the markets set prices that are efficient and appropriate. The market prices of securities may not be indicative of their intrinsic value because of macroeconomic forces like taxes.

Some financial markets are small with little activity, and others, like the New York Stock Exchange (NYSE), trade trillions of dollars of securities daily. The equities (stock) market is a financial market that enables investors to buy and sell shares of publicly traded companies. The primary stock market is where new issues of stocks, called initial public offerings (IPOs), are sold. Any subsequent trading of stocks occurs in the secondary market, where investors buy and sell securities that they already own.

TOPIC 007: FINANCIAL INNOVATION AND THE INVISIBLE HAND

Financial innovation is the process of creating new financial products, services, or processes. Financial innovation has come via advances in financial instruments, technology, and payment systems. Digital technology has helped to transform the financial services industry, changing how we save, borrow, invest, and pay for goods.

While large banks continue to invest in mobile banking, FinTech companies, like Stripe, help small businesses conduct online payments, and investment broker Robinhood seeks to democratize investing and finance. These innovations have increased the number of financial providers available to consumers, borrowers, and businesses.

The invisible hand is a metaphor for the unseen forces that move the free market economy. Through individual self-interest and freedom of production and consumption, the best interest of society, as a whole, are fulfilled. The constant interplay of individual pressures on market supply and demand causes the natural movement of prices and the flow of trade. The term "invisible hand" first appeared in Adam Smith's famous work, The Wealth of Nations, to describe how free markets can incentivize individuals, acting in their own self-interest, to produce what is societally necessary.

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TOPIC 008: TYPES OF FINANCIAL MARKETS

Stock Markets

Perhaps the most ubiquitous of financial markets are stock markets. These are venues where companies list their shares and they are bought and sold by traders and investors. Stock markets, or equities markets, are used by companies to raise capital via an initial public offering (IPO), with shares subsequently traded among various buyers and sellers in what is known as a secondary market.

Stocks may be traded on listed exchanges, such as the New York Stock Exchange (NYSE) or Nasdaq, or else over-the-counter (OTC). Most trading in stocks is done via regulated exchanges, and these play an important role in the economy as both a gauge of the overall health of the economy as well as providing capital gains and dividend income to investors, including those with retirement accounts such as IRAs and 401(k) plans.

Typical participants in a stock market include (both retail and institutional) investors and traders, as well as market makers (MMs) and specialists who maintain liquidity and provide two-sided markets. Brokers are third parties that facilitate trades between buyers and sellers but who do not take an actual position in a stock.

Over-the-Counter Markets

An over-the-counter (OTC) market is a decentralized market—meaning it does not have physical locations, and trading is conducted electronically—in which market participants trade securities directly between two parties without a broker. While OTC markets may handle trading in certain stocks (e.g., smaller or riskier companies that do not meet the listing criteria of exchanges), most stock trading is done via exchanges. Certain derivatives markets, however, are exclusively OTC, and so they make up an important segment of the financial markets. Broadly speaking, OTC markets and the transactions that occur on them are far less regulated, less liquid, and more opaque.

Bond Markets

A bond is a security in which an investor loans money for a defined period at a pre-established interest rate. You may think of a bond as an agreement between the lender and borrower that contains the details of the loan and its payments. Bonds are issued by corporations as well as by municipalities, states, and sovereign governments to finance projects and operations. The bond

market sells securities such as notes and bills issued by the United States Treasury, for example. The bond market also is called the debt, credit, or fixed-income market.

Money Markets

Typically the money markets trade in products with highly liquid short-term maturities (of less than one year) and are characterized by a high degree of safety and a relatively low return in interest. At the wholesale level, the money markets involve large-volume trades between institutions and traders. At the retail level, they include money market mutual funds bought by individual investors and money market accounts opened by bank customers. Individuals may also invest in the money markets by buying short-term certificates of deposit (CDs), municipal notes, or U.S. Treasury bills, among other examples.

Derivatives Markets

A derivative is a contract between two or more parties whose value is based on an agreed-upon underlying financial asset (like a security) or set of assets (like an index). Derivatives are secondary securities whose value is solely derived from the value of the primary security that they are linked to. In and of itself a derivative is worthless. Rather than trading stocks directly, a derivatives market trades in futures and options contracts, and other advanced financial products, that derive their value from underlying instruments like bonds, commodities, currencies, interest rates, market indexes, and stocks.

Futures markets

Futures markets are where futures contracts are listed and traded. Unlike forwards, which trade OTC, futures markets utilize standardized contract specifications, are well-regulated, and utilize clearinghouses to settle and confirm trades. Options markets, such as the Chicago Board Options Exchange (CBOE), similarly list and regulate options contracts. Both futures and options exchanges may list contracts on various asset classes, such as equities, fixed-income securities, commodities, and so on.

Forex Market

The forex (foreign exchange) market is the market in which participants can buy, sell, hedge, and speculate on the exchange rates between currency pairs. The forex market is the most liquid market in the world, as cash is the most liquid of assets. The currency market handles more than $6.6 trillion in daily transactions, which is more than the futures and equity markets combined.

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As with the OTC markets, the forex market is also decentralized and consists of a global network of computers and brokers from around the world. The forex market is made up of banks, commercial companies, central banks, investment management firms, hedge funds, and retail forex brokers and investors.

Commodities Markets

Commodities markets are venues where producers and consumers meet to exchange physical commodities such as agricultural products (e.g., corn, livestock, soybeans), energy products (oil, gas, carbon credits), precious metals (gold, silver, platinum), or "soft" commodities (such as cotton, coffee, and sugar). These are known as spot commodity markets, where physical goods are exchanged for money.

The bulk of trading in these commodities, however, takes place on derivatives markets that utilize spot commodities as the underlying assets. Forwards, futures, and options on commodities are exchanged both OTC and on listed exchanges around the world such as the Chicago Mercantile Exchange (CME) and the Intercontinental Exchange (ICE).

Cryptocurrency Markets

The past several years have seen the introduction and rise of cryptocurrencies such as Bitcoin and Ethereum, decentralized digital assets that are based on blockchain technology. Today, thousands of cryptocurrency tokens are available and trade globally across a patchwork of independent online crypto exchanges. These exchanges host digital wallets for traders to swap one cryptocurrency for another, or for fiat monies such as dollars or euros.

Because the majority of crypto exchanges are centralized platforms, users are susceptible to hacks or fraud. Decentralized exchanges are also available that operate without any central authority. These exchanges allow direct peer-to-peer (P2P) trading of digital currencies without the need for an actual exchange authority to facilitate the transactions. Futures and options trading are also available on major cryptocurrencies.

TOPIC 009: FINANCIAL MARKET RATES

The interest rate is the amount a lender charges a borrower and is a percentage of the principal—the amount loaned. The interest rate on a loan is typically noted on an annual basis known as the annual percentage rate (APR). An interest rate can also apply to the amount earned at a bank or credit union from a savings account or certificate of deposit (CD). Annual percentage yield (APY) refers to the interest earned on these deposit accounts.

Interest is essentially a charge to the borrower for the use of an asset. Assets borrowed can include cash, consumer goods, vehicles, and property. Because of this, an interest rate can be thought of as the "cost of money" - higher interest rates make borrowing the same amount of money more expensive.

Interest rates thus apply to most lending or borrowing transactions. Individuals borrow money to purchase homes, fund projects, launch or fund businesses, or pay for college tuition. Businesses take out loans to fund capital projects and expand their operations by purchasing fixed and long-term assets such as land, buildings, and machinery. Borrowed money is repaid either in a lump sum by a pre-determined date or in periodic installments.

For loans, the interest rate is applied to the principal, which is the amount of the loan. The interest rate is the cost of debt for the borrower and the rate of return for the lender. The money to be repaid is usually more than the borrowed amount since lenders require compensation for the loss of use of the money during the loan period. The lender could have invested the funds during that period instead of providing a loan, which would have generated income from the asset. The difference between the total repayment sum and the original loan is the interest charged.
When the borrower is considered to be low risk by the lender, the borrower will usually be charged a lower interest rate. If the borrower is considered high risk, the interest rate that they are charged will be higher, which results in a higher cost loan.

Simple Interest Rate

If you take out a $300,000 loan from the bank and the loan agreement stipulates that the interest rate on the loan is 4% simple interest, this means that you will have to pay the bank the original loan amount of $300,000 + (4% x $300,000) = $300,000 + $12,000 = $312,000.

The example above was calculated based on the annual simple interest formula, which is:
Simple interest = Principal X interest rate X time
The individual that took out a loan will have to pay $12,000 in interest at the end of the year, assuming it was only a one-year lending agreement. If the term of the loan was a 30-year mortgage, the interest payment will be:
Simple interest = $300,000 X 4% X 30 = $360,000

A simple interest rate of 4% annually translates into an annual interest payment of $12,000. After 30 years, the borrower would have made $12,000 x 30 years = $360,000 in interest payments, which explains how banks make their money.

Compound Interest Rate

Some lenders prefer the compound interest method, which means that the borrower pays even more in interest. Compound interest, also called interest on interest, is applied both to the principal and also to the accumulated interest made during previous periods. The bank assumes that at the end of the first year the borrower owes the principal plus interest for that year. The bank also assumes that at the end of the second year, the borrower owes the principal plus the interest for the first year plus the interest on interest for the first year.

The interest owed when compounding is higher than the interest owed using the simple interest method. The interest is charged monthly on the principal including accrued interest from the previous months. For shorter time frames, the calculation of interest will be similar for both methods. As the lending time increases, however, the disparity between the two types of interest calculations grows.

Using the example above, at the end of 30 years, the total owed in interest is almost $700,000 on a $300,000 loan with a 4% interest rate.

The following formula can be used to calculate compound interest:

Compound interest = $p \times [(1 + \text{interest rate})^n - 1]$

Where:

p = Principal

n = Number of compounding periods

Lesson 03

INTEREST RATES AND RATES OF RETURN

TOPIC 010: RATES OF RETURN ON RISKY ASSETS

A Rate of Return (RoR) is the net gain or loss of an investment over a specified time period, expressed as a percentage of the investment's initial cost.

A rate of return (RoR) can be applied to any investment vehicle, from real estate to bonds, stocks, and fine art. The RoR works with any asset provided the asset is purchased at one point in time and produces cash flow at some point in the future. Investments are assessed based, in part, on past rates of return, which can be compared against assets of the same type to determine which investments are the most attractive. Many investors like to pick a required rate of return before making an investment choice.

The Formula for Rate of Return (RoR)

The formula to calculate the rate of return (RoR) is:

Rate of return = [Initial value (Current value−Initial value)]×100

This simple rate of return is sometimes called the basic growth rate, or alternatively, return on investment (ROI). If you also consider the effect of the time value of money and inflation, the real rate of return can also be defined as the net amount of discounted cash flows (DCF) received on an investment after adjusting for inflation.

Rate of Return (RoR) on Stocks and Bonds

The rate of return calculations for stocks and bonds is slightly different. Assume an investor buys a stock for $60 a share, owns the stock for five years, and earns a total amount of $10 in dividends. If the investor sells the stock for $80, his per-share gain is $80 - $60 = $20. In addition, he has earned $10 in dividend income for a total gain of $20 + $10 = $30. The rate of return for the stock is thus a $30 gain per share, divided by the $60 cost per share, or 50%.

On the other hand, consider an investor that pays $1,000 for a $1,000 par value 5% coupon bond. The investment earns $50 in interest income per year. If the investor sells the bond for $1,100 in premium value and earns $100 in total interest, the investor's rate of return is the $100 gain on the sale, plus $100 interest income divided by the $1,000 initial cost, or 20%.

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Real Rate of Return (RoR) vs. Nominal Rate of Return (RoR)

The simple rate of return is considered a nominal rate of return since it does not account for the effect of inflation over time. Inflation reduces the purchasing power of money, and so $335,000 six years from now is not the same as $335,000 today.

Discounting is one way to account for the time value of money. Once the effect of inflation is taken into account, we call that the real rate of return (or the inflation-adjusted rate of return).

Real Rate of Return (RoR) vs. Compound Annual Growth Rate (CAGR)

A closely related concept to the simple rate of return is the compound annual growth rate (CAGR). The CAGR is the mean annual rate of return of an investment over a specified period of time longer than one year, which means the calculation must factor in growth over multiple periods.

To calculate compound annual growth rate, we divide the value of an investment at the end of the period in question by its value at the beginning of that period; raise the result to the power of one divided by the number of holding periods, such as years; and subtract one from the subsequent result.

Internal Rate of Return (IRR) and Discounted Cash Flow (DCF)

The next step in understanding RoR over time is to account for the time value of money (TVM), which the CAGR ignores. Discounted cash flows take the earnings of an investment and discount each of the cash flows based on a discount rate. The discount rate represents a minimum rate of return acceptable to the investor, or an assumed rate of inflation. In addition to investors, businesses use discounted cash flows to assess the profitability of their investments.

Assume, for example, a company is considering the purchase of a new piece of equipment for $10,000, and the firm uses a discount rate of 5%. After a $10,000 cash outflow, the equipment is used in the operations of the business and increases cash inflows by $2,000 a year for five years. The business applies present value table factors to the $10,000 outflow and to the $2,000 inflow each year for five years.

The $2,000 inflow in year five would be discounted using the discount rate at 5% for five years. If the sum of all the adjusted cash inflows and outflows is greater than zero, the investment is profitable. A positive net cash inflow also means that the rate of return is higher than the 5% discount rate.

The rate of return using discounted cash flows is also known as the internal rate of return (IRR). The internal rate of return is a discount rate that makes the net present value (NPV) of all cash

flows from a particular project or investment equal to zero. IRR calculations rely on the same formula as NPV does and utilizes the time value of money (using interest rates).

When calculating the rate of return, you are determining the percentage change from the beginning of the period until the end.

Example

- $\mathbf{r} = \frac{\textbf{Cash Dividend}}{\textbf{Beginning Price}} + \frac{\textbf{Ending Price of a Share} - \textbf{Beginning Price}}{\textbf{Beginning Price}}$
- r = Dividend Income Component + Price Change Component

Suppose you buy a stock for Rs 100 and hold it for a year. At the end of the year, the stock pays a cash dividend of Rs 5 per share and the price per share is Rs 105 just after the dividend is paid. The one year rate of return, r, is

r = 5% + 5% = 10%

- Suppose you invest in a stock costing Rs. 50. It pays a cash dividend during the year of Rs. 1 and you expect that it's price will be Rs. 60 at year's end. What is your expected rate of return?
- If the stock's price is actually $40 at year's end, what is your realized rate of return?

TOPIC 011: MARKET INDEXES AND MARKET INDEXING

A market index is a hypothetical portfolio of investment holdings that represents a segment of the financial market. The calculation of the index value comes from the prices of the underlying holdings. Some indexes have values based on market-cap weighting, revenue-weighting, float-weighting, and fundamental-weighting. Weighting is a method of adjusting the individual impact of items in an index.

Investors follow different market indexes to gauge market movements. The three most popular stock indexes for tracking the performance of the U.S. market are the Dow Jones Industrial Average (DJIA), S&P 500 Index, and Nasdaq Composite Index. In the bond market, Bloomberg is a leading provider of market indexes with the Bloomberg U.S. Aggregate Bond Index serving as one of the most popular proxies for U.S. bonds. Investors cannot invest directly in an index, so these portfolios are used broadly as benchmarks or for developing index funds.

Some of the market's leading indexes include:

1. S&P 500

2. Dow Jones Industrial Average
3. Nasdaq Composite
4. S&P 100
5. Russell 1000
6. S&P MidCap 400
7. Russell Midcap
8. Russell 2000
9. S&P 600
10. U.S. Aggregate Bond Market
11. Global Aggregate Bond Market

Investors often choose to use index investing over individual stock holdings in a diversified portfolio. Investing in a portfolio of index funds can be a good way to optimize returns while balancing risk. For example, investors seeking to build a balanced portfolio of U.S. stocks and bonds could choose to invest 50% of their funds in an S&P 500 ETF and 50% in a U.S. Aggregate Bond Index ETF.

Investors may also choose to use market index funds to invest in emerging growth sectors. Some popular emerging growth indexes and corresponding exchange-traded funds (ETFs) include the following:

- The iShares Global Clean Energy ETF (ICLN), which tracks the S&P Global Clean Energy Index
- The Reality Shares Nasdaq NexGen Economy ETF (BLCN), which tracks the Reality Shares Nasdaq Blockchain Economy Index
- The First Trust Nasdaq Artificial Intelligence and Robotics ETF (ROBT), which tracks the Nasdaq CTA Artificial Intelligence and Robotics Index

What Are the Major Stock Indexes?

In the United States, the three leading stock indexes are the Dow Jones Industrial Average, the S&P 500, and the Nasdaq Composite. For international markets, the Financial Times Stock Exchange 100 Index and the Nikkei 225 Index are popular proxies for the British and Japanese stock markets, respectively.

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Why Are Indexes Useful to Investors?

Indexes provide investors with a simplified snapshot of a large market sector, without having to examine every single asset in that index. For example, it would be impractical for an ordinary investor to study hundreds of different stock prices in order to understand the changing fortunes of different technology companies. However, a sector-wide index like the NASDAQ-100 Technology Sector Index can show the average trend for the sector.

Indexing

Indexing, broadly, refers to the use of some benchmark indicator or measure as a reference or yardstick. In finance and economics, indexing is used as a statistical measure for tracking economic data such as inflation, unemployment, gross domestic product (GDP) growth, productivity, and market returns.

- Indexing is an investment strategy that seeks to match the investment returns of a specified stock market index.
- In the investment market, indexes exist to represent specific market segments.
- Costs involved must be considered in addition to Average Return Per Anum
- Costs to be considered
- Fund's expense ratio (e.g. advisory fee, distribution charges, operating expenses)
- Portfolio and transaction costs (brokerage and other trading costs)

Passive Investing

1. Indexing is Passive investing technique,
2. Passive investing broadly refers to a buy-and-hold portfolio strategy for long-term investment horizons, with minimal trading in the market.

Active Investing

- Active investing refers to activities entered into by investors or fund managers seeking to rearrange a portfolio of securities. Active investors constantly seek alpha, which is the difference between a return on any actively managed portfolio compared to an index, benchmark, or similar passive investing strategy. It involves on-going buying and selling.

TOPIC 012: INFLATION AND REAL INTEREST RATES

- Nominal interest rate on a bond is a promised amount of money you receive per unit you lend.

- Real rate of return is defined as the nominal interest rate you earn corrected for the change in the purchasing power of money.
- Consumer Price Index (CPI)
- CPI measures average change in prices over time that consumers pay for a basket of goods and services
- In Pakistan, prices of 487 items are accounted for on monthly basis for the calculation of CPI.
- Real Rate= $\frac{\text{Nominal Interest Rate} - \text{Rate of Inflation}}{1+\text{Rate of Inflation}}$
- Example: Suppose nominal rate of return is 8% and inflation is 5%. What will be the real rate of return?
- Real Rate= $\frac{0.08-0.05}{1+0.05} = \frac{0.03}{1.05}$ =0.02857=2.857%

Interest rates represent the cost of borrowing and the return on savings and investing. They're expressed as a percentage of the total amount of a loan or investment. They can be the total return lenders receive when they offer loans or the return people earn when they save and invest. Interest rates can be expressed in nominal or real terms. A nominal interest rate equals the real interest rate plus a projected rate of inflation. A real interest rate reflects the true cost of funds to the borrower and the real yield to the lender or to an investor.

The nominal interest rate is the rate that is advertised by banks, debt issuers, and investment firms for loans and various investments. It is the stated interest rate paid or earned to the lender or by investor. So, if as a borrower, you get a loan of $100 at a rate of 6%, you can expect to pay $6 in interest. The rate has been marked up to take account of inflation.

Nominal Interest Rate = Real Interest Rate + Projected Rate of Inflation

Short-term nominal interest rates are set by central banks. These rates are the basis for other interest rates that are charged by banks and other institutions on, e.g., loans to consumers and credit card balances. Central banks may decide to keep nominal rates at low levels in order to spur economic activity.

Low nominal rates encourage consumers to take on more debt and increase their spending. This was the case following the Great Recession when the U.S. Federal Reserve dropped the federal funds rate to a range of 0% to 0.25%. The rate remained in this range between December 2008 and December 2015.

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EXCERCISE

- Suppose the risk-free nominal interest rate on a one-year Treasury bill is 6% per yearand the expected rate of inflation is 3% per year. What is the expected real rate of return on the T-bill? Why is the T-bill risky in real terms? (=0.03/1.03=0.02913)

TOPIC 013: INTEREST RATE EQUALIZATION

- Competition in financial markets ensures that interest rates on equivalent assets are the same.
- If there are entities that have the ability to borrow and lend on the same terms (maturity and default risk) at different interest rates, then there will be interest-rate arbitrage.

Competition in financial markets ensures that *interest rates* on equivalent assets are the same. Suppose, for instance, that the interest rate the U.S. Treasury currently pays on its one-year T-bills is 4% per year. What interest rate would you expect a major institution such as the World Bank to pay on its one-year dollar-denominated debt securities (assuming they are virtually free of default risk)?

Your answer should be approximately 4% per year.

To see why, suppose that the World Bank offered significantly *less* than 4% per year. Well-informed investors would not buy the bonds issued by the World Bank; instead they would invest in one-year T-bills. Thus, if the World Bank expects to sell its bonds, it must offer at least as high a rate as the U.S. Treasury.

Would the World Bank offer significantly *more* than 4% per year? Assuming that it wants to minimize its borrowing costs, it would offer no more than is necessary to attract investors. Thus, interest rates on any default-free borrowing and lending denominated in dollars with a maturity of one year will tend to be around the same as the 4% per year interest rate on one-year U.S. T-bills.

If there are entities that have the ability to borrow and lend on the *same* terms (e.g., maturity, default risk) at *different* interest rates, then they can carry out **interest-rate arbitrage:** borrowing at the lower rate and lending at the higher rate. Their attempts to expand their activity will bring about an equalization of interest rates.

TOPIC 014: THE FUNDAMENTAL DETERMINANTS OF RATES OF RETURN

There are four main factors that determine rates of return in a market economy:

- *productivity of capital goods*—expected rates of return on mines, dams, roads, bridges, factories, machinery, and inventories;
- *degree of uncertainty regarding the productivity of capital goods;*
- *time preferences of people*—the preference of people for consumption now versus consumption in the future; and
- *risk aversion*—the amount people are willing to give up in order to reduce their exposure to risk.

Productivity of capital goods

- Capital goods are goods produced in the economy that can be used in the production of other goods.
- e.g. mines, roads, canals, dams, factories, machinery, inventories, patents, formulas, brand-name recognition.
- Capital's productivity is represented by rate of return on capital.
- The return on capital is the source of dividends and interest paid to the holders of the stocks or other financial instruments holders.
- The expected return on capital varies over time and space.
- Space: technology, availability of other factors of production.
- The higher the expected rate of return on capital, the higher the level of interest rates in the economy.

Uncertainty about the productivity

- The uncertainties of weather affect agricultural output;
- Dry mines and wells e.g. KEKRA 1 in Pakistan costed Rs. 14 billion
- Machine breakdown
- Change in consumer preference/taste
- Development of substitute
- The greater the degree of uncertainty about the productivity of capital goods, the higher the risk premium on equity securities.

Time preferences of people

- The preferences of people for consumption now and consumption in the future.
- The greater the preference of people for current consumption over future consumption, the higher the interest rate will be.

Risk Aversion

- The greater the degree of risk aversion of the population, the higher the risk premium required, and the lower will be the risk-free rate of interest.

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Lesson 04

FINANCIAL INTERMEDIARIES

TOPIC 015: FINANCIAL INTERMEDIARIES

A financial intermediary is an entity that acts as the middleman between two parties in a financial transaction, such as a commercial bank, investment bank, mutual fund, or pension fund. Financial intermediaries offer a number of benefits to the average consumer, including safety, liquidity, and economies of scale involved in banking and asset management. Although in certain areas, such as investing, advances in technology threaten to eliminate the financial intermediary, disintermediation is much less of a threat in other areas of finance, including banking and insurance.

- Banks
- Other Depository Saving Institutions
- Insurance Companies
- Pension and Retirement Funds
- Mutual Funds
- Investment Banks
- Venture Capital Firms
- Asset Management Firms
- Information Services

TOPIC 016: FINANCIAL INTERMEDIARIES IN PAKISTAN

Financial sector of Pakistan constitute banks, Development Finance Institutions (DFIs), Microfinance Banks (MFBs), Non-banking Finance Companies (NBFCs), insurance companies, Modarabas and other financial intermediaries. Latest asset structure of the financial sector is given below.

Institution	Regulatory Body	Licensed under
Banks	SBP	Section 27 of BCO, 1962
of which: Islamic Banking Institutions	SBP	Section 27 of BCO, 1962
Microfinance Banks	SBP	Microfinance Institutions Ordinance 2001

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Exchange Companies	SBP	Foreign Exchange Regulation Act, 1947
DFIs	SBP	
Nonbank financial institutions:	SECP	
Investment companies	SECP	Companies Ordinance, 1984
Asset Management Companies	SECP	Companies Ordinance, 1984
Mutual Funds and Plans	SECP	Companies Ordinance, 1984
Pension Funds	SECP	Companies Ordinance, 1984
Discretionary & Non-Discretionary Portfolios	SECP	Companies Ordinance, 1984
Real Estate Investment Trust	SECP	Companies Ordinance, 1984
Leasing Companies	SECP	Companies Ordinance, 1984
Modarabas	SECP	Companies Ordinance, 1984
Insurance companies	SECP	Companies Ordinance, 1984

Financial sector of Pakistan predominantly comprises of banks, as they hold the largest share of financial assets as a percentage of GDP.

State Bank of Pakistan (SBP) regulates Banks, DFIs, Exchange Companies and MFBs, while Securities and Exchange Commission of Pakistan (SECP) regulate NBFCs, Insurance Companies and Modaraba Companies.

Discount House

- A discount house is a firm that buys, sells, discounts, and negotiates bills of exchange or promissory notes.
- This is generally performed on a large scale with transactions that also include government bonds and Treasury bills.
- NBP Capital Discount Ltd.

Venture Capital Company

- Government, semi-government, or private firm that provides startup or growth equity capital and/or loan capital to promising ventures for returns that are higher than market interest rates.
- AKD Venture Limited

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- AMZ Ventures Limited
- Pakistan Private Equity Management Limited
- TRG Pakistan Limited

Reinsurance Company (01)

- **Reinsurance** is an insurance that an insurance **company** purchases from another insurance **company** to insulate itself (at least in part) from the risk of a major claims event. Insurance **companies** that accept **reinsurance** refer to the business as 'assumed **reinsurance**'.
- Pakistan Reinsurance Company Limited is a public sector company under the administrative control of the Ministry of Commerce.

TOPIC 017: YIELD CURVE

A yield curve is a line that plots yields (interest rates) of bonds having equal credit quality but differing maturity dates. **The slope of the yield curve** gives an idea of future interest rate changes and economic activity. There are three main shapes of yield curve shapes: normal (upward sloping curve), inverted (downward sloping curve), and flat.

A yield curve is used as a benchmark for other debt in the market, such as mortgage rates or bank lending rates, and it is used to predict changes in economic output and growth. The most frequently reported yield curve compares the three-month, two-year, five-year, 10-year, and 30-year U.S. Treasury debt. Yield curve rates are usually available at the Treasury's interest rate websites by 6:00 p.m. ET each trading day.

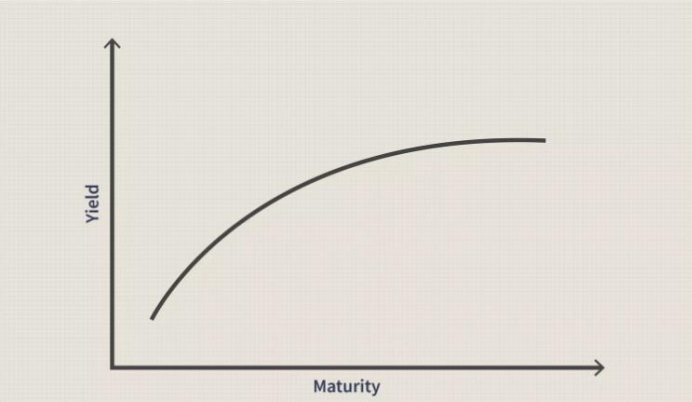
A normal yield curve is one in which longer maturity bonds have a higher yield compared to shorter-term bonds due to the risks associated with time. An inverted yield curve is one in which the shorter-term yields are higher than the longer-term yields, which can be a sign of an upcoming recession. In a flat or humped yield curve, the shorter- and longer-term yields are very close to each other, which is also a predictor of an economic transition.

Types of Yield Curves

A normal or up-sloped yield curve indicates yields on longer-term bonds may continue to rise, responding to periods of economic expansion. A normal yield curve thus starts with low yields for shorter-maturity bonds and then increases for bonds with longer maturity, sloping upwards.

This is the most common type of yield curve as longer-maturity bonds usually have a higher yield to maturity than shorter-term bonds.

For example, assume a two-year bond offers a yield of 1%, a five-year bond offers a yield of 1.8%, a 10-year bond offers a yield of 2.5%, a 15-year bond offers a yield of 3.0%, and a 20-year bond offers a yield of 3.5%. When these points are connected on a graph, they exhibit a shape of a normal yield curve.



A normal yield curve implies stable economic conditions and should prevail throughout a normal economic cycle. A steep yield curve implies strong economic growth in the future—conditions that are often accompanied by higher inflation, which can result in higher interest rates.

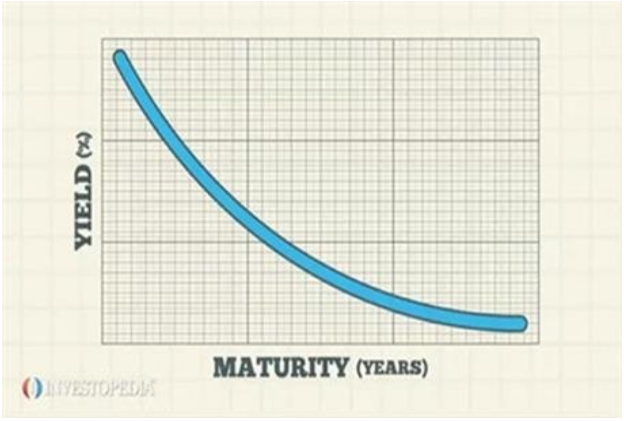
Inverted Yield Curve

An inverted yield curve instead slopes downward and means that short-term interest rates exceed long-term rates. Such a yield curve corresponds to periods of economic recession, where investors expect yields on longer-maturity bonds to become even lower in the future. Moreover, in an economic downturn, investors seeking safe investments tend to purchase these longer-dated bonds over short-dated bonds, bidding up the price of longer bonds driving down their yield.

- An inverted or down-sloped yield curve suggests yields on longer-term bonds may continue to fall, corresponding to periods of economic recession. When investors expect longer-maturity bond yields to become even lower in the future, many would purchase longer-maturity bonds to lock in yields before they decrease further.

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- The increasing onset of demand for longer-maturity bonds and the lack of demand for shorter-term securities lead to higher prices but lower yields on longer-maturity bonds, and lower prices but higher yields on shorter-term securities, further inverting a down-sloped yield curve.



Flat Yield Curve

A flat yield curve is defined by similar yields across all maturities. A few intermediate maturities may have slightly higher yields, which causes a slight hump to appear along the flat curve. These humps are usually for the mid-term maturities, six months to two years.

As the word flat suggests, there is little difference in yield to maturity among shorter and longer-term bonds. A two-year bond could offer a yield of 6%, a five-year bond 6.1%, a 10-year bond 6%, and a 20-year bond 6.05%.

Such a flat or humped yield curve implies an uncertain economic situation. It may come at the end of a high economic growth period that is leading to inflation and fears of a slowdown. It might appear at times when the central bank is expected to increase interest rates. In times of high uncertainty, investors demand similar yields across all maturities.

- A flat yield curve may arise from the normal or inverted yield curve, depending on changing economic conditions. When the economy is transitioning from expansion to slower development and even recession, yields on longer-maturity bonds tend to fall and yields on shorter-term securities likely rise, inverting a normal yield curve into a flat yield curve.
- When the economy is transitioning from recession to recovery and potential expansion, yields on longer-maturity bonds are set to rise and yields on shorter-maturity securities are sure to fall, tilting an inverted yield curve toward a flat yield curve.

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TOPIC 018: ANOMALY IN THE BOND MARKET AND THE RECESSION

Asset prices incorporate people's expectations about the future. That's why, when the bond market displays an unusual pattern called an inverted yield curve, market observers begin to worry about what future event this may foreshadow. Most of the time the yield on long-term debt securities (like 10-year Treasury bonds) is higher than the yield on short-term securities (like 3-month Treasuries). An inverted yield curve refers to a situation in which long-term bonds have a lower yield than their short-term cousins. All of the economic recessions of the past half-century have been preceded by an inverted yield curve. Only once in that time period has an inverted yield curve not been followed by a recession. In early 2006, with 10-year Treasuries yielding only 0.4 percentage points more than bonds maturing in three months, market analysts speculated that the yield curve could invert, with a recession likely to come next.

But what is the linkage between an inverted yield curve and an economic recession? The answer is expectations. Long-term rates are, to a large extent, just a combination of current and future short-term rates. Imagine that you needed a loan today that you would not be able to pay back for ten years. You could issue a single ten-year bond with an interest rate of 7.5%. At the end of ten years, you pay back the principal and interest on the loan. Alternatively, you could issue a 5-year bond with an interest rate of 5%, and in five years you could issue another 5-year bond at a 10% interest rate. In this example, you are indifferent between the two options; your average interest rate is 7.5% either way. But what if you expected rates on 5-year bonds to be 15% five years from now? Then you (and any other borrower) should definitely issue a 10-year bond. Of course, if everyone is issuing 10-year bonds and avoiding series of 5-year bonds, the yield on 10-year bonds will rise, and the yield on 5-year bonds will fall, until all borrowers are indifferent between the two options. Looked at a different way, if long-term rates are not in line with reasonable expectations of future short-term rates, there is an opportunity for arbitrage. Money could be made by selling 10-year bonds and using the proceeds to invest in a series of 5-year bonds. This buying and selling will inevitably result in interest rates that are the same on average, regardless of the maturities of the bonds used.

The final piece of the puzzle? The Federal Reserve. In the United States, the Fed effectively decides short-term rates. An inverted yield curve therefore indicates that the market believes the Fed will lower rates in the future. If the Fed lowers rates, there's a good chance the U.S. economy is not growing very quickly. For the Fed to lower rates significantly (enough to overcome the usually upward-sloping yield curve), the United States is likely to be experiencing a recession. Ultimately it is people's expectations about the future that determine the shape of the yield curve.

Source: "The Long and the Short of It," *The Economist*, January 5, 2006.

- Asset prices depend on the people's expectation about future.
- Sometimes, there is an anomaly when long-term bonds a a lower yield than their short-term cousins.
- All (except one) economic recessions of the past half century have been preceded by inverted yield curve.
- In early 2006, 10-year treasury bonds yield was only 0.4% more than bonds maturing in 3 months.

Reasons

- Expectations
- An inverted yield curve indicates that the market believes the Fed will lower rates in the future. If the Fed lowers interest rates, there is a chance that the economy is not growing very quickly. This may lead to recession.

Yield Comparisons	
	Yields
Treasury 5-year	4.94
10-year	5.05
5-year AAA Bank &Finance	5.58
10-year AAA Bank &Finance	5.91
Source: Bloomberg.com. May 26, 2006.	

TOPIC 019: GERMANY'S INFLATION LINKED BONDS

The S&P Germany Sovereign Inflation-Linked Bond Index is a comprehensive, market value weighted index designed to track the performance of EUR-denominated inflation-linked securities publicly issued by the German government for the domestic market.

- Inflation can be a bondholder's worst enemy unless the return is inflation-protected.
- In the 1920s, the Reichsbank – Germany's central bank started printing too much money.
- This led to inflation.
- In 1922, it was 1,024%
- In 1923, it was 105,700,000,000%
- In 1923, one US dollar was worth about 4.2 billion Reichsmarks
- German bonds issued during WWI became nearly worthless.
- In early 2006, Germany reintroduced inflation-linked bonds.
- As long as inflation remains low, this is one of the cheapest ways for a government to borrow.

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Lesson 05

FINANCIAL REGULATIONS

TOPIC 020: FINANCIAL INFRASTRUCTURE AND REGULATION

Financial infrastructure is a set of institutions, which provides an enabling environment for the effective operation of financial intermediaries. In broad terms, the financial infrastructure encompasses the existing legal and regulatory framework that plays a vital role in determining the structure, growth and the health of financial sector. A safe and efficient financial infrastructure fosters financial stability and is imperative for the successful operation of modern integrated financial markets. On the other hand, a weak financial infrastructure can result in major disruptions to the smooth operation of financial markets, directly exposing market participants to greater financial risk.

- Laws are made to outlaw frauds and to reinforce contracts.
- The laws governing financial system vary from country to country.
- The financial infrastructure consists of the legal and accounting procedures, the organization of trading and clearing facilities, and the regulatory structures that govern the relations among the users of the financial system.
- The financial structure consists of legal and accounting procedures.
- Some regulatory tasks are performed by private sector organizations and some are performed by governmental organizations.

Rules for Trading

- These rules are usually established by organized exchanges and then sometimes given the sanction of law.
- Purpose is standardization.
- It helps in keeping minimum costs.

Accounting Systems

- The discipline that studies reporting of financial information is called accounting.
- Ancient Babylon (around 2000 B.C.)
- Double entry bookkeeping (Italy)

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TOPIC 021: REGULATION BY GOVERNMENT

Government regulation affects the financial services industry in many ways, but the specific impact depends on the nature of the regulation. Increased regulation typically means a higher workload for people in financial services, because it takes time and effort to adapt business practices that follow the new regulations correctly.

While the increased time and workload resulting from government regulation can be detrimental to individual financial or credit services companies in the short term, government regulations can also benefit the financial services industry as a whole in the long term. The Sarbanes-Oxley Act was passed by Congress in 2002 in response to multiple financial scandals involving large conglomerates such as Enron and WorldCom.

The act held senior management of companies accountable for the accuracy of their financial statements, while also requiring that internal controls be established at these companies to prevent future fraud and abuse. Implementing these regulations was expensive, but the act gave more protection to people investing in financial services, which can increase investor confidence and improve overall corporate investment.

- Governmental and Quasi-Governmental Organizations
  - Central Banks
  - They promote public policy objectives by influencing key policy variables such as money supply, bank rate
  - Price stability

Special-Purpose Intermediaries

- This group of organizations includes entities that are set up to encourage specific economic activities by making financing more readily available or by guaranteeing debt instruments of various sorts.
- e.g.
- First Securitization Trust

TOPIC 022: REGIONAL AND WORLD ORGANIZATIONS

- Bank for International Settlements (BIS) in Basel, Switzerland, whose objective is to promote uniformity of banking regulations.
- International Monetary Fund (IMF) and
- International Bank for Reconstruction and Development (World Bank)

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International Monetary Fund (IMF)

- The IMF monitors economic and financial conditions in member countries
- provides technical assistance
- establishes rules for international trade and finance
- provides technical assistance
- establishes rules for international trade and finance
- provides a forum for international consultation, and
- most important, provides resources that permit lengthening the time necessary for individual members to correct imbalances in their payments to other countries.

World Bank

- It finances investment projects in developing countries.
- It raises funds primarily by selling bonds in developed countries and then make loans for projects that must meet certain criteria designed to encourage economic development.

Several international bodies currently exist for the purpose of coordinating the financial policies of national governments. Perhaps the most important is the Bank for International Settlements (BIS) in Basel, Switzerland, whose objective is to promote uniformity of banking regulations.

In addition, two official international agencies operate in the international financial markets to promote growth in trade and finance: the International Monetary Fund (IMF) and the International Bank for Reconstruction and Development (World Bank). The IMF monitors economic and financial conditions in member countries, provides technical assistance, establishes rules for international trade and finance, provides a forum for international consultation, and most important, provides resources that permit lengthening the time necessary for individual members to correct imbalances in their payments to other countries.

The World Bank finances investment projects in developing countries. It raises funds primarily by selling bonds in developed countries and then makes loans for projects that must meet certain criteria designed to encourage economic development.

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Lesson 06

FINANCIAL STATEMENTS

TOPIC 023: FUNCTIONS OF FINANCIAL STATEMENTS

Financial statements are written records that convey the business activities and the financial performance of a company. Financial statements are often audited by government agencies, accountants, firms, etc. to ensure accuracy and for tax, financing, or investing purposes. For-profit primary financial statements include the balance sheet, income statement, statement of cash flow, and statement of changes in equity. Non-profit entities use a similar but different set of financial statements.

Investors and financial analysts rely on financial data to analyze the performance of a company and make predictions about the future direction of the company's stock price. One of the most important resources of reliable and audited financial data is the annual report, which contains the firm's financial statements.

The financial statements are used by investors, market analysts, and creditors to evaluate a company's financial health and earnings potential. The three major financial statement reports are the balance sheet, income statement, and statement of cash flows.

- They provide information to the owners and creditors of the firm about the company's current status and past financial performance.
- Financial statements provides a convenient way for owners and creditors to set performance targets and to impose restrictions on the managers of the firm.
- Financial statements provide convenient templates for financial planning.

TOPIC 024: FINANCIAL STATEMENTS- BALANCE SHEET

- A firm's balance sheet shows its assets (what it owns) and its liabilities (what it owes) at a point in time. The difference between assets and liabilities is the firm's net worth (owner's equity).
- For a corporation, net worth is called stockholder's equity.
- The values of assets, liabilities and net worth carried on a company's published balance sheet are measured at historical acquisition costs in accordance with Generally Accepted Accounting Principles (GAAP).

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Financial Statements- Balance Sheet – Sample

Assets	20x0	20x1	Change
Current assets			
Cash and marketable securities	100.0	120.0	20.0
Receivables	50.0	60.0	10.0
Inventories	150.0	180.0	30.0
Total current assets	300.0	360.0	60.0
Property, plant, and equipment (PP&E)	400.0	490.0	90.0
less Accumulated depreciation	100.0	130.0	30.0
Net PP&E	300.0	360.0	60.0
Total assets	600.0	720.0	120.0
Liabilities and Stockholders' Equity			
Current liabilities			
Accounts payable	60.0	72.0	12.0
Short-term debt	90.0	184.6	94.6
Total current liabilities	150.0	256.6	106.6
Long-term debt (8% interest bonds maturing in 20x7)	150.0	150.0	0.0
Stockholders' equity (1 million shares outstanding)	300.0	313.4	13.4
Paid-in capital	200.0	200.0	0.0
Retained earnings	100.0	113.4	13.4
Total liabilities and equity	600.0	720.0	120.0
Other data: Market price per common share	$200.0	$187.2	–$12.8

All figures, except in the final row, are in millions of U.S. dollars.

The balance sheet provides an overview of a company's assets, liabilities, and shareholders' equity as a snapshot in time. The date at the top of the balance sheet tells you when the snapshot was taken, which is generally the end of the reporting period. Below is a breakdown of the items in a balance sheet.

Assets

Cash and cash equivalents are liquid assets, which may include Treasury bills and certificates of deposit.

Accounts receivables are the amount of money owed to the company by its customers for the sale of its product and service.

Inventory is the goods a company on hand it intends to sell as a course of business. Inventory may include finished goods, work in progress that are not yet finished, or raw materials on hand that have yet to be worked.

Prepaid expenses are costs that have been paid in advance of when they are due. These expenses are recorded as an asset because the value of them has not yet been recognized; should the benefit not be recognized; the company would theoretically be due a refund.

Property, plant, and equipment are capital assets owned by a company for its long-term benefit. This includes buildings used for manufacturing for heavy machinery used for processing raw materials.

Investments are assets held for speculative future growth. These aren't used in operations; they are simply held for capital appreciation.

Trademarks, patents, goodwill, and other intangible assets cannot physically be touched but have future economic (and often long-term benefits) for the company**.**

Liabilities

Accounts payable are the bills due as part of the normal course of operations of a business. This includes the utility bills, rent invoices, and obligations to buy raw materials.

Wages payable are payments due to staff for time worked.

Notes payable are recorded debt instruments that record official debt agreements including the payment schedule and amount.

Dividends payable are dividends that have been declared to be awarded to shareholders but have not yet been paid.

Long-term debt can include a variety of obligations including sinking bond funds, mortgages, or other loans that are due in their entirety in longer than one year. Note that the short-term portion of this debt is recorded as a current liability.

Shareholders' Equity

Shareholders' equity is a company's total assets minus its total liabilities. Shareholders' equity (also known as stockholders' equity) represents the amount of money that would be returned to shareholders if all of the assets were liquidated and all of the company's debt was paid off.

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Retained earnings are part of shareholders' equity and are the amount of net earnings that were not paid to shareholders as dividends.

QUESTION

- What difference would it have made to the end-of-year balance sheet if Company XYZ had issued an additional $50 million in long term debt during the year and added that amount to its holdings in cash and marketable securities?

TOPIC 025: FINANCIAL STATEMENTS - INCOME STATEMENT

- The income statement summarizes the profitability of the firm over a period of time, in this case a year. The income statement is also known as the statement of earnings or the statement of profit and loss.

Unlike the balance sheet, the income statement covers a range of time, which is a year for annual financial statements and a quarter for quarterly financial statements. The income statement provides an overview of revenues, expenses, net income, and earnings per share.

Income Statement- Key Points

- Revenue
- Expenses
- Gains
- Losses
- It does not cover receipts

(money received by the business)
or the cash payments/disbursements
(money paid by the business).

- It starts with sales => net income => and the earnings per share (EPS).
- The process how the net revenue realized by the company gets transformed into net earnings (profit or loss).

Revenue

Operating revenue is the revenue earned by selling a company's products or services. The operating revenue for an auto manufacturer would be realized through the production and sale of autos. Operating revenue is generated from the core business activities of a company.

Non-operating revenue is the income earned from non-core business activities. These revenues fall outside the primary function of the business. Some non-operating revenue examples include:

Interest earned on cash in the bank

Operating Revenue

- Revenue realized through primary activities

Non-Operating Revenue

Revenues realized through secondary, non-core business activities

- The net money made from other activities, e.g. sale of long-term assets.
- These include the net income realized from one-time non-business activities, like a company selling its old transportation van, unused land, or a subsidiary company.

Rental income from a property

Income from strategic partnerships like royalty payment receipts

Income from an advertisement display located on the company's property

Other income is the revenue earned from other activities. Other income could include gains from the sale of long-term assets such as land, vehicles, or a subsidiary.

Expenses

Primary expenses are incurred during the process of earning revenue from the primary activity of the business. Expenses include the cost of goods sold (COGS), selling, general and administrative expenses (SG&A), depreciation or amortization, and research and development (R&D).

Typical expenses include employee wages, sales commissions, and utilities such as electricity and transportation.

Expenses that are linked to secondary activities include interest paid on loans or debt. Losses from the sale of an asset are also recorded as expenses.

Primary Activity Expenses

All expenses incurred for earning the normal operating revenue linked to the primary activity of the business.

1. The cost of goods sold (COGS)
2. General, selling and administrative expenses (GS&A)
3. Depreciation and

4. R&D expenses
- Taxes
- Dividends

Secondary Activity Expenses

- All expenses linked to non-core business activities, like interest paid on loan money.

Losses as Expenses

- All expenses that go towards a loss-making sale of long-term assets, one-time or any other unusual costs, or expenses towards lawsuits.

Gains/Other Income

- The net money made from other activities, e.g. sale of long-term assets.
- These include the net income realized from one-time non-business activities, like a company selling its old transportation van, unused land, or a subsidiary company.

TOPIC 026: STRUCTURE OF INCOME STATEMENT

The main purpose of the income statement is to convey details of profitability and the financial results of business activities; however, it can be very effective in showing whether sales or revenue is increasing when compared over multiple periods.

Investors can also see how well a company's management is controlling expenses to determine whether a company's efforts in reducing the cost of sales might boost profits over time.

- Net Income = (Revenue + Gains) – (Expenses + Losses)
- Example of Company XYZ's Income Statement

Expenses

1. Cost of Goods Sold $110 million

(production+materials+labor)

Gross Margin =

2. General, administrative, and selling (GS&Aa) expense.

Operating income = gross margin-GS&A

GS&A expenses in 2001 – 30 million

Operating income = 60 million

Income Statement - Sample

Sales revenue	$200.0
Cost of goods sold	(110.0)
Gross margin	90.0
General, selling, and administrative expenses	(30.0)
Operating income	60.0
Interest expense	(21.0)
Taxable income	39.0
Income tax	(15.6)
Net income	23.4
Allocation of net income:	
Dividends	10.0
Change in retained earnings	13.4
Earnings per share (1 million shares outstanding)	$23.4

All figures, except those in the final row, are in millions of U.S. dollars.

TOPIC 027: CASH FLOW STATEMENT

- It provides aggregate data regarding all cash inflows a company receives from its ongoing operations and external investment sources. It also includes all cash outflows that pay for business activities and investments during a given period.

Two main aspects

- It focuses attention on what is happening to the firm's cash position over time.
- Financial distress is a condition in which a company or individual cannot generate revenue or income because it is unable to meet or cannot pay its financial obligations. It is not influenced by accrual accounting decisions

The Cash Flow Statement (CFS) measures how well a company generates cash to pay its debt obligations, fund its operating expenses, and fund investments. The cash flow statement complements the balance sheet and income statement.

- The income statement is based on accrual accounting methods hence not every revenue is an inflow of cash and not every expense is an outflow.

- Net income depends on judgments e.g.
  - how to value inventory
  - how to depreciate tangible assets
  - how to amortize its intangible assets

Major Components

1. Cash flow from Operating Activities
2. Cash Flow from Investment Activities
3. Cash Flow from Financing Activities

- The cash flow statement includes cash made by the business through operations, investment, and financing—the sum of which is called net cash flow.
- The first section of the cash flow statement is cash flow from operations, which includes transactions from all operational business activities.
- Cash flow from investment is the second section of the cash flow statement, and is the result of investment gains and losses. For example, cash spent on property, plant, and equipment.
- Cash flow from financing is the final section, which provides an overview of cash used from debt and equity.

Cash Flow Statement - Example

Cash Flow from Operating Activities	
Net income	$23.4
+ Depreciation	+30.0
− Increase in accounts receivable	−10.0
− Increase in inventories	−30.0
+ Increase in accounts payable	+12.0
Total cash flow from Operations	25.4
Cash Flow from Investing Activities	
− Investment in plant and equipment	−90.0
Cash Flow from Financing Activities	
− Dividends paid	−10.0
+ Increase in short-term debt	+94.6
Total change in cash and marketable securities	20.0

TOPIC 028: NOTES TO FINANCIAL STATEMENTS

- Accrual Accounting Methods records revenues and expenses when they are incurred, regardless of when cash is exchanged.
- The term "accrual" refers to any individual entry recording revenue or expense in the absence of a cash transaction.
- An explanation of accounting methods used e.g. for depreciation straight line or accelerated depreciation charges; for inventory costing method LIFO or FIFO.
- Details regarding which specific accounting standards are used.
- Greater detail regarding certain assets or liabilities e.g. the conditions and expiration dates for long- and short-term debt, lease etc.
- information regarding the equity structure of the firm e.g. conditions attached to ownership of shares
- documentation of changes in operations e.g. acquisitions and divestitures

- off-balance sheet items
  e.g. forward contracts, swaps, options for risk management.
-

The CFS allows investors to understand how a company's operations are running, where its money is coming from, and how money is being spent. The CFS also provides insight as to whether a company is on a solid financial footing.

There is no formula, per se, for calculating a cash flow statement. Instead, it contains three sections that report cash flow for the various activities for which a company uses its cash. Those three components of the CFS are listed below.

Operating Activities

The operating activities on the CFS include any sources and uses of cash from running the business and selling its products or services. Cash from operations includes any changes made in cash accounts receivable, depreciation, inventory, and accounts payable. These transactions also include wages, income tax payments, interest payments, rent, and cash receipts from the sale of a product or service.

Investing Activities

Investing activities include any sources and uses of cash from a company's investments in the long-term future of the company. A purchase or sale of an asset, loans made to vendors or received from customers, or any payments related to a merger or acquisition is included in this category.

Also, purchases of fixed assets such as property, plant, and equipment (PPE) are included in this section. In short, changes in equipment, assets, or investments relate to cash from investing.

Financing Activities

Cash from financing activities includes the sources of cash from investors or banks, as well as the uses of cash paid to shareholders. Financing activities include debt issuance, equity issuance, stock repurchases, loans, dividends paid, and repayments of debt.

The cash flow statement reconciles the income statement with the balance sheet in three major business activities.

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Limitations of Financial Statements

Although financial statements provide a wealth of information on a company, they do have limitations.

The statements are open to interpretation, and as a result, investors often draw vastly different conclusions about a company's financial performance.

For example, some investors might want stock repurchases while other investors might prefer to see that money invested in long-term assets. A company's debt level might be fine for one investor while another might have concerns about the level of debt for the company.

When analyzing financial statements, it's important to compare multiple periods to determine if there are any trends as well as compare the company's results to its peers in the same industry.

Last, financial statements are only as reliable as the information being fed into the reports. Too often, it has been documented that fraudulent financial activity or poor control oversight have led to misstated financial statements intended to mislead users. Even when analyzing audited financial statements, there is a level of trust that users must place into the validity of the report and the figures being shown.

What Are the Main Types of Financial Statements?

The three main types of financial statements are the balance sheet, the income statement, and the cash flow statement. These three statements together show the assets and liabilities of a business, its revenues and costs, as well as its cash flows from operating, investing, and financing activities.

What Are the Main Items Shown in Financial Statements?

Depending on the corporation, the line items in a financial statement will differ; however, the most common line items are revenues, costs of goods sold, taxes, cash, marketable securities, inventory, short-term debt, long-term debt, accounts receivable, accounts payable, and cash flows from investing, operating, and financing activities.

What Are the Benefits of Financial Statements?

Financial statements show how a business operates. It provides insight into how much and how a business generates revenues, what the cost of doing business is, how efficiently it manages its cash, and what its assets and liabilities are. Financial statements provide all the detail on how well or poorly a company manages itself.

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How Do You Read Financial Statements?

Financial statements are read in several different ways. First, financial statements can be compared to prior periods to better understand changes over time. For example, comparative income statements report what a company's income was last year and what a company's income is this year. Noting the year-over-year change informs users of the financial statements of a company's health.

Financial statements are also read by comparing the results to competitors or other industry participants. By comparing financial statements to other companies, analysts can get a better sense on which companies are performing the best and which are lagging the rest of the industry.

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Lesson 07

MARKET VALUES AND BOOK VALUES

TOPIC 029: MARKET VALUES VERSUS BOOK VALUES

- The book value means the value of a business according to its books (accounts) that is reflected through its financial statement.
- Formula:

Book value of a company
= Total assets – Total liabilities

Determining the book value of a company is more difficult than finding its market value, but it can also be far more rewarding. Many famous investors, including billionaire Warren Buffett, built their fortunes in part by buying stocks with market valuations below their book valuations. The market value depends on what people are willing to pay for a company's stock. The book value is similar to a firm's net asset value, which jumps around much less than stock prices. Learning how to use the book value formula gives investors a more stable path to achieving their financial goals.

Book value is the net value of a firm's assets found on its balance sheet, and it is roughly equal to the total amount all shareholders would get if they liquidated the company. Market value is the company's worth based on the total value of its outstanding shares in the market, which is its market capitalization.

Market value tends to be greater than a company's book value since market value captures profitability, intangibles, and future growth prospects.

Book value per share is a way to measure the net asset value investors get when they buy a share.

The price-to-book (P/B) ratio is a popular way to compare book and market values, and a lower ratio may indicate a better deal.

The book value literally means the value of a business according to its books or accounts, as reflected on its financial statements. Theoretically, it is what investors would get if they sold all

the company's assets and paid all its debts and obligations. Therefore, book value is roughly equal to the amount stockholders would receive if they decided to liquidate the company.

Book Value Formula

Mathematically, book value is the difference between a company's total assets and total liabilities.
Book value of a company=Total assets−Total liabilities

Example

Suppose that XYZ Company has total assets of $100 million and total liabilities of $80 million. Then, the book valuation of the company is $20 million. If the company sold its assets and paid its liabilities, the net worth of the business would be $20 million.

Total assets cover all types of financial assets, including cash, short-term investments, and accounts receivable. Physical assets, such as inventory, property, plant, and equipment, are also part of total assets. Intangible assets, including brand names and intellectual property, can be part of total assets if they appear on financial statements. Total liabilities include items like debt obligations, accounts payable, and deferred taxes.

Book Value Per Share (BVPS)

- It is total dollar amount of shareholders' equity account of the firm's official balance sheet divided by the number of shares of common stock outstanding.
- It can be used to make a per share comparison.
- Outstanding shares refer to a company's stock currently held by all its shareholders, including share blocks held by institutional investors and restricted shares.

When we divide book value by the number of outstanding shares, we get the book value per share (BVPS). It allows us to make per-share comparisons. Outstanding shares consist of all the company's stock currently held by all its shareholders. That includes share blocks held by institutional investors and restricted shares.

Market Value

The market value represents the value of a company according to the stock market. It is the price an asset would get in the marketplace. In the context of companies, market value is equal to market capitalization. It is a dollar amount computed based on the current market price of the company's shares.

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Market Value Formula

Market value—also known as market cap—is calculated by multiplying a company's outstanding shares by its current market price.

If XYZ Company trades at $25 per share and has 1 million shares outstanding, its market value is $25 million. Financial analysts, reporters, and investors usually mean market value when they mention a company's value.

- It represents the value of a company according to the stock market.
- It is the price an asset would get in the marketplace
- In the context of companies, it represents the market capitalization
- It is the aggregate market value of a company represented as a dollar amount.
- Since it represents the “market” value of a company, it is computed based on the current market price (CMP) of its shares.

As the market price of shares changes throughout the day, the market cap of a company does so as well. On the other hand, the number of shares outstanding almost always remains the same. That number is constant unless a company pursues specific corporate actions. Therefore, market value changes nearly always occur because of per-share price changes.

Market Value Examples

Returning to the examples from before, Microsoft had 7.57 billion shares outstanding at the end of its fiscal year on June 30, 2020.1 On that day, the company's stock closed at $203.51 per share.3 The resulting market cap was about $1,540.6 billion (7.57 billion * $203.51). This market value is over 13 times the value of the company on the books.

Similarly, Walmart had 2.87 billion shares outstanding.4 Its closing price was $114.49 per share at the end of Walmart's fiscal year on January 31, 2020.5 Therefore, the firm's market value was roughly $328.59 billion (2.87 billion * $114.49). That is more than four times Walmart's book valuation of $74.67 billion that we calculated earlier.

It is quite common to see the book value and market value differ significantly. The difference is due to several factors, including the company's operating model, its sector of the market, and the company's specific attributes. The nature of a company's assets and liabilities also factor into valuations.

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TOPIC 030: LIMITATIONS OF BOOK VALUE AND MARKET VALUE

One of the major issues with book value is that companies report the figure quarterly or annually. It is only after the reporting that an investor would know how it has changed over the months.
Book valuation is an accounting concept, so it is subject to adjustments. Some of these adjustments, such as depreciation, may not be easy to understand and assess. If the company has been depreciating its assets, investors might need several years of financial statements to understand its impact. Additionally, depreciation-linked rules and accounting practices can create other issues.

For instance, a company may have to report an overly high value for some of its equipment. That could happen if it always uses straight-line depreciation as a matter of policy.
Book value does not always include the full impact of claims on assets and the costs of selling them. Book valuation might be too high if the company is a bankruptcy candidate and has liens against its assets. What is more, assets will not fetch their full values if creditors sell them in a depressed market at fire-sale prices.

The increased importance of intangibles and difficulty assigning values for them raises questions about book value. As technology advances, factors like intellectual property play larger parts in determining profitability. Ultimately, accountants must come up with a way of consistently valuing intangibles to keep book value up to date.

Market Value Limitations

While market cap represents the market perception of a company's valuation, it may not necessarily represent the real picture. It is common to see even large-cap stocks moving 3 to 5 percent up or down during a day's session. Stocks often become overbought or oversold on a short-term basis, according to technical analysis.

Long-term investors also need to be wary of the occasional manias and panics that impact market values. Market values shot high above book valuations and common sense during the 1920s and the dotcom bubble. Market values for many companies actually fell below their book valuations following the stock market crash of 1929 and during the inflation of the 1970s. Relying solely on market value may not be the best method to assess a stock’s potential.

TOPIC 031: COMPARING BOOK AND MARKET VALUE

- Most investors and traders use both values

Scenario 1: Book value greater than market value

It usually indicates that the market has momentarily lost confidence in the company. It may be due to:

1. Problems with the business,
2. Loss of important business-related lawsuits, or
3. Chances of financial anomalies.

Value investors often prefer such companies hoping that the market perception turns out to be incorrect in the future.

However, there is no guarantee that the price will rise in the future.

Scenario 2: A market value greater than book value

- The stock market is assigning a higher value to the company due to the potential of it and its assets' earnings power.
- It indicates that investors believe the company has excellent future prospects for growth, expansion, and increased profits that will eventually raise the book value of the company.

Scenario 3: Book value equals market value

The market sees no compelling reason to believe the company's assets are better or worse than what is stated on the balance sheet.

The examples given above should make it clear that book and market values are very different. Many investors and traders use both book and market values to make decisions. There are three different scenarios possible when comparing the book valuation to the market value of a company.

Book Value Greater Than Market Value

- Consistently, profitable companies typically have market values greater than book values.
- Most of the companies in the top indexes meet this criterion, e.g. Microsoft and Walmart.
- Growth investors may find such companies promising. However, it may also indicate overvalued or overbought stocks trading at a high price.

It is unusual for a company to trade at a market value that is lower than its book valuation. When that happens, it usually indicates that the market has momentarily lost confidence in the company. It may be due to business problems, loss of critical lawsuits, or other random events. In other

words, the market doesn't believe that the company is worth the value on its books. Mismanagement or economic conditions might put the firm's future profits and cash flows in question.

Value investors actively seek out companies with their market values below their book valuations. They see it as a sign of undervaluation and hope market perceptions turn out to be incorrect. In this scenario, the market is giving investors an opportunity to buy a company for less than its stated net worth. However, there is no guarantee that the price will rise in the future.

Market Value Greater Than Book Value

The market value of a company will usually exceed its book valuation. The stock market assigns a higher value to most companies because they have more earnings power than their assets. It indicates that investors believe the company has excellent future prospects for growth, expansion, and increased profits. They may also think the company's value is higher than what the current book valuation calculation shows.

Profitable companies typically have market values greater than book values. Most of the companies in the top indexes meet this standard, as seen from the examples of Microsoft and Walmart mentioned above. Growth investors may find such companies promising. However, it may also indicate overvalued or overbought stocks trading at high prices.

Book Value Equals Market Value

Sometimes, book valuation and market value are nearly equal to each other. In those cases, the market sees no reason to value a company differently from its assets.

TOPIC 032: PRICE-TO-BOOK (P/B) RATIO

The price per share divided by the book value per share.

- If a company has a P/B of 1, it means that the book value and market value are equal.
- If the market price drops and the P/B ratio becomes less than 1, it means the market value is less than the book value (undervalued).
- If the market price zooms higher and creates a P/B ratio greater than 1, it means market value now exceeds book value (overvalued).
- Since prices change every second, it is possible to track and spot stocks which move from a P/B ratio of less than one to more than one and time the trades to maximize the profits.

The price-to-book (P/B) ratio is a popular way to compare market value and book value. It is equal to the price per share divided by the book value per share.

For example, a company has a P/B of one when the book valuation and market valuation are equal. The next day, the market price drops, so the P/B ratio becomes less than one. That means the market valuation is less than the book valuation, so the market might undervalue the stock. The following day, the market price zooms higher and creates a P/B ratio greater than one. That tells us the market valuation now exceeds book valuation, indicating potential overvaluation. However, the P/B ratio is only one of several ways investors use book value.

- Both book value and market value offer meaningful insights to a company's valuation
- Comparing the two can help investors determine whether a stock is overvalued or undervalued given its assets, liabilities, and its ability to generate income.
- An investor must determine when the book value or market value should be used and when it should be discounted or disregarded in favor of other meaningful parameters when analyzing a company.

TOPIC 033: ACCOUNTING VERSUS ECONOMIC MEASURES OF INCOME

Accounting income or loss recognizes realized gains and losses, and does not recognize unrealized gains and losses. Economic income or loss recognizes all gains and losses, whether realized or unrealized.

Income/Earnings/Profit - Definition

According to John R. Hicks, income is the amount that you could spend during the period while maintaining the wealth with which you started the period.

The accounting definition ignores unrealized gains or losses in the market values of assets and liabilities.

Accounting Profit

It is the net income for a company.

It's the profit after various costs and expenses are subtracted from total revenue or total sales as stipulated by generally accepted accounting principles (GAAP).

- It is the amount of money left over after deducting the explicit costs of running the business.

- Explicit costs are merely the specific amounts that a company pays for those costs in that period – for example, wages.
- Typically, accounting profit or net income is reported on a quarterly and annual basis and is used to measure the financial performance of a company.

Accounting Profit - Costs

- Labor costs, such as wages
- Inventory needed for production
- Raw materials
- Transportation costs
- Sales and marketing costs
- Production costs and overhead

Economic Profit

- Similar to accounting profit in that it deducts explicit costs from revenue. However, economic profit also includes the opportunity costs for taking one action versus another in the period.
- Economic profit is determined by economic principles, not by accounting principles.

Economic Profit - Costs

- Economic profit also uses implicit costs, which are typically the costs of a company's resources.
- Examples of implicit costs include:
    a. Company-owned buildings
    b. Plant and equipment
    c. Self-employment resources

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Lesson 08

RETURN ON ASSETS AND RETURN ON EQUITY

TOPIC 034: FINANCIAL RATIOS

- Financial ratios are powerful tools to help summarize financial statements and the health of a company or enterprise.
- Total Shareholder Return (TSR)

Ratio analysis is a quantitative method of gaining insight into a company's liquidity, operational efficiency, and profitability by studying its financial statements such as the balance sheet and income statement. Ratio analysis is a cornerstone of fundamental equity analysis.

Total Shareholder Return (TSR)

- Total Shareholder Return (TSR) factors in capital gains and dividends when measuring the total return generated by a stock to an investor.
- TSR is the internal rate of return (IRR) of all cash flows to an investor during the holding period of an investment.
- It is the total amount returned to investors.

Total shareholder return is the financial gain that results from a change in the stock's price plus any dividends paid by the company during the measured interval divided by the initial purchase price of the stock.

TSR = { (current price - purchase price) + dividends } ÷ purchase price

Example

Assume that an investor bought 100 shares at $20 and still owns the stock. Company paid out $4.50 in dividends since the investor bought the stock and the current price is $24.

- TSR = { (current price - purchase price) + dividends } ÷ purchase price
- TSR = { ($24 - $20) + $4.50 } ÷ $20 = 0.425 * 100 = 42.5%

Key Points

- There are two basic ways that an Investor makes money in stocks - capital gains and current income (dividends).
- Total Shareholder Return factors in capital gains and dividends when measuring the total return generated by a stock to an investor.
- TSR represents an easily understood figure of the overall financial benefits generated for stockholders.

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Classification of Financial Ratios

Ratio	Formula	Calculation	
Profitability			
Return on sales (*ROS*)	$\frac{EBIT}{Sales}$	$\frac{\$60}{\$200}$	= 30%
Return on assets (*ROA*)	$\frac{EBIT}{Average\ Total\ Assets}$	$\frac{\$60}{(\$600+\$720)/2}$	= 9.1%
Return on equity (*ROE*)	$\frac{Net\ Income}{Stockholders'\ Equity}$	$\frac{\$23.4}{(\$300+\$313.4)/2}$	= 7.6%
Asset Turnover			
Receivables turnover	$\frac{Sales}{Average\ Receivables}$	$\frac{\$200}{(\$50+\$60)/2}$	= 3.6 times
Inventory turnover	$\frac{Cost\ of\ Goods\ Sold}{Average\ Inventory}$	$\frac{\$110}{(\$150+\$180)/2}$	= 0.7 times
Asset turnover	$\frac{Sales}{Average\ Total\ Assets}$	$\frac{\$200}{(\$600+\$720)/2}$	= 0.3 times
Financial Leverage			
Debt	$\frac{Total\ Debt}{Total\ Assets}$	$\frac{\$406.6}{\$720}$	= 57%
Times interest earned	$\frac{EBIT}{Interest\ Expense}$	$\frac{\$60}{\$21}$	= 2.9 times
Liquidity			
Current	$\frac{Current\ Assets}{Current\ Liabilities}$	$\frac{\$360}{\$256.6}$	= 1.4 times
Quick, or acid test	$\frac{Cash+Receivables}{Current\ Liabilities}$	$\frac{\$180}{\$256.6}$	= 0.7 times
Market Value			
Price to earnings	$\frac{Price\ per\ Share}{Earnings\ per\ Share}$	$\frac{\$187.2}{\$23.4}$	= 8.0
Market to book	$\frac{Price\ per\ Share}{Book\ Value\ per\ Share}$	$\frac{\$187.2}{\$313.4}$	= 0.6

TOPIC 035: RETURN ON EQUITY (ROE)

- It is a measure of financial performance calculated by dividing net income by shareholders' equity.
- Shareholders' equity is equal to a company's assets minus its debt, ROE could be thought of as the return on net assets.
- ROE is considered a measure of how effectively management is using a company's assets to create profits.

Formula:

$$\text{ROE} = \frac{\text{Net Income}}{\text{Average Shareholders' Equity}}$$

- Net Income is the amount of income, net of expense, and taxes that a company generates for a given period.
- Average Shareholders' Equity is calculated by adding equity at the beginning of the period. The beginning and end of the period should coincide with that which the net income is earned.
- Net income over the last full fiscal year, or trailing 12 months, is found on the income statement—a sum of financial activity over that period.
- Shareholders' equity comes from the balance sheet—a running balance of a company's entire history of changes in assets and liabilities.

Example

$$\textbf{ROE} = \frac{\textbf{Net Income}}{\textbf{Shareholders' Equity}}$$

$$\textbf{ROE} = \frac{\$23.4}{\$300} = 0.078 = 7.8\%$$

Rule of thumb

- Relatively high or low ROE ratios will vary significantly from one industry group or sector to another.
- When used to evaluate one company to another similar company the comparison will be more meaningful.
- A common shortcut for investors to consider a return on equity near the long-term average of the S&P 500 (14%) as an acceptable ratio and anything less than 10% as poor.

Return on equity (ROE) is a measure of financial performance calculated by dividing net income by shareholders' equity. Because shareholders' equity is equal to a company's assets minus its debt, ROE is considered the return on net assets.

ROE is considered a gauge of a corporation's profitability and how efficient it is in generating profits. The higher the ROE, the more efficient a company's management is at generating income and growth from its equity financing.

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ROE is expressed as a percentage and can be calculated for any company if net income and equity are both positive numbers. Net income is calculated before dividends paid to common shareholders and after dividends to preferred shareholders and interest to lenders.

Return on Equity=Average Shareholders/Equity Net Income
Net income is the amount of income, net expenses, and taxes that a company generates for a given period. Average shareholders' equity is calculated by adding equity at the beginning of the period. The beginning and end of the period should coincide with the period during which the net income is earned.

Net income over the last full fiscal year, or trailing 12 months, is found on the income statement—a sum of financial activity over that period. Shareholders' equity comes from the balance sheet—a running balance of a company's entire history of changes in assets and liabilities.
It is considered best practice to calculate ROE based on average equity over a period because of the mismatch between the income statement and the balance sheet.

TOPIC 036: PROFITABILITY RATIOS

- Profitability ratios are a class of financial metrics that are used to assess a business's ability to generate earnings relative to its revenue, operating costs, balance sheet assets, and shareholders' equity over time, using data from a specific point in time.

Return on Sales (ROS)

- Return on sales (ROS) is a ratio used to evaluate a company's operational efficiency.
- This measure provides insight into how much profit is being produced per dollar of sales.
- An increasing ROS indicates that a company is growing more efficiently, while a decreasing ROS could signal impending financial troubles.
- ROS is very closely related to a firm's operating profit margin.

Formula

$$\textbf{ROS} = \frac{\textbf{Operating Profit}}{\textbf{Net Sales}}$$

Where:

- Operating Profit is calculated as earnings before income and taxes, or EBIT.
- Net sales is the sum of a company's gross sales minus its returns, allowances, and discounts.

- Investors, creditors, and other debt holders rely on this efficiency ratio because it accurately communicates the percentage of operating cash a company makes on its revenue and provides insight into potential dividends, reinvestment potential, and the company's ability to repay debt.
- ROS is used to compare current period calculations with calculations from previous periods. This allows a company to conduct trend analyses and compare internal efficiency performance over time. It is also useful to compare one company's ROS percentage with that of a competing company, regardless of scale.

Example

- ROS is larger if a company's management successfully cuts costs while increasing revenue.
- For example, the company with $50,000 in sales and $30,000 in costs has an operating profit of $20,000 and a ROS of 40% ($20,000 / $50,000).
- If the company's management team wants to increase efficiency, it can focus on increasing sales while incrementally increasing expenses, or it can focus on decreasing expenses while maintaining or increasing revenue.

These ratios convey how well a company can generate profits from its operations. Profit margin, return on assets, return on equity, return on capital employed, and gross margin ratios are all examples of profitability ratios.

For most profitability ratios, having a higher value relative to a competitor's ratio or relative to the same ratio from a previous period indicates that the company is doing well. Profitability ratios are most useful when compared to similar companies, the company's own history, or average ratios for the company's industry.

Gross profit margin is one of the most widely used profitability or margin ratios. Gross profit is the difference between revenue and the costs of production—called cost of goods sold (COGS).

Some industries experience seasonality in their operations. For example, retailers typically experience significantly higher revenues and earnings during the year-end holiday season. Thus, it would not be useful to compare a retailer's fourth-quarter gross profit margin with its first-quarter gross profit margin because they are not directly comparable. Comparing a retailer's fourth-quarter profit margin with its fourth-quarter profit margin from the previous year would be far more informative.

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Examples of Profitability Ratios

Profitability ratios are one of the most popular metrics used in financial analysis, and they generally fall into two categories—margin ratios and return ratios.

Margin ratios give insight, from several different angles, on a company's ability to turn sales into a profit. Return ratios offer several different ways to examine how well a company generates a return for its shareholders.

Some common examples of profitability ratios are the various measures of profit margin, return on assets (ROA), and return on equity (ROE). Others include return on invested capital (ROIC) and return on capital employed (ROCE).

Profit Margin

Different profit margins are used to measure a company's profitability at various cost levels of inquiry, including gross margin, operating margin, pretax margin, and net profit margin. The margins shrink as layers of additional costs are taken into consideration—such as the COGS, operating expenses, and taxes.

Gross margin measures how much a company makes after accounting for COGS. Operating margin is the percentage of sales left after covering COGS and operating expenses. The pretax margin shows a company's profitability after further accounting for non-operating expenses. The net profit margin is a company's ability to generate earnings after all expenses and taxes.

TOPIC 037: RETURN ON ASSETS (ROA)

- Return on assets (ROA) is an indicator of how profitable a company is relative to its total assets.
- ROA gives a manager, investor, or analyst an idea as to how efficient a company's management is at using its assets to generate earnings.
- Return on assets is displayed as a percentage.
- It is calculated by dividing a company's net income by total assets.
- ROA = $\frac{\text{Net Income}}{\text{Total Asset}}$
- Higher ROA indicates more asset efficiency

Example

- Suppose Sam and Fran start shawarma stands. Sam spends $1,500 on the stand and Fran spends $150000 on his unit. Suppose Sam had earned $150 and Fran had earned $1,200. Sam's ROA =$150/1500 = 10%

- Fran's ROA = $1200/$15000 = 8%
- Fran's business is more valuable but Sam has an efficient one.

Example of How to Use Return on Assets—ROA

- ROA is most useful for comparing companies in the same industry, as different industries use assets differently.
- For example, the ROA for service-oriented firms, such as banks, will be significantly higher than the ROA for capital intensive companies, such as construction or utility companies.

Example

- Let's evaluate the return on assets (ROA) for three companies in the retail industry:
- Macy's (M)
- Kohl's (KSS)
- Dillard's (DDS)
- The data in the table is for the trailing twelve months as of Feb. 13, 2019.

Data

Company	Net Income	Total Assets	ROA
Macy's	$1.7 billion	$20.4 billion	8.3%
Kohl's	$996 million	$14.1 billion	7.1%
Dillard's	$243 million	$3.9 billion	6.2%

Explanation

- The data shows that every dollar that Macy's has invested in assets generates 8.3 cents of net income.
- Macy's is better at converting its investment into profits, compared with Kohl's and Dillard's.
- One of management's most important jobs is to make wise choices in allocating its resources.
- It appears Macy's management is more adept than its two peers.

Profitability is assessed relative to costs and expenses and analyzed in comparison to assets to see how effective a company is deploying assets to generate sales and profits. The use of the

term "return" in the ROA measure customarily refers to net profit or net income—the value of earnings from sales after all costs, expenses, and taxes. ROA is net income divided by total assets.

The more assets a company has amassed, the more sales and potential profits the company may generate. As economies of scale help lower costs and improve margins, returns may grow at a faster rate than assets, ultimately increasing ROA.

TOPIC 038: COMPARISON BETWEEN ROA AND ROE

Return on equity (ROE) and return on assets (ROA) are two key measures to determine how efficient a company is at generating profits. The main differentiator between the two is that ROA takes into account leverage/debt, while ROE does not. ROE can be calculated by multiplying ROA by the equity multiplier.

- Both ROA and return on equity (ROE) are measures of how a company utilizes its resources.
  - **ROA =** $\frac{\textbf{Net Income}}{\textbf{Total Asset}}$
  - **ROE =** $\frac{\textbf{Net Income}}{\textbf{Shareholders' Equity}}$
- Essentially, ROE only measures the return on a company's equity, leaving out the liabilities.
- Thus, ROA accounts for a company's debt and ROE does not.
- The more leverage and debt a company takes on, the higher ROE will be relative to ROA.
- The biggest issue with return on assets (ROA) is that it can't be used across industries.
- That's because companies in one industry—such as the technology industry—and another industry like oil drillers will have different asset bases.
- The St. Louis Federal Reserve provides data on US bank ROAs, which have generally hovered around or just above 1% since 1984, the year collection started.

Lesson 09

ASSET TURNOVER RATIOS

TOPIC 039: ASSET TURNOVER RATIO

- It measures the value of a company's sales or revenues relative to the value of its assets.
- This ratio can be used as an indicator of the efficiency with which a company is using its assets to generate revenue.
- The higher the asset turnover ratio, the more efficient a company.
- If a company has a low asset turnover ratio, it indicates it is not efficiently using its assets to generate sales.

The asset turnover ratio measures the efficiency of a company's assets in generating revenue or sales. It compares the dollar amount of sales (revenues) to its total assets as an annualized percentage. Thus, to calculate the asset turnover ratio, divide net sales or revenue by the average total assets.

Receivable turnover ratio

- The accounts receivable turnover ratio is an accounting measure used to quantify a company's effectiveness in collecting its receivables or money owed by clients.
- The ratio shows how well a company uses and manages the credit it extends to customers and how quickly that short-term debt is collected or is paid.
- The receivables turnover ratio is also called the accounts receivable turnover ratio.

Receivable turnover ratio – Formula

Receivable turnover ratio

$$= \frac{\text{Net Credit Sales}}{\text{Average Accounts Receivable}}$$

- Add the value of accounts receivable at the beginning of the desired period to the value at the end of the period and divide the sum by two to get the value for the denominator in the formula.
- Divide the value of net credit sales for the period by the average accounts receivable during the same period.
- Net credit sales are the revenue generated from sales that were done on credit minus any returns from customers.

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Receivable turnover ratio

$$= \frac{\text{Net Credit Sales}}{\text{Average Accounts Receivable}}$$

Receivable turnover ratio

$$= \frac{\$200}{(\$50+\$60)/2} = 3.6 \text{ times}$$

Balance sheet discussed earlier shows Accounts Receivable for 2000 and 2001 as $50 and $60 respectively.

Receivables Turnover Ratio Inferences

- Companies that maintain accounts receivables are indirectly extending interest-free loans to their clients since accounts receivable is money owed without interest.
- If a company generates a sale to a client, it could extend terms of 30 or 60 days, meaning the client has 30 to 60 days to pay for the product.
- The receivables turnover ratio measures the efficiency with which a company collects on their receivables or the credit it had extended to its customers.
- The ratio also measures how many times a company's receivables are converted to cash in a period. The receivables turnover ratio could be calculated on an annual, quarterly, or monthly basis.

TOPIC 040: INVENTORY TURNOVER RATIO

Inventory turnover is a financial ratio showing how many times a company turned over its inventory relative to its cost of goods sold (COGS) in a given period. A company can then divide the days in the period, typically a fiscal year, by the inventory turnover ratio to calculate how many days it takes to sell its inventory, on average.

The inventory turnover ratio can help businesses make better decisions on pricing, manufacturing, marketing, and purchasing. It is one of the efficiency ratios measuring how effectively a company uses its assets.

- Inventory is the account of all the goods a company has in its stock, including raw materials, work-in-progress materials, and finished goods that will ultimately be sold.
- It is a ratio showing how many times a company has sold and replaced inventory during a given period.
- A company can then divide the days in the period by the inventory turnover formula to calculate the days it takes to sell the inventory on hand.

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- Calculating inventory turnover can help businesses make better decisions on pricing, manufacturing, marketing and purchasing new inventory.

Inventory turnover ratio

$$= \frac{\textbf{Sales}}{\textbf{Average Inventory}}$$

where: Average Inventory

=(BI−Ending Inventory)÷2

BI=Beginning Inventory

Days Sales of Inventory (DSI) or Days Inventory

- Days Sales of Inventory (DSI) measures how many days it takes for inventory to turn into sales.
- It is calculated by taking the inverse of the inventory turnover ratio multiplied by 365.
- This puts the figure into a daily context, as follows:

(Average Inventory ÷ Cost of Goods Sold) x 365

Days Sales of Inventory (DSI)

- A lower DSI is ideal since it would translate to fewer days needed to turn inventory into cash. However, DSI values can vary between industries.
- For example, companies that sell groceries, like Kroger supermarkets (KR), have lower days inventory than companies that sell automobiles, like General Motors Co. (GM).

TOPIC 041: INVENTORY TURNOVER RATIO-CALCULATION

- Companies calculate inventory turnover by:
  - Calculating the average inventory, which is done by dividing the sum of beginning inventory and ending inventory by two
  - Dividing sales by average inventory

Inventory Turnover Ratio

$$= \frac{\textbf{Cost of Goods Sold}}{\textbf{Average Inventory}}$$

$$= \frac{\$\,\mathbf{110}}{(\$\mathbf{150}+\$\mathbf{180})/\mathbf{2}} = \mathbf{0.7\ times}$$

Example of Calculating Inventory Turnover

- For the fiscal year ending January 2018, Wal-Mart Stores (WMT) reported annual sales of $500.34 billion, year-end inventory of $43.78 billion, and an annual cost of goods sold (or cost of sales) of $373.40 billion.
- Walmart's **inventory turnover** for the year equals:
- $373.40 billion ÷ $43.78 billion = **8.53**

Days Sales of Inventory (DSI) or Days Inventory

- Its **days inventory** equal:
- (1 ÷ 8.53) x 365 = **42 days**
- This indicates that Walmart sells its entire inventory within a 42-day period, which is quite impressive for such a large, global retailer.
- The inventory turnover ratio is an effective measure of how well a company is turning its inventory into sales.
- The ratio also shows how well management is managing the costs associated with inventory and whether they're buying too much inventory or too little.
- It also shows how well the company sells its goods. If sales are down or the economy is under-performing, it may show up as a lower inventory turnover ratio.
- Usually, a higher inventory turnover ratio is preferred, as it indicates that more sales are being generated given a certain amount of inventory.
- Sometimes a very high inventory ratio could result in lost sales, as there is not enough inventory to meet demand. It is always important to compare the inventory turnover ratio to the industry benchmark to assess if a company is successfully managing its inventory.

Inventory Turnover Formula and Calculation

Inventory Turnover = COGS/Average Value of Inventory

Where COGS=Cost of goods sold

Average value of inventory is used to offset seasonality effects. It is calculated by adding the value of inventory at the end of a period to the value of inventory at the end of the prior period and dividing the sum by 2.

Cost of goods sold (COGS) is also known as cost of sales. Analysts use COGS instead of sales in the formula for inventory turnover because inventory is typically valued at cost, whereas the sales figure includes the company's markup. Some companies may use sales instead of COGS in the calculation, which would tend to inflate the resulting ratio.

TOPIC 042: ASSET TURNOVER RATIO - FORMULA

- Asset turnover is the ratio of total sales or revenue to average assets.
- This metric helps investors understand how effectively companies are using their assets to generate sales.
- Investors use the asset turnover ratio to compare similar companies in the same sector or group.
- It is a tool to see which firms are making the most use of their assets and to identify weaknesses in firms.

The asset turnover ratio measures the value of a company's sales or revenues relative to the value of its assets. The asset turnover ratio can be used as an indicator of the efficiency with which a company is using its assets to generate revenue.

The higher the asset turnover ratio, the more efficient a company is at generating revenue from its assets. Conversely, if a company has a low asset turnover ratio, it indicates it is not efficiently using its assets to generate sales.

Asset Turnover Ratio Formula:

$$\text{Asset Turnover} = \frac{\text{Total Sales}}{\frac{\text{Beginning Assets} + \text{Ending Assets}}{2}}$$

where:

$\text{Total Sales} = \text{Annual sales total}$

$\text{Beginning Assets} = \text{Assets at start of year}$

$\text{Ending Assets} = \text{Assets at end of year}$

- The asset turnover ratio is calculated on an annual basis.
- The total assets number used in the denominator can be calculated by taking the average of assets on the balance sheet at the beginning of the year and at the year's end.
- The higher the asset turnover ratio, the better the company is performing, since higher ratios imply that the company is generating more revenue per dollar of assets.
- The asset turnover ratio tends to be higher for companies in certain sectors than in others.
- Retail and consumer staples, for example, have relatively small asset bases but have high sales volume – thus, they have the highest average asset turnover ratio. Conversely, firms in sectors such as utilities and real estate have large asset bases and low asset turnover.

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- Since this ratio can vary widely from one industry to the next, comparing the asset turnover ratios of a retail company and a telecommunications company would not be very productive.
- Comparisons are only meaningful when they are made for different companies within the same sector.

TOPIC 043: ASSET TURNOVER RATIO – EXAMPLE

- Let's calculate the asset turnover ratio for four companies in the retail and telecommunication-utilities sectors –
  - Walmart Inc. (NYSE: WMT),
  - Target Corporation (NYSE: TGT)
  - AT&T Inc. (NYSE: T), and
  - Verizon Communications Inc. (NYSE: VZ)

for the fiscal year ended 2016.

(in millions)	Walmart	Target	AT&T	Verizon
Beginning Assets	$199,581	$40,262	$402,672	$244,175
Ending Assets	$198,825	$37,431	$403,821	$244,180
Average Total Assets	$199,203	$38,846.50	$403,246.50	$244,177.50
Revenue	$458,873	$69,495	$163,786	$125,980
Asset Turnover Ratio	**2.30**	**1.79**	**0.41**	**0.52**

- AT&T and Verizon have asset turnover ratios of less than one, which is typical for firms in the telecommunications-utilities sector.
- Since these companies have large asset bases, it is expected that they would slowly turn over their assets through sales.
- Clearly, it would not make sense to compare the asset turnover ratios for Walmart and AT&T, since they operate in very different industries.
- But comparing the asset turnover ratios for AT&T and Verizon may provide a better estimate of which company is using assets more efficiently.
- For example, from the table, Verizon turns over its assets at a faster rate than AT&T.

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- For every dollar in assets, Walmart generated $2.30 in sales, while Target generated $1.79.
- Target's turnover may indicate that the retail company was experiencing sluggish sales or holding obsolete inventory. Furthermore, its low turnover may also mean that the company has lax collection methods. The firm's collection period may be too long, leading to higher accounts receivable. Target could also not be using its assets efficiently.

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Lesson 10

FINANCIAL LEVERAGE RATIOS AND LIQUIDITY RATIOS

TOPIC 044: FINANCIAL LEVERAGE RATIOS

A leverage ratio is any one of several financial measurements that look at how much capital comes in the form of debt (loans) or assesses the ability of a company to meet its financial obligations. The leverage ratio category is important because companies rely on a mixture of equity and debt to finance their operations, and knowing the amount of debt held by a company is useful in evaluating whether it can pay off its debts as they come due.

- These ratios highlight the capital structure of the firm, and the extent to which it is burdened with debt.
- Types of financial leverage ratios
  - Debt ratio
  - Times interest earned
  - Debt to Equity Ratio

Debt Ratio

- It measures the capital structure
- It is a financial ratio that measures the extent of a company's leverage.
- It is expressed as a decimal or percentage.
- It can be interpreted as the proportion of a company's assets that are financed by debt.

Formula

- Debt ratio=$\frac{\text{Total Debt}}{\text{Total Assets}}$

A ratio greater than 1 shows that a considerable portion of debt is funded by assets OR the company has more liabilities than assets.

A high ratio also indicates that a company may be putting itself at a risk of default on its loans if interest rates were to rise suddenly.

- A ratio below 1 translates to the fact that a greater portion of a company's assets is funded by equity.
- The debt ratio is also referred to as the debt-to-assets ratio.

Too much debt can be dangerous for a company and its investors. However, if a company's operations can generate a higher rate of return than the interest rate on its loans, then the debt may help to fuel growth. Uncontrolled debt levels can lead to credit downgrades or worse. On the

other hand, too few debts can also raise questions. A reluctance or inability to borrow may be a sign that operating margins are tight.

There are several different ratios that may be categorized as a leverage ratio, but the main factors considered are debt, equity, assets, and interest expenses.

A leverage ratio may also be used to measure a company's mix of operating expenses to get an idea of how changes in output will affect operating income. Fixed and variable costs are the two types of operating costs; depending on the company and the industry, the mix will differ.

Finally, the consumer leverage ratio refers to the level of consumer debt compared to disposable income and is used in economic analysis and by policymakers.

TOPIC 045: FINANCIAL LEVERAGE RATIO-TIMES INTEREST EARNED (TIE)

- It indicates the ability of the firm to cover its interest payments.
- It is a measure of a company's ability to meet its debt obligations based on its current income.
- The formula for a company's TIE number is earnings before interest and taxes (EBIT) divided by the total interest payable on bonds and other debt.
- $\text{TIE} = \frac{\text{EBIT}}{\text{Interest Expense}}$
- The result is a number that shows how many times a company could cover its interest charges with its pretax earnings.
- It is also referred to as the interest coverage ratio.

A better TIE number means a company has enough cash after paying its debts for investment. The TIE ratio is an indication of a company's relative freedom from the constraints of debt. Generating enough cash flow to continue to invest in the business is better than having enough money to stave off bankruptcy.

Example

- Assume, that XYZ Company has $10 million in 4% debt outstanding and $10 million in common stock. The company needs to borrow more loan. The cost of capital for issuing more debt is an annual interest rate of 6%. The company's shareholders expect an annual dividend payment of 8% plus growth in the stock price of XYZ.

- The business decides to issue $10 million in additional debt. Its total annual interest expense will be: (4% X $10 million) + (6% X $10 million), or $1 million annually. The company's EBIT is $3 million.
- This means that the TIE ratio for XYZ Company is 3, or three times the annual interest expense.

The times interest earned (TIE) ratio is a measure of a company's ability to meet its debt obligations based on its current income. The formula for a company's TIE number is earnings before interest and taxes (EBIT) divided by the total interest payable on bonds and other debt.
The result is a number that shows how many times a company could cover its interest charges with its pretax earnings.
TIE is also referred to as the interest coverage ratio.

Obviously, no company needs to cover its debts several times over in order to survive. However, the TIE ratio is an indication of a company's relative freedom from the constraints of debt. Generating enough cash flow to continue to invest in the business is better than merely having enough money to stave off bankruptcy.
A company's capitalization is the amount of money it has raised by issuing stock or debt, and those choices impact its TIE ratio. Businesses consider the cost of capital for stock and debt and use that cost to make decisions.

Assume, for example, that XYZ Company has $10 million in 4% debt outstanding and $10 million in common stock. The company needs to raise more capital to purchase equipment. The cost of capital for issuing more debt is an annual interest rate of 6%. The company's shareholders expect an annual dividend payment of 8% plus growth in the stock price of XYZ.
The business decides to issue $10 million in additional debt. Its total annual interest expense will be: (4% X $10 million) + (6% X $10 million), or $1 million annually. The company's EBIT is $3 million.
This means that the TIE ratio for XYZ Company is 3, or three times the annual interest expense.

TOPIC 046: FINANCIAL LEVERAGE RATIO-DEBT TO EQUITY RATIO

- D/E Ratio

$$=\frac{\text{Total Liabilities}}{\text{Total Shareholders' Equity}}$$

- For example, TKMaxx has $15.53 billion in debt and $4.32 billion in equity (FY 2019).
- The company's D/E ratio is thus:

$$=\frac{\$15.53 \text{ billion}}{\$4.32 \text{ billion}}= 3.59$$

- TKMaxx's liabilities are 359% of shareholders' equity which is very high for a retail company.
- A high debt/equity ratio generally indicates that a company has been aggressive in financing its growth with debt.
- This can result in volatile earnings as a result of the additional interest expense.
- If the company's interest expense grows too high, it may increase the company's chances of a default or bankruptcy.
- Typically, a D/E ratio greater than 2.0 indicates a risky scenario for an investor; however, this yardstick can vary by industry.
- Businesses that require large capital expenditures (CapEx), such as utility and manufacturing companies, may need to secure more loans than other companies.
- It is a good idea to measure a firm's leverage ratios against past performance and with companies operating in the same industry to better understand the data.

TOPIC 047: LIQUIDITY RATIOS

Liquidity ratios are an important class of financial metrics used to determine a debtor's ability to pay off current debt obligations without raising external capital. Liquidity ratios measure a company's ability to pay debt obligations and its margin of safety through the calculation of metrics including the current ratio, quick ratio, and operating cash flow ratio.

- Liquidity Ratios include:
  - Current ratio
  - Quick ratio/acid test

Liquidity is the ability to convert assets into cash quickly and cheaply. Liquidity ratios are most useful when they are used in comparative form. This analysis may be internal or external.

- Internal analysis regarding liquidity ratios involves using multiple accounting periods using the same accounting methods.
- Such comparison helps to track changes in the business.
- A higher liquidity ratio indicates that the company is more liquid and has better coverage of outstanding debts.

- External analysis involves comparing the liquidity ratios of one company to another or an entire industry.
- This information is useful to compare the company's strategic positioning in relation to its competitors when establishing benchmark goals.

For example, internal analysis regarding liquidity ratios involves using multiple accounting periods that are reported using the same accounting methods. Comparing previous periods to current operations allows analysts to track changes in the business. In general, a higher liquidity ratio shows a company is more liquid and has better coverage of outstanding debts.

Limitation

- Liquidity ratio analysis may not be as effective when looking across industries as various businesses require different financing structures.
- Liquidity ratio analysis is less effective for comparing businesses of different sizes in different geographical locations.

TOPIC 048: CURRENT RATIO AND QUICK RATIO

- The current ratio measures a company's ability to pay off its current liabilities (payable within one year) with its current assets such as cash, accounts receivable and inventories. The higher the ratio, the better the company's liquidity position.
- Current Ratio $=\frac{\text{Current Assets}}{\text{Current Liabilities}}$

- The quick ratio measures a company's ability to meet its short-term obligations with its most liquid assets and therefore excludes inventories from its current assets.
- It is also known as the "acid-test ratio."

- Quick Ratio

$$=\frac{\text{C+MS+AR}}{\text{Current Liabilities}}$$

C = Cash and cash equivalents

MS = Marketable securities

AR = Accounts Receivable

CL = Current liabilities

Example

- Consider two hypothetical companies—ABC Inc. and XYZ Co.—with the following assets and liabilities on their balance sheets (figures in millions of dollars). We assume that both companies operate in the same manufacturing sector.

Balance Sheet (in millions of dollars)	ABC Inc.	XYZ Co.
Cash	$5	$1
Marketable Securities	$5	$2
Accounts Receivable	$10	$2
Inventories	$10	$5
Current Assets (a)	$30	$10
Plant and Equipment (b)	$25	$65
Intangible Assets (c)	$20	$0
Total Assets (a + b + c)	$75	$75
Current Liabilities* (d)	$10	$25
Long-Term Debt (e)	$50	$10
Total Liabilities (d + e)	$60	$35
Shareholders' Equity	$15	$40

Calculations - ABC Inc.

- Current ratio $= \frac{\text{Current Assets}}{\text{Current Liabilities}}$

  =$30 / $10 = 3.0
- Quick ratio $= \frac{\text{C+MS+AR}}{\text{Current Liabilities}}$

  = ($30 – $10) / $10 = 2.0
- Debt to equity = $50 / $15 = 3.33
- Debt to assets = $50 / $75 = 0.67

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Calculations - XYZ Co.

- Current ratio = $10 / $25 = 0.40
- Quick ratio = ($10 – $5) / $25 = 0.20
- Debt to equity = $10 / $40 = 0.25
- Debt to assets = $10 / $75 = 0.13

Conclusions

- ABC Inc. has a high degree of liquidity. Based on its current ratio, it has $3 of current assets for every dollar of current liabilities.
- Its quick ratio points to adequate liquidity even after excluding inventories, with $2 in assets that can be converted rapidly to cash for every dollar of current liabilities.
- XYZ Company's current ratio of 0.4 indicates an inadequate degree of liquidity with only 40 cents of current assets available to cover every $1 of current liabilities.
- The quick ratio suggests an even more dire liquidity position, with only 20 cents of liquid assets for every $1 of current liabilities.

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Lesson 11

MARKET VALUE RATIOS

TOPIC 049: MARKET VALUE RATIOS

Market value ratios are used to evaluate the current share price of a publicly-held company's stock. These ratios are employed by current and potential investors to determine whether a company's shares are over-priced or under-priced.

- Market Value ratios measure the relationship between accounting representation of the firm and the market value of the firm.
- Two commonly used market value ratios include:
  - Price to earning ratio
  - Market Price to book ratio

Price-to-Earnings Ratio – P/E Ratio

- The price-to-earnings ratio (P/E ratio) is the ratio for valuing a company that measures its current share price relative to its per-share earnings (EPS). The price-to-earnings ratio is also sometimes known as the price multiple or the earnings multiple.
- P/E ratios are used by investors and analysts to determine the relative value of a company's shares in an apples-to-apples comparison. It can also be used to compare a company against its own historical record or to compare aggregate markets against one another or over time.

Formula

- P/E Ratio

$$= \frac{\text{Market Value Per Share}}{\text{Earnings per share}}$$

- To determine the P/E value, one simply must divide the current stock price by the earnings per share (EPS)
- Sometimes, analysts investigate long term valuation trends and consider the P/E 10 or P/E 30 measures, which average the past 10 or past 30 years of earnings, respectively.
- These measures help to gauge the overall value of a stock index, such as the S&P 500 since these longer term measures can compensate for changes in the business cycle.

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TOPIC 050: MARKET PRICE TO BOOK RATIO

- Companies use the price-to-book ratio (P/B ratio) to compare a firm's market capitalization to its book value. It's calculated by dividing the company's stock price per share by its book value per share (BVPS). The price-to-book ratio is often used by value investors looking for stocks that are underpriced by the market.
- An asset's book value is equal to its carrying value on the balance sheet, and companies calculate it by netting the asset against its accumulated depreciation.
- Book value is also the net asset value of a company calculated as total assets minus intangible assets (patents, goodwill) and liabilities.
- For the initial outlay of an investment, book value may be net or gross of expenses, such as trading costs, sales taxes, and service charges.
- Some people may know this ratio by its less common name, price-equity ratio.

Formula

- Price Equity Ratio
- $= \frac{\text{Market Price Per Share}}{\text{Book value per share}}$

In this equation, book value per share is calculated as follows: (total assets - total liabilities) / number of shares outstanding). Market value per share is obtained by simply looking at the share price quote in the market.

$$P/B\ Ratio = \frac{Market\ Price\ per\ Share}{Book\ Value\ per\ Share}$$

A lower P/B ratio could mean the stock is undervalued. However, it could also mean something is fundamentally wrong with the company. As with most ratios, this varies by industry. The P/B ratio also indicates whether you're paying too much for what would remain if the company went bankrupt immediately.

TOPIC 051: THE RELATIONS AMONG RATIOS

- ROA is a product of two Ratios
- These include:
  - Return on Sales (ROS)
  - Asset Turnover Ratio (ATO)

$$\text{ROA} = \frac{\text{EBIT}}{\text{SALES}} \text{ X } \frac{\text{SALES}}{\text{ASSETS}}$$

- Firms in different sectors with different ROS and ATO can have same ROA
- For example:
- A supermarket chain can have a low profit margin but high turnover ratio, but a high-priced jewelry store can have a high profit margin but low turnover ratio.
- Return on Sales (ROS) is the indicator of operational efficiency.
- Asset Turnover ratio (ATO) also indicates how efficiently the company is deploying its assets to produce revenue.

It is useful to decompose a firm's *ROA* into the product of two ratios as follows:

$$\begin{aligned} ROA &= \frac{EBIT}{\text{Sales}} \times \frac{\text{Sales}}{\text{Assets}} \\ &= \text{Return on Sales} \times \text{Asset Turnover} \\ &= ROS \times ATO \end{aligned}$$

The decomposition of *ROA* into *ROS* and *ATO* highlights the fact that firms in different industries can have vastly different *ROS* and turnover ratios yet the same return on assets. Thus, a supermarket typically has a low profit margin on sales and a high asset turnover, whereas a high-priced jewelry store typically has a high profit margin and a low turnover. Both may have the same *ROA*.

To illustrate, let us take two firms with the same *ROA* of 10% per year. The first is a supermarket chain, the second a public utility company. As Table 3.6 shows, the supermarket chain has a "low" *ROS* of 2% and achieves a 10% *ROA* by "turning over" its assets five times per year. The capital-intensive utility, on the other hand, has a "low" *ATO* of only 0.5 times per year and achieves its 10% *ROA* by having an *ROS* of 20%.

The point here is that a "low" *ROS* or *ATO* ratio need not be a sign of a troubled firm. Each of these ratios must be interpreted in light of industry norms. Even in the same industry systematic differences may exist. For example, a Rolls Royce dealership will almost certainly have a higher margin and lower turnover than a Chevrolet dealership, even though both may have the same *ROA*.

TOPIC 052: THE EFFECTS OF FINANCIAL LEVERAGE

Financial leverage simply means the use of borrowed money. The shareholders of a firm use financial leverage in order to boost their *ROE*, but in so doing they increase the sensitivity of *ROE* to fluctuations in the firm's underlying operating profitability as measured by its *ROA*. In other words, by making use of financial leverage, the firm's shareholders are subjected to financial risk as well as to the operating risk of the firm.

An increase in a firm's financial leverage will increase its *ROE* if and only if its *ROA* exceeds the interest rate on the borrowed funds. This makes intuitive sense. If *ROA* exceeds the borrowing rate, then the firm earns more on the capital it employs than it pays to its creditors. The surplus is thus available to the firm's shareholders, and so raises *ROE*. If, on the other hand, *ROA* is less than the interest rate, the shareholders would have been better off not borrowing at all.

- Financial Leverage means the use of borrowed money.
- The shareholders of a firm use financial leverage to boost their ROE.
- This increases the financial and operating risks.
- Financial leverage has value due to the **interest tax shield**.
- The use of financial leverage also has value when the assets that are purchased with the debt capital earn more than the cost of the debt that was used to finance them.

TOPIC 053: RELATIONSHIP OF ROE, ROA AND LEVERAGE

The exact relationship between *ROE*, *ROA*, and leverage can be summarized in the following equation:

$$ROE = (1 - \text{Tax Rate}) \times \left[ROA + \left(\left(\frac{\text{Debt}}{\text{Equity}} \right) \times (ROA - \text{Interest Rate}) \right) \right]$$

This equation has the following implications. If a firm's *ROA* exceeds the interest rate that it pays to its creditors, then its *ROE* will exceed (1 − tax rate) times *ROA* by an amount that will be greater the higher the debt/equity ratio.

From the perspective of a creditor, an increase in a firm's debt ratio is generally a negative sign. Bond-rating agencies, such as Moody's and Standard and Poors Corporation, will often downgrade a firm's securities if its debt ratio goes up. But from the shareholder's perspective, it might be positive for the firm to increase its debt ratio.

- This equation implies that if a firm's ROA exceeds interest rate that it pays, then ROE will exceed (1-Tax Rate) times ROA plus debt/equity ratio times the difference between ROA and interest rate.

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Lesson 12

THE FINANCIAL PLANNING PROCESS

TOPIC 054: THE FINANCIAL PLANNING PROCESS

Financial Planning is the process of estimating the capital required and determining its competition. It is the process of framing financial policies in relation to procurement, investment and administration of funds of an enterprise.



TOPIC 055: FIRM'S SUSTAINABLE GROWTH RATE

The sustainable growth rate (SGR) is the maximum rate of growth that a company or social enterprise can sustain without having to finance growth with additional equity or debt. In other words, it is the rate at which the company can grow while using its own internal revenue without borrowing from outside sources. The SGR involves maximizing sales and revenue growth without increasing financial leverage. Achieving the SGR can help a company prevent being over-leveraged and avoid financial distress.

First, obtain or calculate the return on equity (ROE) of the company. ROE measures the profitability of a company by comparing net income to the company's shareholders' equity.

Then, subtract the company's dividend payout ratio from 1. The dividend payout ratio is the percentage of earnings per share paid to shareholders as dividends. Finally, multiply the difference by the ROE of the company.

TOPIC 056: WORKING CAPITAL MANAGEMENT

Working capital management is a business strategy designed to ensure that a company operates efficiently by monitoring and using its current assets and liabilities to their most effective use. The efficiency of working capital management can be quantified using ratio analysis. The primary purpose of working capital management is to enable the company to maintain sufficient cash flow to meet its short-term operating costs and short-term debt obligations. A company's working capital is made up of its current assets minus its current liabilities.

Current assets include anything that can be easily converted into cash within 12 months. These are the company's highly liquid assets. Some current assets include cash, accounts receivable, inventory, and short-term investments. Current liabilities are any obligations due within the following 12 months. These include accruals for operating expenses and current portions of long-term debt payments.

TOPIC 057: CASH CONVERSION CYCLE (CCC)

The cash conversion cycle (CCC) is a metric that expresses the time (measured in days) that it takes for a company to convert its investments in inventory and other resources into cash flows from sales. Also called the net operating cycle or simply cash cycle, CCC attempts to measure how long each net input dollar is tied up in the production and sales process before it gets converted into cash received.

This metric takes into account how much time the company needs to sell its inventory, how much time it takes to collect receivables, and how much time it has to pay its bills.
The CCC is one of several quantitative measures that help evaluate the efficiency of a company's operations and management. A trend of decreasing or steady CCC values over multiple periods is a good sign, while rising ones should lead to more investigation and analysis based on other factors. One should bear in mind that CCC applies only to select sectors dependent on inventory management and related operations.

Since CCC involves calculating the net aggregate time involved across the above three stages of the cash conversion life cycle, the mathematical formula for CCC is represented as:
CCC = DIO + DSO – DPO
Where: DIO = Days of inventory outstanding (also known as days sales of inventory)

DSO = Days sales outstanding

DPO = Days payables outstanding

DIO and DSO are associated with the company's cash inflows, while DPO is linked to cash outflow. Hence, DPO is the only negative figure in the calculation. Another way to look at the formula construction is that DIO and DSO are linked to inventory and accounts receivable, respectively, which are considered as short-term assets and are taken as positive. DPO is linked to accounts payable, which is a liability and thus taken as negative.

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Lesson 13

INTEREST RATES AND YIELD TO MATURITY

TOPIC 058: INTEREST RATES

The interest rate is the amount a lender charges a borrower and is a percentage of the principal—the amount loaned. The interest rate on a loan is typically noted on an annual basis known as the annual percentage rate (APR).

An interest rate can also apply to the amount earned at a bank or credit union from a savings account or certificate of deposit (CD). Annual percentage yield (APY) refers to the interest earned on these deposit accounts.

Different debt instruments have very different streams of cash payments (known as cash flows) to the holder, with very different timing. Thus we first need to understand how we can compare the value of one kind of debt instrument with the value of another before we see how interest rates are measured. To do this, we make use of the concept of present value.

Four Types of Credit Instruments

1. Simple loan
2. Fixed-payment loan
3. Coupon bond
4. Discount (zero coupon) bond

Present Value

The concept of present value (or present discounted value) is based on the commonsense notion that a dollar paid to you one year from now is less valuable than a dollar paid to you today. This notion is true because you can deposit a dollar today in a savings account that earns interest and have more than a dollar in one year. Economists use a more formal definition, as explained in this section.

Let's look at the simplest kind of debt instrument, which we will call a simple loan. In this loan, the lender provides the borrower with an amount of funds (called the principal) that must be repaid to the lender at the maturity date, along with an additional payment for the interest. For example, if you made your friend, Jane, a simple loan of $100 for one year, you would require her to repay the principal of $100 in one year's time, along with an additional payment for interest—say, $10. In the case of a simple loan like this one, the interest payment divided by the amount of the loan

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is a natural and sensible way to measure the interest rate. This measure of the so-called simple interest rate, i, is:

$$i = \frac{\$10}{\$100} = 0.10 = 10\%$$

If you make this \$100 loan, at the end of the year you will have \$110, which can be rewritten as

$$\$100 \times (1 + 0.10) = \$110$$

If you then lent out the \$110 at the same interest rate, at the end of the second year you would have

$$\$110 \times (1 + 0.10) = \$121$$

or, equivalently,

$$\$100 \times (1 + 0.10) \times (1 + 0.10) = \$100 \times (1 + 0.10)^2 = \$121$$

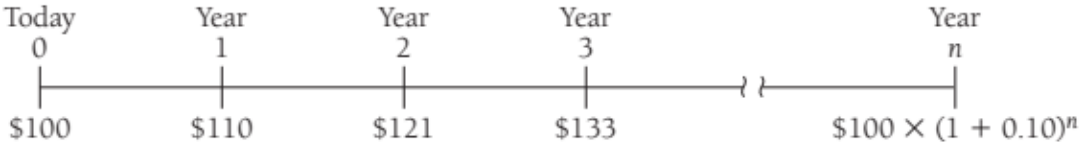
Continuing with the loan again, at the end of the third year you would have

$$\$121 \times (1 + 0.10) = \$100 \times (1 + 0.10)^3 = \$133$$

Generalizing, we can see that at the end of n years, your \$100 would turn into

$$\$100 \times (1 + i)^n$$

The amounts you would have at the end of each year by making the \$100 loan today can be seen in the following timeline:



This timeline clearly indicates that you are just as happy having \$100 today as you will be having \$110 a year from now (of course, as long as you are sure that Jane will pay you back). You are also just as happy having \$100 today as you will be having \$121 two years from now, or \$133 three years from now, or $100 \times (1 + 0.10)^n$ dollars n years from now. The timeline indicates that we can also work backward from future amounts to the present. For example, $\$133 = \$100 \times (1 + 0.10)^3$ three years from now is worth \$100 today, so

$$\$100 = \frac{\$133}{(1 + 0.10)^3}$$

The process of calculating today's value of dollars received in the future, as we have done above, is called *discounting the future*. We can generalize this process by writing today's (present) value of $100 as *PV* and the future cash flow (payment) of $133 as *CF*, and then replacing 0.10 (the 10% interest rate) by *i*. This leads to the following formula:

$$PV = \frac{CF}{(1+i)^n} \qquad (1)$$

Simple Loan

Concept of Present Value

Simple loan of $1 at 10% interest

Years			
1	**2**	**3**	**n**
$1.10	$1.21	$1.33	$1X(1+i)^n

Present Value of Future $1

$=\frac{1}{(1+i)n}$

Interest Rate/Yield to Maturity

Yield to maturity = interest rate that equates today's value with present value of all future payments

1. Simple Loan (i = 10%)

$100 = $110/(1 + i)

$i = \frac{\$110-\$100}{\$100} = \frac{\$10}{\$100} = 0.10$ or 10%

Fixed Payment Loan

Fully Amortized Loan

The lender provides the borrower with an amount of funds, which must be repaid by making the same payment every period.

For example: A person borrowed $1000, a fixed-payment loan might require you to pay $126 every year for 25 years. For example, auto and home loans.

$\$1000 = \frac{\$126}{(1+i)} + \frac{\$126}{(1+i)2} + \ldots + \frac{\$126}{(1+i)25}$

(i = 12%)

- $LV = \frac{FP}{(1+i)} + \frac{FP}{(1+i)} + \ldots + \frac{FP}{(1+i)}$
- LV = Loan value
- FP=Fixed yearly cash flow payment
- n=number of years after maturity

TOPIC 059: YIELD TO MATURITY – COUPON BOND

- A coupon bond, also referred to as a bearer bond or bond coupon, is a debt obligation with coupons attached that represent semiannual interest payments.
- Bondholders receive these coupons during the period between the issuance of the bond and the maturity of the bond.

Strategy

- Equate today's value of the bond with its present value.
- Present value of the bond is calculated as the sum of the present values of all the coupon payments plus the present value of the final payment of the face value of the bond.
- Example: Pakistan Investment Bond

Example

If the face value (FV) of a bond is Rs 1000 with 10 years to maturity and yearly coupon payments of Rs 100 (a 10% coupon rate). Find the present value?

$$P=\frac{100}{(1+i)} + \frac{100}{(1+i)2} + \ldots + \frac{100}{(1+i)10} + \frac{1000}{(1+i)10}$$

Generally, it can be stated as:

$$P=\frac{C}{(1+i)} + \frac{C}{(1+i)2} + \ldots + \frac{C}{(1+i)10} + \frac{F}{(1+i)10}$$

Relationship between Price and Yield to Maturity

Yields to Maturity on a 10 % Coupon Rate Bond Maturing in 10 years (Face Value = Rs 1000)	
Price of Bond	**Yield to Maturity %**
1200	**7.13**
1100	**8.48**
1000	**10.00**
900	**11.75**
800	**13.81**

1. When yield equals coupon rate, bond is at par.
2. Price and yield are negatively related.
3. Yield is greater than coupon rate when bond price is below par value.

Perpetuity

- A perpetual bond, also known as a "consol bond" or "prep," is a fixed income security with no maturity date. This type of bond is often considered a type of equity, rather than debt. One major drawback to these types of bonds is that they are not redeemable.
- Price of perpetuity = $P_c = \frac{C}{iC}$
- Where P_c = price of the perpetuity (consol)
- C = yearly payment
- i_c = yield to maturity of the perpetuity

Perpetuity – Example

What is the yield to maturity on a bond that has a price of Rs 2,000 and pays Rs 100 annually forever?

$P_c = \frac{C}{iC}$ => $iC = \frac{C}{P_c}$

Where C = yearly payment = Rs100

P_c = price of perpetuity = Rs 2000

Thus iC = $\frac{100}{2000}$ = 5%

Coupon Bond To calculate the yield to maturity for a coupon bond, follow the same strategy used for the fixed-payment loan: Equate today's value of the bond with its present value. Because coupon bonds also have more than one cash flow payment, the present value of the bond is calculated as the sum of the present values of all the coupon payments plus the present value of the final payment of the face value of the bond.

The present value of a \$1,000-face-value bond with ten years to maturity and yearly coupon payments of \$100 (a 10% coupon rate) can be calculated as follows: At the end of one year, there is a \$100 coupon payment with a *PV* of $\$100/(1+i)$; at the end of the second year, there is another \$100 coupon payment with a *PV* of $\$100/(1+i)^2$; and so on until, at maturity, there is a \$100 coupon payment with a *PV* of $\$100/(1+i)^{10}$ plus the repayment of the \$1,000 face value with a *PV* of $\$1{,}000/(1+i)^{10}$. Setting today's value of the bond (its current price, denoted by *P*) equal to the sum of the present values of all the cash flow payments for the bond gives

$$P = \frac{\$100}{1+i} + \frac{\$100}{(1+i)^2} + \frac{\$100}{(1+i)^3} + \cdots + \frac{\$100}{(1+i)^{10}} + \frac{\$1{,}000}{(1+i)^{10}}$$

More generally, for any coupon bond,[2]

$$P = \frac{C}{1+i} + \frac{C}{(1+i)^2} + \frac{C}{(1+i)^3} + \cdots + \frac{C}{(1+i)^n} + \frac{F}{(1+i)^n} \qquad (3)$$

where

P = price of the coupon bond
C = yearly coupon payment
F = face value of the bond
n = years to maturity date

In Equation 3, the coupon payment, the face value, the years to maturity, and the price of the bond are known quantities, and only the yield to maturity is not. Hence we can solve this equation for the yield to maturity *i*. As in the case of the fixed-payment loan, this calculation is not easy, so business-oriented software and financial calculators have built-in programs that solve the equation for you.

TOPIC 060: YIELD TO MATURITY- DISCOUNT BOND

- A discount bond is a bond that is issued for less than its par—or face—value.
- Discount bonds may also be a bond currently trading for less than its face value in the secondary market.
- A bond is considered a deep-discount bond if it is sold at a significantly lower price than par value, usually at 20% or more.
- Yield to maturity = i = $\frac{F-P}{P}$

- F = Face value of the discount bond
- P = Current price of the discount bond

Discount Bond – Example

- Suppose a discount bond (a government issued T-bill), which pays a face value of Rs 1000 in one year's time. If the current purchase price of this bill is Rs 900. Equating this price to the present value of the Rs 1000 received in one year.
- Yield to maturity = i = $\frac{F-P}{P}$
- i = $\frac{1000-900}{900}$ =0.111 = 11.1%

Current bond prices and interest rates are negatively related – When the interest rate rises, the price of the bond falls, and vice versa.

Real and Nominal Interest Rates

- Real Interest Rate

The interest rate that is adjusted by subtracting expected changes in the price level (inflation) so that it can represent the true cost of borrowing.

- Nominal Interest Rate

The interest rate before adjusting it for inflation. Nominal can also refer to the advertised or stated interest rate on a loan.

Real Interest Rates

- Ex ante real interest rate is adjusted for expected changes in price level.
- It is considered as an indicator of "real" interest rate.
- The interest rate that is adjusted for actual changes in the price level is called the

ex post real interest rate.

Fisher Equation

- Developed by Irving Fisher.
- The nominal interest rate i equals the real interest rate i_r

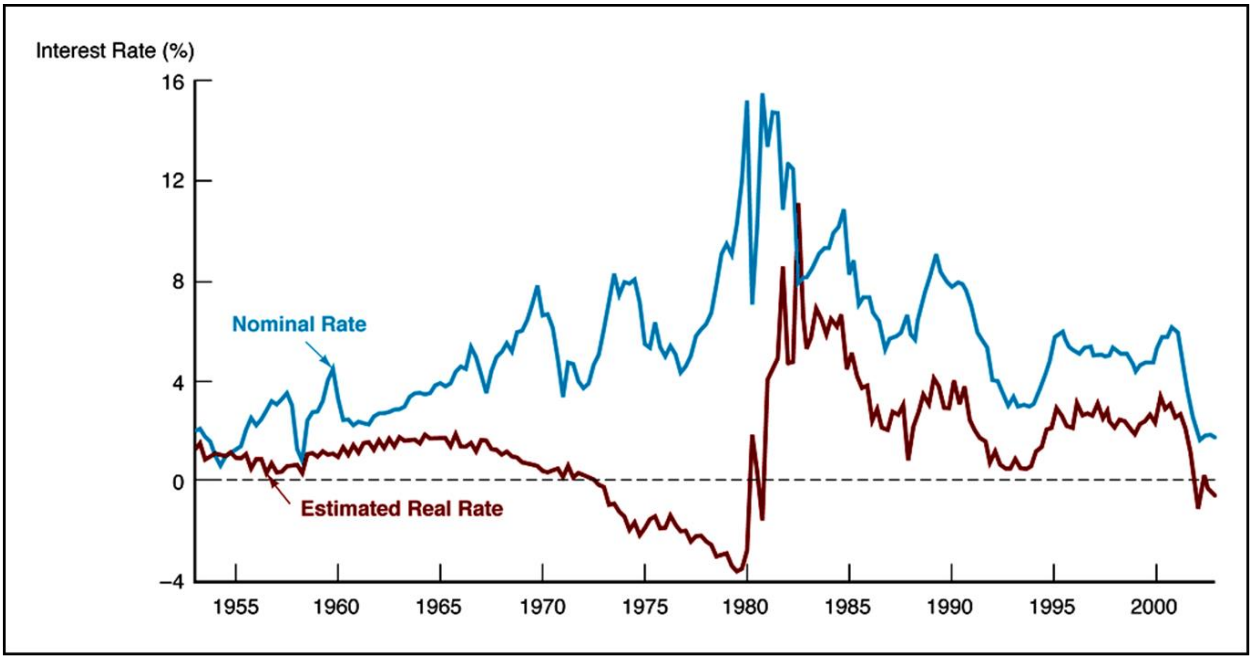
Plus the expected rate of inflation π^e.

$i = i_r + \pi^e$

Rearranging terms, we find the real interest rate:

$i_r = i - \pi^e$

- When the real interest rate is low, there are greater incentives to borrow and fewer incentives to lend.
- It is important to consider the real interest rate for correct decision making.



The yield-to-maturity calculation for a discount bond is similar to that for a simple loan. Let's consider a discount bond such as a one-year U.S. Treasury bill that pays a face value of $1,000 in one year's time but today has a price of $900. As we just saw in the preceding application, the yield to maturity for a one-year discount bond equals the increase in price over the year, $1,000 – $900, divided by the initial price, $900. Hence, more generally, for any one-year discount bond, the yield to maturity can be written as:

$$i = \frac{F - P}{P}$$

Where,

F = face value of the discount bond

P = current price of the discount bond

In other words, the yield to maturity equals the increase in price over the year F – P divided by the initial price P. In normal circumstances, investors earn positive returns from holding these securities and so they sell at a discount, meaning that the current price of the bond is below the face value. Therefore, F – P should be positive, and the yield to maturity should be positive as

well. However, this is not always the case, as extraordinary events in Japan and elsewhere have indicated.

An important feature of this equation is that it indicates that, for a discount bond, the yield to maturity is negatively related to the current bond price. This is the same conclusion that we reached for a coupon bond. The above equation shows that a rise in the bond price—say, from $900 to $950—means that the bond will have a smaller increase in its price at maturity and so the yield to maturity will fall, from 11.1% to 5.3% in our example. Similarly, a fall in the yield to maturity means that the current price of the discount bond has risen.

Summary

The concept of present value tells you that a dollar in the future is not as valuable to you as a dollar today because you can earn interest on a dollar you have today. Specifically, a dollar received n years from now is worth only $1/(1 + i)^n$ today. The present value of a set of future cash flow payments on a debt instrument equals the sum of the present values of each of the future payments. The yield to maturity for an instrument is the interest rate that equates the present value of the future payments on that instrument to its value today. Because the procedure for calculating the yield to maturity is based on sound economic principles, the yield to maturity is the measure that economists think most accurately describes the interest rate.

Our calculations of the yield to maturity for a variety of bonds reveal the important fact that current bond prices and interest rates are negatively related: When the interest rate rises, the price of the bond falls, and vice versa.

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Example

What is the yield to maturity on a one-year, $1,000 Treasury bill with a current price of $900?

Solution
The yield to maturity is 11.1%.
Using the present value formula,

$$PV = \frac{CF}{(1+i)^n}$$

and recognizing that the present value (PV) is the current price of $900, the cash flow in one year is $1,000, and the number of years is 1, we can write:

$$\$900 = \frac{\$1{,}000}{1+i}$$

Solving for i, we get

$$(1+i) \times \$900 = \$1{,}000$$

$$\$900 + \$900i = \$1{,}000$$

$$\$900i = \$1{,}000 - \$900$$

$$i = \frac{\$1{,}000 - \$900}{\$900} = 0.111 = 11.1\%$$

TOPIC 061: INTEREST RATES and Returns

- A return, also known as a financial return, in its simplest terms, is the money made or lost on an investment over some period.
- Interest rate is different from rate of return.
- A coupon bond with Rs 1000-face-value was issued with a coupon rate of 10%.
- It is bought for Rs 1000, held for one year, and then sold for Rs 1200.
- The owner gets two payments
    - i. yearly coupon payment of Rs 100
    - ii. change in its value

(Rs 1200-Rs 1000)

- Rate of Return

$= \frac{100+200}{1000} = \frac{300}{1000}$

= 0.30 or 30%

Hence, the return on a bond will not necessarily equal the interest rate on that bond.
The distinction between interest rate and return can be important.

Many people think that the interest rate on a bond tells them all they need to know about how well off they are as a result of owning it. If Irving the Investor thinks he is well off when he owns a long-term bond yielding a 10% interest rate, and the interest rate then rises to 20%, he will have a rude awakening: As we will shortly see, if he has to sell the bond, Irving will lose his shirt! How well a person does financially by holding a bond or any other security over a particular time period is accurately measured by the security's return, or, in more precise terminology, the rate of return. For any security, the rate of return is defined as the amount of each payment to the owner plus the change in the security's value, expressed as a fraction of its purchase price.

TOPIC 062: RATE OF RETURN - FORMULA

To make this definition clearer, let us see what the return would look like for a $1,000-face-value coupon bond with a coupon rate of 10% that is bought for $1,000, held for one year, and then sold for $1,200. The payments to the owner are the yearly coupon payments of $100, and the change in the bond's value is $1,200 - $1,000 = $200.

Adding these values together and expressing them as a fraction of the purchase price of $1,000 gives us the one-year holding-period return for this bond:

$$\frac{\$100 + \$200}{\$1{,}000} = \frac{\$300}{\$1{,}000} = 0.30 = 30\%$$

You may have noticed something quite surprising about the return that we just calculated: It equals 30%, yet as Table 1 indicates, initially the yield to maturity was only 10%. This discrepancy demonstrates that the return on a bond will not necessarily equal the yield to maturity on that bond. We now see that the distinction between interest rate and return can be important, although for many securities the two may be closely related.

More generally, the return on a bond held from time t to time t + 1 can be written as:

$$R = \frac{C + P_{t+1} - P_t}{P_t}$$

where R = return from holding the bond from time t to time $t + 1$
C = coupon payment
P_t = price of the bond at time t
P_{t+1} = price of the bond at time $t + 1$

A convenient way to rewrite the return formula in Equation 7 is to recognize that it can be split into two separate terms:

$$R = \frac{C}{P_t} + \frac{P_{t+1} - P_t}{P_t}$$

The first term is the current yield i_c (the coupon payment over the purchase price):

$$\frac{C}{P_t} = i_c$$

The second term is the **rate of capital gain**, or the change in the bond's price relative to the initial price:

$$\frac{P_{t+1} - P_t}{P_t} = g$$

Where,

g is the rate of capital gain.

The above equation can be written as:

$R=i_c + g$

Which shows that the return on a bond is the current yield i_c plus the rate of capital gain g. The formula shows that even for a bond for which the current yield i_c is an accurate measure of the yield to maturity, the return can differ substantially from the interest rate. Returns will differ substantially from the interest rate if the price of the bond experiences sizable fluctuations that produce substantial capital gains or losses.

The return on a bond held from time t to time t+1

$$R = \frac{C+P_{t+1}-P_t}{P_t}$$

R = return from holding the bond from time t to time t+1

C = Coupon payment

P_{t+1} = price of the bond at time t+1

P_t = price of the bond at time t

- $R = \frac{C}{P_t} + \frac{P_{t+1} - P_t}{P_t}$
- Let $i_c = \frac{C}{P_t}$ = current yield
- $g = \frac{P_{t+1} - P_t}{P_t}$ = capital gain
- $R = i_c + g$

TOPIC 063: INTEREST RATES, RETURNS AND YEARS TO MATURITY

The finding that the prices of longer-maturity bonds respond more dramatically to changes in interest rates helps explain an important fact about the behavior of bond markets:
Prices and returns for long-term bonds are more volatile than those for shorter-term bonds. Price changes of +20% and -20% within a year, with corresponding variations in returns, are common for bonds that are more than 20 years away from maturity. We now see that changes in interest rates make investments in long-term bonds quite risky. Indeed, the risk level associated with an asset's return that results from interest-rate changes is so important that it has been given a special name, interest rate risk.

Although long-term debt instruments have substantial interest-rate risk, short-term debt instruments do not. Indeed, bonds with a maturity term that is as short as the holding period have no interest-rate risk. We see this for the coupon bond at the bottom of the table given below, which has no uncertainty about the rate of return because it equals the yield to maturity, which is known at the time the bond is purchased.

The key to understanding why there is no interest-rate risk for any bond whose time to maturity matches the holding period is to recognize that (in this case) the price at the end of the holding period is already fixed at the face value. A change in interest rates can then have no effect on the price at the end of the holding period for these bonds, and the return will therefore be equal to the yield to maturity, which is known at the time the bond is purchased.

Table 1: One-year Returns on Different-Maturity 10%-Coupon-Rate Bonds when interest rates rise from 10% to 20%

(1) Years to Maturity When Bond Is Purchased	(2) Initial Current Yield (%)	(3) Initial Price ($)	(4) Price Next Year* ($)	(5) Rate of Capital Gain (%)	(6) Rate of Return [col (2) + col (5)] (%)
30	10	1,000	503	−49.7	−39.7
20	10	1,000	516	−48.4	−38.4
10	10	1,000	597	−40.3	−30.3
5	10	1,000	741	−25.9	−15.9
2	10	1,000	917	−8.3	+1.7
1	10	1,000	1,000	0.0	+10.0

*Calculated with a financial calculator, using Equation 3.

One-Year Returns - Different-Maturity 10%-Coupon-Rate Bonds-

Interest Rates Rise from 10% to 20%;

*Col (4) values are calculated using formula for Price of Coupon bond.

- In the previous table, we have assumed that the interest rate has increased from 10% to 20%
- Increase in interest rates => a fall in bond prices
- It results in capital losses on bonds whose terms to maturity are longer.
- The more distant a bond's maturity date, => the greater is the price change associated with an interest rate change.
- Even if there is a substantial initial interest rate, its return can turn out to be negative if interest rates rise.

TOPIC 064: INTEREST RATE RISK

Interest rate risk is the potential for investment losses that can be triggered by a move upward in the prevailing rates for new debt instruments. If interest rates rise, for instance, the value of a bond or other fixed-income investment in the secondary market will decline. The change in a bond's price given a change in interest rates is known as its duration.

Interest rate risk can be reduced by buying bonds with different durations, or by hedging fixed-income investments with interest rate swaps, options, or other interest rate derivatives.

Interest rate changes can affect many investments, but it impacts the value of bonds and other fixed-income securities most directly. Bondholders, therefore, carefully monitor interest rates and make decisions based on how interest rates are perceived to change over time.

- Prices and returns for long-term bonds are more volatile than those for shorter-term bonds.
- Price changes of +20% and -20% within a year, with corresponding variations in returns, are common for bonds that are more than 20 years away from maturity.
- The risk level associated with an asset's return that results from interest-rate changes is called interest rate risk.
- Long-term debt instruments have substantial interest-rate risk.
- Short-term debt instruments do not have any interest-rate risk.
- Bonds with a maturity term that is as short as the holding period have no interest-rate risk.

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Lesson 14

DEMAND AND SUPPLY ANALYSIS OF BOND MARKET

TOPIC 065: DETERMINANTS OF ASSET DEMAND

Whether to buy and hold an asset?

OR

Whether to buy one asset rather than another?

1. Wealth, the total resources owned by the individual, including all assets

Holding everything else constant, an increase in wealth raises the quantity demanded of an asset

2. Expected return (the return expected over the next period) on one asset relative to alternative assets. An increase in an asset's expected return relative to that of an alternative asset, holding everything else unchanged, raises the quantity demanded of the asset

3. Risk (the degree of uncertainty associated with the return) on one asset relative to alternative assets. Holding everything else constant, if an asset's risk rises relative to that of alternative assets, its quantity demanded will fall.

4. Liquidity (the ease and speed with which an asset can be turned into cash) relative to alternative assets. The more liquid an asset is relative to alternative assets, holding everything else unchanged, the more desirable it is and the greater the quantity demanded will be.

For the demand of an asset, an individual must consider the following factors:

1. Wealth, the total resources owned by the individual, including all assets
2. Expected return (the return expected over the next period) on one asset relative to alternative assets
3. Risk (the degree of uncertainty associated with the return) on one asset relative to alternative assets
4. Liquidity (the ease and speed with which an asset can be turned into cash) relative to alternative assets

Wealth

When we find that our wealth has increased, we have more resources available with which to purchase assets, and so, not surprisingly, the quantity of assets we demand increases. Therefore, the effect of changes in wealth on the quantity demanded of an asset can be summarized as follows: Holding everything else constant, an increase in wealth raises the quantity demanded of an asset.

Expected Returns

The return on an asset (such as a bond) measures how much we gain from holding that asset. When we make a decision to buy an asset, we are influenced by what we expect the return on that asset to be. If an ExxonMobil bond, for example, has a return of 15% half the time and 5% the other half, its expected return (which you can think of as the average return) is 10% (= 0.5 * 15% + 0.5 * 5%).

If the expected return on the ExxonMobil bond rises relative to expected returns on alternative assets, then, holding everything else constant, it becomes more desirable to purchase the ExxonMobil bond, and the quantity demanded increases. This can occur in either of two ways: (1) when the expected return on the ExxonMobil bond rises while the return on an alternative asset—say, stock in Facebook—remains unchanged or (2) when the return on the alternative asset, the Facebook stock, falls while the return on the ExxonMobil bond remains unchanged. To summarize, an increase in an asset's expected return relative to that of an alternative asset, holding everything else unchanged, raises the quantity demanded of the asset.

Risk

The degree of risk or uncertainty of an asset's returns also affects demand for the asset. Consider two assets, stock in Fly-by-Night Airlines and stock in Feet-on-the-Ground Bus Company. Suppose that Fly-by-Night stock has a return of 15% half the time and 5% the other half, making its expected return 10%, while Feet-on-the-Ground stock has a fixed return of 10%. Fly-by-Night stock has uncertainty associated with its returns and so has greater risk than Feet-on-the-Ground stock, whose return is a sure thing.

A risk-averse person prefers stock in Feet-on-the-Ground (the sure thing) to Fly-by Night stock (the riskier asset), even though the stocks have the same expected return, 10%. By contrast, a person who prefers risk is a risk preferrer or risk lover. Most people are risk-averse, especially in their financial decisions: Everything else being equal, they prefer to hold the less risky asset. Hence, holding everything else constant, if an asset's risk rises relative to that of alternative assets, its quantity demanded will fall.

Liquidity

Another factor that affects the demand for an asset is how quickly it can be converted into cash at low costs—its liquidity. An asset is liquid if the market in which it is traded has depth and

breadth, that is, if the market has many buyers and sellers. A house is not a very liquid asset because it may be hard to find a buyer quickly; if a house must be sold to pay off bills, it might have to be sold for a much lower price.

And the transaction costs associated with selling a house (broker's commissions, lawyer's fees, and so on) are substantial. A U.S. Treasury bill, by contrast, is a highly liquid asset. It can be sold in a well-organized market with many buyers, and so it can be sold quickly at low cost. The more liquid an asset is relative to alternative assets, holding everything else unchanged, the more desirable it is and the greater the quantity demanded will be.

TOPIC 066: THEORY OF PORTFOLIO CHOICE

It states that, holding all other factors constant:

1. The quantity demanded of an asset is positively related to wealth.
2. The quantity demanded of an asset is positively related to its expected return relative to alternative assets.
3. The quantity demanded of an asset is negatively related to the risk of its returns relative to alternative assets.
4. The quantity demanded of an asset is positively related to its liquidity relative to alternative assets.

All the determining factors we have just discussed can be assembled into the theory of portfolio choice, which tells us how much of an asset people will want to hold in their portfolios. It states that, holding all other factors constant:

1. The quantity demanded of an asset is positively related to wealth.
2. The quantity demanded of an asset is positively related to its expected return relative to alternative assets.
3. The quantity demanded of an asset is negatively related to the risk of its returns relative to alternative assets.
4. The quantity demanded of an asset is positively related to its liquidity relative to alternative assets.

These results are summarized in the table presented below:

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Response of the Quantity of an Asset Demanded to Changes in Wealth, Expected Returns, Risk, and Liquidity

Variable	Change in Variable	Change in Quantity Demanded
Wealth	↑	↑
Expected return relative to other assets	↑	↑
Risk relative to other assets	↑	↓
Liquidity relative to other assets	↑	↑

Note: Only increases in the variables are shown. The effects of decreases in the variables on the quantity demanded would be the opposite of those indicated in the rightmost column.

TOPIC 067: DEMAND IN THE BOND MARKET

- How are the prices of bonds determined?
- A bond is a fixed income instrument that represents a loan made by an investor to a borrower (corporate or governmental)
- A bond demand curve shows the relationship between the quantity demanded and the price when all other economic variables are held constant.
- Hence
- $i = R^e = \frac{F-P}{P}$
- where i = interest rate = yield to maturity
- R^e = expected return
- F = face value of the discount bond
- P = initial purchase price of the discount bond

Example:

- If the bond sells for $950, the interest rate and expected return are
- $i = R^e = \frac{1000-950}{950}$

  $= 0.053 = 5.3\%$

The analysis of interest-rate determination looks at supply and demand in the bond market so that we can better understand how the prices of bonds are determined. Each bond price is associated with a particular level of the interest rate. Specifically, the negative relationship between bond prices and interest rates means that when a bond's price rises, its interest rate falls, and vice versa.

The first step in our analysis is to obtain a bond demand curve, which shows the relationship between the quantity demanded and the price when all other economic variables are held constant (that is, values of other variables are taken as given). The assumption that all other economic variables are held constant is called ceteris paribus, which means "other things being equal" in Latin.

Demand Curve

To clarify and simplify our analysis, let's consider the demand for one-year discount bonds, which make no coupon payments but pay the owner the $1,000 face value in a year. If the holding period is one year, then, as we saw in Chapter 4, the return on the bonds is known absolutely and is equal to the interest rate as measured by the yield to maturity. This means that the expected return on this bond is equal to the interest rate i, which, using Equation 6 in Chapter 4, is

$$i = R^e = \frac{F - P}{P}$$

where
i = interest rate = yield to maturity
R^e = expected return
F = face value of the discount bond
P = initial purchase price of the discount bond

This formula shows that a particular value of the interest rate corresponds to each bond price. If the bond sells for $950, the interest rate and expected return are

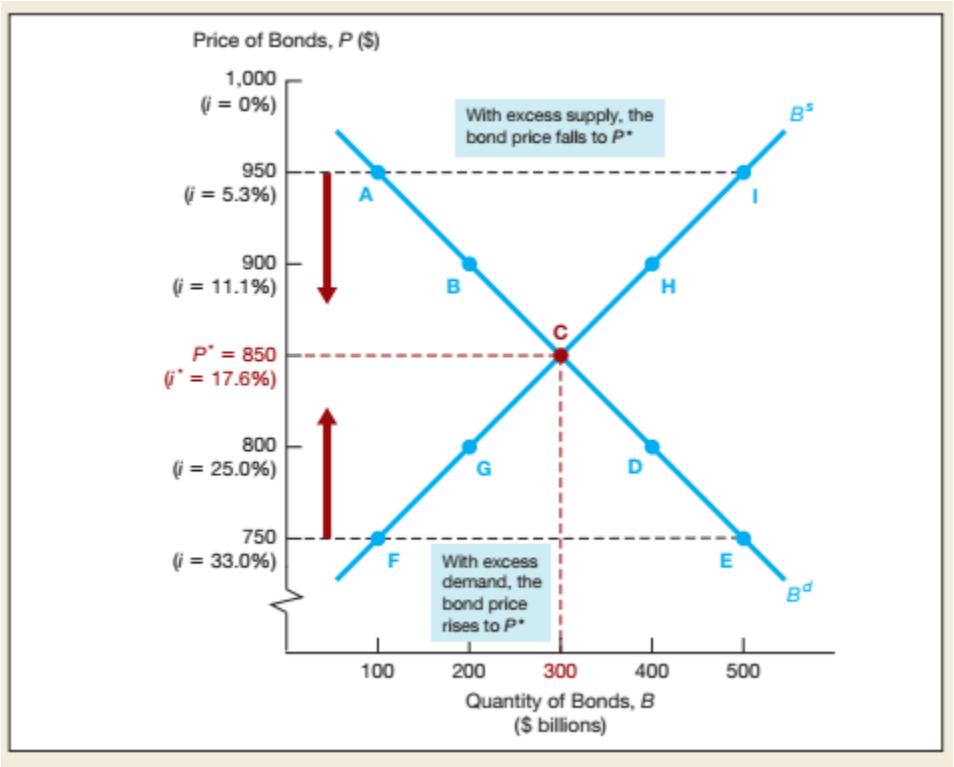
$$\frac{\$1{,}000 - \$950}{\$950} = 0.053 = 5.3\%$$

At this 5.3% interest rate and expected return corresponding to a bond price of $950, let us assume that the quantity of bonds demanded is $100 billion, which is plotted as point A in Figure 1.

At a price of $900, the interest rate and expected return are

$$\frac{\$1{,}000 - \$900}{\$900} = 0.111 = 11.1\%$$

Because the expected return is higher, with all other economic variables (such as wealth, expected returns on other assets, risk, and liquidity) held constant, the quantity demanded of these bonds will be higher, as predicted by portfolio theory. Point B in Figure 1 shows that the quantity of bonds demanded at the price of $900 has risen to $200 billion. Continuing with this reasoning, we see that if the bond price is $850 (interest rate and expected return = 17.6%), the quantity of bonds demanded (point C)



will be greater than at point B. Similarly, at the even lower prices of $800 (interest rate = 25%) and $750 (interest rate = 33.3%), the quantity of bonds demanded will be even higher (points D and E). The curve Bd, which connects these points, is the demand curve for bonds. It has the usual downward slope, indicating that at lower prices of the bond (everything else being equal), the quantity demanded is higher.

TOPIC 068: SUPPLY IN THE BOND MARKET

- A supply curve shows the relationship between the quantity supplied and the price when all other economic variables are held constant.
- It is a positively sloping curve.
- If the interest rate low, it is less costly to borrow by issuing bonds.
- Firms will be willing to borrow more through bond issues, and the quantity of bonds supplied is at the higher level
- If the interest rate is high, it is expensive to borrow by issuing bonds.
- Firms will be willing to borrow more through bond issues, and the quantity of bonds supplied is at the higher level

- The relationship between interest rate and bond price is inverse.

An important assumption behind the demand curve for bonds in Figure 1 is that all other economic variables besides the bond's price and interest rate are held constant. We use the same assumption in deriving a supply curve, which shows the relationship between the quantity supplied and the price when all other economic variables are held constant.

In Figure 1, when the price of the bonds is $750 (interest rate = 33.3%), point F shows that the quantity of bonds supplied is $100 billion for the example we are considering. If the price is $800, the interest rate is the lower rate of 25%. Because at this interest rate it is now less costly to borrow by issuing bonds, firms will be willing to borrow more through bond issues, and the quantity of bonds supplied is at the higher level of $200 billion (point G).

An even higher price of $850, corresponding to a lower interest rate of 17.6%, results in a larger quantity of bonds supplied of $300 billion (point C). Higher prices of $900 and $950 result in even lower interest rates and even greater quantities of bonds supplied (points H and I). The Bs curve, which connects these points, is the supply curve for bonds. It has the usual upward slope found in supply curves, indicating that as the price increases (everything else being equal), the quantity supplied increases.

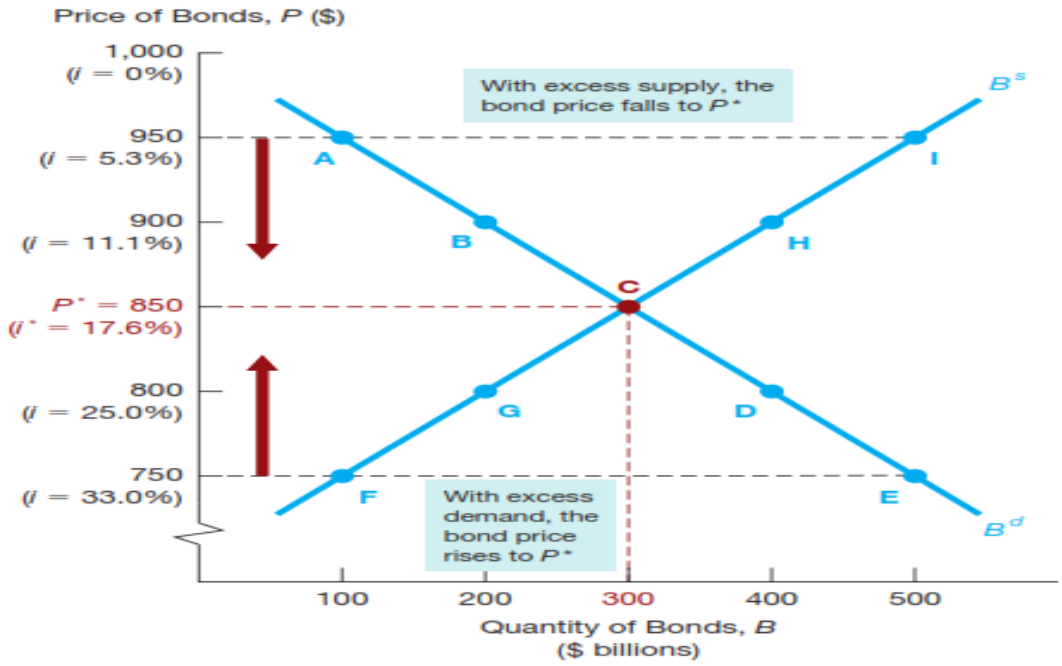
TOPIC 069: EQUILIBRIUM IN THE BOND MARKET

- Market equilibrium occurs when the amount that people are willing to buy (demand) equals the amount that people are willing to sell (supply) at a given price.
- In the bond market, this is achieved when the quantity of bonds demanded equals the quantity of bonds supplied:

  $$Bd = Bs$$

- The asset market approach for understanding behavior in financial markets emphasizes stocks of assets, rather than flows, in determining asset prices.

Figure 1



In economics, market equilibrium occurs when the amount that people are willing to buy (demand) equals the amount that people are willing to sell (supply) at a given price. In the bond market, this is achieved when the quantity of bonds demanded equals the quantity of bonds supplied:

$B^d = B^s$

In Figure 1 (in Topic 10), equilibrium occurs at point C, where the demand and supply curves intersect at a bond price of $850 (interest rate of 17.6%) and a quantity of bonds of $300 billion. The price of P* = 850, where the quantity demanded equals the quantity supplied, is called the equilibrium or market-clearing price. Similarly, the interest rate of i* = 17.6% that corresponds to this price is called the equilibrium or market-clearing interest rate.

The concepts of market equilibrium and equilibrium price or interest rate are useful because the market tends to head toward them. We can see this in Figure 1 by first looking at what happens when we have a bond price that is above the equilibrium price. When the price of bonds is set too high, at, say, $950, the quantity of bonds supplied at point I is greater than the quantity of bonds demanded at point A. A situation like this, in which the quantity of bonds supplied exceeds the quantity of bonds demanded, is called a condition of excess supply. Because people (borrowers) want to sell more bonds than others (lender-savers) want to buy, the price of the bonds will fall, as shown by the downward arrow in the figure at the bond price of $950.

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As long as the bond price remains above the equilibrium price, an excess supply of bonds will continue to be available, and the price of bonds will continue to fall. This decline will stop only when the price has reached the equilibrium price of $850, the price at which the excess supply of bonds has been eliminated.

Now let's look at what happens when the price of bonds is below the equilibrium price. If the price of the bonds is set too low, at, say, $750, the quantity demanded at point E is greater than the quantity supplied at point F. This is called a condition of excess demand. People (lender-savers) now want to buy more bonds than others (borrowers) are willing to sell, so the price of bonds will be driven up, as illustrated by the upward arrow in the figure at the bond price of $750. Only when the excess demand for bonds is eliminated by the bond price rising to the equilibrium level of $850 is there no further tendency for the price to rise.

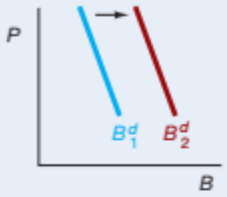
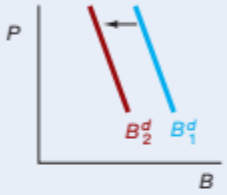
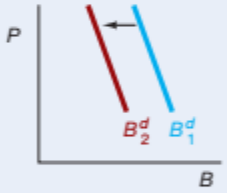
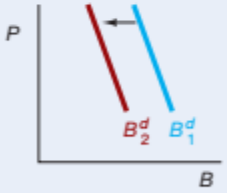
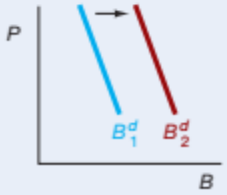
We can see that the concept of equilibrium price is a useful one because it indicates where the market will settle. Because each price on the vertical axis of Figure 1 corresponds to a particular value of the interest rate, the same diagram also shows that the interest rate will head toward the equilibrium interest rate of 17.6%. When the interest rate is below the equilibrium interest rate, as it is when it is at 5.3%, the price of the bond is above the equilibrium price, and an excess supply of bonds results. The price of the bond then falls, leading to a rise in the interest rate toward the equilibrium level. Similarly, when the interest rate is above the equilibrium level, as it is when it is at 33.3%, an excess demand for bonds occurs, and the bond price rises, driving the interest rate back down to the equilibrium level of 17.6%.

TOPIC 070: SHIFTS IN THE DEMAND FOR BONDS

Factors that can cause shift in the demand curve:

1. Wealth
2. Expected returns on bonds relative to alternative assets
3. Risk of bonds relative to alternative assets
4. Liquidity of bonds relative to alternative assets

Factors That Shift the Demand Curve for Bonds

Variable	Change in Variable	Change in Quantity Demanded at Each Bond Price	Shift in Demand Curve
Wealth	↑	↑	
Expected interest rate	↑	↓	
Expected inflation	↑	↓	
Riskiness of bonds relative to other assets	↑	↓	
Liquidity of bonds relative to other assets	↑	↑	

Note: Only increases in the variables are shown. The effects of decreases in the variables on demand would be the opposite of those indicated in the remaining columns.

Figure 1 is a conventional supply and demand diagram with price on the vertical axis and quantity on the horizontal axis. Because the interest rate that corresponds to each bond price is also marked on the vertical axis, this diagram allows us to read the equilibrium interest rate, giving us a model that describes the determination of interest rates. It is important to recognize that a supply and demand diagram like Figure 1 can be drawn for any type of bond because the interest rate

and price of a bond are always negatively related for all kinds of bonds, whether a discount bond or a coupon bond.

An important feature of the analysis here is that supply and demand are always described in terms of stocks (amounts at a given point in time) of assets, not in terms of flows. The asset market approach for understanding behavior in financial markets—which emphasizes stocks of assets, rather than flows, in determining asset prices—is the dominant methodology used by economists, because correctly conducting analyses in terms of flows is very tricky, especially when we encounter inflation.

The theory of portfolio choice, provides a framework for deciding which factors will cause the demand curve for bonds to shift. These factors include changes in the following four parameters:

1. Wealth
2. Expected returns on bonds relative to alternative assets
3. Risk of bonds relative to alternative assets
4. Liquidity of bonds relative to alternative assets

To see how a change in each of these factors (holding all other factors constant) can shift the demand curve, let's look at some examples. (As a study aid, Table 2 summarizes the effects of changes in these factors on the bond demand curve.)

Wealth When the economy is growing rapidly in a business cycle expansion and wealth is increasing, the quantity of bonds demanded at each bond price (or interest rate) increases, as shown in Figure 2. To see how this works, consider point B on the initial demand curve for bonds, Bd 1. With higher wealth, the quantity of bonds demanded at the same price must rise, to point B′. Similarly, for point D, the higher wealth causes the quantity demanded at the same bond price to rise to point D′. Continuing with this reasoning for every point on the initial demand curve Bd 1, we can see that the demand curve shifts to the right from Bd 1 to Bd 2, as indicated by the arrows.

We can conclude that in a business cycle expansion with growing income and wealth, the demand for bonds rises and the demand curve for bonds shifts to the right. Applying the same reasoning, in a recession, when income and wealth are falling, the demand for bonds falls, and the demand curve shifts to the left. Another factor that affects wealth is the public's propensity to save. If households save more, wealth increases and, as we have seen, the demand for bonds rises and

the demand curve for bonds shifts to the right. Conversely, if people save less, wealth and the demand for bonds fall, and the demand curve shifts to the left.

Expected Returns For a one-year discount bond and a one-year holding period, the expected return and the interest rate are identical, so nothing other than today's interest rate affects the expected return. For bonds with maturities of greater than one year, the expected return may differ from the interest rate. A rise in the interest rate on a long-term bond from 10% to 20% would lead to a sharp decline in price and a very large negative return. Hence, if people began to think that interest rates would be higher next year than they had originally anticipated, the expected return today on long-term bonds would fall, and the quantity demanded would fall at each interest rate. Higher expected future interest rates lower the expected return for long-term bonds, decrease the demand, and shift the demand curve to the left.

By contrast, an expectation of lower future interest rates would mean that long-term bond prices would be expected to rise more than originally anticipated, and the resulting higher expected return today would raise the quantity demanded at each bond price and interest rate. Lower expected future interest rates increase the demand for long-term bonds and shift the demand curve to the right (as in Figure 2).

Changes in expected returns on other assets can also shift the demand curve for bonds. If people suddenly become more optimistic about the stock market and begin to expect higher stock prices in the future, both expected capital gains and expected returns on stocks will rise. With the expected return on bonds held constant, the expected return on bonds today relative to stocks will fall, lowering the demand for bonds and shifting the demand curve to the left. An increase in expected return on alternative assets lowers the demand for bonds and shifts the demand curve to the left.

A change in expected inflation is likely to alter expected returns on physical assets (also called real assets), such as automobiles and houses, which affect the demand for bonds. An increase in expected inflation from, say, 5% to 10% will lead to higher prices on cars and houses in the future and hence higher nominal capital gains. The resulting rise in the expected returns today on these real assets will lead to a fall in the expected return on bonds relative to the expected return on real assets today and thus cause the demand for bonds to fall. Alternatively, we can think of the rise in expected inflation as lowering the real interest rate on bonds, and thus the resulting

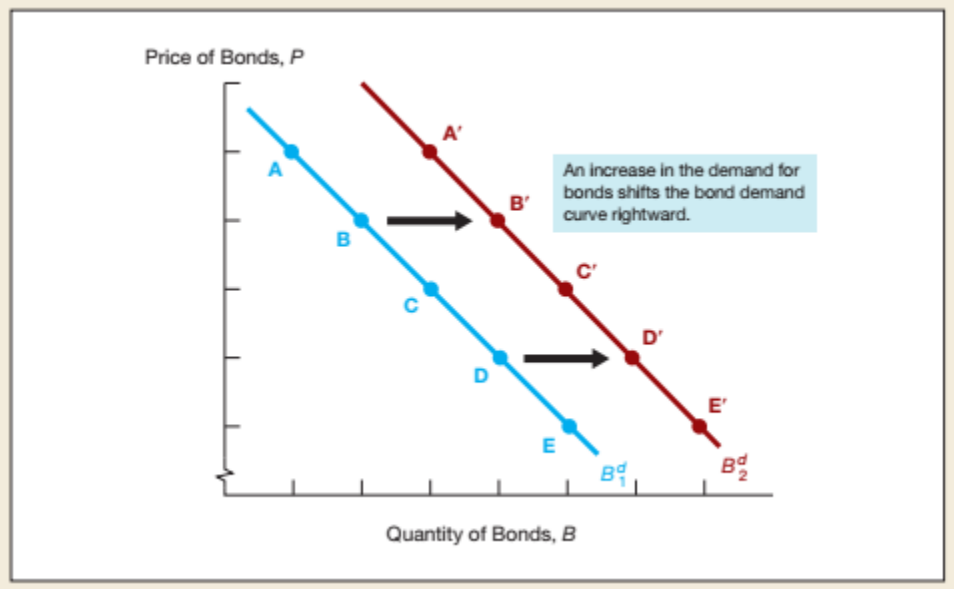
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decline in the relative expected return on bonds will cause the demand for bonds to fall. An increase in the expected rate of inflation lowers the expected return on bonds, causing their demand to decline and the demand curve to shift to the left.

Risk If prices in the bond market become more volatile, the risk associated with bonds increases, and bonds become a less attractive asset. An increase in the riskiness of bonds causes the demand for bonds to fall and the demand curve to shift to the left. Conversely, an increase in the volatility of prices in another asset market, such as the stock market, would make bonds more attractive. An increase in the riskiness of alternative assets causes the demand for bonds to rise and the demand curve to shift to the right (as in Figure 2).

Liquidity If more people started trading in the bond market, and as a result it became easier to sell bonds quickly, the increase in their liquidity would cause the quantity of bonds demanded at each interest rate to rise. Increased liquidity of bonds results in an increased demand for bonds, and the demand curve shifts to the right (see Figure 2). Similarly, increased liquidity of alternative assets lowers the demand for bonds and shifts the demand curve to the left. The reduction of brokerage commissions for trading common stocks that occurred when the fixed-rate commission structure was abolished in 1975, for example, increased the liquidity of stocks relative to bonds, and the resulting lower demand for bonds shifted the demand curve to the left.

Figure 2



TOPIC 071: SHIFTS IN THE SUPPLY OF BONDS

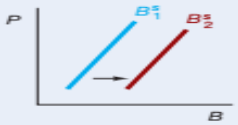
Certain factors can cause the supply curve for bonds to shift. Among these factors are:

1. Expected profitability of investment opportunities

2. Expected inflation

3. Government budget deficits

Factors That Shift the Supply of Bonds

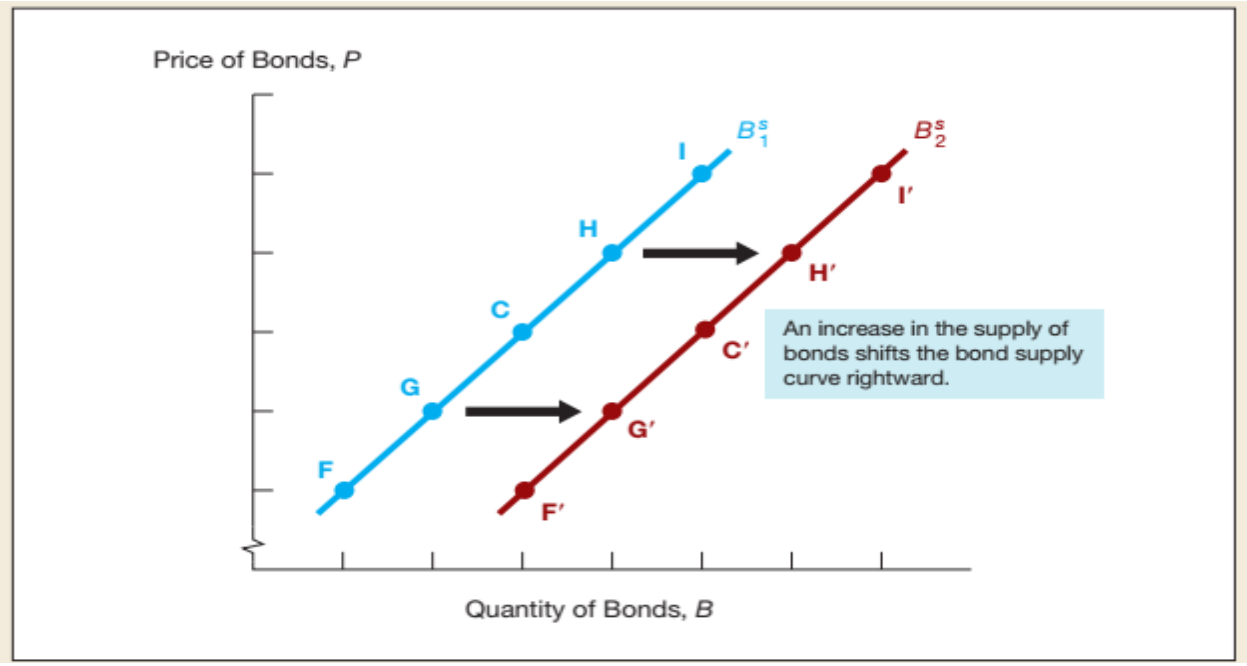
Variable	Change in Variable	Change in Quantity Supplied at Each Bond Price	Shift in Supply Curve
Profitability of investments	↑	↑	
Expected inflation	↑	↑	
Government deficit	↑	↑	

Note: Only increases in the variables are shown. The effects of decreases in the variables on the supply would be the opposite of those indicated in the remaining columns.

We will look at how the supply curve shifts when each of these factors changes (all others remaining constant). (As a study aid, Table 3 summarizes the effects of changes in these factors on the bond supply curve.) bonds, and the resulting lower demand for bonds shifted the demand curve to the left.

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Figure 3



Expected Profitability of Investment Opportunities When opportunities for profitable plant and equipment investments are plentiful, firms are more willing to borrow to finance these investments. When the economy is growing rapidly, as in a business cycle expansion, investment opportunities that are expected to be profitable abound, and the quantity of bonds supplied at any given bond price increases (for example, from G to G′ or H to H′ in Figure 3). **Therefore, in a business cycle expansion, the supply of bonds increases and the supply curve shifts to the right. Likewise, in a recession, when far fewer profitable investment opportunities are expected, the supply of bonds falls and the supply curve shifts to the left.**

Expected Inflation The real cost of borrowing is most accurately measured by the real interest rate, which equals the (nominal) interest rate minus the expected inflation rate. For a given interest rate (and bond price), when expected inflation increases, the real cost of borrowing falls; hence, the quantity of bonds supplied increases at any given bond price. **An increase in expected inflation causes the supply of bonds to increase and the supply curve to shift to the right** (see Figure 3), **and a decrease in expected inflation causes the supply of bonds to decrease and the supply curve to shift to the left.**

Government Budget Deficits The activities of the government can influence the supply of bonds in several ways. The U.S. Treasury issues bonds to finance government deficits, caused by gaps

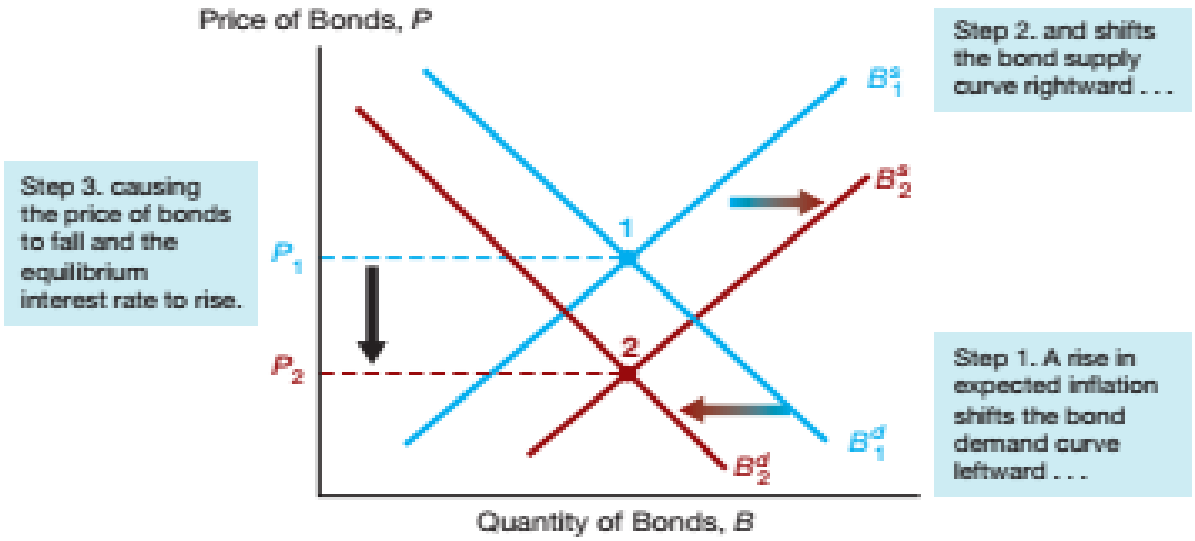
between the government's expenditures and its revenues. When these deficits are large, the Treasury sells more bonds, and the quantity of bonds supplied at each bond price increases. **Higher government deficits increase the supply of bonds and shift the supply curve to the right** (see Figure 3). **On the other hand, government surpluses, as occurred in the late 1990s, decrease the supply of bonds and shift the supply curve to the left.**

State and local governments and other government agencies also issue bonds to finance their expenditures, and this can affect the supply of bonds as well. The conduct of monetary policy involves the purchase and sale of bonds, which in turn influences the supply of bonds. We now can use our knowledge of how supply and demand curves shift to analyze how the equilibrium interest rate can change. The best way to do this is to pursue several applications that are particularly relevant to our understanding of how monetary policy affects interest rates. In studying these applications, keep two things in mind:

1. When we examine the effect of a variable change, remember we are assuming that all other variables are unchanged; that is, we are making use of the ceteris paribus assumption.
2. Remember that the interest rate is negatively related to the bond price, so when the equilibrium bond price rises, the equilibrium interest rate falls. Conversely, if the equilibrium bond price moves downward, the equilibrium interest rate rises.

TOPIC 072: FISHER EFFECT

- If expected inflation rises, the expected return on bonds relative to real assets falls for any given bond price and interest rate.
- The demand for bonds falls, and the demand curve shifts to the left.
- The rise in expected inflation also shifts the supply curve. At any given bond price and interest rate, the real cost of borrowing declines, causing the quantity of bonds supplied to increase and the supply curve to shift to the right.
- When the demand and supply curves shift in response to the rise in expected inflation, the equilibrium moves from point 1 to point 2.
- The equilibrium bond price falls, and, because the bond price is negatively related to the interest rate, this means that the equilibrium interest rate rises.
- When expected inflation rises, interest rates will rise.



The Fisher Effect is an economic theory created by economist Irving Fisher that describes the relationship between inflation and both real and nominal interest rates. The Fisher Effect states that the real interest rate equals the nominal interest rate minus the expected inflation rate. Therefore, real interest rates fall as inflation increases, unless nominal rates increase at the same rate as inflation.

Fisher's equation reflects that the real interest rate can be taken by subtracting the expected inflation rate from the nominal interest rate. In this equation, all the provided rates are compounded.

The Fisher Effect can be seen each time you go to the bank; the interest rate an investor has on a savings account is really the nominal interest rate. For example, if the nominal interest rate on a savings account is 4% and the expected rate of inflation is 3%, then the money in the savings account is really growing at 1%. The smaller the real interest rate, the longer it will take for savings deposits to grow substantially when observed from a purchasing power perspective.

Suppose that expected inflation is initially 5% and the initial supply and demand curves Bs 1 and Bd 1 intersect at point 1, where the equilibrium bond price is P1. If expected inflation rises to 10%, the expected return on bonds relative to real assets falls for any given bond price and interest rate. As a result, the demand for bonds falls, and the demand curve shifts to the left from Bd 1 to Bd 2. The rise in expected inflation also shifts the supply curve. At any given bond price and interest rate, the real cost of borrowing declines,

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causing the quantity of bonds supplied to increase and the supply curve to shift to the right, from Bs1 to Bs2.

When the demand and supply curves shift in response to the rise in expected inflation, the equilibrium moves from point 1 to point 2, at the intersection of Bd 2 and Bs 2. The equilibrium bond price falls from P1 to P2 and, because the bond price is negatively related to the interest rate, this means that the equilibrium interest rate rises. Note that Figure 4 has been drawn so that the equilibrium quantity of bonds remains the same at both point 1 and point 2. However, depending on the size of the shifts in the supply and demand curves, the equilibrium quantity of bonds can either rise or fall when expected inflation rises.

Our supply and demand analysis has led us to an important observation: When expected inflation rises, interest rates will rise. This result has been named the Fisher effect, after Irving Fisher, the economist who first pointed out the relationship of expected inflation to interest rates.

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Lesson 15

RISK STRUCTURE OF INTEREST RATES

TOPIC 073: RISK STRUCTURE OF INTEREST RATES

Bonds with the same maturity have different interest rates due to:

- Default risk
- Liquidity
- Tax considerations
- Term to maturity

Default risk: probability that the issuer of the bond is unable or unwilling to make interest payments or pay off the face value

- Treasury bonds are considered default free (government can raise taxes).
- Risk premium: the spread between the interest rates on bonds with default risk and the interest rates on (same maturity) Treasury bonds.

Response to an Increase in Default Risk on Corporate Bonds

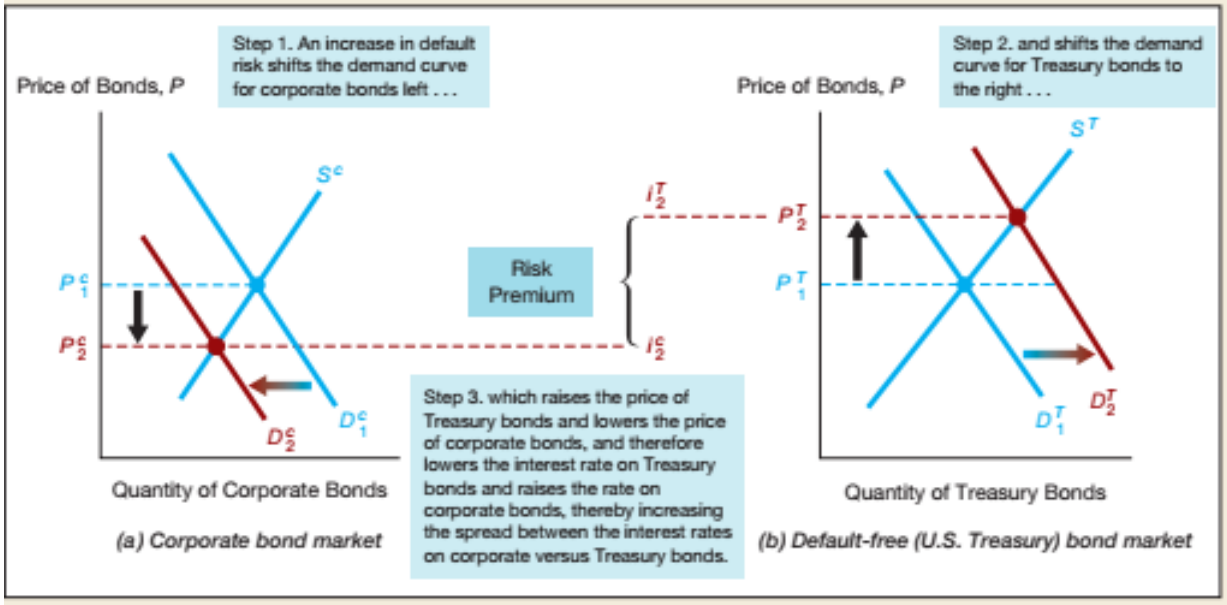


FIGURE 2 **Response to an Increase in Default Risk on Corporate Bonds**

Initially, $P_1^c = P_1^T$, $i_1^c = i_1^T$, and the risk premium is zero. An increase in default risk on corporate bonds shifts the demand curve from D_1^c to D_2^c; simultaneously, it shifts the demand curve for Treasury bonds from D_1^T to D_2^T. The equilibrium price for corporate bonds falls from P_1^c to P_2^c, and the equilibrium interest rate on corporate bonds rises to i_2^c. In the Treasury market, the equilibrium bond price rises from P_1^T to P_2^T, and the equilibrium interest rate falls to i_2^T. The brace indicates the difference between i_2^c and i_2^T, the risk premium on corporate bonds. (Note that because P_2^c is lower than P_2^T, i_2^c is greater than i_2^T.)

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Interest rate behavior for bonds of the same maturity shows two important features: Interest rates on different categories of bonds, although they generally move together, differ from one another in any given year, and the spread (or difference) between the interest rates varies over time. The interest rates on municipal bonds, for example, were higher than those on U.S. government (Treasury) bonds in the late 1930s but lower thereafter. In addition, the spread between the interest rates on Baa corporate bonds (riskier than Aaa corporate bonds) and U.S. government bonds was very large during the Great Depression years from 1930 to 1933, was smaller during the 1940s–1960s, and then widened again afterward, particularly during the global financial crisis from 2007 to 2009. Which factors are responsible for these phenomena?

TOPIC 074: DEFAULT RISK AND RATING AGENCIES

Bond Ratings

- A bond rating is a way to measure the creditworthiness of a bond, which corresponds to the cost of borrowing for an issuer.
- These ratings assign a letter grade to bonds that indicates their credit quality.
- Private independent rating services such as Standard & Poor's, Moody's Investors Service, and Fitch Ratings Inc. evaluate a bond issuer's financial strength, or its ability to pay a bond's principal and interest, in a timely fashion.

Default Risk

One attribute of a bond that influences its interest rate is its risk of default. Default occurs when the issuer of the bond is unable or unwilling to make interest payments when promised or pay off the face value when the bond matures.

Corporations suffering big losses, such as the major airline companies like United, Delta, US Airways, and Northwest in the mid-2000s, and then American Airlines in 2011, might be more likely to suspend interest payments on their bonds. The default risk on their bonds would therefore be quite high. By contrast, U.S. Treasury bonds have usually been considered to have no default risk because the federal government can always increase taxes or print money to pay off its obligations. Bonds like these with no default risk are called default-free bonds.

(However, during the budget negotiations in Congress in 2013, which led to a government shutdown, the Republicans threatened to let Treasury bonds

default, and this threat had an adverse impact on the bond market.) The spread between interest rates on bonds with default risk and interest rates on default free bonds, both of the same maturity, is called the risk premium. The risk premium indicates how much additional interest people must earn to be willing to hold the risky bond. Our supply and demand analysis of the bond market in Chapter 5 can be used to explain why a bond with default risk always has a positive risk premium, and why a higher default risk means a larger risk premium.

To examine the effect of default risk on interest rates, let's look at the supply and demand diagrams for the default-free (U.S. Treasury) and corporate long-term bond markets in Figure 2. To make the diagrams somewhat easier to read, let's assume that initially corporate bonds have the same default risk as U.S. Treasury bonds. In this case, these two bonds have the same attributes (identical risk and maturity); their equilibrium prices and interest rates will initially be equal (Pc 1 = PT 1 and ic 1 = iT 1), and the risk premium on corporate bonds (ic 1 - iT 1) will be zero.

If the possibility of a default increases because a corporation begins to suffer large losses, the default risk on corporate bonds will increase and the expected return on these bonds will decrease. In addition, the corporate bond's return will be more uncertain. The theory of portfolio choice predicts that because the expected return on the corporate bond falls relative to the expected return on the default-free Treasury bond while its relative riskiness rises, the corporate bond is less desirable (holding everything else equal), and demand for it will fall. Another way of thinking about this is that if you were an investor, you would want to hold (demand) a smaller amount of corporate bonds.

The demand curve for corporate bonds in panel (a) of Figure 2 then shifts to the left, from Dc 1 to Dc 2. At the same time, the expected return on default-free Treasury bonds increases relative to the expected return on corporate bonds, while their relative riskiness declines. The Treasury bonds thus become more desirable, and demand rises, as shown in panel (b) by the rightward shift in the demand curve for these bonds from DT 1 to DT 2.

As we can see in Figure 2, the equilibrium price for corporate bonds falls from Pc 1 to Pc 2, and since the bond price is negatively correlated to the interest rate, the equilibrium

interest rate on corporate bonds rises to ic 2. At the same time, however, the equilibrium price for the Treasury bonds rises from PT 1 to PT 2 and the equilibrium interest rate falls to iT 2. The spread between the interest rates on corporate and default-free bonds—that is, the risk premium on corporate bonds—has risen from zero to ic 2 - iT 2.

We can now conclude that a bond with default risk will always have a positive risk premium, and an increase in its default risk will raise the risk premium. Because default risk is so important to the size of the risk premium, purchasers of bonds need to know whether a corporation is likely to default on its bonds. This information is provided by credit-rating agencies, investment advisory firms that rate the quality of corporate and municipal bonds in terms of their probability of default.

TABLE 1 Bond Ratings by Moody's, Standard and Poor's, and Fitch

Rating Agency			
Moody's	S&P	Fitch	Definitions
Aaa	AAA	AAA	Prime Maximum Safety
Aa1	AA+	AA+	High Grade High Quality
Aa2	AA	AA	
Aa3	AA–	AA–	
A1	A+	A+	Upper Medium Grade
A2	A	A	
A3	A–	A–	
Baa1	BBB+	BBB+	Lower Medium Grade
Baa2	BBB	BBB	
Baa3	BBB–	BBB–	
Ba1	BB+	BB+	Noninvestment Grade
Ba2	BB	BB	Speculative
Ba3	BB–	BB–	
B1	B+	B+	Highly Speculative
B2	B	B	
B3	B–	B–	
Caa1	CCC+	CCC	Substantial Risk
Caa2	CCC	—	In Poor Standing
Caa3	CCC–	—	
Ca	—	—	Extremely Speculative
C	—	—	May Be in Default
—	—	D	Default

TOPIC 075: RATING AGENCIES – EXAMPLE

The term credit rating refers to a quantified assessment of a borrower's creditworthiness in general terms or with respect to a particular debt or financial obligation. A credit rating can be

assigned to any entity that seeks to borrow money—an individual, a corporation, a state or provincial authority, or a sovereign government.

Individual credit scores are calculated by credit bureaus such as Experian, Equifax, and TransUnion on a three-digit numerical scale using a form of Fair Isaac Corporation (FICO) credit scoring. Credit ratings for companies and governments are calculated by a credit rating agency such as S&P Global, Moody's, or Fitch Ratings. These rating agencies are paid by the entity seeking a credit rating for itself or one of its debt issues.

Examples

Fitch Ratings

John Knowles Fitch founded the Fitch Publishing Company in 1913, providing financial statistics for use in the investment industry via "The Fitch Stock and Bond Manual" and "The Fitch Bond Book." In 1924, Fitch developed and introduced the AAA through D rating system that has become the basis for ratings throughout the industry.

In the late 1990s, with plans to become a full-service global rating agency, Fitch Ratings merged with IBCA of London, a subsidiary of Fimalac, S.A., a French holding company. Fitch also acquired market competitors Thomson BankWatch and Duff & Phelps Credit Rating Co.

Beginning in 2004, Fitch started to develop operating subsidiaries specializing in enterprise risk management, data services, and finance-industry training with the acquisition of a Canadian company, Algorithmics, and the creation of Fitch Solutions and Fitch Learning.

Moody's Investors Service

John Moody and Company first published Moody's Manual of Industrial and Miscellaneous Securities in 1900. The manual published basic statistics and general information about stocks and bonds of various industries.

From 1903 until the stock market crash of 1907, Moody's Manual was a national publication. In 1909, Moody began publishing Moody's Analyses of Railroad Investments, which added analytical information about the value of securities.

Expanding this idea led to the 1914 creation of Moody's Investors Service, which in the following 10 years would provide ratings for nearly all of the government bond markets at the time. By the 1970s, Moody's began rating commercial paper and bank deposits, becoming the full-scale rating agency that it is today.

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Rating			
Moody's	S&P	Fitch	Definitions
Aaa	AAA	AAA	Prime Maximum Safety
Aa1	AA-	AA-	High Grade High Quality
Aa2	AA	AA	
Aa3	AA-	AA-	
A1	A+	A+	Upper Medium Grade
A2	A	A	
A3	A-	A-	
Baa1	BBB+	BBB+	Lower Medium Grade
Baa2	BBB	BBB	
Baa3	BBB-	BBB-	
Ba1	BB+	BB+	Non Investment Grade
Ba2	BB	BB	Speculative
Ba3	BB-	BB-	
B1	B-	B-	Highly Speculative
B2	B	B	
B3	B-	B-	
Caa1	CCC+	CCC	Substantial Risk
Caa2	CCC	—	In Poor Standing
Caa3	CCC-	—	
Ca	—	—	Extremely Speculative
C	—	—	May be in Default
—	—	DDD	Default
—	—	DD	—
—	D	D	

TOPIC 076: RISK STRUCTURE OF INTEREST RATES – LIQUIDITY

Liquidity

Another attribute of a bond that influences its interest rate is its liquidity. A liquid asset is one that can be quickly and cheaply converted into cash if the need arises. The more liquid an asset is, the more desirable it is (holding everything else constant). U.S.

Treasury bonds are the most liquid of all long-term bonds; because they are so widely traded, they are the easiest to sell quickly, and the cost of selling them is low. Corporate bonds are not as liquid because fewer bonds for any one corporation are traded; thus it can be costly to sell these bonds in an emergency because it might be hard to find buyers quickly.

- Liquidity: the relative ease with which an asset can be converted into cash
- The more liquid an asset is, the more desirable it is (holding everything else constant).
- Treasury bonds are the most liquid of all long-term bonds; because they are so widely traded, they are the easiest to sell quickly, and the cost of selling them is low.
- Corporate bonds are not as liquid because fewer bonds for any one corporation are traded; it can be costly to sell these bonds in an emergency because it might be hard to find buyers quickly.

- The lower liquidity of corporate bonds relative to Treasury bonds increases the spread between the interest rates on these two bonds.

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Lesson 16

RISK STRUCTURE OF INTEREST RATES (CONTINUED)

TOPIC 077: RISK STRUCTURE OF INTEREST RATES - TAX

Income Tax Considerations

The risk structure of interest rates (the relationships among interest rates on bonds with the same maturity) is explained by three factors: default risk, liquidity, and the income tax treatment of a bond's interest payments. As a bond's default risk increases, the risk premium on that bond (the spread between its interest rate and the interest rate on a default-free Treasury bond) rises. The greater liquidity of Treasury bonds also explains why their interest rates are lower than those on less liquid bonds. If a bond has a favorable tax treatment, as do municipal bonds, whose interest payments are exempt from federal income taxes, its interest rate will be lower.

- Interest payments on some government bonds are exempt from income taxes.

Example:

Suppose income tax is 40%.

For every extra 100 rupees of income you earn, you have to pay 40 rupees to the government. If you own a Rs 1,000-face-value Treasury bond that sells for Rs 1,000 and has a coupon payment of Rs 100, you get to keep only Rs 60 of the payment after taxes. Although the bond has a 10% interest rate, you actually earn only 6.0% after taxes.

Summary

- Interest Rates will be LOWER if:
  - default risk is low
  - Liquidity is high, and
  - the income tax is exempted or very low

TOPIC 078: RISK STRUCTURE OF INTEREST RATES- TERM TO MATURITY

We have seen how risk, liquidity, and tax considerations (collectively embedded in the risk structure) can influence interest rates. Another factor that influences the interest rate on a bond is its term to maturity: Bonds with identical risk, liquidity, and tax characteristics may have different interest rates because their times remaining to maturity are different. A plot of the yields on bonds with differing terms to maturity but the same risk, liquidity, and tax considerations is called a yield curve, and it describes the term

structure of interest rates for particular types of bonds, such as government bonds. The Following the Financial News box shows a yield curve for Treasury securities.

- Bonds with identical risk, liquidity, and tax characteristics may have different interest rates because their times remaining to maturity are different.
- A plot of the yields on bonds with differing terms to maturity but the same risk, liquidity, and tax considerations is called a yield curve.
- The three key types of yield curves include normal, inverted and flat. Upward sloping (also known as normal yield curves) is where longer-term bonds have higher yields than short-term ones.
- While normal curves point to economic expansion, downward sloping (inverted) curves point to economic recession.

TOPIC 079: FACTS ABOUT THE YIELD CURVES

Yield curves can be classified as upward-sloping, flat, or downward-sloping (the last sort is often referred to as an inverted yield curve). When yield curves slope upward, which is the most common case, long-term interest rates are above short-term interest rates; when yield curves are flat, short- and long-term interest rates are the same; and when yield curves are inverted, long-term interest rates are below short-term interest rates.

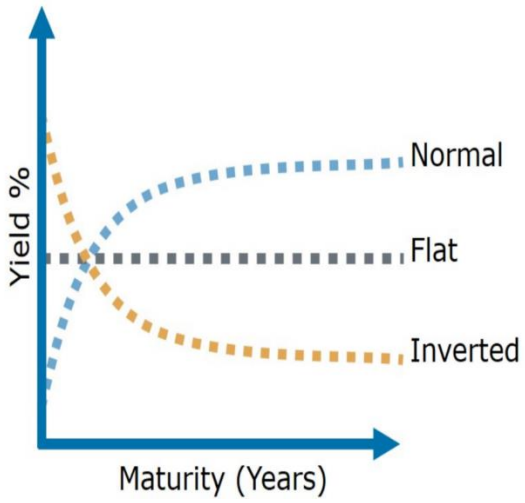
Yield curves can also have more complicated shapes in which they first slope up and then down, or vice versa. Why does the yield curve usually slope upward, but sometimes take on other shapes?

In addition to explaining why yield curves take on different shapes at different times, a good theory of the term structure of interest rates must explain the following three important empirical facts:

1. Interest rates on bonds of different maturities move together over time.

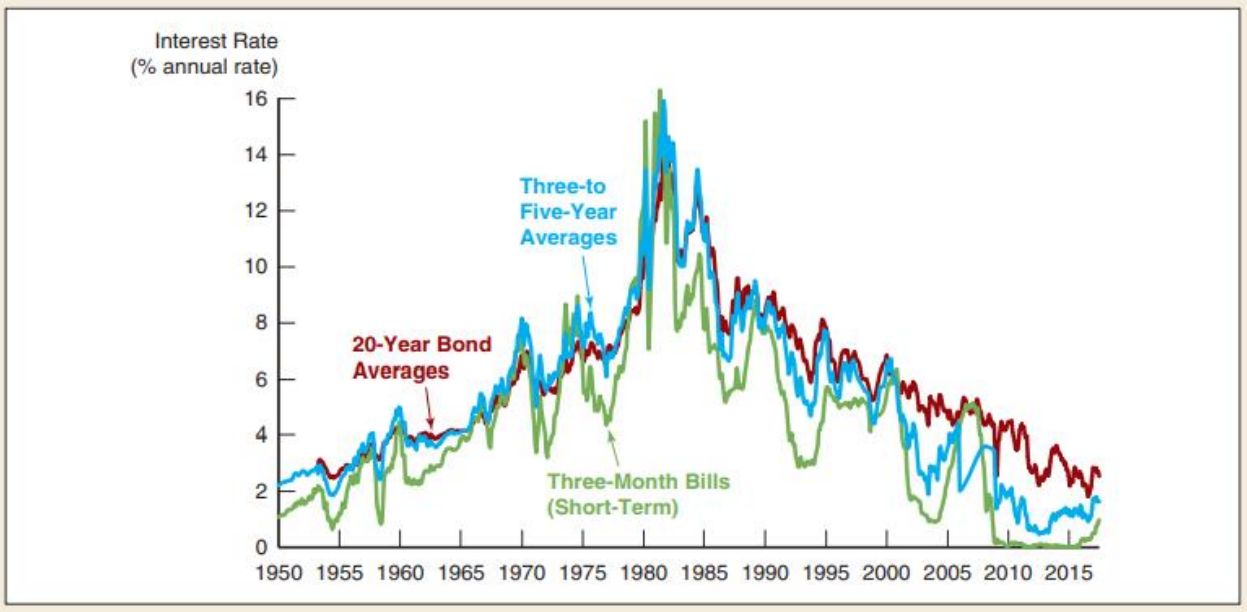
2. When short-term interest rates are low, yield curves are more likely to have an upward slope; when short-term interest rates are high, yield curves are more likely to slope downward and be inverted.

3. Yield curves almost always slope upward.



1. Interest rates on bonds of different maturities move together over time
2. When short-term interest rates are low, yield curves are more likely to have an upward slope; when short-term rates are high, yield curves are more likely to slope downward and be inverted
3. Yield curves almost always slope upward

Movements over Time of Interest Rates on U.S. Government Bonds with Different Maturities



Theories that explain Yield Curve Patterns

Three theories explain the term structure of interest rates—that is, the relationships among interest rates on bonds of different maturities reflected in yield curve patterns

(1) The expectations theory

(2) The segmented markets theory

(3) The liquidity premium theory

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LESSON 17

VARIOUS THEORIES OF TERM STRUCTURE OF INTEREST RATES

TOPIC 080: THE EXPECTATIONS THEORY

- The interest rate on a long-term bond will equal an average of the short-term interest rates that people expect to occur over the life of the long-term bond.
- Buyers of bonds do not prefer bonds of one maturity over another.
- People will not hold any quantity of a bond if its expected return is less than that of another bond with a different maturity
- Bond holders consider bonds with different maturities to be perfect substitutes
- Let the current rate on one-year bond be 6%.
- You expect the interest rate on a one-year bond to be 8% next year.
- Then the expected return for buying two one-year bonds averages (6% + 8%)/2 = 7%.
- The interest rate on a two-year bond must be 7% for you to be willing to purchase it.
- Explains why the term structure of interest rates changes at different times
- Explains why interest rates on bonds with different maturities move together over time (fact 1)
- Explains why yield curves tend to slope up when short-term rates are low and slope down when short-term rates are high (fact 2)
- Cannot explain why yield curves usually slope upward (fact 3)

The expectations theory of the term structure states the following commonsense proposition: The interest rate on a long-term bond will equal the average of the short-term interest rates that people expect to occur over the life of the long-term bond. For example, if people expect that short-term interest rates will be 10%, on average, over the coming five years, the expectations theory predicts that the interest rate on bonds with five years to maturity will be 10%, too.

If short-term interest rates are expected to rise even higher after this five-year period, so that the average short-term interest rate over the coming 20 years is 11%, then the interest rate on 20-year bonds will equal 11% and will be higher than the interest rate on five-year bonds. Thus the expectations theory predicts that interest rates on bonds of different maturities differ because short-term interest rates are expected to have different values at future dates.

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The key assumption behind this theory is that buyers of bonds do not prefer bonds of one maturity over another, so they will not hold any quantity of a bond if its expected return is less than that of another bond with a different maturity. Bonds that have this characteristic are said to be perfect substitutes. In practice, this means that if bonds with different maturities are perfect substitutes, then the expected returns on those bonds must be equal.

To see how the assumption that bonds with different maturities are perfect substitutes leads to the expectations theory, let us consider the following two investment strategies:

1. Purchase a one-year bond, and when it matures in one year, purchase another one year bond.

2. Purchase a two-year bond and hold it until maturity.

Because both strategies must have the same expected return, the interest rate on the two-year bond must equal the average of the two one-year interest rates. For example, let us say that the current interest rate on the one-year bond is 9%, and you expect the interest rate on the one-year bond next year to be 11%. If you pursue the first strategy of buying the two one-year bonds, the expected return over the two years will average out to be (9% + 11%)>2 = 10% per year. You will be willing to hold both the one and two-year bonds only if the expected return per year on the two-year bond equals this return. Therefore, the interest rate on the two-year bond must equal 10%, the average interest rate on the two one-year bonds.

TOPIC 081: SEGMENTED MARKETS THEORY

- This theory sees markets for different-maturity bonds as completely separate and segmented.
- The interest rate on a bond of a particular maturity is then determined by the supply of and demand for that bond.
- The interest rate on a bond of a particular maturity is not affected by expected returns on other bonds with other maturities.
- The key assumption of the segmented markets theory is that bonds of different maturities are not substitutes at all.
- The expected return from holding a bond of one maturity has no effect on the demand for a bond of another maturity.
- This theory is the opposite extreme of the expectations theory, which assumes that bonds of different maturities are perfect substitutes.

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- For example, if you are putting away funds for your young child to go to college in the future, your desired holding period will be much longer, and you will want to hold longer-term bonds.
- The risk-averse investors have short desired holding periods and generally prefer bonds with shorter maturities that have less interest-rate risk.
- The segmented markets theory can explain fact 3, which states that yield curves typically slope upward.
- The demand for long-term bonds is typically relatively lower than that for short-term bonds, long-term bonds will have lower prices and higher interest rates, and hence the yield curve will typically slope upward.

As the name suggests, the segmented markets theory of the term structure sees markets for different-maturity bonds as completely separate and segmented. The interest rate on a bond of a particular maturity is then determined by the supply of and demand for that bond and is not affected by expected returns on other bonds with other maturities.

The key assumption of the segmented markets theory is that bonds of different maturities are not substitutes at all, and so the expected return from holding a bond of one maturity has no effect on the demand for a bond of another maturity. This theory of the term structure is the opposite extreme of the expectations theory, which assumes that bonds of different maturities are perfect substitutes.

TOPIC 082: LIQUIDITY PREMIUM THEORY

The liquidity premium theory of the term structure states that the interest rate on a long-term bond will equal an average of short-term interest rates expected to occur over the life of the long-term bond plus a liquidity premium (also referred to as a term premium) that responds to supply and demand conditions for that bond.
The liquidity premium theory's key assumption is that bonds of different maturities are substitutes, which means that the expected return on one bond does influence the expected return on a bond of a different maturity.

However, the theory allows investors to prefer one bond maturity over another. In other words, bonds of different maturities are

assumed to be substitutes, but not perfect substitutes. Investors tend to prefer shorter-term bonds because these bonds bear less interest-rate risk.

For this reason, investors must be offered a positive liquidity premium to induce them to hold long-term bonds. Thus the expectations theory is modified by adding a positive liquidity premium to the equation that describes the relationship between long- and short-term interest rates. The liquidity premium theory is written as where:

$$i_{nt} = \frac{i_t + i^e_{t+1} + i^e_{t+2} + \ldots + i^e_{t+(n-1)}}{n} + l_{nt}$$

nt is the liquidity (term) premium for the n-period bond at time t, which is always positive and rises with the term to maturity of the bond, n.

- The liquidity premium theory of the term structure states that the interest rate on a long-term bond will equal an average of short-term interest rates expected to occur over the life of the long-term bond plus a liquidity premium (also referred to as a term premium) that responds to supply and demand conditions for that bond.

Key Assumption:

- Bonds of different maturities are substitutes, which means that the expected return on one bond does influence the expected return on a bond of a different maturity.
- However, the theory allows investors to prefer one bond maturity over another.
- Bonds of different maturities are assumed to be substitutes, but not perfect substitutes.
- Investors tend to prefer shorter-term bonds because these bonds bear less interest-rate risk.
- Investors must be offered a positive liquidity premium to induce them to hold longer-term bonds. Thus the expectations theory is modified by adding a positive liquidity premium to the equation that describes the relationship between long- and short-term interest rates.

TOPIC 083: PREFERRED HABITAT THEORY

- Investors have a preference for bonds of one maturity over another.
- They will be willing to buy bonds of different maturities only if they earn a somewhat higher expected return.
- Investors are likely to prefer short-term bonds over longer-term bonds.

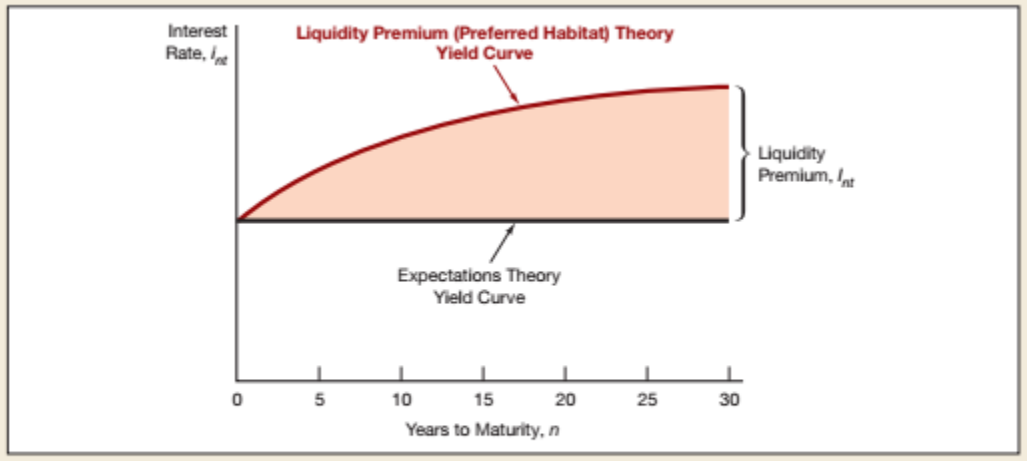
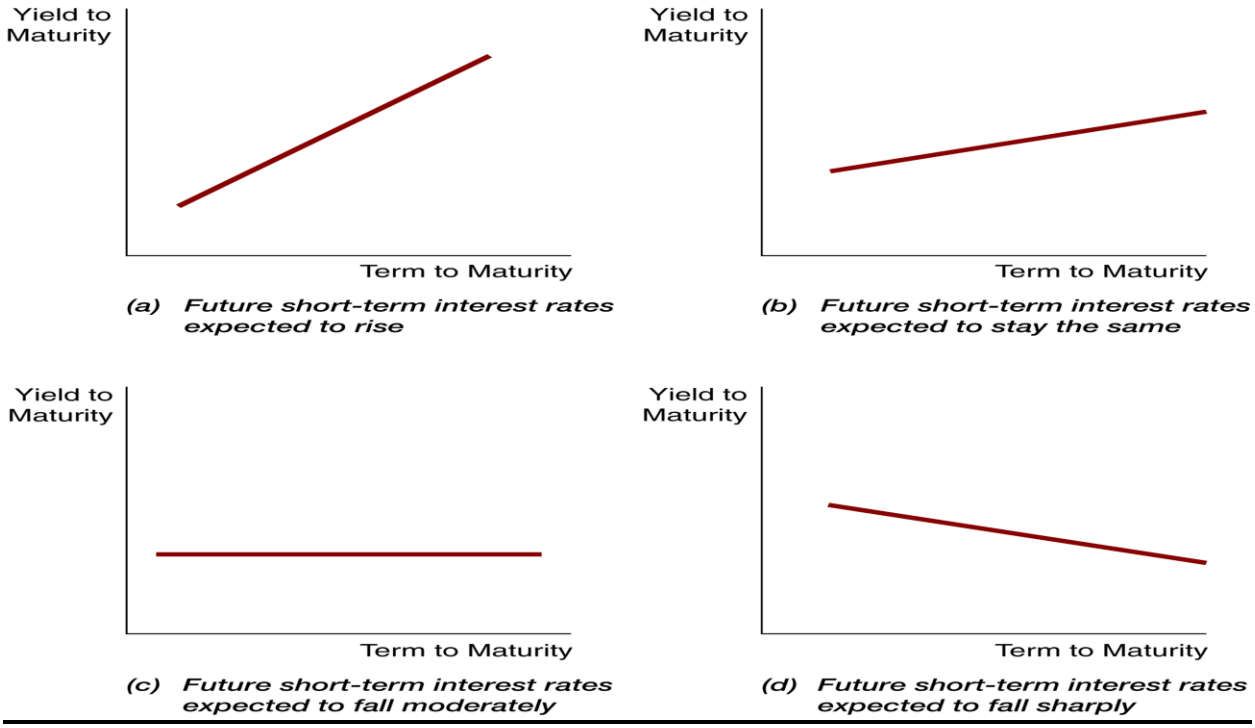


FIGURE 5 **The Relationship Between the Liquidity Premium (Preferred Habitat) and Expectations Theory**

Because the liquidity premium is always positive and grows as the term to maturity increases, the yield curve implied by the liquidity premium and preferred habitat theories is always above the yield curve implied by the expectations theory and has a steeper slope. For simplicity, the yield curve implied by the expectations theory shown here assumes unchanging future one-year interest rates.



It assumes that investors have a preference for bonds of one maturity over bonds of another—a particular bond maturity ("preferred habitat") in which they prefer to invest. Because they prefer bonds of one maturity over bonds of another, they are willing to buy bonds that do not have the preferred maturity (habitat) only if those bonds earn a somewhat higher expected return. Because risk averse investors are likely to prefer the habitat of short-term bonds over that of longer-term bonds, they are willing to hold long-term bonds only if they have higher expected returns. This

reasoning leads to the same Equation 3 implied by the liquidity premium theory, with a term premium that typically rises with maturity.

The relationship between the expectations theory and the liquidity premium and preferred habitat theories is shown in Figure 5. There we see that because the liquidity premium is always positive and typically grows as the term to maturity increases, the yield curve implied by the liquidity premium theory is always above the yield curve implied by the expectations theory and generally has a steeper slope.

(For simplicity, we are assuming that the expectations theory yield curve is flat.) A simple numerical example, similar to the one we used for the expectations hypothesis, further clarifies the liquidity premium and preferred habitat theories given in Equation 3. Again suppose that the one-year interest rates over the next five years are expected to be 5%, 6%, 7%, 8%, and 9%, while investors' preferences for holding short-term bonds means that the liquidity premiums for one- to five-year bonds are 0%, 0.25%, 0.5%, 0.75%, and 1.0%, respectively.

TOPIC 084: INTERPRETING YIELD CURVES, 1980–2017

Figure 7 illustrates several yield curves for U.S. government bonds, for selected dates from 1981 to 2017. What do these yield curves tell us about the public's expectations of future movements of short-term interest rates?

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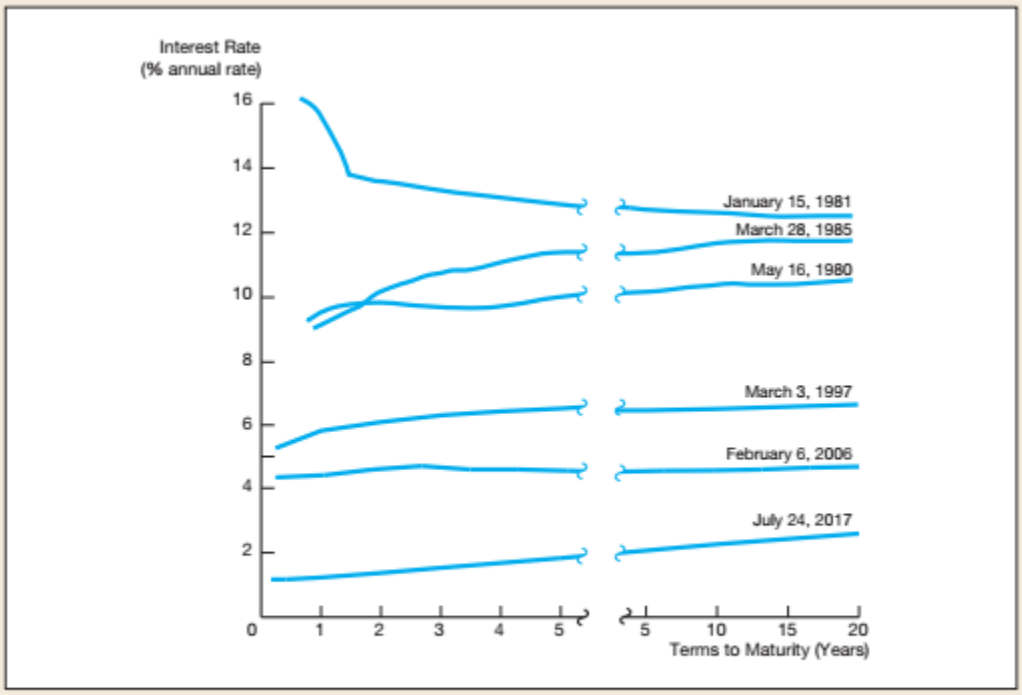


FIGURE 7 **Yield Curves for U.S. Government Bonds**
Yield curves for U.S. government bonds for different dates from 1981 to 2017.
Sources: U.S. Department of the Treasury, https://www.treasury.gov/resource-center/data-chart-center/interest-rates/Pages/TextView.aspx?data=yield.

- Figure shown on the next slide illustrates several yield curves for U.S. government bonds, for selected dates from 1981 to 2017.
- What do these yield curves tell us about the public's expectations of future movements of short-term interest rates?

The steep inverted yield curve that occurred on January 15, 1981 indicated that short-term interest rates were expected to decline sharply in the future. In order for longer-term interest rates, along with their positive liquidity premiums, to be well below short-term interest rates, short-term interest rates must be expected to decline so sharply that their average would be far below the current short-term rate. Indeed, the public's expectations of sharply lower short-term interest rates evident in the yield curve were realized soon after January 15; by March, three-month Treasury bill rates had declined from the 16% level to 13%.

The steep, upward-sloping yield curves that occurred on March 28, 1985, and July 24, 2017, indicated that short-term interest rates were expected to climb in the future. Long-term interest rates are higher than short-term interest rates when short-term interest rates are expected to rise because their average plus the liquidity premium will be higher than the current short-term rate.

The moderately upward-sloping yield curves on May 16, 1980, and March 3, 1997, indicated that short-term interest rates were expected neither to rise nor to fall in the near future. In this case, their average remains the same as the current short-term rate, and the positive liquidity premium for longer-term bonds explains the moderate upward slope of the yield curve. The flat yield curve of February 6, 2006, indicated that short-term interest rates were expected to fall slightly.

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LESSON 18

VARIOUS THEORIES OF TERM STRUCTURE OF INTEREST RATES (CONTINUED)

TOPIC 085: COMMON STOCK

- Common stock is the principal way that listed companies raise equity capital.
- Common stock holders have an ownership interest in the enterprise in the form of a bundle of rights.
- The right to vote at the AGM. The right to be the residual claimant of all cash flows into the company. The right to sell stock.
- Dividends are paid quarterly or six monthly.
- Stockholders may receive dividends from the net earnings of the corporation.
- Dividends are payments made periodically, usually every quarter, to stockholders.
- The board of directors of the firm sets the level of the dividend, usually based on the recommendation of management.
- The term "**common stock**" indicates that the investors in the company do not own any particular assets, but that instead all of the assets are the shared, or **common**, property of all investors. ... When the liquidation happens through bankruptcy, the **common stock** investors typically receive nothing.

Common stock is the principal medium through which corporations raise equity capital. Stockholders—those who hold stock in a corporation—own an interest in the corporation equal to the percentage of outstanding shares they own. This ownership interest gives them a bundle of rights. The most important are the right to vote and to be a residual claimant of all funds flowing into the firm (known as cash flows), meaning that the stockholder receives whatever remains after all other claims against the firm's assets have been satisfied. Stockholders may receive dividends from the net earnings of the corporation. Dividends are payments made periodically, usually every quarter, to stockholders. The board of directors of the firm sets the level of the dividend, usually based on the recommendation of management. In addition, the stockholder has the right to sell the stock.

One basic principle of finance is that the value of any investment is calculated by computing the present value of all cash flows the investment will generate over its life. For example, a commercial building will sell for a price that reflects the net cash flows (rents minus expenses) it

is projected to have over its useful life. Similarly, we value common stock as the value in today's dollars of all future cash flows. The cash flows that a stockholder might earn from stock are dividends, the sales price, or both.

To develop the theory of stock valuation, we begin with the simplest possible scenario: You buy the stock, hold it for one period to get a dividend, then sell the stock. We call this the one-period valuation model.

TOPIC 86: COMPUTING THE PRICE OF A COMMON STOCK

- The value of any investment is calculated by computing the present value of all cash flows the investment will generate over its life.
- We value common stock as the value in today's dollars of all future cash flows.
- The cash flows that a stockholder might earn from stock are dividends, the sales price, or both.
- To answer this question, you need to determine:
- Whether the current price accurately reflects the analyst's forecast.
- To value the stock today, you need to find the present discounted value of the expected cash flows (future payments)
- The discount factor used to discount the cash flows is the required return on investments in equity rather than the interest rate.
- The cash flows consist of one dividend payment plus a final sales price.

Suppose that you have some extra money to invest for one year. After a year, you will need to sell your investment to pay tuition. After watching CNBC or Nightly Business Report on TV, you decide that you want to buy Intel Corp. stock. You call your broker and find that Intel is currently selling for $50 per share and pays $0.16 per year in dividends. The analyst on CNBC predicts that the stock will be selling for $60 in one year. Should you buy this stock?

To answer this question, you need to determine whether the current price accurately reflects the analyst's forecast. To value the stock today, you need to find the present discounted value of the expected cash flows (future payments) using the formula given below. In this equation, the discount factor used to discount the cash flows is the required return on investments in equity rather than the interest rate. The cash flows consist of one dividend payment plus a final sales

price. When these cash flows are discounted back to the present, the following equation computes the current price of the stock:

$$P_0 = \frac{D_1}{(1 + k_e)} + \frac{P_1}{(1 + k_e)} \quad (1)$$

where P_0 = the current price of the stock. The zero subscript refers to time period zero, or the present.
D_1 = the dividend paid at the end of year 1
k_e = the required return on investments in equity
P_1 = the price at the end of year 1; the predicted sales price of the stock

TOPIC 087: THE ONE-PERIOD VALUATION MODEL – EXAMPLE

The One-Period Valuation Model

- Suppose that you have some extra money to invest for one year. You decided that you want to buy Lucky Cement stock.
- You call your broker and find that Lucky Cement share is currently selling for Rs. 50 per share and pays Rs. 0.16 per year in dividends. The analyst predicts that the stock will be selling for Rs. 60 in one year. Should you buy this stock?
- Suppose the expected return on investment is 12% (k = 0.12).
- Dividend is Rs. 0.16 per year (D= 0.16)
- The forecasted share price after the end of one year is Rs. 60 ($\boldsymbol{P_1}$ = Rs. 60).
- $\boldsymbol{P_0} = \frac{\mathbf{0.16}}{\mathbf{1+0.12}} + \frac{\boldsymbol{Rs.}\,\mathbf{60}}{\mathbf{1+0.12}}$
-
- = Rs. 0.14 +Rs 53.57
- = Rs. 53.71

Example

- Market price < Present Value
- Be aware that the stock may be selling for less than Rs. 53.71, because other investors have placed a greater risk on the cash flows or estimated the cash flows to be less than you did.

To see how Equation 1 works, let's compute the price of the Intel stock if, after careful consideration, you decide that you would be satisfied to earn a 12% return on the investment. If you have decided that $k_e = 0.12$, are told that Intel pays \$0.16 per year in dividends ($D_1 = 0.16$), and forecast the share price of \$60 at the end of year 1 ($P_1 = \$60$), you get the following result from Equation 1:

$$P_0 = \frac{0.16}{1 + 0.12} + \frac{\$60}{1 + 0.12} = \$0.14 + \$53.57 = \$53.71$$

TOPIC 088: THE GENERALIZED DIVIDEND VALUATION MODEL

- The value of a stock today is the present value of all future cash flows.
- The only cash flows that an investor will receive are dividends and a final sales price when the stock is ultimately sold in period n.
- The important information is knowing Pn.

Using the present value concept, we can extend the one-period dividend valuation model to any number of periods: The value of a stock today is the present value of all future cash flows. The only cash flows that an investor will receive are dividends and a final sales price when the stock is ultimately sold in period n. The generalized multiperiod formula for stock valuation can be written as

$$P_0 = \frac{D_1}{(1 + k_e)^1} + \frac{D_2}{(1 + k_e)^2} + \ \ldots \ + \frac{D_n}{(1 + k_e)^n} + \frac{P_n}{(1 + k_e)^n} \qquad (2)$$

where P_n = the price of the stock at the end of period n
D_i = the dividend paid at the end of year i

If you tried to use Equation 2 to find the value of a share of stock, you would soon realize that you must first estimate the value the stock will have at some point in the future before you can estimate its value today. In other words, you must find P_n before you can find P_0. However, if P_n is far in the future, it will not affect P_0. For example, the present value of a share of stock that sells for \$50 seventy-five years from now, using a 12% discount rate, is just one cent [$\$50/(1.12^{75}) = \$0.01$]. This reasoning implies that the current value of a share of stock can be calculated as simply the present value of the future dividend stream. The **generalized dividend model** is given in Equation 3. Note that it is the same formula as Equation 2, but without the final sales price:

$$P_0 = \sum_{t=1} \frac{D_t}{(1 + k_e)^t} \qquad (3)$$

Consider the implications of Equation 3 for a moment. The generalized dividend model states that the price of a stock is determined only by the present value of the dividends and that nothing else matters. Many stocks do not pay dividends, so how is it that these stocks have value? *Buyers of the stock expect that the firm will pay dividends someday*. Most of the time a firm institutes dividends as soon as it has completed the rapid growth phase of its life cycle.

TOPIC 089: THE THEORY OF RATIONAL EXPECTATIONS

The analysis of stock price evaluation we outlined in the previous section depends on people's expectations—especially expectations of cash flows. Indeed, it is difficult to think of any sector of the economy in which expectations are not crucial; this is why it is important to examine how expectations are formed. We do so by outlining the theory of rational expectations, currently the most widely used theory to describe the formation of business and consumer expectations.

In the 1950s and 1960s, economists regularly viewed expectations as formed from past experience only. Expectations of inflation, for example, were typically viewed as being an average of past inflation rates. This view of expectation formation, called adaptive expectations, suggests that changes in expectations will occur slowly over time, as data for a variable evolve. So, if inflation had formerly been steady at a 5% rate, expectations of future inflation would be 5%, too. If inflation rose to a steady rate of 10%, expectations of future inflation would rise toward 10%, but slowly: In the first year, expected inflation might rise only to 6%; in the second year, to 7%; and so on.

The adaptive expectations hypothesis has been faulted on the grounds that people use more information than just past data on a single variable to form their expectations of that variable. Their expectations of inflation will almost surely be affected by their predictions of future monetary policy, as well as by current and past monetary policy.

In addition, people often change their expectations quickly in the light of new information. To address these objections to the validity of adaptive expectations, John Muth developed an alternative theory of expectations, called rational expectations, which can be stated as follows: Expectations will be identical to optimal forecasts (the best guess of the future) using all available information.

What exactly does this mean? To explain it more clearly, let's use the theory of rational expectations to examine how expectations are formed in a situation that most of us will encounter at some point in our lifetime: our drive to work. Suppose that if Joe Commuter travels when it is not rush hour, his trip takes an average of 30 minutes. Sometimes his trip takes 35 minutes; other times, 25 minutes; but the average, non rush-hour driving time is 30 minutes. If, however, Joe leaves for work during the rush hour, it takes him, on average, an additional 10 minutes to get to

work. Given that he leaves for work during the rush hour, the best guess of his driving time—the optimal forecast—is 40 minutes.

If the only information available to Joe before he leaves for work related to his driving time is that he is leaving during the rush hour, what does rational expectations theory allow you to predict about Joe's expectations of his driving time? Since the best guess of his driving time, using all available information, is 40 minutes, Joe's expectation should be the same. Clearly, an expectation of 35 minutes would not be rational, because it is not equal to the optimal forecast, or the best guess of the driving time.

Rational Expectations

- In the 1950s and 1960s, economists regularly viewed expectations as formed from past experience only.
- Stock price valuation depends on expectations
- But how are expectations formed?
- One model of expectations formation is the theory of rational expectations.
- John Muth 'Expectations will be identical to optimal forecasts (the best guess of the future) using all available information'

Adaptive Expectations

- Expectations of inflation, for example, were typically viewed as being an average of past inflation rates. This view of expectation formation, called adaptive expectations, suggests that changes in expectations will occur slowly over time, as data for a variable evolve.

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LESSON 19

EFFICIENT MARKET HYPOTHESIS

TOPIC 090: EFFICIENT MARKET HYPOTHESIS

- It is an application of rational expectations to the pricing of stocks and other securities.
- It was developed from economist Eugene Fama's Ph.D. dissertation in the 1960s.
- It is based on the assumption that prices of securities in financial markets fully reflect all available information.
- Return can be calculated using formula:

$$R = \frac{P_{t+1} - P_t + C}{P_t}$$

- R = rate of return on the security held from time t to time t + 1
- Pt + 1 = price of the security at time t + 1 (the end of the holding period)
- Pt = price of the security at time t (the beginning of the holding period)
- C = cash payment (coupon or dividend payments) made in the period t to t + 1
- Expected Return can be calculated using:
- $R^e = \frac{P^e_{t+1} - P_t + C}{P_t}$
- R^e = expected rate of return on the security held from time t to time t + 1
- P^e_{t+1} = expected price of the security at time t + 1 (the end of the holding period)
- The efficient market hypothesis views expectations of future prices as equal to optimal forecasts using all currently available information.
- Simply, we can put it as:

$$R^e = R^*$$

- Equilibrium condition

$$R^e = R^* = R^{of}$$

While monetary economists were developing the theory of rational expectations, financial economists were developing a parallel theory of expectations formation for financial markets. It led them to the same conclusion as that of the rational expectations theorists: Expectations in financial markets are equal to optimal forecasts using all available information. Although financial economists, such as Eugene Fama, later a winner of the Nobel Prize in economics, gave their

theory another name, calling it the efficient market hypothesis, in fact their theory is just an application of rational expectations to the pricing of stocks and other securities.

The efficient market hypothesis is based on the assumption that prices of securities in financial markets fully reflect all available information. The rate of return from holding a security equals the sum of the capital gain on the security (the change in the price), plus any cash payments, divided by the initial purchase price of the security.

$$R = \frac{P_{t+1} - P_t + C}{P_t} \tag{7}$$

where
R = rate of return on the security held from time t to time $t + 1$ (say, the end of 2018 to the end of 2019)
P_{t+1} = price of the security at time $t + 1$, the end of the holding period
P_t = price of the security at time t, the beginning of the holding period
C = cash payment (coupon or dividend payments) made in the period t to $t + 1$

Let's look at the expectation of this return at time t, the beginning of the holding period. Because the current price P_t and the cash payment C are known at the outset, the only variable in the definition of the return that is uncertain is the price next period, P_{t+1}.[5] Denoting expectation of the security's price at the end of the holding period as P^e_{t+1}, the expected return R^e is

$$R^e = \frac{P^e_{t+1} - P_t + C}{P_t}$$

The efficient market hypothesis views expectations of future prices as equal to optimal forecasts using all currently available information. In other words, the market's expectations of future securities prices are rational, so that

$$P^e_{t+1} = P^{of}_{t+1}$$

which in turn implies that the expected return on the security will equal the optimal forecast of the return:

$$R^e = R^{of} \tag{8}$$

The expected return on a security R^e equals the equilibrium return R^*, which equates the quantity of the security demanded to the quantity supplied; that is,

$$R^e = R^* \quad (9)$$

The academic field of finance explores the factors (risk and liquidity, for example) that influence the equilibrium returns on securities. For our purposes, it is sufficient to know that we can determine the equilibrium return and thus determine the expected return with the equilibrium condition.

We can derive an equation to describe pricing behavior in an efficient market by using the equilibrium condition to replace R^e with R^* in the rational expectations equation (Equation 8). In this way, we obtain

$$R^{of} = R^* \quad (10)$$

This equation tells us that ***current prices in a financial market will be set so that the optimal forecast of a security's return using all available information equals the security's equilibrium return***. Financial economists state it more simply: In an efficient market, a security's price fully reflects all available information.

TOPIC 091: RATIONALE BEHIND EFFICIENT MARKET HYPOTHESIS

- Suppose the normal return on ABC common stock is 10% at an annual rate, and its current price Pt is lower than the optimal forecast of tomorrow's price P_{t+1}^{of}, or Pt = 10% and P_{t+1}^{of} = 50%.
- i.e. $R^* < R^{of}$
- Investors will buy more ABC common stocks; Pt will increase relative to its expected future price P_{t+1}^{of}, => R^{of} will fall
- When the current price has risen sufficiently so that $R^{of} = R^*$, the efficient market condition is satisfied, the buying of ABC stock will stop, and the unexploited profit opportunity will disappear.
- A security for which the optimal forecast of the return is -5% and the equilibrium return is 10% ($R^* > R^{of}$) would be a poor investment.
- Investors will sell the security; its price will fall until R^{of} rises to the level of R* and the efficient market condition is satisfied.

$$R^{of} > R^* \rightarrow P_t \uparrow \rightarrow R^{of} \downarrow$$

$$R^{of} < R^* \rightarrow P_t \downarrow \rightarrow R^{of} \uparrow$$

$$\text{until}$$

$$R^{of} = R^*$$

Hence, in an efficient market, all unexploited profit opportunities will be eliminated.

To see why the efficient market hypothesis makes sense, we make use of the concept of arbitrage, in which market participants (arbitrageurs) eliminate unexploited profit opportunities, that is, returns on a security that are larger than what is justified by the characteristics of that security. Arbitrage is of two types: pure arbitrage, in which the elimination of unexploited profit opportunities involves no risk, and the type of arbitrage we discuss here, in which the arbitrageur takes on some risk when eliminating the unexploited profit opportunities.

To see how arbitrage leads to the efficient market hypothesis given a security's risk characteristics, let's look at an example. Suppose the normal return on ExxonMobil common stock is 10% at an annual rate, and its current price Pt is lower than the optimal forecast of tomorrow's price Po t f+ 1, so that the optimal forecast of the return at an annual rate is 50%, which is greater than the equilibrium return of 10%.

We are now able to predict that, on average, ExxonMobil's return will be abnormally high, so there is an unexploited profit opportunity. Knowing that, on average, you can earn an abnormally high rate of return on ExxonMobil stock (because Rof 7 R*), you will buy more, which will in turn drive up the stock's current price Pt relative to its expected future price Po t f+ 1, thereby lowering Rof. When the current price has risen sufficiently so that Rof equals R* and the efficient market condition (Equation 10) is satisfied, the buying of ExxonMobil stock will stop, and the unexploited profit opportunity will disappear.

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Similarly, a security for which the optimal forecast of the return is −5% and the equilibrium return is 10% ($R^{of} < R^*$) would be a poor investment because, on average, the security earns less than the equilibrium return. In such a case, you would sell the security and drive down its current price relative to the expected future price until R^{of} rose to the level of R^* and the efficient market condition was again satisfied. Our discussion can be summarized as follows:

$$R^{of} > R^* \rightarrow P_t\uparrow \rightarrow R^{of}\downarrow$$

$$R^{of} < R^* \rightarrow P_t\downarrow \rightarrow R^{of}\uparrow$$

until

$$R^{of} = R^*$$

Another way to state the efficient market condition is this: ***In an efficient market, all unexploited profit opportunities will be eliminated.***

An extremely important factor in this reasoning is that ***not everyone in a financial market must be well informed about a security or have rational expectations for its price to be driven to the point at which the efficient market condition holds.*** Financial markets are structured so that many participants can play. As long as a few people (often referred to as the "smart money") keep their eyes open for unexploited profit opportunities, they will eliminate the profit opportunities that appear, because in doing so they make a profit. The efficient market hypothesis makes sense because it does not require that everyone in a market be cognizant of what is happening to every security.

TOPIC 092: RANDOM-WALK BEHAVIOR OF STOCK PRICES

- Random Walk describes the movements of a variable whose future values cannot be predicted (are random) because, given today's value, the value of the variable is just as likely to fall as it is to rise.
- Implication of the EMH is that stock prices should approximately follow a random walk.
- Future changes in stock prices should, for all practical purposes, be unpredictable.

Example – Pakistan Stock Exchange

1. Pakistan's equity market expanded at a compound annual growth rate of more than 11% (in terms of US$) since 1992.
2. The KSE 100 increased 3904 points or 8.92% since the beginning of 2021.



Source: https://tradingeconomics.com/pakistan/stock-market

The term random walk describes the movements of a variable whose future values cannot be predicted (are random) because, given today's value, the value of the variable is just as likely to fall as it is to rise. An important implication of the efficient market hypothesis is that stock prices should approximately follow a random walk; that is future changes in stock prices should, for all practical purposes, be unpredictable.

The random-walk implication of the efficient market hypothesis is the one most commonly mentioned in the press because it is the most readily comprehensible to the public. In fact, when people mention the "random-walk theory of stock prices," they are in reality referring to the efficient market hypothesis.

The case for random-walk stock prices can be demonstrated. Suppose people could predict that the price of Happy Feet Corporation (HFC) stock would rise 1% in the coming week. The predicted rate of capital gains and rate of return on HFC stock would then exceed 50% at an annual rate. Since this is very likely to be far higher than the equilibrium rate of return on HFC stock (Rof > R*), the efficient market hypothesis indicates that people would immediately buy this stock and bid up its current price. The action would stop only when the predictable change in the price dropped to near zero, so that Rof = R*.

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Similarly, if people could predict that the price of HFC stock would fall by 1%, the predicted rate of return would be negative (Rof < R*), and people would immediately sell the stock. The current price would fall until the predictable change in the price rose back to near zero, where the efficient market condition again would hold.

The efficient market hypothesis suggests that the predictable change in stock prices will be near zero, leading to the conclusion that stock prices will generally follow a random walk.6 As the Global Box "Should Foreign Exchange Rates Follow a Random Walk?" indicates, the efficient market hypothesis suggests that foreign exchange rates should also follow a random walk.

TOPIC 093: FORMS OF EFFICIENT MARKET HYPOTHESIS (EMH)

There are three tenets to the efficient market hypothesis:

- The weak
- The semi-strong
- The strong

Weak Form Efficiency

- Current stock prices reflect all available information.
- Past performance is irrelevant to what the future holds for the stock.
- Therefore, it assumes that technical analysis can't be used to achieve returns.

Semi-Strong Form Efficiency

- The stock prices are factored into all information that is publicly available.
- No investor can earn excess returns from trading rules based on any publicly available information.
- Therefore, investors can't use fundamental analysis to beat the market and make significant gains.
- Market reaction to new publicly available information is instantaneous and unbiased.
- No over- or under-reaction.
- Hence, ffundamental analysis based on publicly available information shouldn't result in abnormal returns.

Strong Form Efficiency

- No investor can earn excess returns using any information – public or private.
- Even private information! – Insider trading is ineffective
- The market is perfect, and making excessive profits from the market is next to impossible.

Though the efficient market hypothesis (EMH), as a whole, theorizes that the market is generally efficient, the theory is offered in three different versions: weak; semi-strong; and strong. The basic efficient market hypothesis posits that the market cannot be beaten because it incorporates all important determining information into current share prices. Therefore, stocks trade at the fairest value, meaning that they can't be purchased undervalued or sold overvalued. The theory determines that the only opportunity investors have to gain higher returns on their investments is through purely speculative investments that pose a substantial risk.

Weak Form

The three versions of the efficient market hypothesis are varying degrees of the same basic theory. The weak form suggests that today's stock prices reflect all the data of past prices and that no form of technical analysis can be effectively utilized to aid investors in making trading decisions.

Advocates for the weak form efficiency theory believe that if the fundamental analysis is used, undervalued and overvalued stocks can be determined, and investors can research companies' financial statements to increase their chances of making higher-than-market-average profits.

Semi-Strong Form

The semi-strong form efficiency theory follows the belief that because all information that is public is used in the calculation of a stock's current price, investors cannot utilize either technical or fundamental analysis to gain higher returns in the market.

Those who subscribe to this version of the theory believe that only information that is not readily available to the public can help investors boost their returns to a performance level above that of the general market.

Strong Form

The strong form version of the efficient market hypothesis states that all information—both the information available to the public and any information not publicly known—is completely accounted for in current stock prices, and there is no type of information that can give an investor an advantage on the market.

Advocates for this degree of the theory suggest that investors cannot make returns on investments that exceed normal market returns, regardless of information retrieved or research conducted.

TOPIC 094: EVIDENCE IN FAVOR OF EFFICIENT MARKET HYPOTHESIS

- Empirical studies confirm that stock pickers or mutual fund managers cannot outperform the market over a long period of time.
- The EMH states that stock prices reflect all available information so that earnings announcements that are already known will not affect stock prices when the announcements are made.
- Only 'new' news causes stock prices to change.
- Future changes in stock prices should follow a random walk (future changes in prices are unpredictable).

If you follow the stock market, you might have noticed a puzzling phenomenon: When good news about a corporation, such as a particularly favorable earnings report, is announced, the price of the corporation's stock frequently does not rise. The efficient market hypothesis explains this phenomenon.

Because changes in stock prices are unpredictable, when information is announced that has already been expected by the market, stock prices will remain unchanged. The announcement does not contain any new information that would lead to a change in stock prices. If this were not the case, and the announcement led to a change in stock prices, it would mean that the change was predictable.

Because such a scenario is ruled out in an efficient market, **stock prices will respond to announcements only when the information being announced is new and unexpected**. If the news is expected, no stock

price response will occur. This is exactly what the evidence shows: Stock prices do reflect publicly available information.

Sometimes an individual stock price declines when good news is announced. Although this seems somewhat peculiar, it is completely consistent with the workings of an efficient market. Suppose that although the announced news is good, it is not as good as expected. HFC's earnings may have risen by 15%, but if the market expected earnings to rise by 20%, the new information is actually unfavorable, and the stock price declines.

TOPIC 095: EVIDENCE AGAINST EFFICIENT MARKET HYPOTHESIS

- Size effect – Studies show that small firms earn abnormal returns over long periods.
- January effect – studies have confirmed an abnormal price rise from December to January.
- Market overreaction – over/under shooting following 'new' news.
- Excessive volatility – fluctuations in stock prices are greater than the fluctuations in the fundamentals.
- Mean reversion – low returns stock tend to be followed by high returns and vice versa. Stocks that have done poorly in the past tend to do better in the future. But the evidence on this is controversial.
- Lag in effect of 'new' news – stock prices do not always react to news instantly.
- They show some evidence of autocorrelation.
- If capital markets are informationally efficient, why people take different views about the same future?

On October 19, 1987, dubbed "Black Monday," the Dow Jones Industrial Average declined by more than 20%, the largest one-day decline in U.S. history. The collapse of the high-tech companies' share prices from their peaks in March 2000 caused the heavily tech-laden NASDAQ index to fall from about 5,000 in March 2000 to about 1,500 in 2001 and 2002, a decline of well over 60%. These stock market crashes caused many economists to question the validity of the efficient market hypothesis. These economists do not believe that a rational marketplace could have produced such a massive swing in share prices.

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To what degree should these stock market crashes make us doubt the validity of the efficient market hypothesis? Nothing in efficient markets theory rules out large changes in stock prices. A large change in stock prices can result from new information that produces a dramatic decline in optimal forecasts of the future valuation of firms. However, economists are hard pressed to find fundamental changes in the economy that would have caused the Black Monday and tech crashes. One lesson from these crashes is that factors other than market fundamentals probably have an effect on asset prices. There are good reasons to believe that impediments to the proper functioning of financial markets do exist.

Hence these crashes have convinced many economists that the stronger version of market efficiency, which states that asset prices reflect the true fundamental (intrinsic) value of securities, is incorrect. These economists attribute a large role in determination of stock and other asset prices to market psychology and to the institutional structure of the marketplace.

However, nothing in this view contradicts the basic reasoning behind rational expectations or the efficient market hypothesis—that market participants eliminate unexploited profit opportunities. Even though stock market prices may not always solely reflect market fundamentals, as long as market crashes are unpredictable, the basic premises of efficient markets theory hold.

However, other economists believe that market crashes and bubbles suggest that unexploited profit opportunities may exist and that the efficient market hypothesis might be fundamentally flawed. The controversy over the efficient market hypothesis continues.

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LESSON 20

ASYMMETRIC INFORMATION AND THE LEMONS PROBLEM

TOPIC 096: BEHAVIORAL FINANCE

- Doubts about the efficiency of financial markets, triggered by the stock market crash of 1987, led economists such as Nobel Prize winner Robert Shiller to develop a new field of study called behavioral finance.
- It applies concepts from other social sciences, such as anthropology, sociology, and particularly psychology, to explain the behavior of securities prices.
- One of the arguments of EMH is that unexploited profit is eliminated by knowledgeable investors. For this to happen they must engage in short selling.
- Short selling – borrowing the stock from brokers and then sell it in the market with the aim of making a profit by buying the stock back at a lower price.
- Psychologists suggest that people are subject to 'loss aversion'. They are unhappy from losses than happy with equivalent gains. Because the potential losses can be huge from short selling in reality short selling occurs only in special circumstances.
- Psychologists also find that people tend to be overconfident in their own judgements.
- They trade on their beliefs rather than on pure facts.
- Overconfidence and social contagion (fad) explain the creation of speculative bubbles.

As we have seen, the efficient market hypothesis assumes that unexploited profit opportunities are eliminated by "smart money" market participants. But can smart money dominate ordinary investors so that financial markets are efficient? Specifically, the efficient market hypothesis suggests that smart money participants will sell when a stock price goes up irrationally, with the result that the stock price falls back down to a level that is justified by fundamentals. For this to occur, smart money investors must be able to engage in short sales; that is, they must borrow stock from brokers and then sell it in the market, with the aim of earning a profit by buying the stock back again ("covering the short") after it has fallen in price.

Work by psychologists, however, suggests that people are subject to loss aversion:

They are unhappy when they suffer losses than they are happy when they achieve gains. Short sales can result in losses far

in excess of an investor's initial investment if the stock price climbs sharply higher than the price at which the short sale is made (and losses might be unlimited if the stock price climbs to astronomical heights).

Loss aversion can thus explain an important phenomenon: Very little short selling actually takes place. Short selling may also be constrained by rules restricting it, because it seems unsavory for someone to make money from another person's misfortune. The existence of so little short selling can explain why stock prices are sometimes overvalued. That is, the lack of enough short selling means that smart money does not drive stock prices back down to their fundamental value. Psychologists have also found that people tend to be overconfident in their own judgments. As a result, investors tend to believe that they are smarter than other investors. Because investors are willing to assume that the market typically doesn't get it right, they trade on their beliefs rather than on pure facts. This theory may explain why securities markets have such a large trading volume—something that the efficient market hypothesis does not predict.

Overconfidence and social contagion (fads) provide an explanation for stock market bubbles. When stock prices go up, investors attribute their profits to their intelligence and talk up the stock market. This word-of-mouth enthusiasm and glowing media reports then can produce an environment in which even more investors think stock prices will rise in the future. The result is a positive feedback loop in which prices continue to rise, producing a speculative bubble, which finally crashes when prices get too far out of line with fundamentals.10 The field of behavioral finance is a young one, but it holds out hope that we might be able to explain some features of securities markets' behavior that are not well explained by the efficient market hypothesis.

TOPIC 097: BASIC FACTS ABOUT FINANCIAL STRUCTURE

- It includes banks, insurance companies, mutual funds, stock and bond markets, and so on—all of which are regulated by government.
- The financial system channels trillions of dollars per year from savers to people with productive investment opportunities.
- There are EIGHT basic facts of a financial structure.

- Stocks are not the most important source of external financing for businesses.
- Issuing marketable debt and equity securities is not the primary way in which businesses finance their operations.
- Indirect finance, is many times more important than direct finance.
- Financial intermediaries, particularly banks, are the most important source of external funds used to finance businesses.
- The financial system is among the most heavily regulated sectors of the economy.
- Only large, well-established corporations have easy access to securities markets to finance their activities
- Collateral is a prevalent feature of debt contracts for both households and businesses.
- Collateral is property that is pledged to a lender to guarantee payment in the event that the borrower is unable to make debt payments.
- Collateralized debt (secured debt) is the predominant form of household debt and is widely used in business borrowing as well.
- Debt contracts typically are extremely complicated legal documents that place substantial restrictions on the behavior of the borrower.

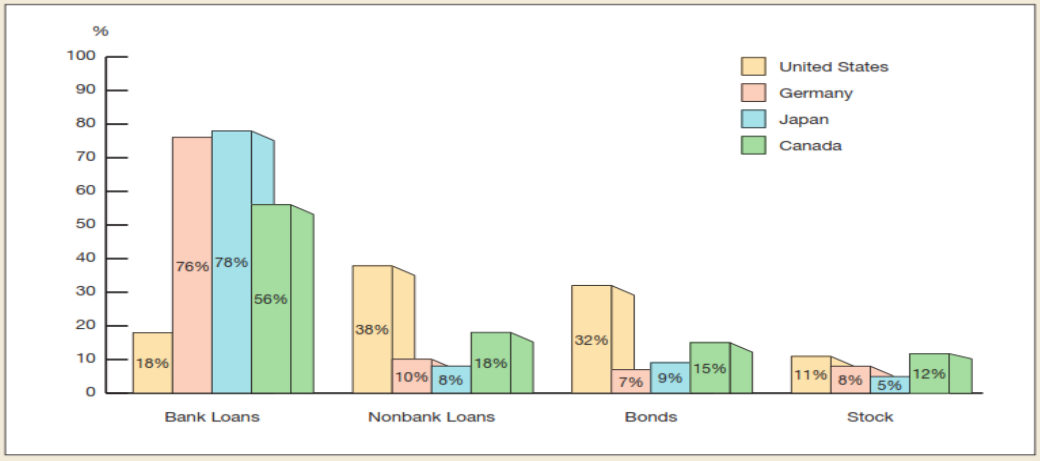


FIGURE 1 **Sources of External Funds for Nonfinancial Businesses: A Comparison of the United States with Germany, Japan, and Canada**

The *Bank Loans* category is made up primarily of loans from depository institutions; the *Nonbank Loans* are primarily loans by other financial intermediaries; the *Bonds* category includes marketable debt securities, such as corporate bonds and commercial paper; and *Stock* consists of new issues of new equity (stock market shares).

Source: Andreas Hackethal and Reinhard H. Schmidt, "Financing Patterns: Measurement Concepts and Empirical Results," Johann Wolfgang Goethe-Universitat Working Paper No. 125, January 2004. The data are for 1970–2000 and are gross flows as percentages of the total, not including trade and other credit data, which are not available.

The financial system is complex in structure and function throughout the world. It includes many different types of institutions: banks, insurance companies, mutual funds, stock and bond markets, and so on—all of which are regulated by government.

The financial system channels trillions of dollars per year from savers to people with productive investment opportunities. If we take a close look at financial structure all over the world, we find eight basic facts, some of which are quite surprising, that we must explain if we are to understand how the financial system works.

The bar chart in Figure 1 shows how American businesses financed their activities using external funds (those obtained from outside the business itself) in the period 1970–2000 and compares U.S. data with data for Germany, Japan, and Canada. The conclusions drawn from this period still hold true today. The Bank Loans category is made up primarily of loans from depository institutions; Nonbank Loans are primarily loans by other financial intermediaries; the Bonds category includes marketable debt securities, such as corporate bonds and commercial paper; and Stock consists of new issues of equity (stock market shares).
Now let's explore the eight facts.

1. Stocks are not the most important source of external financing for businesses. Because so much attention in the media is focused on the stock market, many people have the impression that stocks are the most important source of financing for American corporations. However, as we can see from the bar chart in Figure 1, the stock market accounted for only a small fraction of the external financing of American businesses in the 1970–2000 period: 11%. Similarly, low figures apply for the other countries presented in Figure 1. Why is the stock market less important than other sources of financing in the United States and other countries?

2. Issuing marketable debt and equity securities is not the primary way in which businesses finance their operations. Figure 1 shows that bonds are a far more important source of financing in the United States than stocks are (32% versus 11%). However, stocks and bonds combined (43%), which make up the total share of marketable securities, still supply less than one-half of the external funds needed by U.S. corporations to finance their activities. The fact that issuing marketable securities is not the most important source of financing is true elsewhere in the world as well. Indeed, as we see in Figure 1, other countries have a much smaller share of external financing supplied by marketable securities than does the United States. Why don't businesses use marketable securities more extensively to finance their activities?

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3. Indirect finance, which involves the activities of financial intermediaries, is many times more important than direct finance, in which businesses raise funds directly from lenders in financial markets. Direct finance involves the sale to households of marketable securities, such as stocks and bonds. The 43% share of stocks and bonds used as a source of external financing for American businesses actually greatly overstates the importance of direct finance in our financial system. Since 1970, less than 5% of newly issued corporate bonds and commercial paper, and less than one-third of new issues of stocks, have been sold directly to American households. The rest of these securities have been bought primarily by financial intermediaries such as insurance companies, pension funds, and mutual funds. These figures indicate that direct finance is used in less than 10% of the external funding of American business. Because in most countries marketable securities are an even less important source of finance than in the United States, direct finance is also far less important than indirect finance in the rest of the world. Why are financial intermediaries and indirect finance so important in financial markets? In recent years, however, indirect finance has been declining in importance. Why is this happening?

4. Financial intermediaries, particularly banks, are the most important source of external funds used to finance businesses. As we can see in Figure 1, the primary source of external funds for businesses throughout the world is loans made by banks and other nonbank financial intermediaries, such as insurance companies, pension funds, and finance companies (56% in the United States, but more than 70% in Germany, Japan, and Canada). In other industrialized countries, bank loans are the largest category of sources of external finance (more than 70% in Germany and Japan, and more than 50% in Canada). Thus the data suggest that banks in these countries have the most important role in financing business activities. In developing countries, banks play an even more important role in the financial system than they do in industrialized countries. What makes banks so important to the workings of the financial system? Although banks remain important, their contributions to external funds for businesses have been declining in recent years. What is driving this decline?

5. The financial system is among the most heavily regulated sectors of the economy. The

financial system is heavily regulated in the United States and in all other developed countries. Governments regulate financial markets primarily to promote the provision of information and to ensure the soundness (stability) of the financial system. Why are financial markets so extensively regulated throughout the world?

6. Only large, well-established corporations have easy access to securities markets to finance their activities. Individuals and smaller businesses that are not well established are less likely to raise funds by issuing marketable securities. Instead, they most often obtain their financing from banks. Why do only large, well-known corporations find it easier to raise funds in securities markets?

7. Collateral is a prevalent feature of debt contracts for both households and businesses. Collateral is property that is pledged to a lender to guarantee payment in the event that the borrower is unable to make debt payments. Collateralized debt (also known as secured debt to contrast it with unsecured debt, such as credit card debt, which is not collateralized) is the predominant form of household debt and is widely used in business borrowing as well. The majority of household debt in the United States consists of collateralized loans: Your automobile is collateral for your auto loan, and your house is collateral for your mortgage. Commercial and farm mortgages, for which property is pledged as collateral, make up one-quarter of borrowing by nonfinancial businesses; corporate bonds and other bank loans also often involve pledges of collateral. Why is collateral such an important feature of debt contracts?

8. Debt contracts typically are extremely complicated legal documents that place substantial restrictions on the behavior of the borrower. Many students think of a debt contract as a simple IOU that can be written on a single piece of paper. The reality of debt contracts is far different, however. In all countries, bond or loan contracts typically are long legal documents with provisions (called restrictive covenants) that restrict and specify certain activities that the borrower can engage in. Restrictive covenants are not just a feature of debt contracts for businesses; for example, personal automobile loan and home mortgage contracts include covenants that require the borrower to maintain sufficient insurance on the automobile or house purchased with the loan. Why are debt contracts so complex and restrictive?

TOPIC 098: TRANSACTION COSTS

- Transaction costs are expenses incurred when buying or selling a good or service.
- Transaction costs are the payments that banks and brokers receive from buyers and sellers for their roles.
- Transaction costs are one of the key determinants of net returns.
- Different asset classes have different ranges of transaction costs; investors should select assets with costs that are at the low end of the range for their types.

Say you have $5,000 that you would like to invest, and you are thinking about investing in the stock market. Because you have only $5,000, you can buy only a small number of shares. Even if you use online trading, your purchase is so small that the brokerage commission for buying the stock you pick will be a large percentage of the purchase price of the shares. If, instead, you decide to buy a bond, the problem becomes even worse; the smallest denomination offered on some bonds that you might want to buy is as large as $10,000, and you do not have that much money to invest. You are disappointed and realize that you will not be able to use financial markets to earn a return on your hard-earned savings. You can take some consolation, however, in the fact that you are not alone in being stymied by high transaction costs. This is a fact of life for many of us:

About one-half of American households own any securities.
You also face another problem related to transaction costs. Because you have only a small amount of funds available, you can make only a restricted number of investments, because a large number of small transactions would result in very high transaction costs. That is, you have to put all your eggs in one basket, and your inability to diversify will subject you to a lot of risk.

TOPIC 099: FINANCIAL INTERMEDIARIES AND TRANSACTION COSTS

- **Economies of Scale**
- One solution to the problem of high transaction costs is to bundle the funds of many investors together so that they can take advantage of economies of scale, the reduction in transaction costs per dollar of investment as the size (scale) of transactions increases.

- **Expertise**
- Financial intermediaries are also better able to develop expertise, e. g. in computer technology. Low transaction costs enable financial intermediaries to provide their customers with liquidity services. For example, money market mutual funds pay shareholders relatively high interest rates.

Financial intermediaries, an important part of the financial structure, have evolved to reduce transaction costs and allow small savers and borrowers to benefit from the existence of financial markets.

Economies of Scale One solution to the problem of high transaction costs is to bundle the funds of many investors together so that they can take advantage of economies of scale, the reduction in transaction costs per dollar of investment as the size (scale) of transactions increases. Bundling investors' funds together reduces transaction costs for each individual investor. Economies of scale exist because the total cost of carrying out a transaction in financial markets increases only a little as the size of the transaction grows. For example, the cost of arranging a purchase of 10,000 shares of stock is not much greater than the cost of arranging a purchase of 50 shares of stock.

The presence of economies of scale in financial markets helps explain the development of financial intermediaries and why financial intermediaries have become such an important part of our financial structure. The clearest example of a financial intermediary that arose because of economies of scale is a mutual fund. A mutual fund is a financial intermediary that sells shares to individuals and then invests the proceeds in bonds or stocks. Because it buys large blocks of stocks or bonds, a mutual fund can take advantage of lower transaction costs. These cost savings are then passed on to individual investors after the mutual fund has taken its cut in the form of management fees for administering their accounts.

An additional benefit for individual investors is that a mutual fund is large enough to purchase a widely diversified portfolio of securities.
The increased diversification for individual investors reduces their risk, making them better off. Economies of scale are also important in lowering the costs of resources that financial institutions need to accomplish their tasks, such as computer technology. Once a large mutual fund has

invested a lot of money in setting up a telecommunications system, for example, the system can be used for a huge number of transactions at a low cost per transaction.

Expertise Financial intermediaries are also better able to develop expertise that can be used to lower transaction costs. Their expertise in computer technology, for example, enables them to offer their customers convenient services such as check-writing privileges on their accounts and toll-free numbers that customers can call for information on how well their investments are doing. Low transaction costs enable financial intermediaries to provide their customers with liquidity services, which are services that make it easier for customers to conduct transactions. Money market mutual funds, for example, not only pay shareholders relatively high interest rates but also allow them to write checks for convenient bill paying.

TOPIC 100: ASYMMETRIC INFORMATION

Asymmetric Information

- A situation that arises when one party's insufficient knowledge about the other party involved in a transaction makes it impossible for the first party to make accurate decisions when conducting the transaction—is an important aspect of financial markets.

Adverse Selection

- It occurs before a transaction takes place. The parties who are most likely to produce an undesirable outcome are also the ones most likely to want to engage in the transaction. For example, big risk takers or outright crooks are often the most eager to take out a loan because they know they are unlikely to pay it back.

Moral Hazard

- It arises after the transaction occurs. The lender runs the risk that the borrower will engage in activities that are undesirable from the lender's point of view, because such activities make it less likely that the loan will be paid back.

Moral Hazard – Example

- Once borrowers have obtained a loan, they may take on big risks (which have possible high returns but also run a greater risk of default) because they are playing with someone else's money.

- Moral hazard lowers the probability that the loan will be repaid, lenders may decide that they would rather not make a loan.

The presence of transaction costs in financial markets partly explains why financial intermediaries and indirect finance play such an important role in financial markets (fact 3). To understand financial structure more fully, however, we turn to the role of information in financial markets.

Asymmetric information—a situation that arises when one party's insufficient knowledge about the other party involved in a transaction makes it impossible for the first party to make accurate decisions when conducting the transaction—is an important aspect of financial markets. For example, managers of a corporation know whether they are honest and usually have better information about how well their business is doing than stockholders do.

Adverse selection is an asymmetric information problem that occurs before a transaction occurs: Potential bad credit risks are the ones who most actively seek out loans. Thus the parties who are most likely to produce an undesirable outcome are also the ones most likely to want to engage in the transaction. For example, big risk takers or outright crooks are often the most eager to take out a loan because they know they are unlikely to pay it back. Because adverse selection increases the chances that a loan might be made to a bad credit risk, lenders might decide not to make any loans, even though good credit risks can be found in the marketplace.

Moral hazard arises after the transaction occurs: The lender runs the risk that the borrower will engage in activities that are undesirable from the lender's point of view, because such activities make it less likely that the loan will be paid back. For example, once borrowers have obtained a loan, they may take on big risks (which have possible high returns but also run a greater risk of default) because they are playing with someone else's money. Because moral hazard lowers the probability that the loan will be repaid, lenders may decide that they would rather not make a loan. The analysis of how asymmetric information problems affect economic behavior is called agency theory.

TOPIC 101: THE LEMONS PROBLEM

- Nobel Prize winner George Akerlof.
- It is called the “lemons problem” because it resembles the problem created by “lemons,” that is, bad cars, in the used-car market.
- What is the Lemons Problem?
- The lemons problem refers to issues that arise regarding the value of an investment or product due to asymmetric information possessed by the buyer and the seller.
- If investors can’t distinguish between good and bad securities, they are only willing to pay only average of good and bad securities’ values.
- Result: Good securities are undervalued and firms won’t issue them; bad securities overvalued, so too many issued.
- Few bonds are likely to sell in this market, so it will not be a good source of financing.
- Investors won’t want to buy bad securities, so market won’t function well.
- Less asymmetric information for well known firms, so smaller lemons problem.

A particular aspect of the way the adverse selection problem interferes with the efficient functioning of a market was outlined in a famous article by Nobel Prize winner George Akerlof. It is called the “lemons problem” because it resembles the problem created by “lemons,” that is, bad cars, in the used-car market. Potential buyers of used cars are frequently unable to assess the quality of a car; that is, they can’t tell whether a particular used car is one that will run well or a lemon that will continually give them grief.

The price that a buyer pays must therefore reflect the average quality of the cars in the market, somewhere between the low value of a lemon and the high value of a good car. The owner of a used car, by contrast, is more likely to know whether the car is a peach or a lemon. If the car is a lemon, the owner is more than happy to sell it at the price the buyer is willing to pay, which, being somewhere between the value of a lemon and that of a good car, is greater than the lemon’s value.

However, if the car is a peach, that is, a good car, the owner knows that the car is undervalued at the price the buyer is willing to pay, and so the owner may not want to sell it. As a result of this adverse selection problem, fewer good used cars will come to the market. Because the average quality of a used

car available in the market will be low, and because very few people want to buy a lemon, there will be few sales. The used-car market will function poorly, if at all.

Lemons in the Stock and Bond Markets

A similar lemons problem arises in securities markets—that is, the debt (bond) and equity (stock) markets. Suppose that our friend Irving the Investor, a potential buyer of securities such as common stock, can't distinguish between good firms with high expected profits and low risk and bad firms with low expected profits and high risk. In this situation, Irving will be willing to pay only a price that reflects the average quality of firms issuing securities—a price that lies between the value of securities from bad firms and the value of those from good firms. If the owners or managers of a good firm have better information than Irving and know that they have a good firm, then they know that their securities are undervalued and will not want to sell them to Irving at the price he is willing to pay.

The only firms willing to sell Irving securities will be bad firms (because his price is higher than the securities are worth). Our friend Irving is not stupid; he does not want to hold securities in bad firms, and hence he will decide not to purchase securities in the market. In an outcome similar to that in the used-car market, this securities market will not work very well because few firms will sell securities in it to raise capital.

The analysis is similar if Irving considers purchasing a corporate debt instrument in the bond market rather than an equity share. Irving will buy a bond only if its interest rate is high enough to compensate him for the average default risk of the good and bad firms trying to sell the debt. The knowledgeable owners of a good firm realize that they will be paying a higher interest rate than they should, so they are unlikely to want to borrow in this market.

Only the bad firms will be willing to borrow, and because investors like Irving are not eager to buy bonds issued by bad firms, they will probably not buy any bonds at all. Few bonds are likely to sell in this market, so it will not be a good source of financing. The analysis we have just conducted explains fact 2—why marketable securities are not the primary source of financing for businesses in any country in the world. It also partly explains fact 1—why stocks are not the most important source of financing for American businesses. The presence of the lemons problem keeps securities markets

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such as the stock and bond markets from being effective in channeling funds from savers to borrowers.

TOPIC 102: TOOLS TO HELP SOLVE ADVERSE SELECTION (LEMONS) PROBLEM

- If purchasers of securities can distinguish good firms from bad, they will pay the full value of securities issued by good firms, and good firms will sell their securities in the market.
- The securities market will then be able to move funds to the good firms that have the most productive investment opportunities.
- Private Production and Sale of Information
- Reduce asymmetric information by furnishing the people supplying funds with more details about the individuals or firms seeking to finance their investment activities.
- Private companies that collect and produce information distinguishing good firms from bad firms and then sell it to the saver-lenders.
- Examples:
- Standard and Poor's, Moody's, and Value Line gather information on firms' balance sheet positions and investment activities, publish these data, and sell them to subscribers (individuals, libraries, and financial intermediaries involved in purchasing securities).

In the absence of asymmetric information, the lemons problem goes away. If buyers know as much about the quality of used cars as sellers, so that all involved can tell a good car from a bad one, buyers will be willing to pay full value for good used cars. Because the owners of good used cars can now get a fair price, they will be willing to sell them in the market. The market will have many transactions and will perform its intended job of channeling good cars to people who want them. Similarly, if purchasers of securities can distinguish good firms from bad, they will pay the full value of securities issued by good firms, and good firms will sell their securities in the market. The securities market will then be able to move funds to the good firms that have the most productive investment opportunities.

Private Production and Sale of Information

The solution to the adverse selection problem in financial markets is to reduce asymmetric information by furnishing the people supplying funds with more details about the individuals or firms seeking to finance their investment activities. One way for saver-lenders to get this information is through

private companies that collect and produce information distinguishing good firms from bad firms and then sell it to the saver-lenders. In the United States, companies such as Standard and Poor's, Moody's, and Value Line gather information on firms' balance sheet positions and investment activities, publish these data, and sell them to subscribers (individuals, libraries, and financial intermediaries involved in purchasing securities).

TOPIC 103: FREE RIDER PROBLEM

- The free-rider problem occurs when people who do not pay for information take advantage of the information that other people have paid for.
- The free-rider problem suggests that the private sale of information is only a partial solution to the lemons problem.
- Suppose you have just purchased information that tells you which firms are good and which are bad.
- You believe that this purchase is worthwhile because you can make up the cost of acquiring this information, by purchasing the securities of good firms that are undervalued.
- When a (free-riding) investor sees you buying certain securities, he buys right along with you, even though he has not paid for any information.
- If many other investors act as he did, the increased demand for the undervalued good securities causes their low price to be bid up immediately to reflect the securities' true value.
- Due to free rider problem adverse selection (the lemons problem) will still interfere with the efficient functioning of securities markets.

The system of private production and sale of information does not completely solve the adverse selection problem in securities markets, however, because of the free-rider problem. The free-rider problem occurs when people who do not pay for information take advantage of the information that other people have paid for. The free-rider problem suggests that the private sale of information is only a partial solution to the lemons problem. To see why, suppose you have just purchased information that tells you which firms are good and which are bad.

You believe that this purchase is worthwhile because you can make up the cost of acquiring this information, and then some, by purchasing the securities of good firms that are undervalued. However, when our savvy

(free-riding) investor Irving sees you buying certain securities, he buys right along with you, even though he has not paid for any information. If many other investors act as Irving does, the increased demand for the undervalued good securities causes their low price to be bid up immediately to reflect the securities' true value. Because of all these free riders, you can no longer buy the securities for less than their true value. Now, because you will not gain any profit from purchasing the information, you realize that you never should have paid for the information in the first place. If other investors come to the same realization, private firms and individuals may not be able to sell enough of this information to make it worth their while to gather and produce it.

The weakened ability of private firms to profit from selling information will mean that less information is produced in the marketplace, so adverse selection (the lemons problem) will still interfere with the efficient functioning of securities markets.

LESSON 21

ADVERSE SELECTION AND MORAL HAZARD

TOPIC 104: GOVERNMENT REGULATION TO INCREASE INFORMATION

- The government could, for instance, produce information to help investors distinguish good from bad firms and provide it to the public free of charge.
- This solution, however, would involve the government releasing negative information about firms, a practice that might be politically difficult.
- A second possibility (and one followed by Pakistan, and most governments throughout the world) is for the government to regulate securities markets in a way that encourages firms to reveal honest information about themselves so that investors can determine how good or bad the firms are.
- Securities Exchange Commission of Pakistan (SECP)
- Government regulation lessens the adverse selection problem, it does not eliminate it entirely
- A lot more is involved in knowing the quality of a firm than statistics alone can provide.
- Bad firms have an incentive to make themselves look like good firms because this enables them to fetch a higher price for their securities.
- Bad firms will slant the information they are required to transmit to the public, thus making it harder for investors to sort out the good firms from the bad.

The free-rider problem prevents the private market from producing enough information to eliminate all the asymmetric information that leads to adverse selection. Could financial markets benefit from government intervention? The government could, for instance, produce information to help investors distinguish good from bad firms and provide it to the public free of charge. This solution, however, would involve the government releasing negative information about firms, a practice that might be politically difficult.

A second possibility (and one followed by the United States and most governments throughout the world) is for the government to regulate securities markets in a way that encourages firms to reveal honest information about themselves so that investors can determine how good or bad the firms are. In the United States, the Securities and Exchange Commission (SEC) is the government agency that requires firms selling their securities

to undergo independent audits, in which accounting firms certify that the firm is adhering to standard accounting principles and disclosing accurate information about sales, assets, and earnings. Similar regulations are found in other countries. However, disclosure requirements do not always work well, as the collapse of Enron and accounting scandals at other corporations, such as WorldCom and Parmalat (an Italian company), suggest (see the FYI box, "The Enron Implosion").

The asymmetric information problem of adverse selection in financial markets helps explain why financial markets are among the most heavily regulated sectors of the economy (fact 5). Government regulation aimed at increasing the information available to investors is necessary to reduce the adverse selection problem, which interferes with the efficient functioning of securities (stock and bond) markets.

Although government regulation lessens the adverse selection problem, it does not eliminate it entirely. Even when firms provide information to the public about their sales, assets, or earnings, they still have more information than investors: A lot more is involved in knowing the quality of a firm than statistics alone can provide. Furthermore, bad firms have an incentive to make themselves look like good firms because this enables them to fetch a higher price for their securities. Bad firms will slant the information they are required to transmit to the public, thus making it harder for investors to sort out the good firms from the bad.

TOPIC 105: TOOLS TO REDUCE ADVERSE SELECTION-FINANCIAL INTERMEDIATION

- A financial intermediary, e.g. a bank, becomes an expert in producing information about firms.
- It can sort out good credit risks from bad ones.
- It then can acquire funds from depositors and lend them to the good firms.

Role of Financial Intermediaries

- The bank is able to lend mostly to good firms, it is able to earn a higher return on its loans than the interest it has to pay to its depositors.
- A bank can avoid the free-rider problem by primarily making private loans, rather than by purchasing securities that are traded in the open market.

So far we have seen that private production of information and government regulation to encourage provision of information lessen, but do not eliminate, the adverse selection problem in financial markets. How, then, can the financial structure help promote the flow of funds to people with productive investment opportunities when asymmetric information exists? A clue is provided by the structure of the used-car market. An important feature of the used-car market is that most used cars are not sold directly by one individual to another. An individual who considers buying a used car might pay for privately produced information by subscribing to a magazine like Consumer Reports to find out if a particular make of car has a good repair record.

Nevertheless, reading Consumer Reports does not solve the adverse selection problem, because even if a particular make of car has a good reputation, the specific car someone is trying to sell could be a lemon. The prospective buyer might also bring the used car to a mechanic for a once-over. But what if the prospective buyer doesn't know a mechanic who can be trusted or the mechanic charges a high fee to evaluate the car? Because these roadblocks make it hard for individuals to acquire enough information about used cars, most used cars are not sold directly by one individual to another. Instead, they are sold by an intermediary, a used-car dealer who purchases used cars from individuals and resells them to other individuals.

Used-car dealers produce information in the market by becoming experts in determining whether a car is a peach or a lemon. Once a dealer knows that a car is good, the dealer can sell it with some form of a guarantee: either an explicit guarantee such as a warranty or an implicit guarantee in which the dealer stands by its reputation for honesty. People are more likely to purchase a used car because of a dealer guarantee, and the dealer is able to sell the used car at a higher price than he or she paid for it. Thus the dealer profits from the production of information about automobile quality. If dealers purchase and then resell cars on which they have produced information, they avoid the problem of other people freeriding on the information they produced.

Just as used-car dealers help solve adverse selection problems in the automobile market, financial intermediaries play a similar role in financial markets. A financial intermediary, such as a bank, becomes an expert in producing information about firms

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so that it can sort out good credit risks from bad ones. It then can acquire funds from depositors and lend them to the good firms. Because the bank is able to lend mostly to good firms, it is able to earn a higher return on its loans than the interest it has to pay to its depositors. The resulting profit that the bank earns gives it the incentive to engage in this information production activity.

An important element of the bank's ability to profit from the information it produces is that it avoids the free-rider problem by primarily making private loans, rather than by purchasing securities that are traded in the open market. Because a private loan is not traded, other investors cannot watch what the bank is doing and bid down the loan's interest rate to the point that the bank receives no compensation for the information it has produced.

The bank's role as an intermediary that holds mostly nontraded loans is the key to its success in reducing asymmetric information in financial markets. Our analysis of adverse selection indicates that financial intermediaries in general—and banks in particular, because they hold a large fraction of nontraded loans—should play a greater role in moving funds to corporations than securities markets do. Our analysis thus explains facts 3 and 4: why indirect finance is so much more important than direct finance and why banks are the most important source of external funds for financing businesses.

Our analysis also explains the greater importance of banks, as opposed to securities markets, in the financial systems of developing countries. As we have seen, better information about the quality of firms lessens asymmetric information problems, making it easier for firms to issue securities. Information about private firms is harder to collect in developing countries than in industrialized countries; therefore, the smaller role played by securities markets leads to a greater role for financial intermediaries such as banks. As a corollary to our analysis, as information about firms becomes easier to acquire, the role of banks should decline. A major development in the past 30 years in the United States has been huge improvements in information technology. Thus our analysis suggests that the lending role of financial institutions such as banks in the United States should have declined, and this is exactly what has occurred.

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Our analysis of adverse selection also explains fact 6, which questions why large firms are more likely to obtain funds from securities markets, a direct route, rather than from banks and financial intermediaries, an indirect route. The better known a corporation is, the more information about its activities is available in the marketplace. Thus it is easier for investors to evaluate the quality of the corporation and determine whether it is a good firm or a bad one. Because investors have fewer worries about adverse selection when dealing with well-known corporations, they are more willing to invest directly in their securities. Thus, in accordance with adverse selection, a pecking order for firms that can issue securities should exist. Hence we have an explanation for fact 6: The larger and more established a corporation is, the more likely it will be to issue securities to raise funds.

TOPIC 106: ADVERSE SELECTION, COLLATERAL AND NET WORTH

- Adverse selection interferes with the functioning of financial markets if a lender suffers a loss when a borrower is unable to make loan payments and defaults on the loan.
- Collateral (property promised to the lender if the borrower defaults), reduces the consequences of adverse selection because it reduces the lender's losses in the event of a default.
- Lenders are more willing to make loans secured by collateral, and borrowers are willing to supply collateral because the reduced risk for the lender makes it more likely that the loan will be made, perhaps even at a better loan rate.
- The presence of adverse selection in credit markets thus explains why collateral is an important feature of debt contracts.

Adverse selection interferes with the functioning of financial markets only if a lender suffers a loss when a borrower is unable to make loan payments and thereby defaults on the loan. Collateral, property promised to the lender if the borrower defaults, reduces the consequences of adverse selection because it reduces the lender's losses in the event of a default. If a borrower defaults on a loan, the lender can sell the collateral and use the proceeds to make up for the losses on the loan. For example, if you fail to make your mortgage payments, the lender can take the title to your house, auction it off, and use the receipts to pay off the loan.

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TOPIC 107: ADVERSE SELECTION AND NET WORTH

- Net worth (also called equity capital), is the difference between a firm's assets (what it owns or is owed) and its liabilities (what it owes), can perform a similar role to that of collateral.
- If a firm has a high net worth, then even if it engages in investments that lead to negative profits and so defaults on its debt payments.
- The lender can take title to the firm's net worth, sell it off, and use the proceeds to recoup some of the losses from the loan.
- In addition, the more net worth a firm has in the first place, the less likely it is to default, because the firm has a cushion of assets that it can use to pay off its loans.
- Hence, when firms seeking credit have high net worth, the consequences of adverse selection are less important and lenders are more willing to make loans.
- This concept lies behind the often-heard lament, "Only the people who don't need money can borrow it!"

Lenders are thus more willing to make loans secured by collateral, and borrowers are willing to supply collateral because the reduced risk for the lender makes it more likely that the loan will be made, perhaps even at a better loan rate. The presence of adverse selection in credit markets thus explains why collateral is an important feature of debt contracts (fact 7).

Net worth (also called equity capital), the difference between a firm's assets (what it owns or is owed) and its liabilities (what it owes), can perform a similar role to that of collateral. If a firm has a high net worth, then even if it engages in investments that lead to negative profits and so defaults on its debt payments, the lender can take title to the firm's net worth, sell it off, and use the proceeds to recoup some of the losses from the loan.

In addition, the more net worth a firm has in the first place, the less likely it is to default, because the firm has a cushion of assets that it can use to pay off its loans. Hence, when firms seeking credit have high net worth, the consequences of adverse selection are less important and lenders are more willing to make loans. This concept lies behind the often-heard lament, "Only the people who don't need money can borrow it!

TOPIC 108: MORAL HAZARD AND THE CHOICE BETWEEN DEBT AND EQUITY

- Equity contracts, such as common stock, are claims to a share in the profits and assets of a business.
- Equity contracts are subject to a particular type of moral hazard called the principal–agent problem.
- Managers own only a small fraction of the firm they work for, the stockholders who own most of the firm's equity (called the principals) are not the same people as the managers of the firm.
- The managers are the agents of the owners. This separation of ownership and control involves moral hazard.
- The managers in control (the agents) may act in their own interest rather than in the interest of the stockholder-owners (the principals) because the managers have less incentive to maximize profits than the stockholder-owners do.

Moral hazard is the asymmetric information problem that occurs after a financial transaction takes place, when the seller of a security may have incentives to hide information and engage in activities that are undesirable for the purchaser of the security. Moral hazard has important consequences for whether a firm finds it easier to raise funds with debt than with equity contracts.

Moral Hazard in Equity Contracts: The Principal–Agent Problem

Equity contracts, such as common stock, are claims to a share in the profits and assets of a business. Equity contracts are subject to a particular type of moral hazard called the principal–agent problem. When managers own only a small fraction of the firm they work for, the stockholders who own most of the firm's equity (called the principals) are not the same people as the managers of the firm. Thus the managers are the agents of the owners. This separation of ownership and control involves moral hazard, in that the managers in control (the agents) may act in their own interest rather than in the interest of the stockholder-owners (the principals) because the managers have less incentive to maximize profits than the stockholder-owners do.

To understand the principal–agent problem more fully, suppose that your friend Steve asks you to become a silent partner in his ice-cream store. The store setup requires an initial investment of $10,000, and Steve has only $1,000. So you purchase

an equity stake (stock shares) for $9,000, which entitles you to 90% of the ownership of the firm, while Steve owns only 10%. If Steve works hard to make tasty ice cream, keeps the store clean, smiles at all the customers, and hustles to wait on tables quickly, after all expenses (including Steve's salary) have been paid, the store will make $50,000 in profits per year, of which Steve will receive 10% ($5,000) and you will receive 90% ($45,000).

But if Steve doesn't provide quick and friendly service to his customers, uses the $50,000 in income to buy artwork for his office, and even sneaks off to the beach while he should be at the store, the store will not earn any profit. Steve can earn the additional $5,000 (his 10% share of the profits) over his salary only if he works hard and forgoes unproductive investments (such as art for his office). Steve might decide that the extra $5,000 just isn't enough to make him expend the effort to be a good manager. If Steve feels this way, he does not have enough incentive to be a good manager and will end up with a beautiful office, a good tan, and a store that doesn't show any profits. Because the store won't show any profits, Steve's decision not to act in your interest will cost you $45,000 (your 90% of the profits had he chosen to be a good manager instead).

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LESSON 22

THE PRINCIPAL-AGENT PROBLEM

TOPIC 109: THE PRINCIPAL–AGENT PROBLEM – EXAMPLE

- Suppose that your friend Steve asks you to become a silent partner in his ice-cream store.
- The store setup requires an initial investment of $10,000, and Steve has only $1,000.
- You purchase an equity stake (stock shares) for $9,000, which entitles you to 90% of the ownership of the firm, while Steve owns only 10%.
- Suppose Steve:
- works hard to make tasty ice cream,
- keeps the store clean,
- smiles at all the customers, and hustles to wait on tables quickly
- After all expenses (including Steve's salary) have been paid, the store will make $50,000 in profits per year, of which
- Steve will receive 10% ($5,000) and you will receive 90% ($45,000).

The moral hazard arising from the principal–agent problem might be even worse if Steve is not totally honest. Because his ice-cream store is a cash business, Steve has the incentive to pocket $50,000 in cash and tell you that the profits were zero. He now gets a return of $50,000, and you get nothing.
Further proof that the principal–agent problem created by equity contracts can be severe is provided by past scandals involving corporations such as Enron and Tyco International, in which managers were found to have diverted corporate funds for their own personal use. In addition to pursuing personal benefits, managers might also pursue corporate strategies (such as the acquisition of other firms) that enhance their personal power but do not increase the corporation's profitability.

The principal–agent problem would not arise if the owners of a firm had complete information about what the managers were up to and could prevent wasteful expenditures or fraud. The principal–agent problem, which is an example of moral hazard, arises only because a manager, such as Steve, has more information about his activities than the stockholder does—that is, information is asymmetric. The principal–agent problem also would not occur if Steve alone owned the store, and ownership and control were not separate. If this were the case, Steve's hard work and avoidance of

unproductive investments would yield him a profit (and extra income) of $50,000, an amount that would make it worth his while to be a good manager.

TOPIC 110: TOOLS TO HELP SOLVE THE PRINCIPAL–AGENT PROBLEM

Production of Information: Monitoring

- The principal–agent problem arises because managers have more information about their activities and actual profits than stockholders do.
- One way for stockholders to reduce this moral hazard problem is to engage in information production.
- Monitoring the firm's activities by auditing the firm frequently and checking on what the management is doing.
- Monitoring process can be expensive (time and money).

Costly state verification

- It makes the equity contract less desirable and explains in part why equity is not a more important element in our financial structure.

One way for stockholders to reduce this moral hazard problem is for them to engage in a particular type of information production: monitoring the firm's activities by auditing the firm frequently and checking on what the management is doing. The problem is that the monitoring process can be expensive in terms of time and money, as reflected in the name economists give it: costly state verification. Costly state verification makes the equity contract less desirable and explains in part why equity is not a more important element in our financial structure. As with adverse selection, the free-rider problem decreases the amount of private information production that would reduce the moral hazard (principal–agent) problem.

In this example, the free-rider problem decreases monitoring. If you know that other stockholders are paying to monitor the activities of the company you hold shares in, you can take a free ride on their activities. Then you can use the money you save by not engaging in monitoring to vacation on a Caribbean island. If you can do this, though, so can other stockholders. Perhaps all the stockholders will go to the islands, and no one will spend any resources on monitoring the firm. The moral hazard problem for shares of common stock will then be severe, making it hard for firms to issue

them to raise capital (providing an additional explanation for fact 1).

TOPIC 111: PRINCIPAL–AGENT PROBLEM: GOVERNMENT REGULATION

- Governments everywhere have laws to force firms to adhere to standard accounting principles that make profit verification easier.
- Examples
  - International financial reporting standards (IFRS)
  - The Accounting and Auditing Organization for Islamic Financial Institutions standards (AAOIFI)

Accounting Standards

- International financial reporting standards (IFRS) is a set of accounting rules and standards that determine how accounting events should be reported in your business's financial statements.
- The Accounting and Auditing Organization for Islamic Financial Institutions (AAOIFI) maintains and promotes Shari'ah standards for Islamic financial institutions and the overall industry.
- The Government also pass laws to impose stiff criminal penalties on people who commit the fraud of hiding and stealing profits.
- These measures can be only partly effective.

As with adverse selection, the government has an incentive to try to reduce the moral hazard problem created by asymmetric information, which provides another reason why the financial system is so heavily regulated (fact 5). Governments everywhere have laws to force firms to adhere to standard accounting principles that make profit verification easier. They also pass laws to impose stiff criminal penalties on people who commit the fraud of hiding and stealing profits. However, these measures can be only partly effective. Catching this kind of fraud is not easy; fraudulent managers have the incentive to make it very hard for government agencies to find or prove fraud.

TOPIC 112: PRINCIPAL–AGENT PROBLEM: FINANCIAL INTERMEDIATION

- Financial intermediaries have the ability to avoid the free-rider problem in the face of moral hazard
- One financial intermediary that helps reduce the moral hazard arising from the principal–agent problem is the venture capital firm.

Venture Capital Firm

- Venture capital firms pool the resources of their partners and use the funds to help budding entrepreneurs start new businesses.
- In exchange for supplying the venture capital, the firm receives an equity share in the new business.
- Verification of earnings and profits is so important in eliminating moral hazard
- Venture capital firms usually insist on having several of their own people participate as members of the managing body— the board of directors—of the new business so that they can keep a close watch on the new firm's activities.
- When a venture capital firm supplies start-up funds, the equity in the new firm is private—that is, not marketable to anyone except the venture capital firm.
- Other investors are unable to take a free ride on the venture capital firm's verification activities.
- As a result of this arrangement, the venture capital firm is able to garner the full benefits of its verification activities and is given the appropriate incentives to reduce the moral hazard problem.
- Venture capital firms have been important in the development of the high-tech sector in the United States, which has resulted in job creation, economic growth, and increased international competitiveness.

Financial intermediaries have the ability to avoid the free-rider problem in the face of moral hazard, and this is another reason why indirect finance is so important (fact 3). One financial intermediary that helps reduce the moral hazard arising from the principal–agent problem is the venture capital firm. Venture capital firms pool the resources of their partners and use the funds to help budding entrepreneurs start new businesses. In exchange for supplying the venture capital, the firm receives an equity share in the new business.

Because verification of earnings and profits is so important in eliminating moral hazard, venture capital firms usually insist on having several of their own people participate as members of the managing body—the board of directors—of the new business so that they can keep a close watch on the new firm's activities. When a venture capital firm supplies start-up funds, the equity in the new firm is private—that is, not marketable to anyone except the venture capital

firm.3 Thus other investors are unable to take a free ride on the venture capital firm's verification activities. As a result of this arrangement, the venture capital firm is able to garner the full benefits of its verification activities and is given the appropriate incentives to reduce the moral hazard problem. Venture capital firms have been important in the development of the high-tech sector in the United States, which has resulted in job creation, economic growth, and increased international competitiveness.

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Lesson 23

RISK MANAGEMENT AND ECONOMIC DECISIONS

TOPIC 113: PRINCIPLES OF RISK MANAGEMENT

Risk implies future uncertainty about deviation from expected earnings or expected outcome. Risk measures the uncertainty that an investor is willing to take to realize a gain from an investment. We will explain how risk affects financial decision making. We will also explain the conceptual framework for the management of risk. In these topics, we also explain how the financial system facilitates the efficient allocation of risk bearing.

Three analytical pillars of finance

- The time value of money
- Valuation, and
- Risk Management

Uncertainty and Risk

- Uncertainty exists whenever one does not know for sure what will happen in the future
- Risk (in financial terms) is the chance that an outcome or investment's actual gains will differ from an expected outcome or return.
- Risk includes the possibility of losing some or all of an original investment.
- Every risky situation is uncertain but there can be uncertainty without risk

Example

- Risk and uncertainty in a dinner party and a one-dish party
- In case of Dinner Party, there is uncertainty (less or more people might come) and risk (food shortage or surplus).
- In case of One-Dish Party, there is uncertainty (less or more people might come) but no risk of shortage or surplus of food.

TOPIC 114: RISK MANAGEMENT

- In the financial world, risk management is the process of identification, analysis, and acceptance or mitigation of uncertainty in investment decisions.
- Risk management occurs when an investor or fund manager analyzes and attempts to quantify the potential for losses in an investment, such as a moral hazard, and then takes the appropriate action (or inaction) given the fund's investment objectives and risk tolerance.

- The appropriateness of a risk-management decision should be judged in the light of the information available at the time the decision is made.
- Risk is inseparable from return in the investment world.
- A variety of ways exist to ascertain risk.
- One of the most common is standard deviation, a statistical measure of dispersion around a central tendency.

It is possible and prudent to manage investing risks by understanding the basics of risk and how it is measured. Learning the risks that can apply to different scenarios and some of the ways to manage them holistically will help all types of investors and business managers to avoid unnecessary and costly losses.

In the financial world, risk management is the process of identification, analysis, and acceptance or mitigation of uncertainty in investment decisions. Essentially, risk management occurs when an investor or fund manager analyzes and attempts to quantify the potential for losses in an investment, such as a moral hazard, and then takes the appropriate action (or inaction) given the fund's investment objectives and risk tolerance.

Risk is inseparable from return. Every investment involves some degree of risk, which is considered close to zero in the case of a U.S. T-bill or very high for something such as emerging-market equities or real estate in highly inflationary markets. Risk is quantifiable both in absolute and in relative terms. A solid understanding of risk in its different forms can help investors to better understand the opportunities, trade-offs, and costs involved with different investment approaches.

Risk management occurs everywhere in the realm of finance. It occurs when an investor buys U.S. Treasury bonds over corporate bonds, when a fund manager hedges his currency exposure with currency derivatives, and when a bank performs a credit check on an individual before issuing a personal line of credit. Stockbrokers use financial instruments like options and futures, and money managers use strategies like portfolio diversification, asset allocation and position sizing to mitigate or effectively manage risk.

Inadequate risk management can result in severe consequences for companies, individuals, and the economy. For example, the subprime mortgage meltdown in 2007 that helped trigger the Great Recession stemmed from bad risk-management decisions, such as lenders who extended mortgages to individuals with poor credit; investment firms who bought, packaged, and resold these mortgages; and funds that invested excessively in the repackaged, but still risky, mortgage-backed securities (MBSs).

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TOPIC 115: RISK EXPOSURE

- Risk exposure is the measure of potential future loss resulting from a specific activity or event.
- An analysis of the risk exposure for a business often ranks risks according to their probability of occurring multiplied by the potential loss if they do.
- The riskiness of an asset or a transaction cannot be assessed in isolation or in the abstract.
- The purchase of sale of a particular asset may add to your risk exposure; in another, the same transaction may be risk reducing.

Speculators and Hedgers

- Speculators are the investors who take positions that increase their exposure to certain risks in the hope of increasing their wealth.
- Hedgers take positions to reduce their exposures.
- The same person can be a speculator on some exposures and a hedger on others.

Risk exposure in any business or an investment is the measurement of potential future loss due to a specific event or business activity and is calculated as the probability of the event multiplied by the expected loss due to the risk impact.

What is a Speculator?

A speculator utilizes strategies and typically a shorter time frame in an attempt to outperform traditional longer-term investors. Speculators take on risk, especially with respect to anticipating future price movements, in the hope of making gains that are large enough to offset the risk.

Speculators that take on excessive risk typically don't last long. Speculators exert control over long-term risks by employing various strategies such as position sizing, stop loss orders, and monitoring the statistics of their trading performance. Speculators are typically sophisticated risk-taking individuals with expertise in the markets in which they are trading.

Hedgers are primary participants in the futures markets. A hedger is any individual or firm that buys or sells the actual physical commodity. Many hedgers are producers, wholesalers, retailers or manufacturers and they are affected by changes in commodity prices, exchange rates, and interest rates.

TOPIC 116: RISK AND ECONOMIC DECISIONS

- Financial decisions expose us to various risk exposures
- Saving, Investment and financing decisions are significantly influenced by the presence of risk, and therefore, are risk-management decisions.

- The ultimate function of a financial system is to help Implement optimal consumption and resource allocation of households.
- Economic organizations such as firms and governments exist primarily to facilitate the achievement of that ultimate function.
- To have a better understanding of how risk influences economic decisions, it is important to explain it in terms of:

1. Households
2. Firms, and
3. Government.

Some financial decisions, such as how much insurance to buy against risk exposures, relate exclusively to the management of risk. But many general resource allocation decisions, are also significantly influenced by the presence of risk and therefore, are partly risk-management decisions.

For example, some households saving is motivated by the desire for the increased security that comes from owning assets that can cover unanticipated expenses in the future. Economists call this precautionary saving.

TOPIC 117: RISKS FACING HOUSEHOLDS

Five major categories of risk exposures for households:

1. Sickness, disability, and death
2. Unemployment risk
3. Consumer-durable asset risk in case of fire, theft, technological change
4. Liability risk

- The risk that others will have a financial claim against you because they suffer a loss for which you can be held responsible.
- For example, you cause a car accident and are required to pay for the injury and damage caused.

5. Financial-asset risk

- The risk arising from holding different kinds of financial assets such as equities or fixed-income securities denominated in one or more currencies.

- The underlying sources of financial-asset risk are the uncertainties faced by the firms, governments, or other economic organizations that have issued these securities.

There can be many types of risks which households face but there are five major categories of risk that the households face the most.

Sickness, disability, and death: Unexpected sickness or accidental injuries can impose large costs on people because of the need for treatment and care and because of the loss of income caused by the inability to work.

Unemployment risk: This is the risk of losing one's job.

Consumer-durable asset risk: This is the risk of loss arising from ownership of a house, car, etc. Losses can occur due to hazards such as fire or theft, etc.

Liability risk: This is the risk that others will have a financial claim against you because they suffer a loss for which you can be held responsible. For example, you cause a car accident through reckless driving and are required to cover the cost to others of personal injury and property damage.

Financial-asset risk: This is the risk arising from holding different kinds of financial assets such as equities or fixed-income securities denominated in one or more currencies. The underlying sources of financial-asset risk are the uncertainties face by the firms, governments, or their economic organizations that have issued these securities.

The risks faced by households influence virtually all of their economic decisions.

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Lesson 24

RISKS FACING FIRMS AND ROLE OF GOVERNMENT

TOPIC 118: RISKS FACING FIRMS

- Running a business comes with many types of risk.
- Some of these potential hazards can destroy a business, while others can cause serious damage that is costly and time-consuming to repair.
- Despite the risks implicit in doing business, CEOs and risk management officers can anticipate and prepare, regardless of the size of their business.

Types of Risks facing Firms

1. Physical Risks
2. Location Risks
3. Human Risks
4. Technology Risks
5. Strategy Risks

Some of the major categories of risk, firms are exposed to include:

Production risk: this is the risk that machines will break down, that deliveries of raw materials will not arrive on time, that workers will not show up for work, or that a new technology will make the firm's existing equipment obsolete.

Price risk of outputs: This is the risk that the demand for the baked goods produced by the bakery will unpredictably change because of an unanticipated shift in consumer preferences (e.g., celery becomes a popular substitute for bread at restaurants and therefore, the market price of baked goods might fall.

Price risk of inputs: This is the risk that the prices of some of the inputs of the bakery will change unpredictably. Flour can become more expensive, or wage rates rise. If the bakery borrows money to finance its operations at a floating interest rate, it is exposed to risk that interest rates might rise.

TOPIC 119: PHYSICAL AND LOCATION RISKS FACING FIRMS

Building risks are the most common type of physical risk. Think fires or explosions.

- Fire alarms
- Emergency numbers
- Emergency exits

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Physical Risks

Hazardous material risk is present where spills or accidents are possible. The risk from hazardous materials can include:

- Acid
- Gas
- Toxic fumes
- Toxic dust or filings
- Poisonous liquids or waste

Location Risks

- Among the location hazards facing a business are nearby fires, storm damage, floods, hurricanes or tornados, earthquakes, and other natural disasters.

Building risks are the most common type of physical risk. Think fires or explosions. To manage building risk, and the risk to employees, it is important that organizations do the following:

Make sure all employees know the exact street address of the building to give to a 911 operator in case of emergency.

Make sure all employees know the location of all exits.

Install fire alarms and smoke detectors.

Install a sprinkler system to provide additional protection to the physical plant, equipment, documents and, of course, personnel.

Inform all employees that in the event of emergency their personal safety takes priority over everything else. Employees should be instructed to leave the building and abandon all work-associated documents, equipment and/or products.

Fire department hazardous material units are prepared to handle these types of disasters. People who work with these materials, however, should be properly equipped and trained to handle them safely.

Organizations should create a plan to handle the immediate effects of these risks. Government agencies and local fire departments provide information to prevent these accidents. Such agencies can also provide advice on how to control them and minimize their damage if they occur.

Among the location hazards facing a business are nearby fires, storm damage, floods, hurricanes or tornados, earthquakes, and other natural disasters. Employees should be familiar with the streets leading in and out of the neighborhood on all sides of the place of business. Individuals should keep sufficient fuel in their vehicles to drive out of and away from the area. Liability or

property and casualty insurance are often used to transfer the financial burden of location risks to a third-party or a business insurance company.

There are other business risks associated with location that are not directly related to hazards, such as city planning. For example, a gas station exists on a major road, and as a result of its location, it receives plenty of business. City planning can eventually restructure the area around the gas station. The city may close the road the gas station is on, build other infrastructure that would make the gas station inaccessible, or overall just not take the gas station into consideration with any redevelopment. This would leave the gas station with no traffic to serve.

TOPIC 120: HUMAN AND TECHNOLOGY RISKS FACING FIRMS

Human Risks

- Illness or injury
- Alcohol and drug abuse are major risks to personnel in the workforce. Employees suffering from alcohol or drug abuse should be urged to seek treatment, counseling, and rehabilitation if necessary.
- Embezzlement, theft and fraud
- A system of double-signature requirements for checks, invoices, and payables verification can help prevent embezzlement and fraud. Stringent accounting procedures may discover embezzlement or fraud. A thorough background check before hiring personnel can uncover previous offenses in an applicant's past

Technology Risks

- A power outage is perhaps the most common technology risk. Auxiliary gas-driven power generators are a reliable back-up system to provide electricity for lighting and other functions.
- Computers may be kept up and running with high-performance back-up batteries.
- Establish offline and online data back-up systems to protect critical documents.
- Telephone and communications failure are relatively uncommon, risk managers may consider providing emergency-use company cell phones to personnel whose use of the phone or internet is critical to their business.

Alcohol and drug abuse are major risks to personnel in the workforce. Employees suffering from alcohol or drug abuse should be urged to seek treatment, counseling, and rehabilitation if necessary. Some insurance policies may provide partial coverage for the cost of treatment.

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Protection against embezzlement, theft and fraud may be difficult, but these are common crimes in the workplace. A system of double-signature requirements for checks, invoices, and payables verification can help prevent embezzlement and fraud. Stringent accounting procedures may discover embezzlement or fraud. A thorough background check before hiring personnel can uncover previous offenses in an applicant's past. While this may not be grounds for refusing to hire an applicant, it would help HR to avoid placing a new hire in a critical position where the employee is open to temptation.

Illness or injury among the workforce is a potential problem. To prevent loss of productivity, assign and train backup personnel to handle the work of critical employees when they are absent due to a health-related concern.

Technology Risks

A power outage is perhaps the most common technology risk. Auxiliary gas-driven power generators are a reliable back-up system to provide electricity for lighting and other functions. Manufacturing plants use several large auxiliary generators to keep a factory operational until utility power is restored.

Computers may be kept up and running with high-performance back-up batteries. Power surges may occur during a lightning storm (or randomly), so organizations should furnish critical business systems with surge-protection devices to avoid the loss of documents and the destruction of equipment.

Cloud storage is another source of risks nowadays. The process involves backing up data with Amazon Web Services, for example, using Azure, IBM, and Oracle, for instance. This is a huge undertaking that should be considered given the reliance on cloud-based data to run most businesses now. It is important to establish both offline and online data backup systems to protect critical documents.

Although **telephone and communications failure** are relatively uncommon, risk managers may consider providing emergency-use company cell phones to personnel whose use of the phone or internet is critical to their business.

TOPIC 121: STRATEGIC RISKS FACING FIRMS

- Strategic risks are those that arise from the fundamental decisions that directors take concerning an organization’s objectives.
- Essentially, strategic risks are the risks of failing to achieve these business objectives.

Strategic Risks – Types

1. Strategic Business risks

2. Strategic Non-Business risks

Strategic Business Risks

Risks that derive from the decisions that the board takes about the products or services that the organization supplies. They include risks associated with developing and marketing those products or services, economic risks affecting product sales and costs, and risks arising from changes in the technological environment which impact on sales and production.

Strategic Non-Business Risks

- Risks that do not derive from the products or services supplied, e.g. risks associated with the long-term sources of finance used.
- Competitor actions will affect risk levels in product markets, and technological developments may mean that production processes, or products, quickly become out-of-date.

Strategy risks are not altogether undesirable. Financial institutions such as banks or credit unions take on strategy risk when lending to consumers, while pharmaceutical companies are exposed to strategy risk through research and development for a new drug. Each of these strategy-related risks is inherent in an organization's business objectives. When structured efficiently, the acceptance of strategy risks can create highly profitable operations.

Companies exposed to substantial strategy risk can mitigate the potential for negative consequences by creating and maintaining infrastructures that support high-risk projects. A system established to control the financial hardship that occurs when a risky venture fails often includes diversification of current projects, healthy cash flow, or the ability to finance new projects in an affordable way, and a comprehensive process to review and analyze potential ventures based on future return on investment.

TOPIC 122: ROLE OF GOVERNMENT IN RISK MANAGEMENT

- Governments at all levels play an important role in managing risks by:
  - Preventing them or
  - Redistributing them.
- Governments provide relief in case of natural disasters and various human-caused hazards, e.g. war, pollution

Governments at all levels play an important role in managing risks either by preventing them or redistributing them. People often rely on government to provide protection and financial relief from natural disasters and various human-caused hazards, including war and pollution of the environment. An argument in favor of an activist role for government in economic development is that government can readily spread the risk of an investment in infrastructure among all of the taxpayers within its jurisdiction. Government managers often use the markets and other channels of the financial system to implement their own risk-management policies in much the same way that managers of firms and other non-governmental economic organizations do.

As is the case with these other organizations, however, all risks are ultimately borne by the people. Whether government offers insurance against the risk of natural disasters or insurance against default on bank deposits, it is not free. The government either charges the insured parties a price sufficient to cover the costs of these insurance services, or taxpayers pay the claims.

TOPIC 123: THE RISK-MANAGEMENT PROCESS

- The risk-management process can be broken into five steps:

1. Risk identification
2. Risk assessment
3. Selection of risk-management techniques
4. Implementation
5. Review

Risk Identification

- The process of figuring out what the most important risk exposures are for the unit of analysis, i.e. a household, a firm, or some other entity.
- Sometimes the households or firms are completely unaware of the possible risks they are exposed to.
- Effective risk identification requires that one take the perspective of the entity as a whole and consider the totality of uncertainties affecting it.
- For example, future prices of raw material, stocks, foreign-exchange rate, price risks, quantity risks
- Maintain a checklist of the entity's potential exposures and the relations among them.
- Detailed knowledge about the economics of the industry in which the firm competes, the technology of the firm, and its sources of supply.

The **risk-management process** is a systematic attempt to analyze and deal with risk. The process can be broken down into five steps:

- *Risk identification*
- *Risk assessment*
- *Selection of risk-management techniques*
- *Implementation*
- *Review*

Risk identification consists of figuring out what the most important risk exposures are for the unit of analysis, be it a household, a firm, or some other entity. Households or firms are sometimes not aware of all of the risks to which they are exposed. For example, a person who has never missed a day of work because of illness or injury may give little thought to the risk of disability. Buying insurance against disability risk might make sense, but may not even be considered.

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Lesson 25

RISK ASSESSMENT

TOPIC 124: RISK ASSESSMENT

- The quantification of the costs associated with the risks that have been identified in the first step of risk-management.
- For better assessment of risk, actuaries are employed.
- Actuaries are professionals specially trained in mathematics and statistics to gather and analyze data and estimate the probabilities of illness, accidents and other such risks.
- In case of financial-asse risks, households and firms often seek expert advice in assessing their exposures and
- In quantifying the trade-offs between the risks and rewards of investing in various categories of assets, such as stocks and bonds, they typically turn to professional investment advisors, mutual funds, or other financial intermediaries an service firms that help them make those assessments.

Risk assessment is the quantification of the costs associated with the risks that have been identified in the first step of risk management. For example, consider a single woman who has just graduated from college and started a job. When she was in college, she was covered by her parents' health insurance policy, but now she has no health insurance coverage. She, therefore, identifies illness as a major risk exposure. To assess her exposure, information is needed. How likely is it that someone of her age and health status will get sick? What is the cost of treatment?

Clearly, she needs information, and information may be costly to gather. One of the main functions of insurance companies is to provide this kind of information. They employ actuaries, who are professional specially trained in mathematics and statistics, to gather and analyze data and estimate the probabilities of illness, accidents, and other such risks.\In the realm of final-asset risks, households and firms often need expert advice in assessing their exposures and in quantifying the trade-offs between the risks and rewards of investing in various categories of assets, such as stocks and bonds. They typically turn to professional investment advisors, mutual funds, or other financial intermediaries and service firms that help them make those assessments.

TOPIC 125: RISK MANAGEMENT TECHNIQUES

There are FOUR basic techniques available for reducing risk:

1. Risk avoidance
2. Loss prevention and control
3. Risk retention
4. Risk transfer

Risk avoidance

- A conscious decision of not to be exposed to a particular risk.
- People may decide to avoid the risks of going into certain professions and firms may avoid certain lines of business because they are considered too risky.
- It is not always feasible to avoid risk, e. g. in case of illness.

Loss prevention and control

- Actions taken to reduce the likelihood or the severity of losses.
- Such actions can be taken prior to, concurrent with, or after a loss occurs

Risk Retention

- Absorbing the risk and covering losses out of one's own resources.
- This sometimes happens by default.
- For example, when may decide to bear all expenses if gets ill instead of buying insurance.

Risk transfer

- Transferring the risk to others.
- It can be done by:
  - Selling the risky asset to somebody else
  - Buying insurance.

TOPIC 126: THE THREE DIMENSIONS OF RISK TRANSFER

The three dimensions of risk transfer include **hedging, insuring and diversifying**. Each of these dimensions are discussed in the subsequent topics in detail.

- Financial system plays the greatest role in transferring risk
- Risk transfer has three dimensions
  - Hedging
  - Insuring
  - Diversifying

Hedging

- Hedging is a strategy that tries to limit risks in financial assets.
- One is said to hedge a risk when the action taken to reduce one's exposure to a loss also causes one to give up the possibility of gain.

- Example: Farmers sell their crop before harvest to hedge against price fluctuations.

Insuring

- Insuring means paying premium.
- Example: Car insurance by paying premium of Rs 30000 per month for two years Suppose car price is 20 lacs.

Difference between Insuring and Hedging

- There is a fundamental difference between insuring and hedging.
- When you hedge, you eliminate the risk of loss by giving up the potential for gain.
- When you insure, you pay a premium to eliminate the risk of loss and retain the potential for gain.

Diversifying

- It means holding similar amounts of many risky assets instead of concentrating all your investment in only one.
- It limits your exposure to the risk of any single asset.

TOPIC 127: RISK TRANSFER AND ECONOMIC EFFICIENCY

- Institutional arrangements for the transfer of risk contribute to economic efficiency in two fundamental ways:
  - They reallocate existing risks to those most willing to bear them
  - They cause a reallocation of resources to production and consumption according to the new distribution of risk bearing.

Efficient Bearing of Existing Risks

- Suppose there are two investors; a widow and a graduating student. Each has Rs 100,000 to invest. The widow is a more conservative investor, and the graduating student is aggressive.
- The widow has all her wealth in the form of shares left by her husband. The college student has all her wealth in the form of bank certificate of deposit (CD).
- Both will be better off if they could swap their assets.

Institutional arrangements for the transfer of risk contribute to economic efficiency in two fundamental ways:

1. They reallocate existing risks to those most willing to bear them, and

2. They cause a reallocation of resources to production and consumption in accordance with the new distribution of risk bearing.

Efficient Bearing of Existing Risks

It means that a conservative investor or a risk avertor investor sells his risky investments to those who are risk lovers and buys financial investors that are considered to be less risky.

Risk and Resource Allocation

- The ability to reallocate risks facilitate the undertakings of valuable projects that might not otherwise be undertaken because they are too risky.
- The ability to pool and share risks can lead to an increase in innovations and the development of new products.

We first investigate how the facility to reallocate risks among people can make everyone better off. Consider the hypothetical case of two investors in very different economic circumstances. The first is a retired widow who has an accumulated "nest egg" of $100,000, which is her sole source of income. The second is a college student who has $100,000 and who anticipates a good stream of earnings in the future after graduating from college.

Typically, the widow is assumed to be a more conservative investor and the student more aggressive. That is, we would expect the widow to be concerned primarily with the safety of her stream of investment income, whereas we might expect the student to be willing to bear more risk in exchange for a higher expected return.

Suppose the widow currently holds all of her wealth in the form of a portfolio of stocks left to her by her recently deceased husband, and suppose the college student has all of her wealth in a bank certificate of deposit (CD) that her parents started for her years before. Both would be better off if they could somehow swap their assets so that the widow winds up holding the CD and the student the stock portfolio.

One of the most important functions of the financial system is to facilitate such transfers of risk. One way for this transfer of risk to take place is for the widow to simply sell her stocks and for the student to buy them. Typically, several financial intermediaries would be involved in the process. For example, the widow might have her stocks in an account at a brokerage firm. She gives the broker an order to sell them and to invest the proceeds of the sale in a bank CD. The student, on the other hand, cashes in her CD at the bank and buys the stocks through her broker.

In this set of transactions no immediate change occurs in the wealth of either party except that each must pay the cost of carrying out the transactions (i.e., broker fees and bank charges). The student and the widow each had $100,000 of assets before, and immediately afterwards they each have $100,000 (less the broker fees and bank charges). The sole purpose and result of the transactions is to allow each party to hold the portfolio of assets offering them the combination of risk and expected return most attractive for their circumstances.

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Risk and Resource Allocation

Now let us consider how the ability to reallocate risks facilitates the undertaking of valuable projects that might not otherwise be undertaken because they are too risky. The ability to pool and share risks can lead to an increase in inventive activity and the development of new products.

For example, consider the case of creating new pharmaceuticals. The research and development effort that goes into the discovery, testing, and production of new drugs requires enormous amounts of investment extended over a considerable period of time. The return on that investment is highly uncertain. Even if an individual investor had the wealth necessary to finance the development of a new drug, risk aversion might deter him from doing so on his own.

TOPIC 128: INSTITUTIONS FOR RISK MANAGEMENT

- Over the centuries, various economic organizations and contractual arrangements have evolved to facilitate a more efficient allocation of risk bearing both by expanding the scope of diversification and by permitting greater specialization in the management of risk.
- Examples
    - Insurance companies
    - Futures markets

Debt and equity securities are designed to differ in the risks of the business they carry.

- New discoveries in telecommunications, information processing, and finance theory have significantly lowered the cost of achieving greater global diversification and specialization in the bearing of risks.
- However, increased volatility of exchange rates, interest rates, and commodity prices have increased the demand for ways to manage risk.

Over the centuries, various economic organizations and contractual arrangements have evolved to facilitate a more efficient allocation of risk bearing both by expanding the scope of diversification and by permitting greater specialization in the management of risk. Insurance companies and futures markets are examples of institutions whose primary economic function is to further these ends.

The allocation of risks is also an important consideration in the design of securities. Debt and equity securities issued by firms are intentionally designed to differ in the risks of the

business that they carry. By choosing to invest in either the debt or the equity securities of a firm or some combination of them, people can thus select the kinds of risk they wish to bear.

In the last few decades, the rate of introduction of innovations that facilitate risk management has greatly accelerated because of changes on both the supply side and the demand side of the markets for risk bearing. New discoveries in telecommunications, information processing, and finance theory have significantly lowered the costs of achieving greater global diversification and specialization in the bearing of risks. At the same time, increased volatility of exchange rates, interest rates, and commodity prices has increased the demand for ways to manage risk. Thus, the rapid and widespread development of futures, options, and swap contracts starting in the 1970s and 1980s can largely be explained as market responses to these cost and demand factors.

But the theoretical ideal of complete markets for allocating risks can never be fully achieved because in the real world there are a number of limiting factors that can never be overcome entirely. Two key categories of factors limiting the efficient allocation of risks are *transactions costs* and *incentive problems.*

Transactions costs include the costs of establishing and running institutions such as insurance companies or securities exchanges and the costs of writing and enforcing contracts. These institutions will not come into existence unless the pecuniary benefits from their creation exceed the costs.

Lesson 26

PROBABILITY DISTRIBUTION OF RETURNS

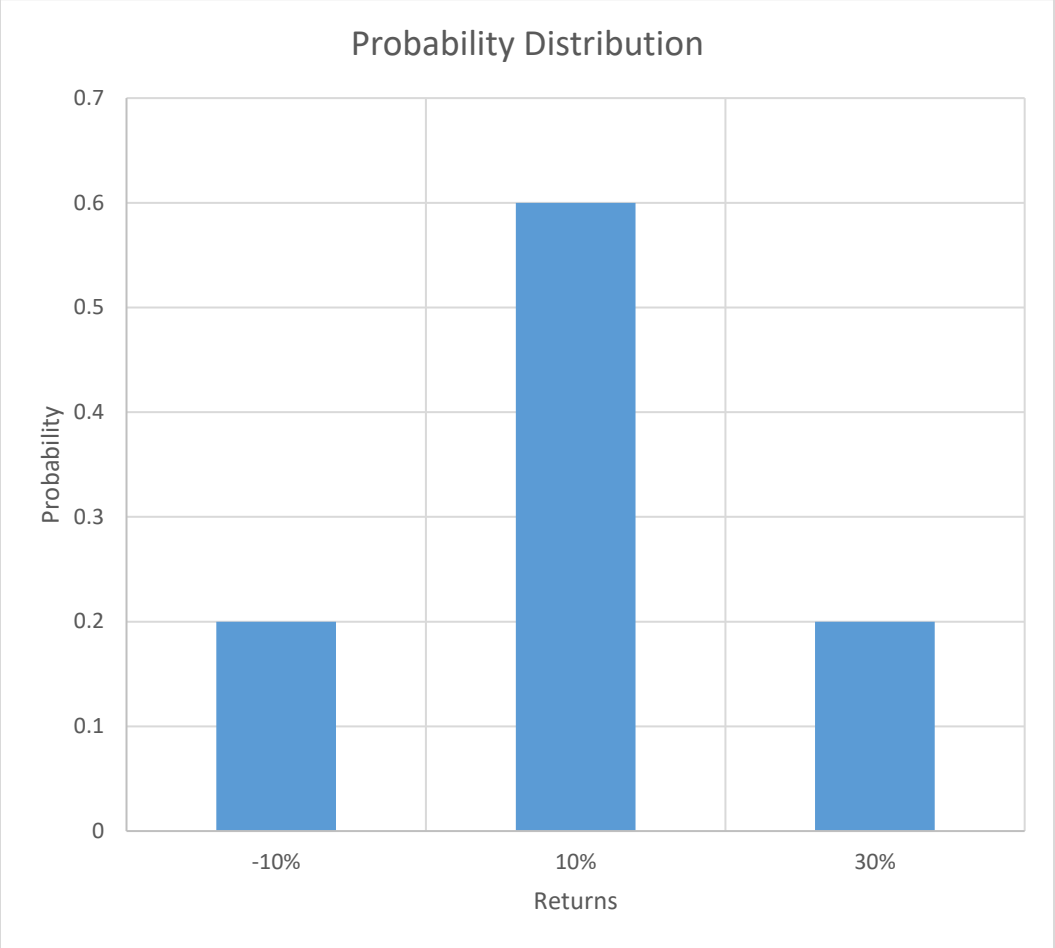
TOPIC 129: PROBABILITY DISTRIBUTION OF RETURNS

- A probability distribution is a statistical function that describes all the possible values and likelihoods that a random variable can take within a given range.
- It's shape depends on the distribution's mean (average), standard deviation, skewness, and kurtosis.
- Investors use probability distributions to anticipate returns on assets such as stocks over time and to hedge their risk.

Probability Distribution – Example

- Suppose there is stock called GENCO

Probability Distribution of Rate of Return on Genco		
State of the Economy	**Rate of Return on Genco**	**Probability**
Strong	**30%**	**0.20**
Normal	**10%**	**0.60**
Weak	**-10%**	**0.20**



Portfolio theory is defined as quantitative analysis for optimal risk management. Whether the unit of analysis is a household, a firm, or some other economic organization, applying portfolio theory consists of formulating and evaluating the trade-offs between the benefits and costs of risk reduction in order to find an optimal course of action.

For households, consumption and risk preferences are taken as given. Preferences do change over time, but the mechanisms and reasons for those changes are not addressed by the theory. Instead, portfolio theory addresses the problem of how to choose among financial alternatives so as to maximize their given preferences. In general, the optimal choice involves evaluating the trade-off between receiving a higher expected return and taking greater risk.

The early formal models in portfolio theory were developed to deal with this class of risk-management decision.[4] These models use **probability distributions** to quantify the trade-off between risk and expected return. An asset portfolio's expected return is identified with the **mean** of the distribution, and its risk with the **standard deviation.**

These concepts are developed more fully in the next section.

TOPIC 130: EXPECTED RATE OF RETURN

The Total Rate of Return

= Dividend Income Component +Price Change Component

$= \frac{Cash\ Dividend}{P_0} + \frac{P_1 - P_0}{P_0}$

- Suppose you expect the dividend component to be 3%, and the price-change component to be 7%. The Expected Rate of Return
- = r = 3% +7% = 10%
- The expected rate of return (the mean) is defined as the sum over all possible outcomes of each possible rate of return multiplied by the respective probability of its happening.
- $E(r) = P_1r_1+P_2r_2+\ldots+ P_nr_n$
- $E(r) = \sum_{i=1}^{n} P_i r_i$

Expected Rate of Return – Example

Probability Distribution of Rate of Return on Genco		
State of the Economy	Rate of Return on Genco	Probability
Strong	30%	0.20
Normal	10%	0.60
Weak	-10%	0.20

Using the above data, the expected rate of return is calculated as:

E(r) = 0.2X30%+0.60X10%+0.2X(- 10%)

= 0.1 or 10%

Consider the case of Genco stock. Suppose you buy shares of Genco Corporation at a price of $100 per share and intend to hold it for a year. As shown in chapter 2, the *total* rate of return can be decomposed into the sum of a dividend-income component and a price-change component:

$$r = \frac{\text{Cash Dividend}}{\text{Beginning Price}} + \frac{\text{Ending Price of a Share} - \text{Beginning Price}}{\text{Beginning Price}}$$

$$r = \text{Dividend-Income Component} + \text{Price-Change Component}$$

In Genco's case, suppose that you expect the dividend component to be 3%, and the price-change component to be 7% so that the *expected* rate of return is 10%:

$$r = 3\% + 7\% = 10\%$$

A commonly used measure of the riskiness of an asset such as Genco stock is **volatility.**[5] Volatility is related to the range of possible rates of return from holding the stock and to their likelihood of occurring. *A stock's volatility is larger, the wider the range of possible outcomes and the larger the probabilities of those returns at the extremes of the range.*

The **expected rate of return** (the mean) is defined as the sum over all possible outcomes of each possible rate of return multiplied by the respective probability of its happening:

$$E(r) = P_1 r_1 + P_2 r_2 + \cdots + P_n r_n$$

$$E(r) = \sum_{i=1}^{n} P_i r_i \qquad \textbf{(10.1)}$$

Where E(r) is the expected rate of return, P_i is the probability of event I, and i=1, 2, 3, …., n. r_i is the return of an outcome i.

TOPIC 131: EXPECTED RATE OF RETURN – MICROSOFT EXCEL

- Enter the data for returns and their probabilities of occurrence
- Calculate (return X probability)
- Find the sum
- The value of sum is the expected rate of return

TOPIC 132: VOLATILITY

- It is related to the range of possible rates of return from holding the stock and to their likelihood of occurring.

- If the range of possible outcomes is wide, and the probabilities of returns at the extremes is high, the stock's volatility will be higher.

Standard Deviation as a Measure of Risk

- Standard deviation is a measure of the risk that an investment will fluctuate from its expected return.
- The smaller an investment's standard deviation, the less volatile it is, thus less risky the investment is.

The volatility of a stock's return was shown to depend on the range of possible outcomes and on the probabilities of extreme values occurring. We use standard deviation to measure the volatility of a stock's probability distribution of returns. The formula used to measure standard deviation is

$$\sigma = \sqrt{\sum_{i=1}^{n} P_i (r_i - E(r))^2}$$

TOPIC 133: CALCULATION OF EXPECTED RETURN AND STANDARD DEVIATION OF EXPECTED RATE OF RETURN

Suppose Risco and Genco are two stocks with the following details:

Probability Distribution of the Two Stocks

Economy	Rate of Return on Risco	Rate of Return on Genco	Probability
Weak	-30%	-10%	0.2
Strong	10%	10%	0.6
Normal	50%	30%	0.2

The standard deviation for Risco stock is:

- $\sigma = \sqrt{[(0.2)\,(0.5 - 0.1)^2 + (0.6)\,(0.1 - 0.1)^2 + (0.2)\,(-0.3 - 0.1)^2]}$

  = 25.30%

The standard deviation for Genco stock is:

- $\sigma = \sqrt{[(0.2)\,(-0.1 - 0.1)^2 + (0.6)\,(0.1 - 0.1)^2 + (0.2)\,(0.3 - 0.1)^2]}$

  = 12.65%

Let us take an investment A, which has a 20% probability of giving a 15% return on investment, a 50% probability of generating a 10% return, and a 30% probability of resulting in a 5% loss. This is an example of calculating a discrete probability distribution for potential returns.

The expected return on investment A would then be calculated as follows:

Expected Return of A = 0.2(15%) + 0.5(10%) + 0.3(-5%)

(That is, a 20%, or .2, probability times a 15%, or .15, return; plus a 50%, or .5, probability times a 10%, or .1, return; plus a 30%, or .3, probability of a return of negative 5%, or -.5)

= 3% + 5% – 1.5%

= 6.5%

Therefore, the probable long-term average return for Investment A is 6.5%.

Calculating Expected Return of a Portfolio

Calculating expected return is not limited to calculations for a single investment. It can also be calculated for a portfolio. The expected return for an investment portfolio is the weighted average of the expected return of each of its components. Components are weighted by the percentage of the portfolio's total value that each accounts for. Examining the weighted average of portfolio assets can also help investors assess the diversification of their investment portfolio.

To illustrate the expected return for an investment portfolio, let's assume the portfolio is comprised of investments in three assets – X, Y, and Z. $2,000 is invested in X, $5,000 invested in Y, and $3,000 is invested in Z. Assume that the expected returns for X, Y, and Z have been calculated and found to be 15%, 10%, and 20%, respectively. Based on the respective investments in each component asset, the portfolio's expected return can be calculated as follows:

Expected Return of Portfolio = 0.2(15%) + 0.5(10%) + 0.3(20%)

= 3% + 5% + 6%

= 14%

Thus, the expected return of the portfolio is 14%.

Standard Deviation of the Expected Return of A = $\sqrt{\mathbf{0.2^2(15\%) + 0.5^2(10\%) + 0.3^2(-5\%)}}$

=0.1628

I have used the data followed for calculation of expected rate of return.

Standard deviation is unit free measure.

Lesson 27

HEDGING RISK USING FORWARD AND FUTURES CONTRACTS

TOPIC 134: USING FORWARD AND FUTURES CONTRACTS TO HEDGE RISK

Three ways of transferring risk

- Hedging
- Insuring
- Diversifying

Hedging

- Hedging is a risk management strategy employed to offset losses in investments by taking an opposite position in a related asset.
- The reduction in risk provided by hedging also typically results in a reduction in potential profits.
- Hedging is a risk management strategy employed to offset losses in investments by taking an opposite position in a related asset.
- The reduction in risk provided by hedging also typically results in a reduction in potential profits.
- Hedging requires one to pay money for the protection it provides, known as the premium.
- Hedging strategies typically involve derivatives, such as options and futures contracts.

Hedging techniques generally involve the use of financial instruments known as derivatives. Two of the most common derivatives are options and futures. With derivatives, you can develop trading strategies where a loss in one investment is offset by a gain in a derivative.

Forward Contract

- Hedging strategies typically involve derivatives, such as options and futures contracts.
- Anytime two parties agree to exchange some item in the future at a prearranged price they are entering into a forward contract.

Main Features of Forward Contracts

1. Two parties agree to exchange some item in the future at a price specified now – the forward price.
2. The price for immediate delivery of the item is called the spot price.
3. No money is paid in the present by either party to the other.
4. The face value of the contract is the quantity of the item specified in the contract times the forward price

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5. The party who agrees to buy the specified item is said to take a long position, and the party who agrees to sell the item is said to take a short position.

Futures Contract

- It is a standardized forward contract that is traded on some organized exchange.
- The exchange interposes itself between the buyer and the seller, so that each has a separate contract with the exchange.
- Standardization: the terms of futures contract are the same for all contracts

Forward and futures contracts are derivatives arrangements that involve two parties who agree to buy or sell a specific asset at a set price by a certain date in the future. Buyers and sellers can mitigate the risks associated with price movements down the road by locking in the purchase/sale price in advance.

A forward contract is an arrangement that is made over-the-counter (OTC) and settles just once at the end of the contract. Both parties involved in the agreement negotiate the exact terms of the contract. It is privately negotiated and comes with a degree of default risk since the counterparty is responsible for remitting payment.

Futures contracts, on the other hand, are standardized contracts that trade on stock exchanges. As such, they are settled on a daily basis. These arrangements come with fixed maturity dates and uniform terms. There is very little risk with futures, as they guarantee payment on the agreed-upon date.

Futures are derivative financial contracts that obligate parties to buy or sell an asset at a predetermined future date and price. The buyer must purchase or the seller must sell the underlying asset at the set price, regardless of the current market price at the expiration date.

Underlying assets include physical commodities and financial instruments. Futures contracts detail the quantity of the underlying asset and are standardized to facilitate trading on a futures exchange. Futures can be used for hedging or trade speculation.

TOPIC 135: FORWARD CONTRACT – EXAMPLE

- A forward contract can often reduce the risks faced by both the buyer and the seller.
- Suppose a farmer has planted his fields with wheat.
- It is now a month before harvest time.
- Farmer wants to eliminate risk due to uncertainty about its future price.
- Suppose, there is a baker, who also want to minimize risk

- Risk is due to uncertainty about the future price of wheat.
- He has a large investment in the bakery.
- The farmer and the baker agree to a certain forward price that the baker will pay the farmer at the time of delivery.
- They sign a contract that the farmer will deliver a specified quantity of wheat to the baker at the forward price *regardless of what the spot price turns out to be at the delivery date*.
- Suppose the size of the farmer's crop is 400,000 kg (10,000 maunds)
- The forward price for delivery a month from now is Rs. 2000 per 40 kg.
- After a month, the farmer will deliver 400,000 kg wheat and will get Rs. 20 million in return.
- With this agreement, both parties eliminate the risk associated with the uncertainty about the spot price of wheat at the delivery date.
- They both are hedging their exposures.

Consider the following example of a forward contract. Assume that an agricultural producer has two million bushels of corn to sell six months from now and is concerned about a potential decline in the price of corn. It thus enters into a forward contract with its financial institution to sell two million bushels of corn at a price of $4.30 per bushel in six months, with settlement on a cash basis.

In six months, the spot price of corn has three possibilities:

1. **It is exactly $4.30 per bushel.** In this case, no monies are owed by the producer or financial institution to each other and the contract is closed.
2. **It is higher than the contract price, say $5 per bushel.** The producer owes the institution $1.4 million, or the difference between the current spot price and the contracted rate of $4.30.
3. **It is lower than the contract price, say $3.50 per bushel.** The financial institution will pay the producer $1.6 million, or the difference between the contracted rate of $4.30 and the current spot price.

TOPIC 136: HEDGING PRICE RISK WITH FUTURES CONTRACTS

Example:

- The size of the farmer's crop is 400,000 kg (10,000 maunds)
- To hedge his exposure to the price risk, he takes a short position in a one-month futures contract for his total crop at a futures price of Rs. 2000 per 40 kg.

- After a month, the farmer will deliver 400,000 kg wheat and will get Rs. 20 million in return.

Farmer's Transaction	Spot Price of Wheat on Delivery Date	
	Rs. 1500 per maund	Rs. 2000 per maund
Proceeds from sale of wheat to distributor	Rs. 15 million	Rs. 20 million
Cash flow from the futures contract	Rs. 5 million paid to farmer	0
Total Receipts	Rs. 20 million	Rs. 20 million

Baker's Transaction	Spot Price of Wheat on Delivery Date	
	Rs. 1500 per maund	Rs. 2000 per maund
Cost of wheat bought from supplier	Rs. 15 million	Rs. 20 million
Cash flow from the futures contract	Rs. 5 million paid by baker	0
Total Outlays	Rs. 20 million	Rs. 20 million

Futures can be used to hedge the price movement of the underlying asset. Here, the goal is to prevent losses from potentially unfavorable price changes rather than to speculate. Many companies that enter hedges are using—or in many cases producing—the underlying asset.
For example, corn farmers can use futures to lock in a specific price for selling their corn crop. By doing so, they reduce their risk and guarantee they will receive the fixed price. If the price of corn decreased, the farmer would have a gain on the hedge to offset losses from selling the corn at the market. With such a gain and loss offsetting each other, the hedging effectively locks in an acceptable market price.

- The farmer is able to eliminate the price risk he faces from owning the wheat by taking a short position in a futures contract, effectively selling the wheat for future delivery at the futures price.

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- The baker too is able to eliminate risk by taking long position in the futures market for wheat.

TOPIC 137: HEDGING FOREIGN-EXCHANGE RISK WITH SWAP CONTRACTS

Hedging Foreign-Exchange Risk with Swap Contracts

- Hedging can be done by using swap contracts
- A swap contract consists of two parties exchanging (or "swapping") a series of cash flows at specified intervals over a specified period of time.

Swap Contracts

- The swap payments are based on an agreed principal amount (the notional amount).
- There is no immediate payment of money.
- A swap contract could call for the exchange of anything.

Swap Contracts –Types

- Two most common and most basic types of swaps:
  - interest rate swaps
  - currency swaps

Swaps Market

- Swaps are not exchange-traded instruments.
- They are customized contracts that are traded in the over-the-counter (OTC) market between private parties.
- Firms and financial institutions dominate the swaps market, with few (if any) individuals ever participating.

Swaps Market- Risk

- Swaps occur on the OTC market, hence there is always the risk of a counterparty defaulting on the swap.

Currency risk is the financial risk that arises from potential changes in the exchange rate of one currency in relation to another. And it's not just those trading in the foreign exchange markets that are affected. Adverse currency movements can often crush the returns of a portfolio with heavy international exposure, or diminish the returns of an otherwise prosperous international business venture. Companies that conduct business across borders are exposed to currency risk when income earned abroad is converted into the money of the domestic country, and when payables are converted from the domestic currency to the foreign currency.

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The currency swap market is one way to hedge that risk. Currency swaps not only hedge against risk exposure associated with exchange rate fluctuations, but they also ensure receipt of foreign monies and achieve better lending rates.

A currency swap is a financial instrument that involves the exchange of interest in one currency for the same in another currency.

Currency swaps comprise two notional principals that are exchanged at the beginning and end of the agreement. These notional principals are predetermined dollar amounts, or principal, on which the exchanged interest payments are based. However, this principal is never actually repaid: It's strictly "notional" (which means theoretical). It's only used as a basis on which to calculate the interest rate payments, which do change hands.

TOPIC 138: HEDGING FOREX RISK WITH SWAP CONTRACTS-EXAMPLE

- Suppose you have computer software business in US, and a German company wants to acquire the right to produce and market your software in Pakistan.
- The German company agrees to pay you € 100,000 each year for the next 10 years for these rights.
- If you want to hedge the risk of fluctuations in the dollar value of your expected stream of revenues, you can enter a CURRENCY SWAP now to exchange your future stream of euros for a future stream of dollars at a set of forward exchange rates specified now.
- Suppose the dollar/euro exchange rate is currently $1.30 per euro and this exchange rate also applies to all forward contracts covering the next 10 years. Notional amount in your swap is € 100,000 per year.
- By entering in the swap contract, you lock in a dollar revenue of $130,000 per year.
- Each year on the settlement date you will receive (or pay) an amount of cash equal to € 100,000 times the difference between the forward rate and the actual spot rate at that time.

Items traded through Swap Agreements

- The international swap market began in the early 1980s.
- In addition to currency and interest-rate swaps, many other items, i.e. returns on different stock indexes, bushels of wheat and barrels of oil are also exchanged through swap agreements.

A U.S. company (Party A) is looking to open up a €3 million plant in Germany, where its borrowing costs are higher in Europe than at home. Assuming a 0.6 euro/USD exchange rate, the company can borrow €3 million at 8% in Europe or $5 million at 7% in the U.S. The company borrows the

$5 million at 7% and then enters into a swap to convert the dollar loan into euros. Party B, the counterparty of the swap may likely be a German company that requires $5 million in U.S. funds. Likewise, the German company will be able to attain a cheaper borrowing rate domestically than abroad—let's say that the Germans can borrow at 6% within from banks within the country's borders.

Now, let's take a look at the physical payments made using this swap agreement. At the outset of the contract, the German company gives the U.S. Company the €3 million needed to fund the project, and in exchange for the €3 million, the U.S. Company provides the German counterparty with $5 million.

Subsequently, every six months for the next three years (the length of the contract), the two parties will swap payments. The German firm pays the U.S. company the sum that's the result of $5 million (the notional amount paid by the U.S. company to the German firm at initiation), multiplied by 7% (the agreed-upon fixed rate), over a period expressed as .5 (180 days ÷ 360 days). This payment would amount to $175,000 ($5 million x 7% x .5). The U.S. company pays the Germans the result of €3 million (the notional amount paid by the Germans to the U.S. company at initiation), multiplied by 6% (the agreed-upon fixed rate), and .5 (180 days ÷ 360 days). This payment would amount to €90,000 (€3 million x 6% x .5).

The two parties would exchange these fixed two amounts every six months. Three years after initiation of the contract, the two parties would exchange the notional principals. Accordingly, the U.S. company would "pay" the German company €3 million and the German company would "pay" the U.S. company $5 million.

TOPIC 139: HEDGING SHORTFALL RISK BY MATCHING ASSETS TO LIABILITIES

- Insurance companies and other financial intermediaries that sell insured savings plans and other insurance contracts need to assure their customers that the product they are buying is free of DEFAULT RISK.
- One way to assure customers about the risk of contract default is for insurance companies to hedge their liabilities in the financial markets by investing in assets that match the characteristics of their liabilities.

Example:

- Suppose that an insurance company sells a customer a guaranteed investment contract that promises to pay $1,000 five years from now for a one-time premium today of $783.53. (@ 5% p.a.).

- The insurance company can hedge this customer liability by buying a default-free zero coupon bond with a face value of $1,000 issued by the government.
- The insurance company will be matching assets to liabilities
- The insurance company will buy the government bond for less that $783.53

Insurance companies and other financial intermediaries that sell insured savings plans and other insurance contracts need to assure their customers that the product they are buying is free of default risk. One way to assure customers about the risk of contract default is for insurance companies to hedge their liabilities in the financial markets by investing in assets that match the characteristics of their liabilities.

For example, suppose that an insurance company sells a customer a guaranteed investment contract that promises to pay $1,000 five years from now for a one-time premium today of $783.53. (This implies that the customer is earning an interest rate of 5% per year.) The insurance company can *hedge* this customer liability by buying a default-free zero-coupon bond with a face value of $1,000 issued by the government.

The insurance company is *matching assets to liabilities*. In order to earn a profit on this set of transactions, the insurance company has to be able to buy the five-year government bond for less than $783.53. (In other words, the interest rate on the five-year government bond must be greater than 5% per year.) If instead of hedging its liability by buying a bond, the insurance company invests the premium in a portfolio of stocks, then there will be a risk of a shortfall—the value of the stocks in five years may turn out to be less than the $1,000 promised to the customer.

Many financial intermediaries pursue hedging strategies that involve matching their assets to their liabilities. In each case, the objective is to reduce the risk of a shortfall. The nature of the hedging instrument varies with the type of customer liability.

Thus, if a savings bank has customer liabilities that are short-term deposits earning an interest rate that floats, the appropriate hedging instrument is a floating-rate bond, or a strategy of "rolling over" short-term bonds. Another way the bank might hedge its floating-rate deposit liabilities is to invest in long-term fixed-rate bonds and enter into a swap contract to swap the fixed rate it receives on its bonds for a floating rate.

TOPIC 140: MINIMIZING THE COST OF HEDGING

- There may be multiple ways for hedging risk.
- A rational manager will choose the one that costs the least.
- You also have to consider the transaction costs (broker fees, the amount of effort and time involved, etc.)
- Another important point to be considered is the desired level of reduction of risk.

If there are two or more ways to hedge against risk, the investor will analyze the cost of hedging and choose the method that has less cost.

This can be explained with the help of the example presented below.

For example, suppose that you live in Boston and are planning to move to Tokyo a year from now for an extended visit. You have found a wonderful apartment there and have agreed to buy it for 10.3 million yen, which you will pay to the apartment's current owner at the time you move in. You have just sold your condominium in Boston for $100,000, and plan to use that money to pay for the apartment in Tokyo. You have invested the money in one-year U.S. Treasury bills at an interest rate of 3%, so you know that you will have $103,000 a year from now.

The dollar/yen exchange rate is currently $0.01 per yen (or 100 yen per dollar). If it remains unchanged for a year, you will have exactly the 10.3 million yen you need to pay for the Tokyo apartment a year from now. But you discover that in the past year the dollar/yen exchange rate has fluctuated quite a bit. It was as low as $0.008 per yen and as high as $0.011. You are, therefore, concerned that, one year from now, your $103,000 may not buy enough yen to pay for the Tokyo apartment.

If the exchange rate is $0.008 per yen in a year, you will receive 12.875 million yen for your $103,000 ($103,000/$0.008 per yen)—enough to buy the apartment and some nice furnishings. If, however, the exchange rate is $0.012 per yen a year from now, then you will get only 8.583 million yen ($103,000/$0.012 per yen), and you will be 1.717 million yen short of the purchase price you agreed to a year earlier.

Suppose that there are two ways you can eliminate your exposure to the risk of a rise in the dollar price of the yen. One way is to get the owner of the Tokyo apartment to sell it to you for a price fixed in U.S. dollars. The other way is by entering a forward contract for yen with a bank.

Let us compare the costs to you of the two methods of hedging the foreign-exchange risk. Suppose that in our example the bank's forward price is $0.01 per yen. By entering into a forward contract with the bank to exchange your $103,000 in a year at $0.01 per yen, you can completely eliminate your risk. No matter what happens to the dollar/yen exchange rate over the next year, you will have the 10.3 million yen you need to buy your apartment in Tokyo one year from now.

Now consider the other alternative of negotiating a fixed price in U.S. dollars with the owner of the apartment. If the owner of the Tokyo apartment is willing to sell you the apartment for a dollar price less than $103,000, then that is a better deal than entering into a forward contract with the bank.

On the other hand, if the owner of the Tokyo apartment demands a dollar price higher than $103,000, then you are better off setting the price in yen (10.3 million yen) and entering into a forward contract with the bank to exchange the yen for dollars at the forward price of $0.01 per yen. You also have to consider the transaction costs (broker fees, the amount of time and effort involved, etc.) associated with each method of hedging the risk.

The important point to recognize in this example is that *the mechanism chosen to implement the hedge should be the one that minimizes the cost of achieving the desired reduction of risk.*

Lesson 28

INSURANCE VERSUS HEDGING AND THE DIVERSIFICATION PRINCIPLE

TOPIC 141: INSURING VERSUS HEDGING

- There is fundamental difference between insuring and hedging.
- When you hedge, you eliminate the risk of loss by giving up the potential for gain.
- When you insure, you pay a premium to eliminate the risk of loss and retain the potential for gain.

Example

- You are planning a trip from Lahore to London a year from now.
- You make your flight reservations now; you can buy the ticket now or close to the day of your travel. If you decide to lock in the price today, you have hedges against the risk of loss.
- It costs you nothing to do so, but you have given up the possibility of paying less than the amount you have paid for the ticket.
- Alternatively, the airline may offer you that if you pay $20 now, you will have to pay $1000 whenever you buy the ticket.
- This is an example of buying insurance.

Insurance and hedging both reduce your exposure to financial risk, but they do so in different ways. Insurance typically involves paying someone else to bear risk, while hedging involves making an investment that offsets risk.

Insurance Shifts Risk

Buying an insurance policy that protects your home against fire does not guarantee that your home won't burn down. Having auto insurance doesn't mean you won't crash your car, and life insurance won't keep you from dying. What insurance does is shift potential financial losses from you to someone else. If your house burns down or your car gets totaled, you don't have to pay to replace it because the insurance company does.

Hedging Offsets Risk

Hedging reduces uncertainty, which is really just another word for risk. For a simple example, say you do a lot of business with Europe, and you've discover that you lose money if the exchange rate rises above $1.50 per euro. So you buy a series of options contracts that give you the right to buy euros for, say, $1.40 per euro. Those options offset your risk from rising exchange rates.

If the rate never rises about $1.40, then you just let the options expire. But if the rate tops $1.40, then you've locked in an exchange rate that offsets the increase and protects your profit. The options, therefore, are hedges.

TOPIC 142: BASIC FEATURES OF INSURANCE CONTRACTS

- Four important features of insurance contracts include:

1. Exclusions
2. Caps
3. Deductibles, and
4. Copayments

Exclusions

- Exclusions are losses that might seem to meet the conditions for coverage under the insurance contracts but are specifically excluded.
- For example, in case of life insurance, case of suicide will be excluded.

Caps

- Caps are limits placed on compensation for particular losses covered under an insurance contract.
- For example, if a health insurance policy is capped at Rs. 5 lacs, it means the insurance company will pay no more than this amount for the treatment of an illness.

Deductibles

- A deductible is an amount of money that the insured party must pay out of his or her own resources before receiving any compensation from the insurer.
- For example, if your car insurance policy has a Rs. 1000 deductible for damage due to accidents, you must pay the first Rs. 1000 in repair costs and the insurer will only pay for the amount in excess of Rs. 1000.

Copayments

- A copayment feature means that the insured party must cover a fraction of the loss. For example, an insurance policy might stipulate that the copayment is 20% of any loss, and the insurance company pays the other 80%.
- Copayments are similar to deductibles. The difference is in the way the paying part is computed.

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Why Deductibles and Copayments are Imposed?

- To create incentives for the insured party to control losses.
- Example:
  - In case of health insurance, the insured pays part of the doctor's fee

Insurance policies can contain both deductibles and copayments.

The FOUR important features of insurance contracts are EXCLUSIONS AND CAPS, DEDUCTIBLES, AND COPAYMENTS.

EXCLUSIONS AND CAPS

Exclusions are losses that might seem to meet the conditions for coverage under the insurance contracts but are specifically excluded.

Caps are limits placed on compensation for particular losses covered under an insurance contract. For example, for a damaged car insurance company pays only up to 2 lacs for the repair.

Deductibles

An insurance deductible is a specific amount you must spend before your insurance policy pays for some or all of your claims.

Insurance companies use deductibles to ensure policyholders have skin in the game and will share the cost of any claims.

Deductibles cushion against financial stress caused by catastrophic loss or an accumulation of small losses all at once for an insurer.

In addition to premiums, individuals must meet health insurance deductibles and may also be required for other costs like copays and coinsurance, depending on their plans.

The general rule is that policies with higher premiums come with lower deductibles while those with lower premiums tend to have higher deductibles.

Copayment or simply a copay

A copay is a fixed out-of-pocket amount paid by an insured for covered services. It is a standard part of many health insurance plans. Insurance providers often charge co-pays for services such as doctor visits or prescription drugs.

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Copays are a specified dollar amount rather than a percentage of the bill, and they usually paid at the time of service. Not all medical services ask you for a copay. For example, some insurance companies do not require a copay for annual physicals.

TOPIC 143: FINANCIAL GUARANTEES

- Financial guarantees are insurance against credit risk, which is the risk that the other party to a contract into which you have entered will default.
- A loan guarantee is a contract that obliges the guarantor to make the promised payment on a loan if the borrower fails to do so.

Example

- Banks and other issuers of credit cards guarantee to merchants that they will stand behind all customer purchases made their credit cards.
- Credit card issuers provide merchants with insurance against credit risk.

Financial Guarantors

- Banks, insurance companies and governments offer guarantees on a broad spectrum of financial instruments ranging from credit cards to interest-rate and currency swaps.

A financial guarantee is an agreement that guarantees a debt will be repaid to a lender by another party if the borrower defaults. Essentially, a third party acting as a guarantor promises to assume responsibility for a debt should the borrower be unable to keep up on its payments to the creditor. Guarantees can also come in the form of a security deposit or collateral. The types vary, ranging from corporate guarantees to personal ones.

Some financial agreements may require the use of a financial guarantee before they can be executed. In many cases, a guarantee is a legal contract that promises repayment of a debt to a lender. This agreement takes place when a guarantor agrees to take on the financial responsibility if the original debtor defaults on their financial obligation or goes insolvent. All three parties must sign the agreement in order for it to go into effect.

Guarantees may take on the form of a security deposit. Common in the banking and lending industries, this is a form of collateral provided by the debtor that can be liquidated if the debtor defaults.

For instance, a secured credit card requires the borrower—usually someone with no credit history—to put down a cash deposit for the amount of the credit line.

Financial guarantees act just like insurance and are very important in the financial industry. They allow certain financial transactions, especially those that wouldn't normally take place, to go through, permitting, for instance, high-risk borrowers to take out loans and other forms of credit. In short, they mitigate the risk associated with lending to high-risk borrowers and extending credit during times of financial uncertainty.

TOPIC 144: CAPS AND FLOORS ON INTEREST RATES

Caps on Interest Rates

- A cap is a limit on the interest rates a variable-rate credit product can charge.
- The cap limits the interest levels that borrowers have to pay in rising rate environments.
- Adjustable-rate mortgages (ARMs) typically have a rate cap to limit how much interest homebuyers pay for a home loan.

Variable-Rate Capped Products

- Products with a capped interest rate have a variable rate structure that includes an indexed rate and a spread.
- An indexed rate is based on the lowest rate creditors are willing to offer.
- The spread or margin is based on a borrower’s credit profile and determined by the underwriter.

Who is an Underwriter?

- An underwriter is any party that evaluates and assumes another party's risk for a fee, which often takes the form of a commission, premium, spread, or interest.
- Agents and brokers represent both consumers and insurance companies, while underwriters work for insurance companies.

Advantage of Putting a Cap on Interest Rate

- If a product has a capped rate, then the interest rate will rise with increases in the indexed rate until it reaches a specified cap.
- The cap is advantageous for borrowers because it limits the level of interest, they have to pay in a rising rate environment.

Floors on Interest Rates

- A floor sets a base level of interest that a borrower must pay and also sets a base level of interest that a lender or investor can expect to earn.
- A floor benefits the lender or credit investor in a falling rate environment. Limiting the interest base level, however, requires a borrower to pay a specified floor interest rate even when the current market rate is lower.

Caps and Floors on Interest Rates

- Variable interest rate products can have both a cap and a floor, which sets a base level of interest that a lender or investor can expect to earn.

An interest rate cap is a type of interest rate derivative in which the buyer receives payments at the end of each period in which the interest rate exceeds the agreed strike price. An example of a cap would be an agreement to receive a payment for each month the LIBOR rate exceeds 2.5%. Similarly an interest rate floor is a derivative contract in which the buyer receives payments at the end of each period in which the interest rate is below the agreed strike price.

Caps and floors can be used to hedge against interest rate fluctuations. For example, a borrower who is paying the LIBOR rate on a loan can protect himself against a rise in rates by buying a cap at 2.5%. If the interest rate exceeds 2.5% in a given period the payment received from the derivative can be used to help make the interest payment for that period, thus the interest payments are effectively "capped" at 2.5% from the borrowers' point of view.

TOPIC 145: OPTIONS AS INSURANCE

- The term option refers to a financial instrument that is based on the value of underlying securities such as stocks.
- An options contract offers the buyer the opportunity to buy or sell—depending on the type of contract they hold—the underlying asset.

Futures and Options

- Unlike futures, the holder is not required to buy or sell the asset if they decide against it.
- Options are versatile financial products. These contracts involve a buyer and seller, where the buyer pays a premium for the rights granted by the contract.

Call and Put Options

- Call options allow the holder to buy the asset at a stated price within a specific timeframe.
- Put options, on the other hand, allow the holder to sell the asset at a stated price within a specific timeframe.
- Each call option has a bullish buyer and a bearish seller while put options have a bearish buyer and a bullish seller.
- The fixed price specified in an option contract is called the option's strike price or exercise price.
- The date after which an option can no longer be exercised is called its expiration date or maturity date.
- If an option can be exercised on the expiration date only, it is called a European-type option.
- If it can be exercised at any time up to and including the expiration date it is called an American-type option.

Options started as insurance policies for either long or short stock. A put option gives the buyer the right to sell a set stock at a set price on or before a set date. This means that no matter how low a stock goes, the investor has the right to sell the stock for the agreed upon price.

Why would an investor purchase insurance on the stock instead of using the stop-loss? The answer to that question will be specific to each investor, but things like tax considerations when selling, dividend payments, or belief in the company through a rough patch are all very valid reasons that an investor might buy insurance on the stock.

There is a special set of terms associated with option contracts:

- An option to *buy* the specified item at a fixed price is a **call;** an option to *sell* is a **put.**
- The fixed price specified in an option contract is called the option's **strike price** or **exercise price.**
- The date after which an option can no longer be exercised is called its **expiration date** or *maturity* date.

If an option can be exercised on the expiration date only, it is called a *European-type* option. If it can be exercised at any time up to and including the expiration date it is called an *American-type* option.

Lesson 29

THE DIVERSIFICATION PRINCIPLE

TOPIC 146: THE DIVERSIFICATION PRINCIPLE

- Diversifying means splitting an investment among many risky assets instead of concentrating it all in only one.
- The diversification principle states that by diversifying across risky assets people can sometimes achieve a reduction in their overall risk exposure with no reduction in their expected return.

Diversification with Uncorrelated Risks

- Suppose you are planning to invest $100,000 in a medicine business.
- Assume there is a 0.5 probability of success for each medicine and 0.5 probability of failure.
- If the medicine gets successful, you will get $400,000. If it fails, you will get nothing.

Probability Distribution for Investment in a Single Medicine

Outcome	Probability (Pi)	Payoff (XI)	Rate of Return) rj
Medicine fails	0.5	0	-100%
Medicines succeeds	0.5	$400,000	300%

- The cost of developing a medicine is $100,000. The rate of return is the payoff minus the cost divided by the cost.

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Diversification with Two Drugs

Outcome	Probability (Pi)	Payoff (Xi)	Rate of Return (r_j)
No drugs succeed	0.25	0	-100%
One drug succeeds	0.5	$200,0000	100%
Both drugs succeed	0.25	$400,000	300%

- The cost of developing a medicine is $100,000. The rate of return is the payoff minus the cost divided by the cost.

Advantages of Diversification

- Thus, by diversifying and holding a portfolio of two drugs you reduce the probability of losing your entire investment to only one-half of what it would be without diversification.
- The probability of winding up with $400,000 has fallen from 0.5 to 0.25.

If one were to poll investors and investment professionals to determine their ideal investment outcome, the vast majority would no doubt agree: It's a double-digit total return in all economic environments, each and every year.

Naturally, they would also agree that the worst-case scenario is an overall decrease in asset value. But despite this knowledge, very few achieve this desired outcome; and many, indeed, encounter the worst-case scenario—losses. The reasons for this are diverse: misallocation of assets, pseudo-diversification, hidden correlation, weighting imbalance, false returns, and underlying devaluation.

The solution, however, could be simpler than you would expect. In this article, we will show how to achieve true diversification through asset class selection, rather than stock picking and market timing.

Many investors do not truly understand effective diversification, often believing they are fully diversified after spreading their investment across large-, mid- or small-cap stocks; energy, financial, health care or technology stocks; or even investing in emerging markets. In reality, however, they have merely invested in multiple sectors of the equities asset class and are prone to the rise and fall within that market.

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TOPIC 147: THE DIVERSIFICATION PRINCIPLE- PROBABILITY DISTRIBUTION

For diversification, the probabilities of all the possible outcomes are considered in the form of a probability distribution (discussed earlier). Expected rate of returns and standard deviations are estimated to decide how diversification will be done.

- The formula for the expected payoff is:
- $E(X) = \sum_{i=1}^{n} P_i \, X_i$
- Expected Payoff – One drug ➔ E(X) = 0.5 (0)+0.5(400,000) = $200,000
- Standard Deviation:
- $\sigma = \sqrt{0.5(0 - 200{,}000)^2 + 0.5(400{,}000 - 200{,}000)^2}$
- = $200,000
- Expected Payoff – Two drugs ➔
- E(X) = 0.25 (0) + 0.5(200,000)+0.25(400,000) = $200,000
- Standard Deviation:
- $\sigma =$ $\sqrt{0.25(0 - 200{,}000)^2 + 0.5(200,\ 000 - 200{,}000)^2 + 0.25(400{,}000 - 200{,}000)^2}$
- = $141,421

Advantage of Diversification

- When we diversify between two uncorrelated drugs the expected payoff remains $200,000, but the standard deviation of the rate of return falls from $200, 000 to $141,421.
- If the number of drugs in the portfolio increases further:
- The expected payoff stays the same, but
- The standard deviation declines in proportion to the square root of the number of drugs:
- $\sigma_{portfolio} = \$200{,}000/\sqrt{N}$

TOPIC 148: NON-DIVERSIFIABLE RISK

- Many important risks are positively correlated with each other because they get affected by the same underlying economic factors.

Diversifiable and Non-diversifiable Risks

- The part of portfolio volatility that can be eliminated by adding more stocks is the diversifiable risk, and
- The part that remains no matter how many stocks are added is the non-diversifiable risk.

What accounts for Non-diversifiable Risk?

- If an event occurs that affects many firms, such as an unanticipated downturn in general economic conditions, then many stocks will be affected.
- The risk of loss arising from such events is called market risk.

Firm-Specific Risk

- Random events that affect the prospects of only one firm, such as a lawsuit, a strike or a new product failure, give rise to random losses that are uncorrelated across stocks and can, be diversified away.
- Such risk is called firm-specific risk.

International Diversification

- By combining stocks of firms located in different countries it is possible to reduce the risk of one's stock portfolio, but there is a limit to this risk reduction.
- International diversification can improve the prospects for risk reduction for people around the world, a significant amount of risk remains for even the best-diversified global stock portfolio.

Non-diversifiable risk can be referred to a risk which is common to a whole class of assets or liabilities. The investment value might decline over a specific period of time only due to economic changes or other events which affect large sections of the market. However, diversification and asset allocation can provide protection against non-diversifiable risk as different sections of the market have a tendency to underperform at different times. Non-diversifiable risk can also be referred as market risk or systematic risk.

Putting it simple, risk of an investment asset (real estate, bond, stock/share, etc.) which cannot be mitigated or eliminated by adding that asset to a diversified investment portfolio can be delineated as non-diversifiable risks. Moreover, this is the risk you are exposed to in an individual investment. This risk type is involved in almost every investment, i.e. uncertainty of market moving up or down and the particular movement of the investment.

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Understanding non-diversifiable risk

Being unavoidable and non-compensating for exposure to such risks, non-diversifiable risk can be taken as the significant section of an asset's risk attributable to market factors affecting all firms. The main reasons for this risk type include inflation, war, political events, and international incidents. Moreover, it cannot be purged through diversification.

Factors responsible for non-diversifiable risk

Non-diversifiable risk is an outcome of factors influencing the complete market like changes in investment policy, foreign investment policy, alterations in taxation clauses, altering of socio-economic parameters, global security threats and measures, etc. the non-diversifiable risk is not under the investors' control and is also difficult to be mitigated to a large extent.

However, non-diversifiable risks are identified through the analysis and estimation of the statistical relationships between the different asset portfolios of the company through different techniques, including principal components analysis. There is no specific method that can be used to handle the non-diversifiable risk. This is due to their impact which is reflected on the entire market.

TOPIC 149: DIVERSIFICATION AND THE COST OF INSURANCE

- The cost of insuring a diversified portfolio of risks against a loss is almost always less than the cost of insuring against each risk separately.

Example

- Suppose you are investing $100,000 in drug stocks.
- If you invest $50,000 in each of the two drug's stocks you will have to pay separately for insurance.
- But for the portfolio as a whole, you will have to pay for one insurance.
- The more diversified are the risks in a portfolio of a given size, the less it will cost to insure the portfolio's total value against a loss.

The cost of insuring a diversified portfolio of risks against a loss is almost always less than the cost of insuring against each risk separately. To see why, let us return to the biotechnology example of section 11.10.1. You are investing $100,000 in drug stocks. The success or failure of each drug is independent of the others.

Let us assume that you have decided to invest $50,000 in each of two drug stocks and, therefore, face the probability distribution in Table 11.4. For each separate drug stock there is a probability of 0.5 that you will lose 100% of your investment in that drug. But for the portfolio as a whole there is a probability of 0.25 that you will lose 100% of your $100,000 investment.

If you insure *each* $50,000 stock investment against a loss, it will cost more than insuring the $100,000 portfolio of *both* stocks against a loss. To see this, suppose that the cost of insurance equals the expected amount that the insurance company will pay you. Then the cost of an insurance policy on the total portfolio would be the probability of a loss times the magnitude of the loss:

$$0.25 \times \$100,000 = \$25,000$$

The cost of insuring each of the two separate investments would be the probability of both stocks losing (0.25) times $100,000 plus the probability of only one losing (0.5) times $50,000:

$$0.25 \times \$100,000 + 0.5 \times \$50,000 = \$50,000$$

So insuring each stock separately would cost *twice* as much as insuring the portfolio of two stocks. To protect against loss of wealth, you do not need separate insurance on each stock. If only one of the two drugs fails, then the profit from the successful drug will more than offset the loss from the failed one, so that your total wealth will be $200,000. You only need insurance against the risk that *both* drugs will fail (see Box 11.2). This example suggests the following general proposition:

> The more diversified are the risks in a portfolio of a given size, the less it will cost to insure the portfolio's total value against a loss.

Lesson 30

THE PROCESS OF PERSONAL PORTFOLIO SELECTION

TOPIC 150: THE PROCESS OF PERSONAL PORTFOLIO SELECTION

- The process of Portfolio Selection describes how people should invest their wealth.
- A person's wealth includes all of his or her assets (stocks, bonds, shares, houses, pension benefits, insurance policies, etc.)
- There is no single portfolio selection strategy that is best for all people.
- It is a process of trading off risk and expected return to find the best portfolio of assets and liabilities.
- There are however some principles that are to be followed for portfolio selection
- For example, the principle of diversification.

Determinants of Personal Portfolio Selection

- The Life Cycle
- Time Horizons
- Risk Tolerance
- The Role of Professional Asset Managers

A portfolio is a collection of financial investments like stocks, bonds, commodities, cash, and cash equivalents, including closed-end funds and exchange traded funds (ETFs). People generally believe that stocks, bonds, and cash comprise the core of a portfolio. Though this is often the case, it does not need to be the rule. A portfolio may contain a wide range of assets including real estate, art, and private investments.

You may choose to hold and manage your portfolio yourself, or you may allow a money manager, financial advisor, or another finance professional to manage your portfolio.

One of the key concepts in portfolio management is the wisdom of diversification—which simply means not putting all of your eggs in one basket. Diversification tries to reduce risk by allocating investments among various financial instruments, industries, and other categories. It aims to maximize returns by investing in different areas that would each react differently to the same event. There are many ways to diversify.

How you choose to do it is up to you. Your position in the life cycle, time horizon, yYour goals for the future, your appetite for risk, and your personality are all factors in deciding how to build your

portfolio. It may also get affected by the type of adice you will get from your professional asset manager.

Regardless of your portfolio's asset mix, all portfolios should contain some degree of diversification, and reflect the investor's tolerance for risk, return objectives, time horizon, and other pertinent constraints, including tax position, liquidity needs, legal situations, and unique circumstances.

TOPIC 151: PERSONAL PORTFOLIO SELECTION – LIFE CYCLE

- In portfolio selection the best strategy depends on an individual's personal circumstances (age, family status, occupation, income, wealth, etc.)
- For a young couple recently starting a family it may be optimal to buy a house and take out a mortgage loan.
- For an older couple about to retire, it may be optimal to sell their house and invest the proceeds in some asset that will provide a steady stream of income for as long as they live.
- People of the same age, with the same income and wealth, may have different perspectives on buying a house or buying insurance.
- The same is true of investing in stocks, bonds, and other securities.
- There is no single portfolio that is best for all people.

In portfolio selection the best strategy depends on an individual's personal circumstances (age, family status, occupation, income, wealth, etc.). For some people holding a particular asset may add to their total risk exposure, but for others the same asset may be risk reducing. An asset that is risk reducing at an early stage in the life cycle may not be at a later stage.

For a young couple starting a family it may be optimal to buy a house and take out a mortgage loan. For an older couple about to retire it may be optimal to sell their house and invest the proceeds in some asset that will provide a steady stream of income for as long as they live.

TOPIC 152: PERSONAL PORTFOLIO SELECTION – TIME HORIZONS

Time Horizon

- In formulating a plan for portfolio selection you begin by determining your goals and time horizon.
- The planning horizon is the total length of time for which one plans.

- The longest time horizon would typically correspond to the retirement goal and would be the balance of one's lifetime.
- Thus for a 25 year old who expects to live to the age of 85, the planning horizon will be 60 years.
- There are shorter planning horizons also.
- For example, if you have a ten years old child and you plan to pay for his education when he reaches age 22, the planning horizon for this goal is 12 years.

Decision Horizon

- The decision horizon is the length of time between decisions to revise the portfolio.
- The length of the decision horizon is controlled by the individual within certain limits.
- Some people review their portfolios at regular intervals, e.g. once in a month, etc. or once a year

Strategy

- Portfolio decisions you make today are influenced by what you think might happen tomorrow. A plan that takes account of future decisions in making current decisions is called a strategy.

Investors should consider how long they have to invest when building a portfolio. In general, investors should be moving toward a conservative asset allocation as their goal date approaches, to protect the portfolio's earnings up to that point.

For example, a conservative investor might favor a portfolio with large-cap value stocks, broad-based market index funds, investment-grade bonds, and a position in liquid, high-grade cash equivalents.

TOPIC 153: PERSONAL PORTFOLIO SELECTION – RISK TOLERANCE

Risk Tolerance

- Risk tolerance is a measure of how much of a loss an investor is willing to endure within their portfolio.
- A person's age, investment goals, income, and comfort level all play into determining their risk tolerance.
- It looks at how much market risk—stock volatility, stock market swings, economic or political events, regulatory, or interest rate changes—an investor can tolerate, considering that all of these factors might cause their portfolio to slide.

Risk tolerance assessments

- Risk tolerance assessments for investors are done through risk-related surveys or questionnaires.
- As an investor, you may also want to review historical worst-case returns for different asset classes to get an idea of how much money you would feel comfortable losing if your investments have a bad year or bad series of years.

Factors affecting Risk Tolerance

These include:

- the time horizon you have to invest,
- your future earning capacity, and
- the presence of other assets such as a home, pension, Social Security, or an inheritance.

Levels of Risk Tolerance

- An aggressive investor, or someone with higher risk tolerance, is willing to risk more money for the possibility of better returns than a conservative investor, who has lower tolerance.
- A person with moderate risk tolerance sits in the balance between an aggressive and conservative investor.

Risk tolerance is the degree of risk that an investor is willing to endure given the volatility in the value of an investment. An important component in investing, risk tolerance often determines the type and amount of investments that an individual chooses.

Greater risk tolerance is often synonymous with investment in stocks, equity funds, and exchange-traded funds (ETFs), while lower risk tolerance is often associated with the purchase of bonds, bond funds, and income funds.

TOPIC 154: PORTFOLIO SELECTION – ROLE OF PROFESSIONAL ASSET MANAGERS

- Most people have neither the knowledge nor the time to carry out portfolio optimization.
- They hire an investment advisor to do it for them or they buy finished product from a financial intermediary.

Finished Products

- Finished products include:

– Various investment accounts and mutual funds offered by banks, securities firms, investment companies, and insurance companies.

How Financial Institutions develop Products?

- Financial intermediaries decide what asset choices to offer to households.
- The selection is done by examining the quantitative trade-off between risk and expected return.

Objective of Professional Asset Managers

- To find the portfolio that offers investors the highest expected rate of return for any degree of risk they are willing to tolerate.

An asset manager manages assets on behalf of someone else, making important investment decisions that will help the client's portfolio grow. An asset manager also ensures the client's investment doesn't depreciate and that exposure to risk is mitigated. Doing this means watching the market, keeping up to date with research and trends, and staying current with political, financial, and economic news. The advice of the asset manager plays an important role and can shape the portfolio selection process to a great extent.

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Lesson 31

PORTFOLIO OPTIMIZATION

TOPIC 155: PORTFOLIO OPTIMIZATION

- It is done in two steps:
  - 1. Find the combination of risky assets, and
  - 2. mix this optimal risky-asset portfolio and a riskless asset.

The single risky-asset portfolio is composed of many risky assets chosen in an optimal way.

- A riskless asset is defined as a security that offers a perfectly predictable rate of return in terms of account selected for the analysis and the investor's decision horizon.
- When no specific investor is identified, the riskless asset refers to an asset that offers a predictable rate of return over the trading horizon (i.e., the shortest possible decision horizon).

Example

- If Pakistani Rupee is taken as a unit of account and trading horizon is a day, the riskless rate is the interest rate on Treasury bill maturing the next day.

Optimization is the process of making a trading system more effective by adjusting the variables used for technical analysis. A trading system can be optimized by reducing certain transaction costs or risks, or by targeting assets with greater expected returns. The single risky-asset portfolio is composed of many risky assets chosen in an optimal way.

Portfolio optimization is often done as a two-step process:

1. Find the optimal combination of risky assets, and
2. Mix this optimal risky-asset portfolio with the riskless asset

A riskless asset is defined as a security that offers a perfectly predictable rate of return in terms of account selected for the analysis and the investor's decision horizon.

When no specific investor is identified, the riskless asset refers to an asset that offers a predictable rate of return over the trading horizon (i.e., the shortest possible decision horizon).

If Pakistani Rupee is taken as a unit of account and trading horizon is a day, the riskless rate is the interest rate on Treasury bill maturing the next day.

TOPIC 156: FRAMEWORK USED BY PROFESSIONAL PORTFOLIO MANAGERS

- The framework used by professional portfolio managers for examining the quantitative trade-off between risk and expected return.

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Combining the Riskless Asset and a Single Risky Asset

- Suppose you have Rs. 100,000 to invest. You are choosing between a riskless asset with an interest rate of 0.06 per year and a risky asset with an expected rate of return of 0.14 per year and a standard deviation of 0.2. How much of your Rs 100,000 should you invest in the risky asset?

How Portfolio Composition is Determined?

- The portfolio composition for any point lying on the trade-off line in the figure shown above, are based on the points listed in the table discussed earlier.

Portfolio management is the art and science of selecting and overseeing a group of investments that meet the long-term financial objectives and risk tolerance of a client, a company, or an institution.

Some individuals do their own investment portfolio management. That requires a basic understanding of the key elements of portfolio building and maintenance that make for success, including asset allocation, diversification, and rebalancing. Most people have neither the knowledge nor the time to carry out portfolio optimization.

They hire an investment advisor to do it for them or they buy finished product from a financial intermediary. Finished products include:

Various investment accounts and mutual funds offered by banks, securities firms, investment companies, and insurance companies.

The objective of the professional asset manager is to find the portfolio that offers investors the highest expected rate of return for any degree of risk they are willing to tolerate.

The Professional Portfolio Managers use risk-reward tradeoff to decide between (among) various investment options.

1. The risk-return tradeoff is an investment principle that indicates that the higher the risk, the higher the potential reward.
2. To calculate an appropriate risk-return tradeoff, investors must consider many factors, including overall risk tolerance, the potential to replace lost funds and more.
3. Investors consider the risk-return tradeoff on individual investments and across portfolios when making investment decisions.

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When an investor considers high-risk-high-return investments, the investor can apply the risk-return tradeoff to the vehicle on a singular basis as well as within the context of the portfolio as a whole. Examples of high-risk-high return investments include options, penny stocks and leveraged exchange-traded funds (ETFs). Generally speaking, a diversified portfolio reduces the risks presented by individual investment positions. For example, a penny stock position may have a high risk on a singular basis, but if it is the only position of its kind in a larger portfolio, the risk incurred by holding the stock is minimal.

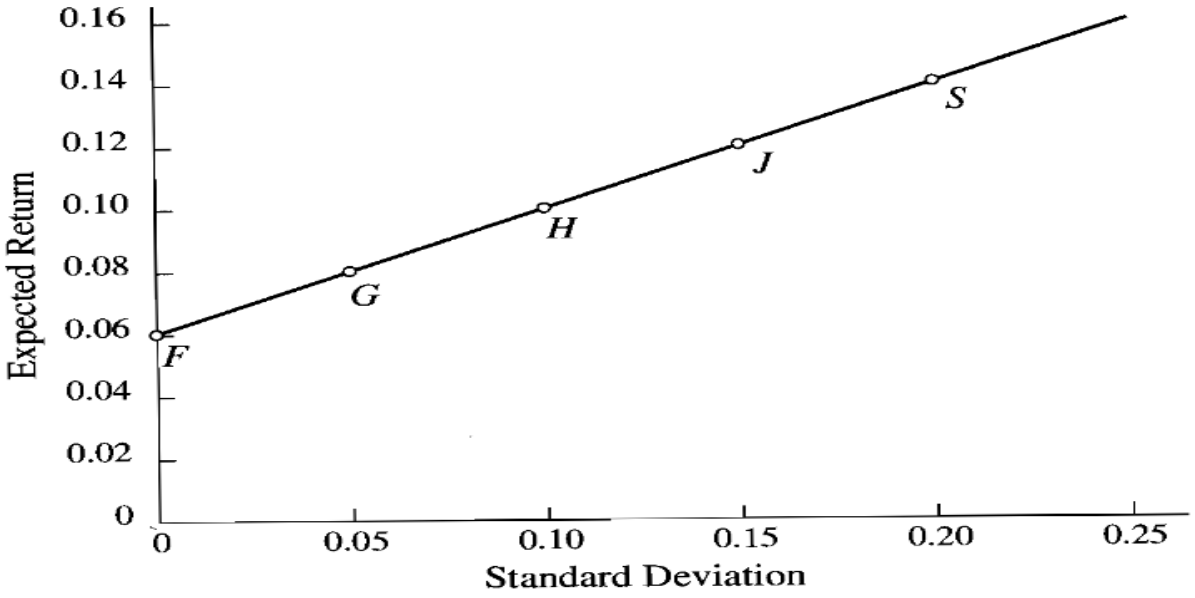
Risk-Return Tradeoff at the Portfolio Level

That said, the risk-return tradeoff also exists at the portfolio level. For example, a portfolio composed of all equities presents both higher risk and higher potential returns. Within an all-equity portfolio, risk and reward can be increased by concentrating investments in specific sectors or by taking on single positions that represent a large percentage of holdings. For investors, assessing the cumulative risk-return tradeoff of all positions can provide insight on whether a portfolio assumes enough risk to achieve long-term return objectives or if the risk levels are too high with the existing mix of holdings.

The Professional Portfolio Managers use quantitative analysis of risk and reward tradeoff to do the portfolio management.

Suppose you have Rs. 100,000 to invest. You are choosing between a riskless asset with an interest rate of 0.06 per year and a risky asset with an expected rate of return of 0.14 per year and a standard deviation of 0.2. How much of your Rs 100,000 should you invest in the risky asset?

Portfolio (1)	Proportion Invested in the Risky Asset (2)	Proportion Invested in Riskless Asset (3)	Expected Rate of Return E(r) (4)	Standard Deviation σ (5)
F	0	100%	0.06	0.00
G	25%	75%	0.08	0.05
H	50%	50%	0.10	0.10
I	75%	25%	0.12	0.15
J	100%	0%	0.14	0.20



The Risk-Reward Trade-Off Line (Capital Allocation Line)

The portfolio composition for any point lying on the trade-off line in the figure shown above, not only the points listed in the table discussed earlier.

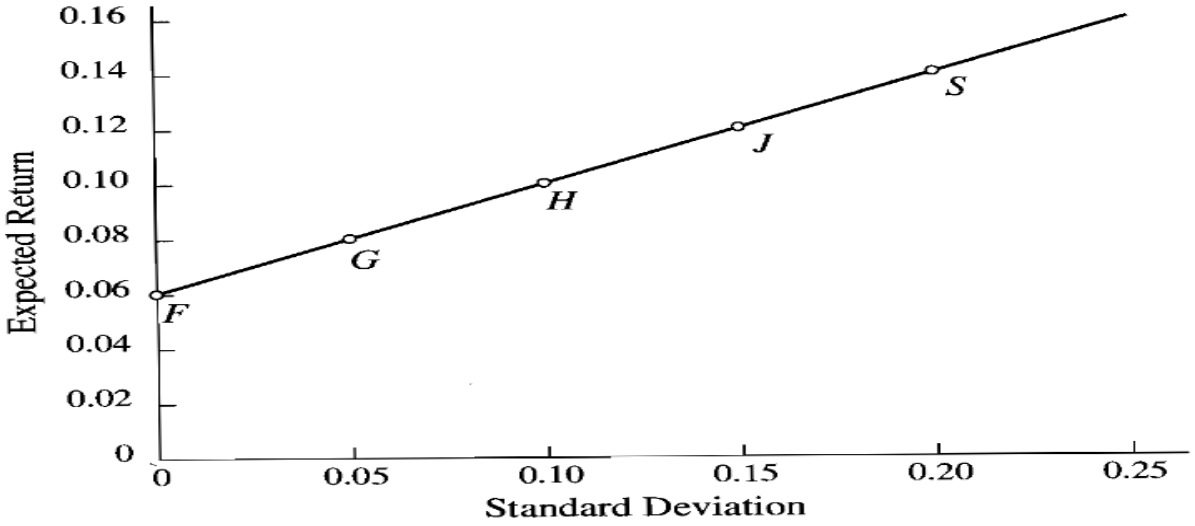
TOPIC 157: FORMULA FOR THE TRADE-OFF LINE

Step 1: Relate the portfolio's expected return to the proportion invested in the risky asset.

E(r) = w E(r_s) +(1-w)r_f

= r_f+ w [E(r_s) - r_f]

The Risk-Reward Trade-Off Line



Example: Suppose w denote the proportion of the PKR 100,000 investment to be allocated to the risky asset. Then, (1-w) will be invested in riskless asset.

Substituting 0.06 for r_f **and 0.14 for** (r_s)

The Expected Rate of Return E(r) will be: E(r) = 0.06+w(0.14-0.06) = 0.06 + 0.08 w

Interpretation

The portfolio is expected to earn a risk premium which depends on:

1. The risk premium on the risky asset E(r_s) - r_f, which is 0.14-0.06=0.08 in the above example
2. The proportion of the portfolio invested in the risky asset i.e. w

TOPIC 158: PORTFOLIO COMPOSITION AND AN EXPECTED RATE OF RETURN

To find portfolio composition corresponding to an expected rate of return we use the following formula:

E(r) = r_f+ w [E(r_s) - r_f]

Suppose, a person wants to know the portfolio composition corresponding to an expected rate of return of 0.09, risk premium is 0.08 and risk free interest is 0.06. Substituting these values in the formula discussed above, we get:

0.09 = 0.06 +0.08w

w = ($\mathbf{0.09 - 0.06}$)/($\mathbf{0.08}$) = 0.375

Thus, the portfolio mix is 37.5% risky asset and 62.5% riskless asset.

TOPIC 159: PORTFOLIO COMPOSITION AND STANDARD DEVIATION

After determining the portfolio composition, Professional Portfolio Managers relate the portfolio standard deviation to the proportion invested in the risky asset.

If σ_s is the standard deviation of the risky asset, the portfolio's standard deviation is:

$\sigma=\sigma_s w$

Example: Using the data given in the example discussed earlier

w = 0.375

$\sigma_s=0.2$

$\sigma=\sigma_s w$

= 0.2X0.375 = 0.075

Thus, the portfolio standard deviation is 0.075.

Relate the portfolio expected rate of return to its standard deviation.

Rearranging $\sigma=\sigma_s w$ for w,

w = σ/σ_s

Substituting value of w in:

E(r) = w E(r_s) +(1-w)r_f

= r_f+ w [E(r_s) - r_f], we get

E(r) = r_f +(E(r_s) − r_f)/σ_s σ

The Sharpe (Reward-to-Volatility) Measure

E(r) = r_f +(E(r_s) − r_f)/σ_s σ

S = Sharpe Ratio

"E(" r_s ") − " r_f = Portfolio Risk Premium

σ_s = Standard deviation of Portfolio Excess Return

Example: Suppose r_f= 0.06; $\sigma_s=0.2$;

E(r_s) - r_f=0.14-0.06=0.08

"E(r) =" r_f +("E(" r_s ") − " r_f)/σ_s σ

=0.06+ "0.08" /(0.2) σ =0.06+0.4 σ

The slope of the trade-off line measures the extra expected return the market offers for each unit of extra risk an investor is willing to bear.

TOPIC 160: ACHIEVING A TARGET EXPECTED RETURN

Find the portfolio corresponding to an expected rate of return of 0.11 per year. What is its standard deviation?

Expected Rate of Return = 0.11

Substituting the given values in the equation

E(r) = w E(r_s) +(1-w)r_f

= r_f+ w [E(r_s) - r_f]

and solving for w

0.11 = 0.06+0.08w

w = ($\mathbf{0.11-0.06}$)/($\mathbf{0.08}$) = 0.625

Thus, the portfolio mix is 62.5% risky asset and 37.5% riskless asset.

To find standard deviation, we substitute w= 0.625 in

σ=σ_s w = 0.2X0.625=0.125

Thus, the portfolio standard deviation is 0.125.

Lesson 32

PORTFOLIO EFFICIENCY

TOPIC 161: PORTFOLIO EFFICIENCY

- An efficient portfolio is defined as the portfolio that offers the investor the highest possible expected rate of return at a specified level of risk. Investors must examine their own risk-return preferences to determine where they should invest in the efficient frontier.

Efficient Frontier

- The line that connects all these efficient portfolios is known as the efficient frontier.
- The efficient frontier represents those portfolios that have the maximum rate of return for every given level of risk.
- The last thing investors want is a portfolio with a low expected return and a high level of risk.

Modern Portfolio Theory (MPT)

- Investors must examine their own risk-return preferences to determine where they should invest in the efficient frontier.
- This concept was first formulated by Harry Markowitz in 1952. This concept is called the modern portfolio theory (MPT).

Overall Portfolio's Risk and Return

- According to MPT, an investment's risk and return characteristics should not be viewed alone. Instead, they should be evaluated by how the investment affects the overall portfolio's risk and return.

Inefficient Portfolio (1/2)

- An inefficient portfolio exposes an investor to a higher degree of risk than necessary to achieve a target return.
- For example, a portfolio of high-yield bonds expected to provide only the risk-free rate of return would be said to be inefficient.
- An investor could achieve the same return by purchasing Treasury bills, which are considered among the safest investments in the world (rather than high-yield bonds, which are, by definition, rated as risky investments).

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Portfolio efficiency is achieved by applying Modern Portfolio Theory (MPT). This concept was first formulated by Harry Markowitz in 1952. According to MPT, an investment's risk and return characteristics should not be viewed alone. Instead, they should be evaluated by how the investment affects the overall portfolio's risk and return.

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An investor could achieve the same return by purchasing Treasury bills, which are considered among the safest investments in the world (rather than high-yield bonds, which are, by definition, rated as risky investments).

TOPIC 162: EFFICIENT FRONTIER

- The efficient frontier theory was introduced by Nobel Laureate Harry Markowitz in 1952 and is a cornerstone of modern portfolio theory (MPT). The efficient frontier rates portfolios (investments) on a scale of return (y-axis) versus risk (x-axis).
- The compound annual growth rate (CAGR) of an investment is commonly used as the return component while standard deviation (annualized) depicts the risk metric.
- The efficient frontier graphically represents portfolios that maximize returns for the risk assumed. Returns are dependent on the investment combinations that make up the portfolio.
- The less synchronized the securities (lower covariance), the lower the standard deviation. If this mix of optimizing the return versus risk paradigm is successful, then that portfolio should line up along the efficient frontier line.

The line that connects all these efficient portfolios is known as the efficient frontier.

The efficient frontier represents those portfolios that have the maximum rate of return for every given level of risk.

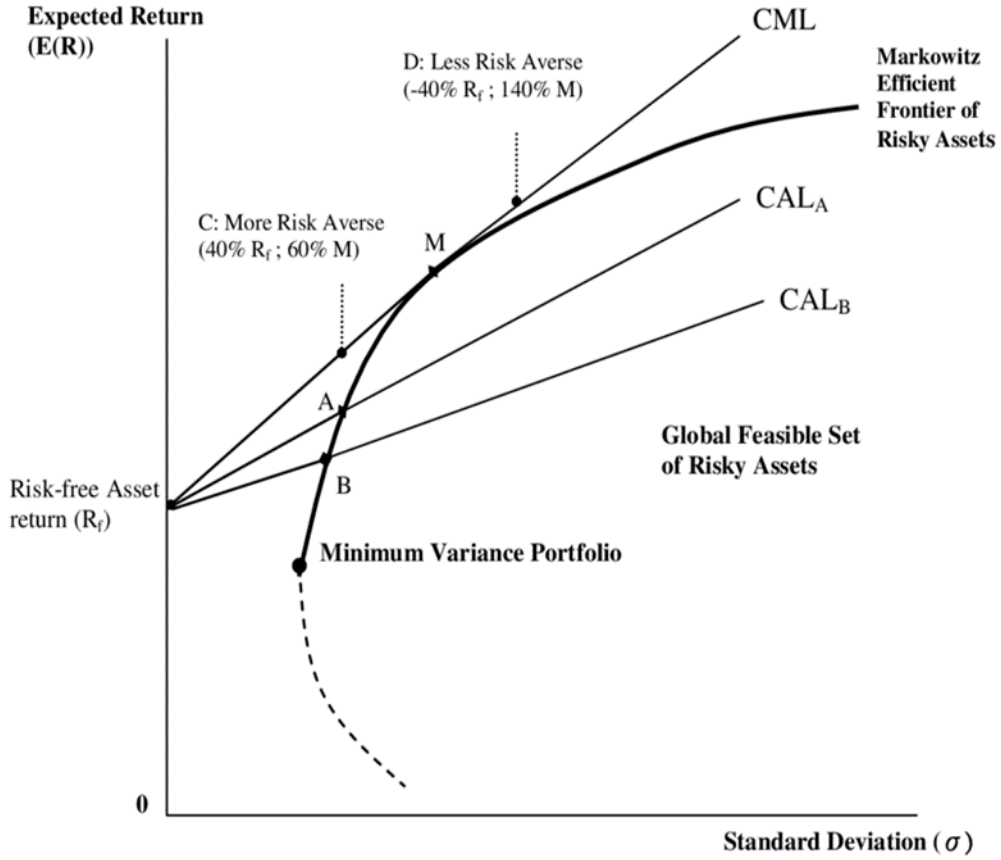
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The less synchronized the securities (lower covariance), the lower the standard deviation. If this mix of optimizing the return versus risk paradigm is successful, then that portfolio should line up along the efficient frontier line.

The ultimate capital allocation line (CAL) tangent to the optimal risky portfolio is termed the capital market line (CML), which offers the highest possible expected return for any given level of risk, and the lowest possible risk for any given level of expected return.



- The ultimate capital allocation line (CAL) tangent to the optimal risky portfolio is termed the capital market line (CML), which offers the highest possible expected return for any given level of risk, and the lowest possible risk for any given level of expected return.

TOPIC 163: EFFICIENT DIVERSIFICATION WITH MANY RISKY ASSETS

- Portfolio risk depends on the correlation between the returns of the assets in the portfolio
- Covariance and the correlation coefficient provide a measure of the returns on two assets to vary.

Two Asset Portfolio Return

- Portfolios of Two Risky Assets

Expected Rate of Return of the Portfolio

E(r) = w E(r1) + (1-w) E(r2)

Efficient Diversification with Many Risky Assets

Variance = $\boldsymbol{\sigma^2}$

= $\boldsymbol{w^2\ \sigma_1^2 + 〖(1-w)〗^2\ \sigma_2^2 + 2w(1-w)\rho_{(1,2)}\ \sigma_1\ \sigma_2}$

Covariance = $\boldsymbol{\rho_{(1,2)}\ \sigma_1\ \sigma_2}$

$\boldsymbol{\rho_{(1,2)}}$ = $\boldsymbol{(Cov\ (r_1, r_{(2}))}$" ")/($\boldsymbol{\sigma_1\ \sigma_2}$)

If Asset 2 is risk free $\boldsymbol{\sigma_2}$=0

Correlation Coefficients: Possible Values

$\boldsymbol{\rho}$ is correlation coefficient.

Range of values for $\boldsymbol{\rho_{(1,2)}}$

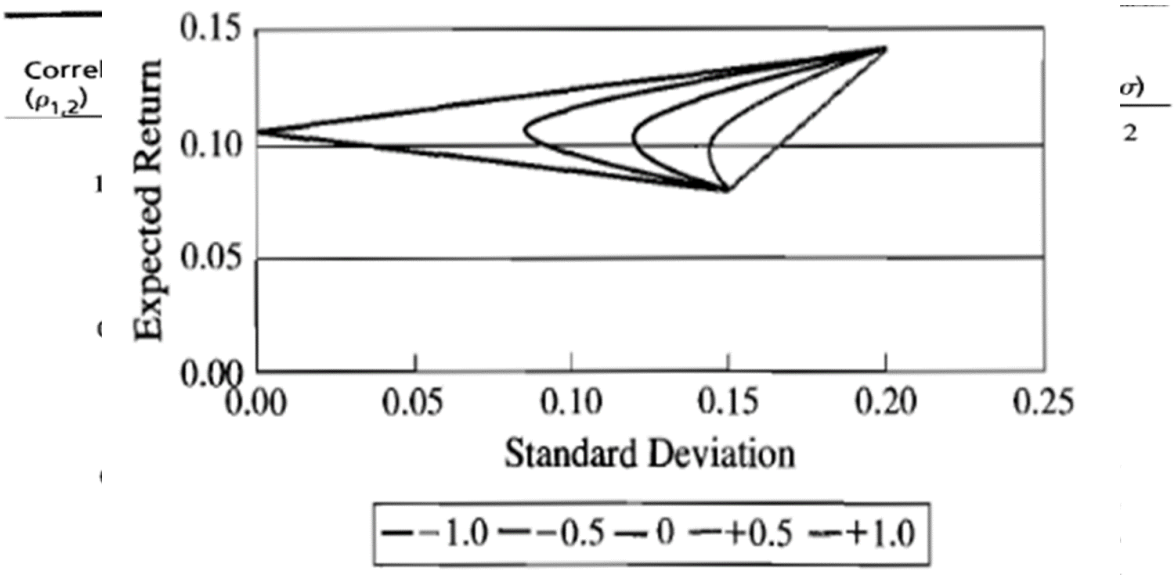
$-1.0 < \boldsymbol{\rho_{(1,2)}} < 1.0$

If $\boldsymbol{\rho_{(1,2)}}$ = 1.0, the securities would be perfectly positively correlated

If $\boldsymbol{\rho_{(1,2)}}$ = - 1.0, the securities would be perfectly negatively correlated

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Risk Reward Tradeoff between two risky assets



−0.50	25.00%	75.00%	9.50%	9.76%
	40.54%	59.46%	10.43%	*8.54%
	50.00%	50.00%	11.00%	9.01%
	75.00%	25.00%	12.50%	13.52%
−1.00	25.00%	75.00%	9.50%	6.25%
	42.86%	57.14%	10.57%	*0.00%
	50.00%	50.00%	11.00%	2.50%
	75.00%	25.00%	12.50%	11.25%
	100.00%	0.00%	14.00%	20.00%

Risk Reward Tradeoff Curves: Risky Assets

Minimum -Variance Portfolio

Risky Asset 1 =36%

Risky Asset 2 = 64%

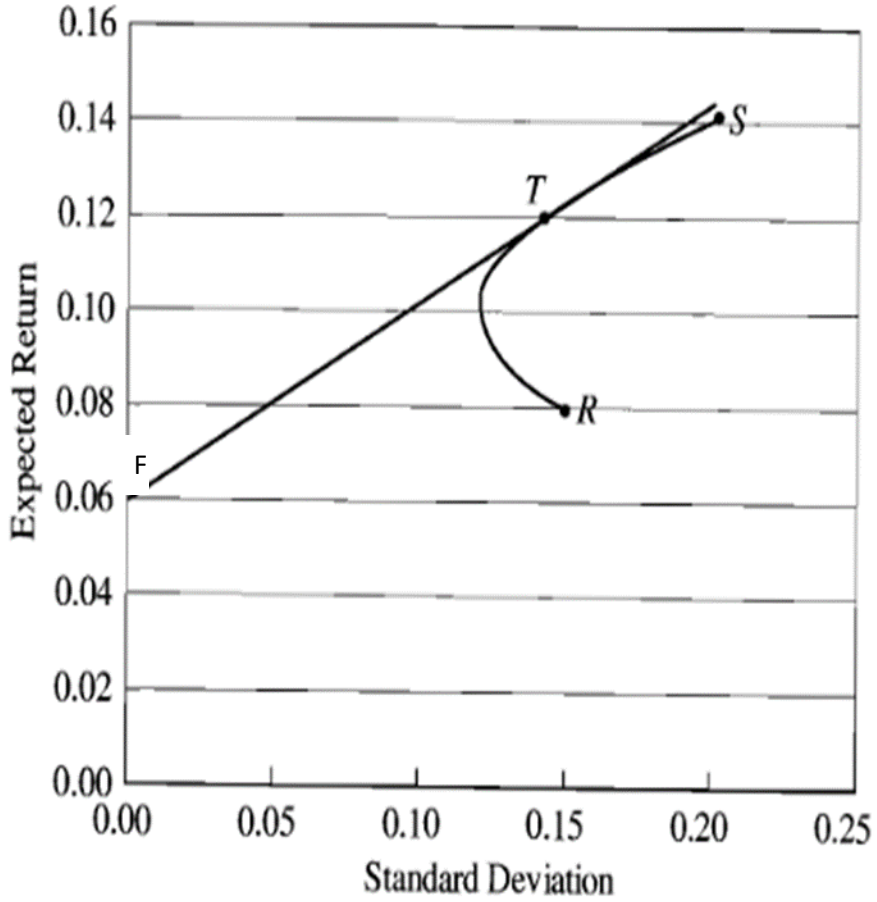
Expected Return = 10.15%

Standard Deviation = 12%

Correlation Coefficient = 0.00

TOPIC 164: OPTIMAL COMBINATION OF RISKY ASSETS

- The optimal combinations result in lowest level of risk for a given return
- The optimal trade-off is described as the efficient frontier.
- For Graphical Analysis, the optimal combination is obtained by drawing a tangent on the efficiency frontier.



- The point corresponding to the point of tangency is the risky portfolio, indication the optimal combination of risky assets.
- Optimal = Most efficient portfolios

Graphical Illustration

The point corresponding to the point of tangency is the risky portfolio, indication the optimal combination of risky assets.

Optimal = Most efficient portfolios

Formula for Efficient Combination

$$w_1 = \frac{[E(r_1) - r_f]\sigma_2^2 - [E(r_2) - r_f]\rho_{1,2}\sigma_1\sigma_2}{[E(r_1) - r_f]\sigma_2^2 + [E(r_2) - r_f]\sigma_1^2 - [E(r_1) - r_f + E(r_2) - r_f]\rho_{1,2}\sigma_1\sigma_2}$$

$$w_2 = 1 - w_1$$

Where w1 and w2 are the proportions of the risky assets.

Selection of the Preferred Portfolio

- Any point between F and T
- Depending on:
- Planning Horizon
- Life Cycle
- Risk Tolerance

Expected Rate of Return of the Portfolio

- To calculate a portfolio's expected return, an investor needs to calculate the expected return of each of its holdings, as well as the overall weight of each holding.
- The basic expected return formula involves multiplying each asset's weight in the portfolio by its expected return, then adding all those figures together.

Problem with Expected Return

- The expected return is usually based on historical data and is therefore not guaranteed.
- Since the market is volatile and unpredictable, calculating the expected return of a security is more guesswork than definite. So it could cause inaccuracy in the resultant expected return of the overall portfolio.
- Expected returns do not paint a complete picture, so making investment decisions based on them alone can be dangerous.

Standard Deviation of the Rate of Return of the Portfolio

- It is calculated to judge the realized performance of a portfolio manager.

- In a large fund with multiple managers with different styles of investing, a CEO or head portfolio manager might calculate the risk of continuing to employ a portfolio manager who deviates too far from the mean in a negative direction.
- This can go the other way as well, and a portfolio manager who outperforms their colleagues and the market can often expect a hefty bonus for their performance.

TOPIC 165: SELECTION OF THE PREFERRED PORTFOLIO

Optimal Combination of Risky Assets

- The optimal risky portfolio is found at the point where the CAL is tangent to the efficient frontier. This asset weight combination gives the best risk-to-reward ratio, as it has the highest slope for CAL.

$$w_1 = \frac{[E(r_1) - r_f]\sigma_2^2 - [E(r_2) - r_f]\rho_{1,2}\sigma_1\sigma_2}{[E(r_1) - r_f]\sigma_2^2 + [E(r_2) - r_f]\sigma_1^2 - [E(r_1) - r_f + E(r_2) - r_f]\rho_{1,2}\sigma_1\sigma_2}$$

$$w_2 = 1 - w_1$$

- *w1* and *w2* are the proportions of the risky assets.

Selection of the Preferred Portfolio

- Any point between F and T
- Depending on:
- Planning Horizon
- Life Cycle
- Risk Tolerance
- To find the optimal combination of risky assets, we do not need to know about investor's wealth or preferences.
- The composition of this portfolio depends on the expected rate of returns and standard deviations of Risky Assets 1 and 2 and the correlation between them.

Preferred Portfolio will be a portfolio shown by any point between F and T. Depending on:

- **Planning Horizon**
- **Life Cycle**

- **Risk Tolerance**

To calculate a portfolio's expected return, an investor needs to calculate the expected return of each of its holdings, as well as the overall weight of each holding.

The basic expected return formula involves multiplying each asset's weight in the portfolio by its expected return, then adding all those figures together.

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To find the optimal combination of risky assets, we do not need to know about investor's wealth or preferences.

The composition of this portfolio depends on the expected rate of returns and standard deviations of Risky Assets 1 and 2 and the correlation between them.

TOPIC 166: PORTFOLIOS OF MANY RISKY ASSETS

- When there are many risky assets, we use two-step method of portfolio construction.
- First, we consider portfolios constructed from the risky assets only
- Second, we find the tangency portfolio of risky assets to combine with the riskless asset

Point of Tangency and CAL

- The straight line connecting the riskless asset and the tangency point representing the optimal combination of risky assets is the best risk-reward trade-off line achievable.

- People have thousands of assets to choose from but they make their choices from a list of a few final products offered by financial intermediaries such as bank accounts, stock and bond mutual funds, and real estate.
- Intermediaries make use of the latest advances in financial technology but the basic mean-variance approach is still the dominant method.

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Lesson 33

CAPITAL ASSET PRICING MODEL (CAPM)

TOPIC 167: CAPITAL ASSET PRICING MODEL (CAPM)

- The Capital Asset Pricing Model is an equilibrium theory that is based on the theory of portfolio selection.
- It was developed in 1964 by William F. Sharpe.
- The Capital Asset Pricing Model (CAPM) describes the relationship between systematic risk and expected return for assets, particularly stocks. CAPM is widely used throughout finance for pricing risky securities and generating expected returns for assets given the risk of those assets and cost of capital.

Assumptions

1. Investors are rational, mean-variance optimizers.
2. Investors use identical input lists, referred to as homogeneous expectations.
3. All assets are publicly traded (short positions are allowed) and investors can borrow or lend at a common risk-free rate.

Market Portfolio

- A portfolio that holds all assets in proportion to their observed market values is called the market portfolio.

Example

- Suppose there are only three assets:
- GM stocks; Toyota stocks and the risk-free asset.
- Total market values of each at current prices are $66 billion of GM, $22 billion of Toyota, and $12 billion of the risk-free asset.
- Total market value of all assets is $100 billion.
- The CAPM says that in equilibrium any investor's relative holdings of risky assets will be the same as in the market portfolio.
- In our example, all investors will hold GM and Toyota stock in the proportion of 3 to 1 (i.e., 66/22).

The Capital Asset Pricing Model is an equilibrium theory that is based on the theory of portfolio selection. It was developed in 1964 by William F. Sharpe. The Capital Asset Pricing Model (CAPM) describes the relationship between systematic risk, or the general perils of investing, and expected return for assets, particularly stocks.

It is a finance model that establishes a linear relationship between the required return on an investment and risk. The model is based on the relationship between an asset's beta, the risk-free rate (typically the Treasury bill rate), and the equity risk premium, or the expected return on the market minus the risk-free rate.

CAPM evolved as a way to measure this systematic risk. It is widely used throughout finance for pricing risky securities and generating expected returns for assets, given the risk of those assets and cost of capital.

Assumptions

Investors are rational, mean-variance optimizers.

Investors use identical input lists, referred to as homogeneous expectations.

All assets are publicly traded (short positions are allowed) and investors can borrow or lend at a common risk-free rate.

The formula for calculating the expected return of an asset, given its risk, is as follows:

$ERi=Rf+\beta i(ERm-Rf)$

where: ERi=expected return of investment Rf=risk-free rate

βi=beta of the investment

$(ERm-Rf)$=market risk premium

Investors expect to be compensated for risk and the time value of money. The risk-free rate in the CAPM formula accounts for the time value of money. The other components of the CAPM formula account for the investor taking on additional risk.

The goal of the CAPM formula is to evaluate whether a stock is fairly valued when its risk and the time value of money are compared with its expected return. In other words, by knowing the individual parts of the CAPM, it is possible to gauge whether the current price of a stock is consistent with its likely return.

A portfolio that holds all assets in proportion to their observed market values is called the market portfolio.

Example: Suppose there are only three assets:

GM stocks; Toyota stocks and the risk-free asset.

Total market values of each at current prices are $66 billion of GM, $22 billion of Toyota, and $12 billion of the risk-free asset.

Total market value of all assets is $100 billion.

The CAPM says that in equilibrium any investor's relative holdings of risky assets will be the same as in the market portfolio.

In our example, all investors will hold GM and Toyota stock in the proportion of 3 to 1 (i.e., 66/22).

TOPIC 168: CAPITAL MARKET LINE (CML)

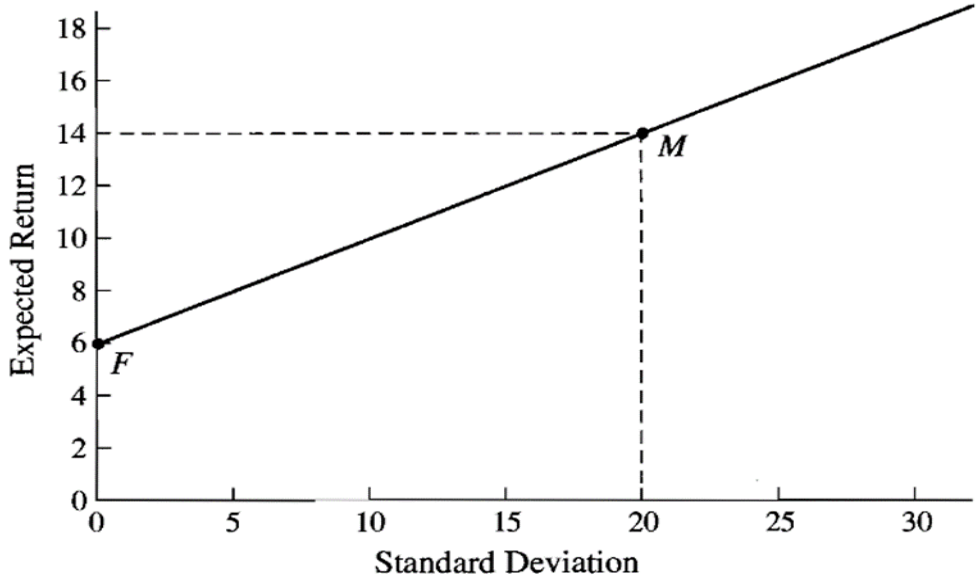
- The capital market line (CML) represents portfolios that optimally combine risk and return.
- CML is a special case of the capital allocation line (CAL) where the risk portfolio is the market portfolio. Thus, the slope of the CML is the Sharpe ratio of the market portfolio.
- The intercept point of CML and efficient frontier would result in the most efficient portfolio called the tangency portfolio.
- As a generalization, buy assets if Sharpe ratio is above CML and sell if Sharpe ratio is below CML.

CAPM and CML

- The CAPM says that in equilibrium, the CML represents the best risk-reward combinations available to all investors.

CML's formula

- $$E(r) = r_f + \frac{E(r_M) - r_f}{\sigma_M}\ \sigma$$
- **Slope of CML** is the risk premium on the market portfolio divided by its standard deviation.



Capital Market Line

TOPIC 169: DETERMINANTS OF THE RISK PREMIUM ON THE MARKET PORTFOLIO

The size of the risk premium of the market portfolio depends on:

- the aggregate risk aversion of investors and
- the volatility of the market return

- ✓ To be induced to accept the risk of the market portfolio, investors must be offered an expected rate of return that exceeds the risk-free rate of return.
- ✓ The greater the average degree of risk aversion, the higher the risk premium needed.

Equilibrium Risk Premium

- ✓ In the CAPM, the equilibrium risk premium on the market portfolio is equal to the variance of the market portfolio times a weighted average of the degree of risk aversion of the holders of wealth (A);

- $E(r_M) - r_f = A\,\sigma_M^2$

Risk Premium on Market Portfolio

- ✓ A = index of the degree of risk aversion in the economy

Example

Suppose the standard deviation of the market portfolio is 0.2, and the average degree of risk aversion is 2. The risk premium on the market portfolio will be:

- $E(r_M) - r_f = A\,\sigma_M^2 = 2(0.04)$

= 0.08

TOPIC 170: ACTIVE AND PASSIVE INVESTING

- Active investing requires a hands-on approach, typically by a portfolio manager or other so-called active participant.
- Passive investing involves less buying and selling and often results in investors buying index funds or other mutual funds.

Active Investing

- The goal of active money management is to beat the stock market's average returns and take full advantage of short-term price fluctuations.
- It involves a much deeper analysis and the expertise to know when to pivot into or out of a particular stock, bond, or any asset.

- A portfolio manager usually oversees a team of analysts who look at qualitative and quantitative factors, then gaze into their crystal balls to try to determine where and when that price will change.

Active investing, as its name implies, takes a hands-on approach and requires that someone act in the role of a portfolio manager. The goal of active money management is to beat the stock market's average returns and take full advantage of short-term price fluctuations. It involves a much deeper analysis and the expertise to know when to pivot into or out of a particular stock, bond, or any asset. A portfolio manager usually oversees a team of analysts who look at qualitative and quantitative factors, then gaze into their crystal balls to try to determine where and when that price will change.

Active investing requires confidence that whoever is managing the portfolio will know exactly the right time to buy or sell. Successful active investment management requires being right more often than wrong.

Passive Investing

If you are a passive investor, you invest for the long haul. Passive investors limit the amount of buying and selling within their portfolios, making this a very cost-effective way to invest. The strategy requires a buy-and-hold mentality. That means resisting the temptation to react or anticipate the stock market's every next move.

The prime example of a passive approach is to buy an index fund that follows one of the major indices like the S&P 500 or Dow Jones Industrial Average (DJIA). Whenever these indices switch up their constituents, the index funds that follow them automatically switch up their holdings by selling the stock that's leaving and buying the stock that's becoming part of the index. This is why it is such a big deal when a company becomes big enough to be included in one of the major indices: It guarantees that the stock will become a core holding in thousands of major funds.

- Although both styles of investing are beneficial, passive investments have garnered more investment flows than active investments.
- Historically, passive investments have earned more money than active investments.
- Active investing has become more popular than it has in several years, particularly during market upheavals.

TOPIC 171: BETA AND RISK PREMIUMS ON INDIVIDUAL SECURITIES

- ✓ The risk of security A is larger than the risk of security B, if in equilibrium the expected return on A exceeds the expected return on B.
- ✓ In other words, along CML, among optimal (efficient) portfolios, the larger the standard deviation of its return, the larger the equilibrium expected return E(r), and therefore, the larger the risk.
- ✓ The risk of an efficient portfolio is measured by σ

The measure of a security's risk is its beta, which is defined as:

$\beta_j = \sigma_{jM} / (\sigma_M^2)$

Where

σ_{jM} denotes the covariance between the return on security j and the return on the market portfolio.

Equilibrium

According to CAPM, in equilibrium, the risk premium on any asset is equal to its beta times the risk premium on the market portfolio, i.e.

$E(r_j) - r_f = \beta_j [E(r_M) - r_f]$

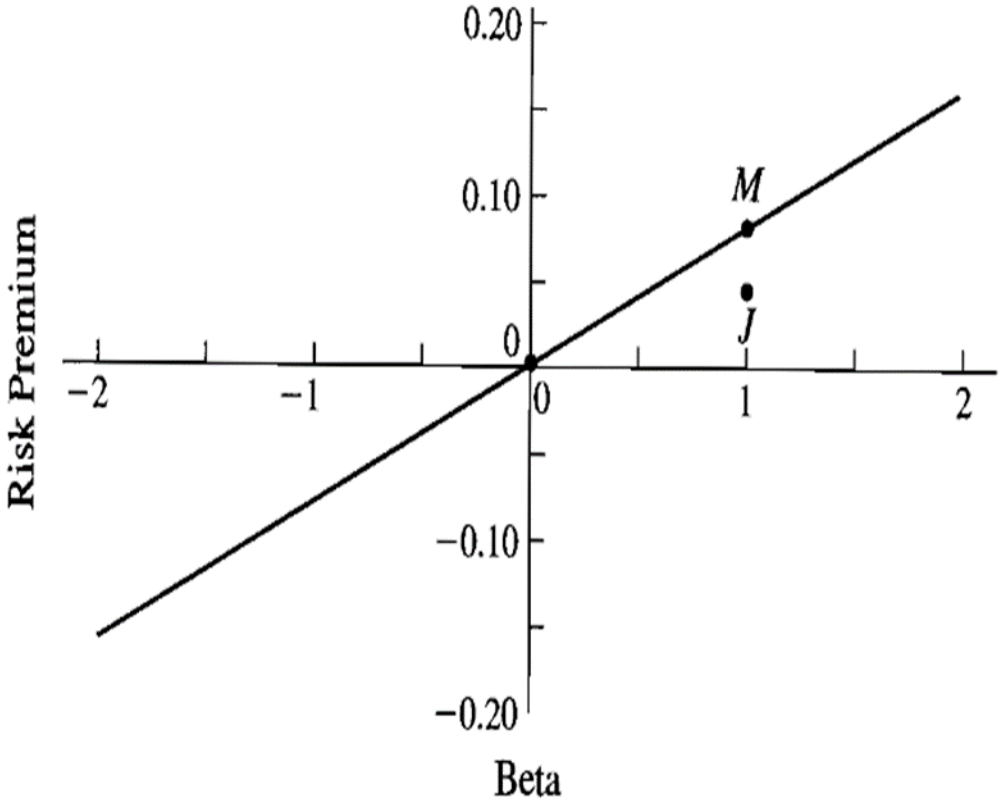
TOPIC 172: SECURITY MARKET LINE

- The security market line is an investment evaluation tool derived from the CAPM—a model that describes risk-return relationship for securities
- It is based on the assumption that investors need to be compensated for both the
- time value of money (TVM) and
- The corresponding level of risk associated with any investment, referred to as the risk premium.

Security Market Line – Equation

- $E(r_j) - r_f = \beta_j [E(r_M) - r_f]$
- Slope of SML is the risk premium.
- If any security is not on SML, e.g. it is at J, its expected return is "too low" to support equilibrium.
- The existence of such situation contradicts CAPM and implies that the market is not in equilibrium.

The security market line (SML) is a line drawn on a chart that serves as a graphical representation of the capital asset pricing model (CAPM)—which shows different levels of systematic, or market



risk, of various marketable securities, plotted against the expected return of the entire market at any given time.

Also known as the "characteristic line," the SML is a visualization of the CAPM, where the x-axis of the chart represents risk (in terms of beta), and the y-axis of the chart represents expected return. The market risk premium of a given security is determined by where it is plotted on the chart relative to the SML.

The security market line is an investment evaluation tool derived from the CAPM—a model that describes risk-return relationship for securities—and is based on the assumption that investors need to be compensated for both the time value of money (TVM) and the corresponding level of risk associated with any investment, referred to as the risk premium.

If any security is not on SML, e.g. it is at J, its expected return is “too low” to support equilibrium. The existence of such situation contradicts CAPM and implies that the market is not in equilibrium.

Lesson 34

USING THE CAPM IN PORTFOLIO SELECTION

TOPIC 173: USING THE CAPM IN PORTFOLIO SELECTION

- ✓ The CAPM implies that the market portfolio of risky assets is an efficient portfolio.
- ✓ It means that an investor follows a passive portfolio selection strategy of combining a market index fund and the risk-free asset.

Passive Portfolio Strategy

- ✓ Diversify your holdings of risky assets in the proportions of the market portfolio, and
- ✓ Mix this portfolio with the risk-free asset to achieve a desired risk-reward combination.

Example

- ✓ Suppose you have $1 million to invest.
- ✓ You want to invest in three asset classes, i.e. stocks, bonds and the risk-free asset.
- ✓ The net relative supply of first two is 60% and 40%.
- ✓ The composition of the portfolio will follow this ratio.

Process

- ✓ The investors will compare the rate of return earned on the managed portfolio to the rate of return attainable by simply mixing the market portfolio and risk free asset in proportions that would have produced the same volatility
- ✓ The volatility of the managed portfolio will be computed for the same period. Then the average rate of return will be estimated and compared with the managed portfolio's average rate of return. Managed portfolio is the benchmark portfolio.
- ✓ Then the average rate of return will be estimated and compared with the managed portfolio's average rate of return.
- ✓ Managed portfolio is the benchmark portfolio.

TOPIC 174: INDEXING IN FINANCIAL MARKETS

- An index is a method to track the performance of a group of assets in a standardized way.
- Indexes typically measure the performance of a basket of securities intended to replicate a certain area of the market. These could be a broad-based index that captures the entire market, such as the Standard & Poor's 500 Index or KSE100 Index.

- Indexes can also be more specialized, such as indexes that track a particular industry or segment.
- Index providers have numerous methodologies for constructing investment market indexes.
- Index fund management, aims to duplicate the return of a particular market index or benchmark.
- Managers buy the same stocks that are listed on the index, using the same weighting that they represent in the index.
- CAPM contributed to the rise in the use of indexing–assembling a portfolio of shares to mimic a particular market or asset class–by risk-averse investors.
- This is largely due to CAPM's message that it is only possible to earn higher returns than those of the market as a whole by taking on higher risk (beta).
- If a fund manager is underperforming the S&P 500 over the long term, for example, it will be hard to entice investors into the fund.

What is alpha?

- Alpha is a measure of the active return on an investment, the performance of that investment compared with a suitable market index.

Indexing, broadly, refers to the use of some benchmark indicator or measure as a reference or yard stick. In finance and economics, indexing is used as a statistical measure for tracking economic data such as inflation, unemployment, gross domestic product (GDP) growth, productivity, and market returns.

Indexing may also refer to passive investment strategies that replicate benchmark indexes. Index investing has become increasingly popular over the past decades.
Indexes typically measure the performance of a basket of securities intended to replicate a certain area of the market. These could be a broad-based index that captures the entire market, such as the Standard & Poor's 500 Index or KSE100 Index.

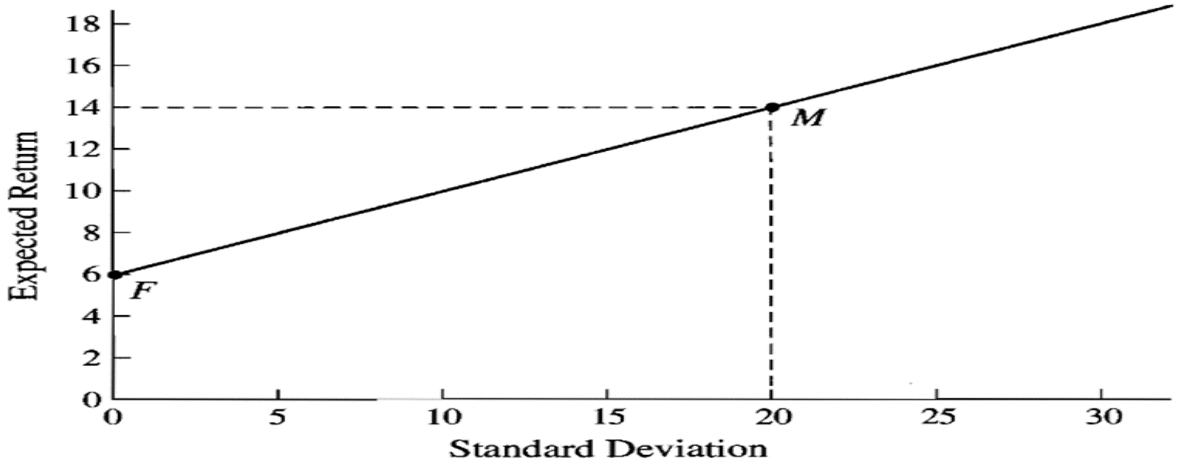
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If a fund manager is underperforming the S&P 500 over the long term, for example, it will be hard to entice investors into the fund.

TOPIC 175: ALPHA FUND AND THE SECURITY MARKET LINE



Alpha is a measure of the active return on an investment, the performance of that investment compared with a suitable market index. The capital market line (CML) shows the rates of return for a specific portfolio. The security market line (SML) represents the market's risk and return at a given time, and shows the expected returns of individual assets. The measure of risk in the CML is the standard deviation of returns (total risk), the risk measure in the SML is systematic risk or beta.

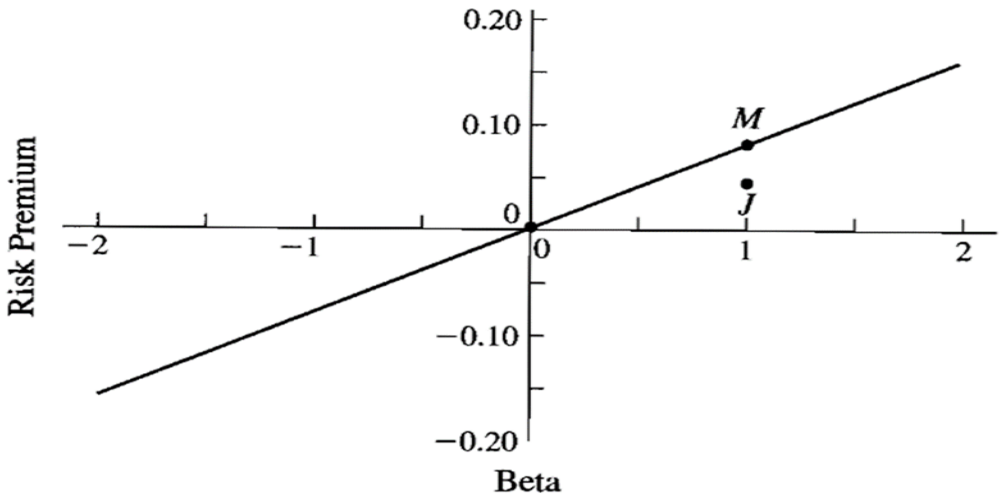
CML's equation

- $\mathbf{E(r) = r_f + \frac{E(r_M) - r_f}{\sigma_M}\ \sigma}$

SML's equation

- $\mathbf{E(r_j) - r_f = \beta_j [E(r_M) - r_f]}$

Capital Market Line



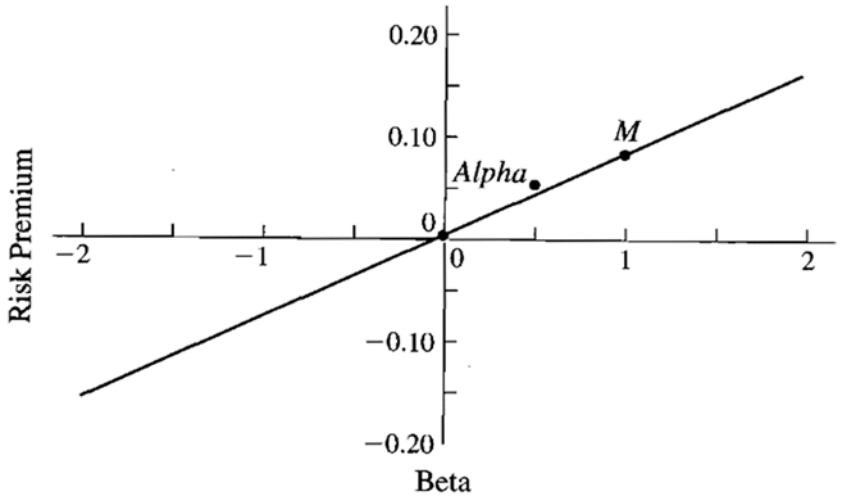
Security Market Line

Alpha is used to determine by how much the realized return of the portfolio varies from the required return, as determined by the CAPM.

$\alpha = R - [r_f + \beta_j [E(r_M) - r_f]]$

R=realized return of portfolio

$E(r_M)$ = Expected Market Return



Alpha Fund and the Security Market Line

Point alpha represents the alpha fund. In the figure above, alpha lies above SML. Alpha Fund's alpha is measured as the vertical distance between alpha and SML.

- The capital market line (CML) shows the rates of return for a specific portfolio
- The security market line (SML) represents the market's risk and return at a given time, and shows the expected returns of individual assets.
- The measure of risk in the CML is the standard deviation of returns (total risk), the risk measure in the SML is systematic risk or beta.

TOPIC 176: VALUATION AND REGULATING RATES OF RETURN

Risk premiums derived from the CAPM are used to:

1. Portfolio Selection
2. Discounted Cash Flow (DCF) Valuation Models
3. Capital Budgeting decisions of firms.
4. Establish 'fair' rates of return on invested capital in regulated firms

Discounted Cash Flow (DCF) Valuation Models

The price of a share is sometimes viewed as the present value of all expected future dividends discounted at the market capitalization rate

$$P_0 = \frac{D_1}{(1+k)} + \frac{D_1}{(1+k)^2} + \frac{D_1}{(1+k)^3} + \cdots$$

$$= \sum_{t=1}^{\infty} \frac{D_t}{(1+k)^t}$$

- D_t= the expected dividend per share in period t
- k = the risk-adjusted discount rate (expected rate of return)

Analysts use CAPM to calculate the value of k.

TOPIC 177: COST OF CAPITAL

- The firm's cost of capital is a weighted average of its cost of equity capital and debt.
- Practitioners use a CAPM based method to estimate the cost of equity capital.

Example

- Suppose you want to compute your firm's cost of equity capital.
- You compute the beta of your company's stock and find it to be 1.1.
- The current risk free rate is 0.06 per year
- You assume that the market premium is 0.08 per year.
- The equilibrium expected rate of return on the stock of your company will be:
- E(r)= $r_f + \beta\ [E(r_M) - r_f]$
- = 0.06 +1.1 (0.08)

- = 0.148
- Thus, 14.8% per year is the cost of equity capital.

Cost of capital is a company's calculation of the minimum return that would be necessary in order to justify undertaking a capital budgeting project, such as building a new factory.

The term cost of capital is used by analysts and investors, but it is always an evaluation of whether a projected decision can be justified by its cost. Investors may also use the term to refer to an evaluation of an investment's potential return in relation to its cost and its risks.

Many companies use a combination of debt and equity to finance business expansion. For such companies, the overall cost of capital is derived from the weighted average cost of all capital sources. This is known as the weighted average cost of capital (WACC).

The concept of the cost of capital is key information used to determine a project's hurdle rate. A company embarking on a major project must know how much money the project will have to generate in order to offset the cost of undertaking it and then continue to generate profits for the company.

Cost of capital, from the perspective of an investor, is an assessment of the return that can be expected from the acquisition of stock shares or any other investment. This is an estimate and might include best- and worst-case scenarios. An investor might look at the volatility (beta) of a company's financial results to determine whether a stock's cost is justified by its potential return.

The concept of the cost of capital is key information used to determine a project's hurdle rate. A company embarking on a major project must know how much money the project will have to generate in order to offset the cost of undertaking it and then continue to generate profits for the company.

Practitioners use a CAPM based method to estimate the cost of equity capital.

Example

Suppose you want to compute your firm’s cost of equity capital. You compute the beta of your company’s stock and find it to be 1.1. The current risk free rate is 0.06 per year. You assume that the market premium is 0.08 per year.

The equilibrium expected rate of return on the stock of your company will be:

E(r)= $\boldsymbol{r_f}$ + $\boldsymbol{\beta}$ [$\boldsymbol{E(r_M}$)− $\boldsymbol{r_f}$]

= 0.06 +1.1 (0.08)

= 0.148

Thus, 14.8% per year is the cost of equity capital.

TOPIC 178: REGULATION AND COST-PLUS PRICING

- ✓ Regulators may use the CAPM to establish a “fair” rate of return on invested capital for public utilities and other firms subject to price regulations.

Example

- ✓ A commission regulating an electric power company may have to establish a price that the company is allowed to charge its customers for electricity.
- ✓ The commission will do so by computing the cost of producing the electricity, including an allowance for the cost of capital.

Lesson 35

MODIFICATIONS AND ALTERNATIVES TO THE CAPM

TOPIC 179: MODIFICATIONS AND ALTERNATIVES TO THE CAPM

- Recent research on asset pricing indicated that the original simple version of the CAPM needs to be modified.
- It was suggested that the market portfolio used in CAPM were incomplete and inadequate representations of the true market portfolio.
- Market imperfections such as borrowing costs and constraints, shortsale restrictions and costs, different tax treatments for various assets and the non-tradability of important assets like human capital are not contemplated in CAPM.
- CAPM must consider more realistic and complicated factors that effect the optimal portfolio selection process.
- Two famous alternatives to CAPM are:

1. The multifactor Intertemporal Capital Asset Pricing Model (ICAPM).
2. Arbitrage Pricing Theory (APT)

TOPIC 180: FORWARD AND FUTURES MARKETS

- A forward market is an over-the-counter marketplace that sets the price of a financial instrument or asset for future delivery.
- Forward markets are used for trading a range of instruments, but the term is primarily used with reference to the foreign exchange market.
- It can also apply to markets for securities and interest rates as well as commodities.
- A forward market leads to the creation of forward contracts.

Forward and Futures Markets

- While forward contracts like futures contracts may be used for both hedging and speculation, but they are different.
- Forward contracts can be customized to fit a customer's requirements, while futures contracts have standardized features in terms of their contract size and maturity.
- Forwards are executed between banks or between a bank and a customer
- Futures are done on an exchange, which is a party to the transaction. The flexibility of forwards contributes to their attractiveness in the foreign exchange market.

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Forward Markets – Pricing

- Prices in the forward market are interest-rate based.
- In the foreign exchange market, the forward price is derived from the interest rate differential between the two currencies
- In interest rate forwards, the price is based on the yield curve to maturity.

How A Forward Market Works

A forward market leads to the creation of forward contracts. While forward contracts—like futures contracts—may be used for both hedging and speculation, there are some notable differences between the two. Forward contracts can be customized to fit a customer's requirements, while futures contracts have standardized features in terms of their contract size and maturity.

Forwards are executed between banks or between a bank and a customer; futures are done on an exchange, which is a party to the transaction. The flexibility of forwards contributes to their attractiveness in the foreign exchange market.

Pricing

Prices in the forward market are interest-rate based. In the foreign exchange market, the forward price is derived from the interest rate differential between the two currencies, which is applied over the period from the transaction date to the settlement date of the contract. In interest rate forwards, the price is based on the yield curve to maturity.

Foreign Exchange Forwards

Interbank forward foreign exchange markets are priced and executed as swaps. This means that currency A is purchased vs. currency B for delivery on the spot date at the spot rate in the market at the time the transaction is executed. At maturity, currency A is sold vs. currency B at the original spot rate plus or minus the forward points; this price is set when the swap is initiated.

The interbank market usually trades for straight dates, such as a week or a month from the spot date. Three- and six-month maturities are among the most common, while the market is less liquid beyond 12 months. Amounts are commonly $25 million or more and can range into the billions.

Customers, both corporations and financial institutions such as hedge funds and mutual funds, can execute forwards with a bank counter-party either as a swap or an outright transaction. In an outright forward, currency A is bought vs. currency B for delivery on the maturity date, which can be any business day beyond the spot date. The price is again the spot rate plus or minus the

forward points, but no money changes hands until the maturity date. Outright forwards are often for odd dates and amounts; they can be for any size.

A futures market is an auction market in which participants buy and sell commodity and futures contracts for delivery on a specified future date. Futures are exchange-traded derivatives contracts that lock in future delivery of a commodity or security at a price set today.
Examples of futures markets are the New York Mercantile Exchange (NYMEX), the Chicago Mercantile Exchange (CME), the Chicago Board of Trade (CBoT), the Cboe Options Exchange (Cboe), and the Minneapolis Grain Exchange.

Originally, such trading was carried on through open outcry and the use of hand signals in trading pits, located in financial hubs such as New York, Chicago, and London. Throughout the 21st century, like most other markets, futures exchanges have become mostly electronic.
In order to understand fully what a futures market is, it's important to understand the basics of futures contracts, the assets traded in these markets.
Futures contracts are made in an attempt by producers and suppliers of commodities to avoid market volatility. These producers and suppliers negotiate contracts with an investor who agrees to take on both the risk and reward of a volatile market.

Futures markets or futures exchanges are where these financial products are bought and sold for delivery at some agreed-upon date in the future with a price fixed at the time of the deal. Futures markets are for more than simply agricultural contracts, and now involve the buying, selling and hedging of financial products and future values of interest rates.
Futures contracts can be made or "created" as long as open interest is increased, unlike other securities that are issued. The size of futures markets (which usually increase when the stock market outlook is uncertain) is larger than that of commodity markets and is a key part of the financial system.

Large futures markets run their own clearinghouses, where they can both make revenue from the trading itself and from the processing of trades after the fact. Some of the biggest futures markets that operate their own clearinghouses include the Chicago Mercantile Exchange, the ICE, and Eurex. Other markets like Cboe have outside clearinghouses (Options Clearing Corporation) settle trades.

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Most all futures markets are registered with the Commodity Futures Trading Commission (CFTC), the main U.S. body in charge of regulation of futures markets. Exchanges are usually regulated by the nation's regulatory body in the country in which they are based.

Example

If a coffee farm sells green coffee beans at $4 per pound to a roaster, and the roaster sells that roasted pound at $10 per pound and both are making a profit at that price, they'll want to keep those costs at a fixed rate. The investor agrees that if the price for coffee goes below a set rate, the investor agrees to pay the difference to the coffee farmer.

If the price of coffee goes higher than a certain price, the investor gets to keep profits. For the roaster, if the price of green coffee goes above an agreed rate, the investor pays the difference and the roaster gets the coffee at a predictable rate. If the price of green coffee is lower than an agreed-upon rate, the roaster pays the same price and the investor gets the profit.

TOPIC 181: DISTINCTIONS BETWEEN FORWARD AND FUTURE CONTRACTS

1. Two parties agree to exchange some item in the future at a delivery price specified now.
2. The forward price is defined as the delivery price that makes the current market value of the contract zero.
3. No money is paid in the present by either party to the other.
4. The face value of the contract is the quantity of the item specified in the contract times the forward price.
5. The party who agrees to buy the specified item is said to take a long position, and the party who agrees to sell the item is said to take a short position.

Who benefits from a forward contract?

- If the spot price on the contract maturity date is higher than the forward price, the party who is long makes money.
- If the spot price on the contract maturity date is lower than the forward price, the party who is short makes money.
- By contrast, futures contracts are standardized contracts that are traded on exchanges. The exchange specifies the exact commodity, the contract size, and where and when delivery will be made.

Distinctions between Forward and Future Contracts

Futures	Forwards
Traded on Exchange	Privately negotiated
Standardized, having an exchange-specified contract unit, expiration, tick size, and notional value	Customized
Actively traded	Non-transferrable
No counterparty risk, since payment is guaranteed by the exchange clearing house	Credit default risk, since it is privately negotiated, a dependent on the counterparty for payment
Regulated	Not regulated

Forward and futures contracts are derivatives arrangements that involve two parties who agree to buy or sell a specific asset at a set price by a certain date in the future. Buyers and sellers can mitigate the risks associated with price movements down the road by locking in the purchase/sale price in advance.

A forward contract is an arrangement that is made over-the-counter (OTC) and settles just once at the end of the contract. Both parties involved in the agreement negotiate the exact terms of the contract. It is privately negotiated and comes with a degree of default risk since the counterparty is responsible for remitting payment.

Futures contracts, on the other hand, are standardized contracts that trade on stock exchanges. As such, they are settled on a daily basis. These arrangements come with fixed maturity dates and uniform terms. There is very little risk with futures, as they guarantee payment on the agreed-upon date.

The forward contract is a privately-negotiated agreement between a buyer and seller to trade an asset at a future date at a specified price. As such, they don't trade on an exchange. Because of the nature of the contract, forward contracts have more flexible terms and conditions, including the number of units of the underlying asset and what exactly will be delivered, among other factors. Forwards have one settlement date: the end of the contract.

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Many hedgers use forward contracts to cut down on the volatility of an asset's price. Since the terms are set when it is executed, a forward contract is not subject to price fluctuations. That means if two parties agree to the sale of 1,000 ears of corn at $1 each (for a total of $1,000), the terms cannot change even if the price of corn goes down to 50 cents per ear. It also ensures that delivery of the asset or cash settlement (if specified) will take place.

Because of the nature of these contracts, forwards are not readily available to retail investors. The market for them is often hard to predict. That's because the agreements and their details are generally kept between the buyer and seller, and are not made public. Since they are private agreements, there is a high degree of counterparty risk, which means there may be a chance that one party will default.

Like forwards, futures contracts involve the agreement to buy and sell an asset at a specific price at a future date. The futures contract, however, has some differences from the forward contract. These contracts are marked-to-market (MTM) daily, which means that daily changes are settled day by day until the end of the contract. The futures market is highly liquid, giving investors the ability to enter and exit whenever they choose to do so.

These contracts are frequently used by speculators, who bet on the direction in which an asset's price will move, they are usually closed out prior to maturity and delivery usually never happens. In this case, a cash settlement usually takes place.

Because they are traded on an exchange, they have clearing houses that guarantee the transactions. This drastically lowers the probability of default to almost never. Contracts are available on stock exchange indexes, commodities, and currencies. The most popular assets for futures contracts include crops like wheat and corn, and oil and gas.

Key Differences

One of the things that set forward contracts from futures contracts is how they're regulated. Forward contracts aren't regulated at all while futures are overseen by a central government body. The agency that provides oversight and regulation of futures contracts is the Commodity Futures Trading Commission (CFTC). The CFTC was established in 1974 to regulate the derivatives market, to ensure the markets run efficiently, and to protect the interests of investors by preventing fraud and manipulation.

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Guarantees for each contract are also provided by different parties. Since forwards are privately negotiated, they provide the guarantee to settle the contract. Futures, on the other hand, have an institutional guarantee provided by the clearinghouses that back them. Unlike forwards, where there is no guarantee until the contract settles, futures require a deposit or margin. This acts as collateral to cover the risk of default.

TOPIC 182: THE ECONOMIC FUNCTIONS OF FUTURE MARKETS

- To facilitate the reallocation of exposure to commodity price risk among market participants.
- To provide information to the producers, distributors, and consumers of commodities who must decide how much wheat to sell (consume) and how much to store for the future.

Example

Suppose that it is one month before the next harvest. A wheat distributor is left with one ton of wheat in storage. The spot price of wheat is $2 per bushel. The futures price for delivery a month from now is F.

The distributor can hedge its exposure to price change by either

1. Selling the wheat in the spot market for $2 per bushel and delivering it immediately, or
2. Selling short a futures contract at a price of F and delivering the wheat a month from now.

Suppose the distributor's cost of physically storing the wheat is 10 cents per bushel per month. The distributor will store the wheat only if F > $2.10. If the storage cost is 15 cents, the distributor will sell the wheat immediately if F= $2.10.

Hence, distributor will choose to carry wheat for the next month only if

$$C_j < F - S$$

C_j is cost of carry for distributor j

Spot price

F – S => Spread

TOPIC 183: THE ROLE OF SPECULATORS

- Speculators are risk-takers who bet on the short- and mid-term future direction of movements in an asset without having any other stake in that asset.

- Speculators provide the markets with liquidity, aid in price discovery, and take on risk that other market participants wish to unload.
- In commodities markets, speculators also keep markets efficient and stave off shortages of goods by bidding them up when prices fall and financing the middlemen who link supply chains.
- A speculator should thus not be confused with the middleman or broker.
- Speculators often get a bad rep, especially when headlines report a crash in stocks, a spike in oil prices, or a currency's value is shattered in short order. This is because the media often confounds speculation with manipulation.
- Manipulation is fraudulent and unethical, many times leading to extensive economic damage; whereas speculation, while risky for the speculator, performs several important functions that keep our markets and economy healthy.
- Speculation in the commodities markets keeps not only financial markets, but also our supermarket shelves and food supply chains running smoothly.

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Lesson 36

COMMODITY SPOT AND FUTURES PRICES

TOPIC 184: RELATION BETWEEN COMMODITY SPOT AND FUTURES PRICES

- The futures price of a commodity is set in advance between producer and buyer.
- The spot price is the commodity's value when it is ready for delivery.
- The difference in the two values is where arbitrage traders make their money.

Formula

- The futures price cannot exceed the spot price by more than the cost of carry:
- $F-S \leq C$
- The cost of carry can vary over time and across market participants.

Futures Prices Converge upon Spot Prices

- It's a fairly safe bet that the price of a future will inch toward its spot price as the delivery month of a futures contract approaches, and it could even match the price. This is a very strong trend that occurs regardless of the contract's underlying asset.

- The convergence depends on:
    - Arbitrage interest and
    - The law of supply and demand.

The spot price is the current price in the marketplace at which a given asset—such as a security, commodity, or currency—can be bought or sold for immediate delivery. While spot prices are specific to both time and place, in a global economy the spot price of most securities or commodities tends to be fairly uniform worldwide when accounting for exchange rates. In contrast to the spot price, a futures price is an agreed upon price for future delivery of the asset.
Spot prices are most frequently referenced in relation to the price of commodity futures contracts, such as contracts for oil, wheat, or gold. This is because stocks always trade at spot. You buy or sell a stock at the quoted price, and then exchange the stock for cash.

A futures contract price is commonly determined using the spot price of a commodity, expected changes in supply and demand, the risk-free rate of return for the holder of the commodity, and the costs of transportation or storage in relation to the maturity date of the contract. Futures contracts with longer times to maturity normally entail greater storage costs than contracts with nearby expiration dates.

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Spot prices are in constant flux. While the spot price of a security, commodity, or currency is important in terms of immediate buy-and-sell transactions, it perhaps has more importance in regard to the large derivatives markets. Options, futures contracts, and other derivatives allow buyers and sellers of securities or commodities to lock in a specific price for a future time when they want to deliver or take possession of the underlying asset. Through derivatives, buyers and sellers can partially mitigate the risk posed by constantly fluctuating spot prices.

The Relationship Between Spot Prices and Futures Prices

The difference between spot prices and futures contract prices can be significant. Futures prices can be in contango or backwardation. Contango is when futures prices fall to meet the lower spot price. Backwardation is when futures prices rise to meet the higher spot price. Backwardation tends to favor net long positions since futures prices will rise to meet the spot price as the contract get closer to expiry. Contango favors short positions, as the futures lose value as the contract approaches expiry and converges with the lower spot price.

Futures markets can move from contango to backwardation, or vice versa, and may stay in either state for brief or extended periods of time. Looking at both spot prices and futures prices is beneficial to futures traders.

Example

An asset can have different spot and futures prices. For example, gold may have a spot price of $1,000 while its futures price may be $1,300. Similarly, the price for securities may trade in different ranges in the stock market and the futures market. For example, Apple Inc. (AAPL) may trade at $200 in the stock market but the strike price on its options may be $150 in the futures market, reflecting pessimistic trader perceptions of its future.

TOPIC 185: EXTRACTING INFORMATION FROM COMMODITY FUTURES PRICES

Sometimes the future prices can provide information about investor expectations of spot prices in the future.

Information?

There can be two possibilities:

Case 1. No wheat is in storage

Case 2. Wheat is in storage

Case 1. No wheat is in storage

There is stock out.

$F-S < C$

It will be a strict inequality.

The forward price will provide information about the expected future spot price that is not extractable from the current spot price.

Case 2. Wheat is in storage

No further inference about the future expected spot price is possible beyond that extractable from the current spot price.

F-S = C

It will be an equality.

The forward price is completely specified by knowing the cost of carry and the spot price.

TOPIC 186: FORWARD-SPOT PRICE PARITY FOR GOLD

Suppose you are planning to invest in an ounce of gold for the next year.

You can do it in two ways:

Buy gold at the current spot price, S

Put it in storage

At the end of year 1, sell it at a price of S1.

In this case, your rate of return will be:

$r_g=(S_1-S)/S$ - b

b = storage cost

Example

Suppose the spot price of gold is $300 and storage costs are 2% per year, your rate of return is:

$r_g=(S_1-300)/300$ - 0.02

Another way is to take the same $300 and invest in synthetic gold (by paying the spot price).

At the same time by taking a long position in a gold forward contract with a delivery date of after a year at a forward price of F.

The rate of return for synthetic gold is:

- $\widehat{r_g} = \frac{S_1-F}{S} + r$

Suppose the risk free return r is 8%, return on synthetic gold will be:

- $\widehat{r_g} = \frac{S_1-F}{300} + 0.08$

By the Law of One Price, these two equivalent investments must offer the same return.

Hence by equating the two equations, we get:

$(\boldsymbol{S_1-F})/\boldsymbol{S}$ +r =$(\boldsymbol{S_1-S})/\boldsymbol{S}$ – b

Solving it for F, we get:

F=(1+r+b)S

Hence, the forward price for delivery of gold in one year should be:

F =(1+r+b)S = (1+0.08+0.02)X300

= 1.10X300=$330

TOPIC 187: THE IMPLIED COST OF CARRY

- Cost of carry refers to costs associated with the carrying value of an investment.
- It includes financial costs, such as the interest costs on bonds, interest on loans used to make an investment, and any storage costs involved in holding a physical asset.

The Implied Cost of Carry

Implied Cost of Carry = F –S

Carrying Cost can be calculated by using the following formula:

F = S(1+r+s)

$(\boldsymbol{F-S})/\boldsymbol{S}$ = r+s

Implied Cost of Storing:

s = $(\boldsymbol{F-S})/\boldsymbol{S}$ – r

Example

Suppose the spot price of gold is $300 per ounce, the one-year forward price is $330, and the risk-free interest rate is 8%.

Implied Cost of Carry:

F-S = $300 - $330 = $30 per ounce

year, your rate of return is:

Implied Storage Cost:

s=(($\boldsymbol{F-S}$))/$\boldsymbol{S}$ - r = 0.10-0.08=0.02

or 2% per year

Lesson 37

FINANCIAL FUTURES AND FORWARD PRICE

TOPIC 188: FINANCIAL FUTURES

- Futures are derivative financial contracts obligating the buyer to purchase an asset or the seller to sell an asset at a predetermined future date and set price.
- A futures contract allows an investor to speculate on the direction of a security, commodity, or financial instrument.
- Futures are used to hedge the price movement of the underlying asset to help prevent losses from unfavorable price changes.

Types of Futures

- Commodity futures such as crude oil, natural gas, corn, and wheat
- Stock index futures such as the PSX 100 Index
- Currency futures including those for the dollar and the British pound
- Precious metal futures for gold and silver

The concept of futures trading can seem fairly simple when we are explaining physical commodities such as agricultural products, metals or crude oil. But things start to get more complex when we enter the intangible world of financial futures.

This is where we get into the business of financial instruments – trading numbers and figures, calculations and data, percentages and indexes. The products may seem abstract, but they are integral to the world's economies because they enable governments, businesses and financial institutions to manage the costs of doing business.

Three types of financial futures markets

1. THE FOREIGN CURRENCY MARKET

If you buy products or services in other countries, you must manage the risk of fluctuations in foreign exchange rates.

2. THE INTEREST RATE MARKET

If you lend or borrow money, you must manage the risk of shifting interest rates to ensure a steady level of access to your capital.

3. THE EQUITY INDEX MARKET

If you invest in stocks, you may want to manage the risk of price changes reflected in underlying equity indexes such as the S&P 500 and the Dow Jones Industrial Average.

TOPIC 189: FINANCIAL FUTURES – EXAMPLE

- A trader wants to speculate on the price of crude oil by entering into a futures contract in May with the expectation that the price will be higher by year-end.
- The December crude oil futures contract is trading at $50 and the trader locks in the contract.

Example

- The investor now has a position worth $50,000 of crude oil (1,000 x $50 = $50,000).
- The trader will only need to pay a fraction of that amount up-front—the initial margin that they deposit with the broker.
- From May to December, the price of oil fluctuates as does the value of the futures contract. If oil's price gets too volatile, the broker may ask for additional funds to be deposited into the margin account—a maintenance margin.

Example

- Suppose in December, the price of crude oil has risen to $65, and the trader sells the original contract to exit the position.
- The net difference is cash-settled, and they earn $15,000, less any fees and commissions from the broker ($65 - $50 = $15 x 1000 = $15,000).
- However, if the price oil had fallen to $40 instead, the investor would have lost $10,000 ($40 - $50 = negative $10 x 1000 = negative $10,000).

TOPIC 190: IMPLIED RISKLESS RATE

- In practice, the risk-free rate of return does not truly exist, as every investment carries at least a small amount of risk. To calculate the real risk-free rate, subtract the inflation rate from the yield of the Treasury bond matching your investment duration.
- The interest rate on a three-month Treasury bill (T-bill) is often used as the risk-free rate for investors.
- The three-month Treasury bill is a useful proxy because the market considers there to be virtually no chance of the government defaulting on its obligations.
- The large size and deep liquidity of the market contribute to the perception of safety. However, a foreign investor whose assets are not denominated in PKR incurs currency risk when investing in Treasury bills. The risk can be hedged via currency forwards and options but affects the rate of return.

TOPIC 191: FORWARD PRICE AND FUTURE SPOT PRICE

- If a stock that pays no dividend and offers a positive risk premium to investors, forward price is not a forecast of the expected future spot price.
- Suppose that the risk premium on S&P stock is 7% and the riskless interest rate is 8%. The expected rate of return on S&P is 15 %.
- If the current spot price is $100 per share, the expected spot price one year from now will be $115.
- In the absence of any dividends, the ending spot price must be 15% higher than the beginning spot price.

- **Expected Rate of Return on S&P =** $\frac{Ending\ Price - Beginning\ Price}{Beginning\ Price}$
- $\bar{r}_{SP} = \frac{\overline{S_1} - S}{S} = 0.15$
- $\overline{S_1} = 1.15S = 1.15X100 = 115$

Important Point

- The forward-spot price parity relation says that the forward price of the S&P for delivery in one year must be $108.
- It means only risk premium is to be taken.

The precise meanings of the terms "forward rate" and "spot rate" are somewhat different in different markets. In general, a spot rate refers to the current price or bond yield, while a forward rate refers to the price or yield for the same product or instrument at some point in the future.

In commodities futures markets, a spot rate is the price for a commodity being traded immediately, or "on the spot". A forward rate is the settlement price of a transaction that will not take place until a predetermined date.

In bond markets, the forward rate refers to the effective yield on a bond, commonly U.S. Treasury bills, and is calculated based on the relationship between interest rates and maturities.

If a stock that pays no dividend and offers a positive risk premium to investors, forward price is not a forecast of the expected future spot price.

Suppose that the risk premium on S&P stock is 7% and the riskless interest rate is 8%. The expected rate of return on S&P is 15 %.

If the current spot price is $100 per share, the expected spot price one year from now will be $115.

In the absence of any dividends, the ending spot price must be 15% higher than the beginning spot price.

Example

The forward-spot price parity relation says that the forward price of the S&P for delivery in one year must be $108.

It means only risk premium is to be taken.

TOPIC 192: FORWARD-SPOT PRICE-PARITY RELATION WITH CASH PAYOUTS

Suppose everyone expects the stock to pay a cash dividend of D per share at the end of the year.

Price of Stock= S = $(\boldsymbol{D}+\boldsymbol{F})/((\mathbf{1}+\boldsymbol{r}))$

F= S(1+r) – D

F = S + rS – D

F>S if D <rS

TOPIC 193: IMPLIED DIVIDENDS

Implied Dividend can be calculated using:

- $\overline{\boldsymbol{D}} = \boldsymbol{S}(\mathbf{1} + \boldsymbol{r})\boldsymbol{F}$

Example

Suppose S=$100; r =0.08; F=$103

Putting these values in the above mentioned formula, we get:

= 100(1.08) -103

= $5

Lesson 38

FOREIGN-EXCHANGE PARITY RELATION

TOPIC 194: FOREIGN-EXCHANGE PARITY RELATION

- Foreign-Exchange Parity is also known as Interest Rate Parity (IRP).
- Interest rate parity (IRP) is a theory according to which the interest rate differential between two countries is equal to the differential between the forward exchange rate and the spot exchange rate.
- Parity is used by forex traders to find arbitrage opportunities.

Interest Rate Parity is:

Where:

F=Forward Price

S = Spot Price

ic = interest rate in country c

ib = interest rate in country b

- $$F = S \times \left(\frac{1+i_c}{1+i_b}\right)$$
- The difference between the forward rate and spot rate is known as swap points.
- If this difference (forward rate minus spot rate) is positive, it is known as a forward premium;
- A negative difference is termed a forward discount.
- A currency with lower interest rates will trade at a forward premium in relation to a currency with a higher interest rate. For example, the U.S. dollar typically trades at a forward premium against the Canadian dollar. Conversely, the Canadian dollar trades at a forward discount versus the U.S. dollar.

TOPIC 195: THE ROLE OF EXPECTATIONS IN DETERMINING EXCHANGE RATES

- Expectations Hypothesis states that the forward price of a currency is equal to its expected future spot price.
- If $S1$ denotes the spot dollar price of the pound one year from now and ($S1$) is the expected future spot price, then

$$F=E(S1)$$

TOPIC 196: PRICING OF SWAP CONTRACTS

Derivatives contracts can be divided into two general families:

- $1 m/1.08+$1.04m/(1.08)^2 = 100,000,000 F(1/1.08+1/ 1.08^2)
- $F=\frac{\$1\ m/1.08+\$1.04m/(1.08)^2}{100,000,000\ (1/1.08+1/1.08^2)}$
- = $0.010192307 per yen

1. Contingent claims (e.g., options)
2. Forward claims, which include exchange-traded futures, forward contracts, and swaps

A swap is an agreement between two parties to exchange sequences of cash flows for a set period of time. Usually, at the time the contract is initiated, at least one of these series of cash flows is determined by a random or uncertain variable, such as an interest rate, foreign exchange rate, equity price, or commodity price.

Swap Contracts

- Swaps are derivative instruments that represent an agreement between two parties to exchange a series of cash flows over a specific period of time. Swaps offer great flexibility in designing and structuring contracts based on mutual agreement.

Pricing of Swap Contracts

- Consider a yen-dollar swap.
- Suppose the contract is of two years, with a notional principal of 100 million yens.
- Suppose that the one-year forward price of the yen is $0.01 and two year forward price is $0.0104.
- In the first year it is $1 million and in the second it is $1.04 million.
- But the currency swap calls for a single swap exchange rate to apply in both years.
- To calculate this, we use the following formula
- Suppose riskless dollar interest rate is 8% per year. Let F be the swap rate.
- F will be calculated using:
- $1 m/1.08+$1.04m/(1.08)^2 = 100,000,000 F(1/1.08+1/1.08^2)
- $F=\frac{\$1\ m/1.08+\$1.04m/(1.08)^2}{100,000,000\ (1/1.08+1/1.08^2)}$
- = $0.010192307 per yen
-

Conceptually, one may view a swap as either a portfolio of forward contracts or as a long position in one bond coupled with a short position in another bond. This article will discuss the two most common and most basic types of swaps: interest rate and currency swaps.
Consider a yen-dollar swap.

TOPIC 197: FOREIGN EXCHANGE MARKET

- The foreign exchange market (also known as forex, FX, or the currencies market) is an over-the-counter (OTC) global marketplace that determines the exchange rate for currencies around the world. Participants in these markets can buy, sell, exchange, and speculate on the relative exchange rates of various currency pairs.

- Foreign exchange markets are made up of banks, forex dealers, commercial companies, central banks, investment management firms, hedge funds, retail forex dealers, and investors.

- An exchange rate is the price of one currency in terms of another.
- There are two kinds of exchange rate transactions:
- -Spot transactions, involve the immediate exchange of bank deposits.
- -Forward transactions, involve the exchange of bank deposits at some specified future date.

The foreign exchange market—also called forex, FX, or currency market—was one of the original financial markets formed to bring structure to the burgeoning global economy. In terms of trading volume, it is, by far, the largest financial market in the world. Aside from providing a venue for the buying, selling, exchanging, and speculation of currencies, the forex market also enables currency conversion for international trade settlements and investments.
Currencies are always traded in pairs, so the "value" of one of the currencies in that pair is relative to the value of the other. This determines how much of country A's currency country B can buy, and vice versa. Establishing this relationship (price) for the global markets is the main function of the foreign exchange market. This also greatly enhances liquidity in all other financial markets, which is key to overall stability.

The value of a country's currency depends on whether it is a "free float" or "fixed float." Free-floating currencies are those whose relative value is determined by free-market forces, such as supply-demand relationships.

A fixed float is where a country's governing body sets its currency's relative value to other currencies, often by pegging it to some standard. Free-floating currencies include the U.S. dollar, Japanese yen, and British pound, while examples of fixed floating currencies include the Panamanian Balboa and the Saudi Riyal.

One of the most unique features of the forex market is that it is comprised of a global network of financial centers that transact 24 hours a day, closing only on the weekends. As one major forex hub closes, another hub in a different part of the world remains open for business. This increases the liquidity available in currency markets, which adds to its appeal as the largest asset class available to investors.

The most liquid trading pairs are, in descending order of liquidity:

- EUR/USD
- USD/JPY
- GBP/USD

There are three main forex markets: the spot forex market, the forward forex market, and the futures forex market.

Spot Forex Market: The spot market is the immediate exchange of currencies at the current exchange. On the spot. This makes up a large portion of the total forex market and involves buyers and sellers from across the entire spectrum of the financial sector, as well as those individuals exchanging currencies.

Forward Forex Market: The forward market involves an agreement between the buyer and seller to exchange currencies at an agreed-upon price at a set date in the future. No exchange of actual currencies takes place, just the value. The forward market is often used for hedging.

Futures Forex Market: The futures market is similar to the forward market, in that there is an agreed price at an agreed date. The primary difference is that the futures market is regulated and happens on an exchange. This removes the risk found in other markets. Futures are also used for hedging.

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TOPIC 198: IMPORTANCE OF FOREIGN EXCHANGE

- The exchange rate plays an important role in a country's trade performance. Whether determined by exogenous shocks or by policy, the relative valuations of currencies and their volatility often have important repercussions on international trade, the balance of payments and overall economic performance.

When a currency increases in value, it experiences appreciation; when it falls in value and is worth fewer U.S. dollars, it undergoes depreciation.

At the beginning of 1999, for example, the euro was valued at 1.18 dollars; on Jun 19, 2017, it was valued at 1.12 dollars.

The euro depreciated by 6%

(1.12 - 1.18)/1.18= -0.06 = -6%

Appreciation

Equivalently, we could say that the U.S. dollar, which went from a value of 0.85 euro per dollar at the beginning of 1999 to a value of 0.89 euro per dollar on June 19, 2017, appreciated by 6%: (0.89 - 0.85)/0.85 = 0.06 = 6%.

Foreign Exchange

- When a country's currency appreciates, the country's goods abroad become more expensive, and foreign goods in that country become cheaper (holding domestic prices constant in both countries).
- Conversely, when a country's currency depreciates, its goods abroad become cheaper, and foreign goods in that country become more expensive.

TOPIC 199: HOW IS FOREIGN EXCHANGE TRADED?

- The foreign exchange market is where currencies are traded. Currencies are important because they allow us to purchase goods and services locally and across borders. International currencies need to be exchanged to conduct foreign trade, business, travel, etc.

Where is Forex Traded?

- Forex is traded primarily via three venues: spot markets, forwards markets, and futures markets. The spot market is the largest of all three markets because it is the "underlying" asset on which forwards and futures markets are based.

How to trade FOREX?

- The first step to forex trading is to educate yourself about the market's operations and terminology.
- Next, develop a trading strategy based on your finances and risk tolerance.
- Finally, open a brokerage account.
- It can be done online.

TOPIC 200: EXCHANGE RATES IN THE LONG RUN - LAW OF ONE PRICE

- The exchange rate is determined in the long run by prices, which are determined by the relative supply of money across countries and the relative real demand of money across countries.

Law of One Price

- The law of one price is an economic concept that states that the price of an identical asset or commodity will have the same price globally, regardless of location, when certain factors are considered.

The law of one price takes into account a frictionless market, where there are no transaction costs, transportation costs, or legal restrictions, the currency exchange rates are the same, and that there is no price manipulation by buyers or sellers. The law of one price exists because differences between asset prices in different locations would eventually be eliminated due to the arbitrage opportunity.

The arbitrage opportunity would be achieved whereby a trader would purchase the asset in the market it is available at a lower price and then sell it in the market where it is available at a higher price. Over time, market equilibrium forces would align the prices of the asset.

The law of one price is the foundation of purchasing power parity. Purchasing power parity states that the value of two currencies is equal when a basket of identical goods is priced the same in both countries. It ensures that buyers have the same purchasing power across global markets.

In reality, purchasing power parity is difficult to achieve, due to various costs in trading and the inability to access markets for some individuals.

The formula for purchasing power parity is useful in that it can be applied to compare prices across markets that trade in different currencies. As exchange rates can shift frequently, the formula can be recalculated on a regular basis to identify mispricings across various international markets.

Example of the Law of One Price

If the price of any economic good or security is inconsistent in two different free markets after considering the effects of currency exchange rates, then to earn a profit, an arbitrageur will purchase the asset in the cheaper market and sell it in the market where prices are higher. When the law of one price holds, arbitrage profits such as these will persist until the price converges across markets.

For example, if a particular security is available for $10 in Market A but is selling for the equivalent of $20 in Market B, investors could purchase the security in Market A and immediately sell it for $20 in Market B, netting a profit of $10 without any true risk or shifting of the markets.

As securities from Market A are sold on Market B, prices on both markets should change in accordance with the changes in supply and demand, all else equal. Increased demand for these securities in Market A, where it is relatively cheaper, should lead to an increase in its price there. Conversely, increased supply in Market B, where the security is being sold for a profit by the arbitrageur, should lead to a decrease in its price there. Over time, this would lead to a balancing of the price of the security in the two markets, returning it to the state suggested by the law of one price.

Violations of the Law of One Price

In the real world, the assumptions built into the law of one price frequently do not hold, and persistent differentials in prices for many kinds of goods and assets can be readily observed.

1. Transportation Cost

When dealing in commodities, or any physical good, the cost to transport them must be included, resulting in different prices when commodities from two different locations are examined.

2. Transaction Cost

These are the payments that banks and brokers receive for their roles.

In case of buying and selling real estate, transaction costs include the agent's commission and closing costs, such as title search fees, appraisal fees, and government fees. It also include the time and labor associated with transporting goods across long distances.

3. Legal Restrictions

Legal barriers to trade, such as tariffs, capital controls, or in the case of wages, immigration restrictions, can lead to persistent price differentials rather than one price. These will have a similar effect to transportation and transaction costs, and might even be thought of as a type of transaction cost.

4. Market Structure

The number of buyers and sellers (and the ability of buyers and sellers to enter the market) can vary between markets, market concentration and ability of buyers and sellers to set prices can vary as well.

- A seller who enjoys a high degree of market power due to natural economies of scale in a given market might act like a monopoly price setter and charge a higher price. This can lead to different prices for the same good in different markets.

Lesson 39

THEORY OF PURCHASING POWER PARITY

TOPIC 201: THEORY OF PURCHASING POWER PARITY

- PPP is an economic theory that compares different countries' currencies through a "basket of goods" approach.
- According to this concept, two currencies are in equilibrium—known as the currencies being at par—when a basket of goods is priced the same in both countries, taking into account the exchange rates.

Formula

- Relative version of Purchasing Power Parity
- S =$\frac{P_1}{P_2}$
- S = Exchange rate of currency 1 to currency 2
- P_1=Cost of good X in Currency 1
- P_2 = Cost of good X in Currency 2

Example

- Suppose it costs $10 to buy a shirt in the U.S., and it costs €8.00 to buy an identical shirt in Germany. To make a comparison, we must first convert the €8.00 into U.S. dollars.
- If the exchange rate was such that the shirt in Germany costs $15.00, the PPP would, therefore, be 15/10, or 1.5.
- In other words, for every $1.00 spent on the shirt in the U.S., it takes $1.50 to obtain the same shirt in Germany buying it with the euro.

One popular macroeconomic analysis metric to compare economic productivity and standards of living between countries is purchasing power parity (PPP). PPP is an economic theory that compares different countries' currencies through a "basket of goods" approach.

According to this concept, two currencies are in equilibrium—known as the currencies being at par—when a basket of goods is priced the same in both countries, taking into account the exchange rates.

Relative version of Purchasing Power Parity

S =$\boldsymbol{P1/P2}$

S = Exchange rate of currency 1 to currency 2

$\boldsymbol{P1}$=Cost of good X in Currency 1

P_2= Cost of good X in Currency 2

Drawbacks of Purchasing Power Parity

Since 1986, The Economist has playfully tracked the price of McDonald's Corp.'s (MCD) Big Mac hamburger across many countries. Their study results in the famed "Big Mac Index". In "Burgernomics"—a prominent 2003 paper that explores the Big Mac Index and PPP—authors Michael R. Pakko and Patricia S. Pollard cited the following factors to explain why the purchasing power parity theory is not a good reflection of reality.

Transport Costs

Goods that are unavailable locally must be imported, resulting in transport costs. These costs include not only fuel but import duties as well. Imported goods will consequently sell at a relatively higher price than do identical locally sourced goods.

Tax Differences

Government sales taxes such as the value-added tax (VAT) can spike prices in one country, relative to another.

Government Intervention

Tariffs can dramatically augment the price of imported goods, where the same products in other countries will be comparatively cheaper.

Non-Traded Services

The Big Mac's price factors input costs that are not traded. These factors include such items as insurance, utility costs, and labor costs. Therefore, those expenses are unlikely to be at parity internationally.

Market Competition

Goods might be deliberately priced higher in a country. In some cases, higher prices are because a company may have a competitive advantage over other sellers. The company may have a monopoly or be part of a cartel of companies that manipulate prices, keeping them artificially high.

The Bottom Line

While it's not a perfect measurement metric, purchase power parity does allow for the possibility of comparing pricing between countries that have differing currencies.

TOPIC 202: FACTORS THAT AFFECT EXCHANGE RATES IN THE LONG RUN

1. Relative Price Levels

If the demand for a domestic good relative to a foreign good increases, the domestic currency will appreciate and vice versa.

2. Trade Barriers

- Tariffs, quotas and other restrictions.
- Increasing trade barriers causes a country's currency to appreciate in the long run.

3. Preferences for Domestic versus Imported Goods

- Increased demand for a country's exports causes its currency to appreciate in the long run.
- Pakistan imports in July 2021: $5,601 million
- Pakistan exports in Aug 2021: $2,248 million

4. Productivity

In the long run, as a country becomes more productive relative to other countries, its currency appreciates.

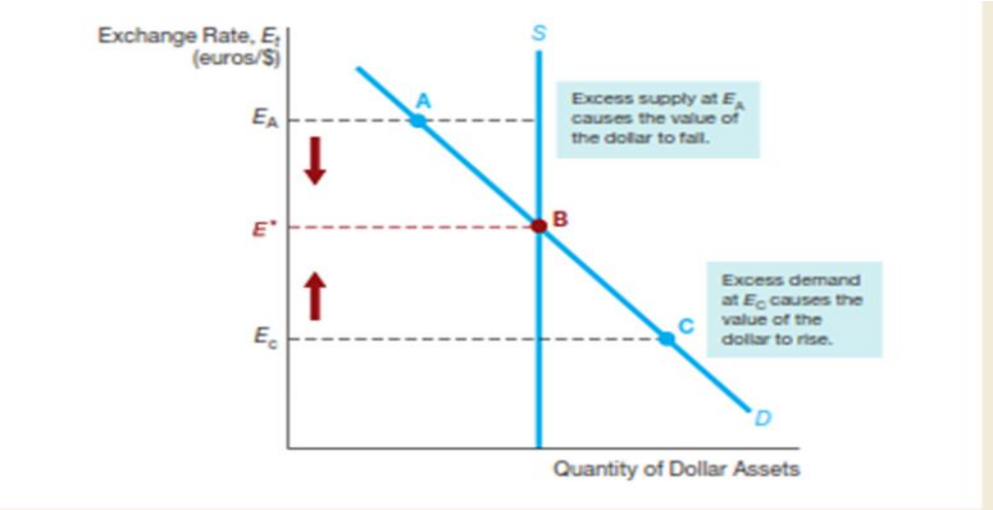
TOPIC 203: EXCHANGE RATES IN THE SHORT RUN

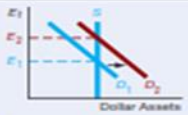
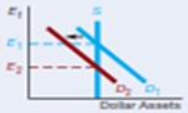
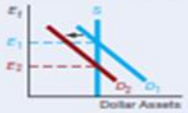
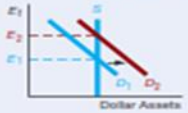
- Traders and institutions buy and sell currencies 24 hours a day during the week. For a trade to occur, one currency must be exchanged for another. To buy British Pounds (GBP), another currency must be used to buy it. Whatever currency is used will create a currency pair.
- If U.S. dollars (USD) are used to buy GBP, the exchange rate is for the GBP/USD pair.
- If the USD/CAD currency pair is 1.33, that means it costs 1.33 Canadian dollars for 1 U.S. dollar.
- In USD/CAD, the first currency listed (USD) always stands for one unit of that currency.

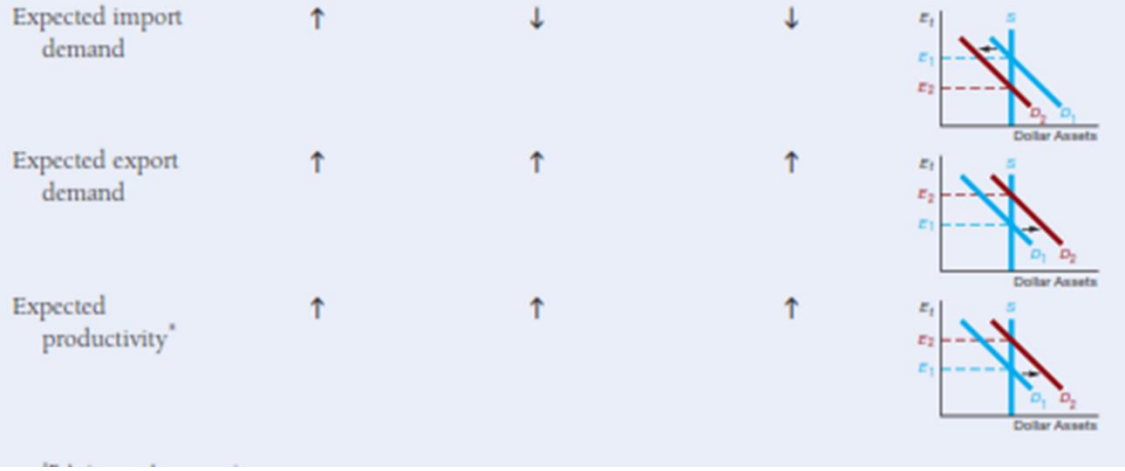
Equilibrium in Foreign Exchange Market

Factors that Shift the Demand Curve for Domestic Assets

Factors that Shift the Demand Curve for Domestic Assets (continued)



Factor	Change in Factor	Change in Quantity Demanded of Domestic Assets at Each Exchange Rate	Response of Exchange Rate, E_t	
Domestic interest rate, i^D	↑	↑	↑	
Foreign interest rate, i^F	↑	↓	↓	
Expected domestic price level*	↑	↓	↓	
Expected trade barriers*	↑	↑	↑	



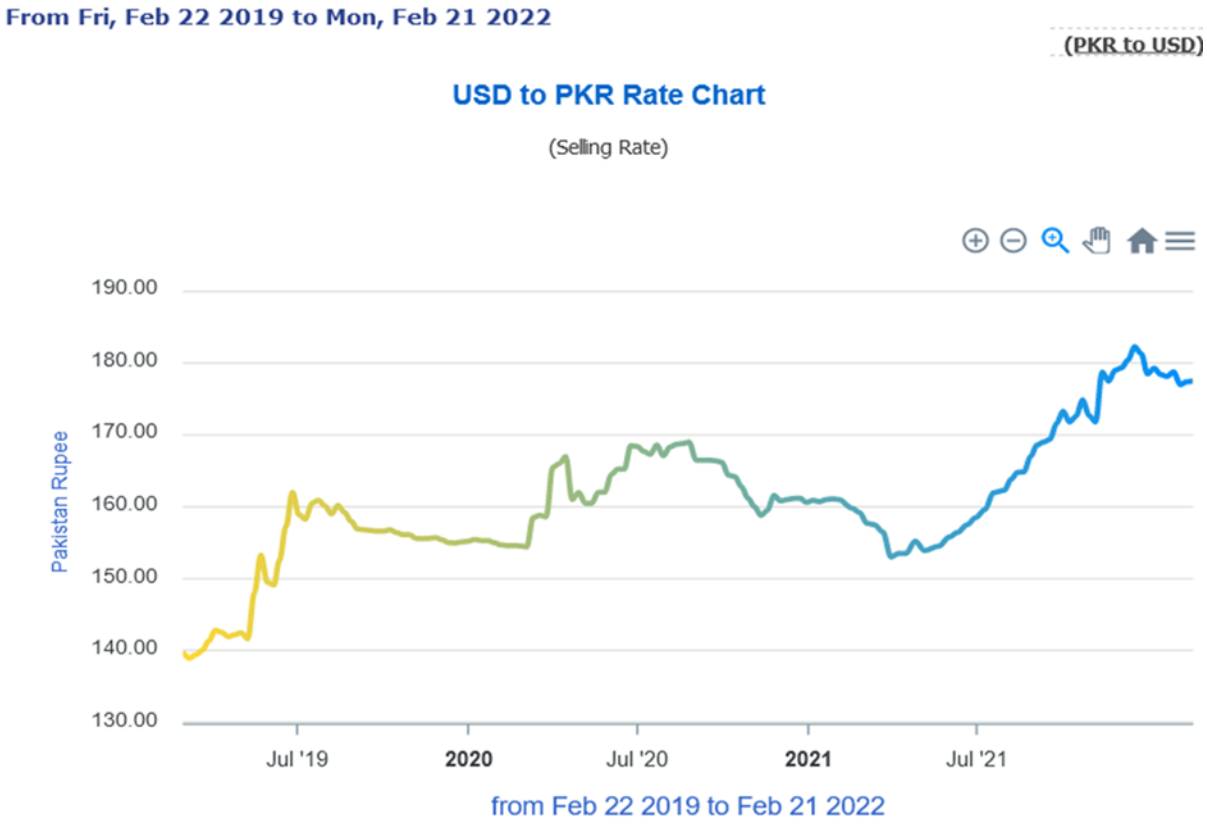
TOPIC 204: PAKISTAN'S EXCHANGE RATE

The broad-based up surge in global commodity prices, COVID-19 vaccine imports, and demand-side pressures, all contributed to the rising imports. While the PKR exchange rate acted as a shock absorber and depreciated, in response to external payments pressure. Amidst the challenging present global environment, Pakistan's external account also came under strain during Jul-Mar FY2022 and the trade deficit widened substantially over last year. On one hand, export receipts and workers' remittances both reached record-high levels during the nine-month period. On the other hand, however, the import payments also registered a sizable, broad-based increase. As a result, the current account deficit widened considerably over last year. These payment pressures manifested on the interbank PKR-USD exchange rate, which depreciated 14.1 percent during Jul-Mar FY2022. The SBP's FX reserves also came under pressure from Q2 onwards, dropping US$ 5.9 billion during Jul-Mar FY2022 to US$ 11.4 billion by end March 2022. SBP has announced to embark on the market based flexible exchange rate regime from May 2019. This has helped SBP not only to build foreign exchange reserves from US$ 7.3 billion in June 2019 to US$16.4 billion at end March 2022 but also to reduce size of its forward swap book from US$ 8.0 billion in June 2019 to US$ 4.4 billion at end January 2022.

Despite the uncertainties of the COVID-19, market-based exchange rate regime helped the external sector to record marked improvement during FY2021.The current account deficit fell to US$1.9 billion (0.6 percent of the GDP) in FY21 from US$4.4 billion (1.7 percent of GDP) in FY20. This is the lowest deficit in 10 years with all-time high exports (US$25.6 billion) and workers remittances (US$29.4 billion). With the sharp rise in global commodity prices amid supply chain disruptions, however, current account deficit widened to US$12.1 billion during Jul-Feb FY2022.

Exchange rate is continuing to play its role of shock absorber and as of 28th March 2022 is depreciated by around 13.5 percent since end June 2021. This depreciation together with other policy actions are expected to contain the current account deficit in the rest of FY2022 and FY2023. Despite adverse terms of trade shock, current account deficit (CAD) narrowed sharply to US$0.5 billion in February 2022, almost one fifth of US$ 2.5 billion in January 2022. This is a broad-based improvement as indicated by reduction of deficits in balances of goods and services, primary income and increase in secondary income.

To analyze the fluctuations in the Pakistan US Dollar Exchange Rate in the past three years see the following trend diagram.



TOPIC 205: BEHAVIORAL FINANCE

- Behavioral finance, a subfield of behavioral economics, proposes that psychological influences and biases affect the financial behaviors of investors and financial practitioners.
- Moreover, influences and biases can be the source for the explanation of all types of market anomalies and specifically market anomalies in the stock market, such as severe rises or falls in stock price. As behavioral finance is such an integral part of investing, the Securities and Exchange Commission has staff specifically focused on behavioral finance.

Behavioral finance can be analyzed from a variety of perspectives. Stock market returns are one area of finance where psychological behaviors are often assumed to influence market outcomes and returns but there are also many different angles for observation. The purpose of the classification of behavioral finance is to help understand why people make certain financial choices and how those choices can affect markets.

Within behavioral finance, it is assumed that financial participants are not perfectly rational and self-controlled but rather psychologically influential with somewhat normal and self-controlling tendencies. Financial decision-making often relies on the investor's mental and physical health. As an investor's overall health improves or worsens, their mental state often changes. This impacts their decision-making and rationality towards all real-world problems, including those specific to finance.

One of the key aspects of behavioral finance studies is the influence of biases. Biases can occur for a variety of reasons. Biases can usually be classified into one of five key concepts. Understanding and classifying different types of behavioral finance biases can be very important when narrowing in on the study or analysis of industry or sector outcomes and results.

TOPIC 206: OVERCONFIDENCE AND INVESTMENT

Overconfidence has two components:

1. overconfidence in the quality of your information,
2. and your ability to act on said information at the right time for maximum gain.

- ✓ Studies show that overconfident traders trade more frequently and fail to appropriately diversify their portfolios.
- ✓ One study analyzed trades from 10,000 clients at a large discount brokerage firm. The study sought to ascertain if frequent trading led to higher returns.
- ✓ The study found that the purchased stocks underperformed the sold stocks by 5% over one year and 8.6% over two years. In other words, the more active the retail investor, the less money they make.

How to avoid this bias?

- Trade less and invest more.

- Understand that by entering into trading activities you're trading against computers, institutional investors, and others around the world with better data and more experience than you.
- By increasing your time frame, mirroring indexes, and taking advantage of dividends, you will likely build wealth over time. Resist the urge to believe that your information and intuition are better than others in the market.

After backing out tax-loss trades and others to meet liquidity needs, the study found that the purchased stocks underperformed the sold stocks by 5% over one year and 8.6% over two years. In other words, the more active the retail investor, the less money they make.

This study was replicated several times in multiple markets and the results were always the same. The authors concluded that traders are, "basically paying fees to lose money."

How to Avoid This Bias

Trade less and invest more. Understand that by entering into trading activities you're trading against computers, institutional investors, and others around the world with better data and more experience than you. The odds are overwhelmingly in their favor. By increasing your time frame, mirroring indexes, and taking advantage of dividends, you will likely build wealth over time. Resist the urge to believe that your information and intuition are better than others in the market.

TOPIC 207: LOSS AVERSION

- Loss aversion refers to a phenomenon where a real or potential loss is perceived by individuals as psychologically or emotionally more severe than an equivalent gain.
- For instance, the pain of losing $100 is often far greater than the joy gained in finding the same amount.
- This overwhelming fear of loss can cause investors to behave irrationally and make bad decisions, such as holding onto a stock for too long or too little time.
- Investors can avoid psychological traps by adopting a strategic asset allocation strategy, thinking rationally, and not letting emotion get the better of them.

Loss aversion occurs when investors place a greater weighting on the concern for losses than the pleasure from market gains. In other words, they're far more likely to try to assign a higher priority to avoiding losses than making investment gains.

As a result, some investors might want a higher payout to compensate for losses. If the high payout isn't likely, they might try to avoid losses altogether even if the investment's risk is acceptable from a rational standpoint.

Applying loss aversion to investing, the so-called disposition effect occurs when investors sell their winners and hang onto their losers. Investors' thinking is that they want to realize gains quickly. However, when an investment is losing money, they'll hold onto it because they want to get back to even or their initial price. Investors tend to admit they are correct about an investment quickly (when there's a gain).

However, investors are reluctant to admit when they made an investment mistake (when there's a loss). The flaw in disposition bias is that the performance of the investment is often tied to the entry price for the investor. In other words, investors gauge the performance of their investment based on their individual entry price disregarding fundamentals or attributes of the investment that may have changed.

TOPIC 208: INERTIA

- Inertia is the endurance of a stable state associated with inaction and the concept of status quo bias (Madrian & Shea 2001). Behavioral nudges can either work *with* people's decision inertia (e.g. by setting defaults) or *against* it (e.g. by giving warnings)(Jung, 2019).
- In social psychology, the term inertia is sometimes also used in relation to a persistence in (or commitments to) attitudes and relationships.

In behavioral economics, inertia is the endurance of a stable state associated with inaction and the concept of status quo bias (Madrian & Shea 2001). Behavioral nudges can either work with people's decision inertia (e.g. by setting defaults) or against it (e.g. by giving warnings)(Jung, 2019). In social psychology, the term inertia is sometimes also used in relation to a persistence in (or commitments to) attitudes and relationships.

TOPIC 209: FRAMING

- Framing bias occurs when people make a decision based on the way the information is presented, as opposed to just on the facts themselves. The same facts presented in two different ways can lead to people making different judgments or decisions.
- For example, an equity research report may come with a lot of opinion and bias included in the research.

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- Try to remove any editorial/judgmental comments and look at only the key numbers and underlying assumptions driving the valuation. Then arrive at your own conclusions, rather than being swayed by how the information is presented to you.

Choices can be presented in a way that highlights the positive or negative aspects of the same decision, leading to changes in their relative attractiveness. This technique was part of Tversky and Kahneman's development of prospect theory, which framed gambles in terms of losses or gains (Kahneman & Tversky, 1979). Different types of framing approaches have been identified, including risky choice framing (e.g. the risk of losing 10 out of 100 lives vs the opportunity to save 90 out of 100 lives), attribute framing (e.g. beef that is 95% lean vs 5% fat), and goal framing (e.g. motivating people by offering a $5 reward vs imposing a $5 penalty) (Levin et al., 1998).

The concept of framing also has a long history in political communication, where it refers to the informational emphasis a communicator chooses to place in a particular message. In this domain, research has considered how framing affects public opinions of political candidates, policies, or broader issues (Busby et al., 2018).

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Lesson 40

MODERN AND BEHAVIORAL PORTFOLIO THEORIES

TOPIC 210: MODERN AND BEHAVIORAL PORTFOLIO THEORIES

- Modern portfolio theory (MPT) was developed by Harry Markowitz
- It identifies how a rational actor would construct a diversified portfolio across several asset classes in order to maximize expected return for a given level of risk preference.
- The resulting theory constructed an "efficient frontier," or the best possible portfolio mix for any risk tolerance.
- Modern portfolio theory then uses this theoretical limit to identify optimal portfolios through a process of mean-variance optimization (MVO).
- It means that emotion and psychology play a role when investors make decisions, sometimes causing them to behave in unpredictable or irrational ways.
- Behavioral economists Shefrin and Statman developed the behavioral portfolio theory based on mental accounts.
- They view behavioral portfolios as being formed of a layered pyramid, with each layer a separate mental account.
- The base layers represent assets designed to provide ‘protection from poverty,’ which results in conservative investments designed to avoid loss.
- Higher layers represent ‘hopes for riches’ and are invested in risky assets in the hope of high returns.
- An individual investor can simultaneously display risk-averse and risk-tolerant behavior, depending on which mental account they are thinking about.
- This model can help explain why individuals can buy at the same time both ‘insurance’ and ‘lottery tickets’ such as a handful of small-cap stocks.
- The theory also suggests that investors treat each layer in isolation and don’t consider the relationship between the layers.

Modern portfolio theory (MPT) and behavioral finance represent differing schools of thought that attempt to explain investor behavior. Perhaps the easiest way to think about their arguments and positions is to think of modern portfolio theory as how the financial markets would work in the ideal world, and to think of behavioral finance as how financial markets work in the real world.

Having a solid understanding of both theory and reality can help you make better investment decisions.

Modern Portfolio Theory

Modern portfolio theory is the basis for much of the conventional wisdom that underpins investment decision making. Many core points of modern portfolio theory were captured in the 1950s and1960s by the efficient market hypothesis put forth by Eugene Fama of the University of Chicago.

According to Fama's theory, financial markets are efficient, investors make rational decisions, market participants are sophisticated, informed and act only on available information. Since everyone has the same access to that information, all securities are appropriately priced at any given time. If markets are efficient and current, it means that prices always reflect all information, so there's no way you'll ever be able to buy a stock at a bargain price.

Other snippets of conventional wisdom include the theory that the stock market will return an average of 8% per year (which will result in the value of an investment portfolio doubling every nine years), and that the ultimate goal of investing is to beat a static benchmark index. In theory, it all sounds good. The reality can be a bit different.

Modern portfolio theory (MPT) was developed by Harry Markowitz during the same period to identify how a rational actor would construct a diversified portfolio across several asset classes in order to maximize expected return for a given level of risk preference. The resulting theory constructed an "efficient frontier," or the best possible portfolio mix for any risk tolerance. Modern portfolio theory then uses this theoretical limit to identify optimal portfolios through a process of mean-variance optimization (MVO).

Behavioral Finance

Despite the nice, neat theories, stocks often trade at unjustified prices, investors make irrational decisions, and you would be hard-pressed to find anyone who owns the much-touted "average" portfolio generating an 8% return every year like clockwork.

So what does all of this mean to you? It means that emotion and psychology play a role when investors make decisions, sometimes causing them to behave in unpredictable or irrational ways. This is not to say that theories have no value, as their concepts do work—sometimes.

Perhaps the best way to consider the differences between theoretical and behavioral finance is to view the theory as a framework from which to develop an understanding of the topics at hand, and to view the behavioral aspects as a reminder that theories don't always work out as expected.

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Accordingly, having a good background in both perspectives can help you make better decisions. Comparing and contrasting some of the major topics will help set the stage.

Education can be put to work on behalf of your portfolio in a logical way, yet with your eyes wide open to the degree of illogical factors that influence not only investors' actions, but security prices as well. By paying attention, learning the theories, understanding the realities and applying the lessons, you can make the most of the bodies of knowledge that surround both traditional financial theory and behavioral finance.

TOPIC 211: MANAGING DIVERSIFICATION

1. Naïve diversification

Evidence suggests that investors use 'naïve' rules of thumb for portfolio construction in the absence of better information.

One such rule has been dubbed the '1/n' approach, where investors allocate equally to the range of available asset classes or funds ('n' stands for the number of options available).

This approach ignores the specific risk-return characteristics of the investments and the relationships between them.

2. Investing in the familiar

Investors have been documented to prefer investing in familiar assets. Investors associate familiarity with low risk. This manifests itself in home bias – high portfolio weights in assets from an investor's own country.

Institutional and individual investors around the globe tend to bias portfolios towards familiar local markets and away from international markets. In these cases, the danger is one of inadequate diversification.

TOPIC 212: USING OR MISUSING INFORMATION

Researchers have documented a number of biases in the way in which we filter and use information when making decisions.

Anchoring

Decisions can be 'anchored' by the way information is presented.

Example: A salesman can offer a very high price to start negotiations that is objectively well above fair value. Yet, because the high price is an anchor, the final selling price will also tend to be higher than if the salesman had offered a fair or low price to start.

Availability bias

Some evidence suggests that recently observed or experienced events strongly influence decisions. Psychologists refer to this as the 'availability bias.'

Researchers found that investors are more likely to be fearful of a stock market crash when one has occurred in the recent past.

Representativeness Bias

It reflects the case where decisions are made based on a situation's superficial characteristics rather than a detailed evaluation of reality. It means that decisions are based on stereotypes.

Example: Investors assume that shares in a high-profile, well-managed company will automatically be a good investment.

TOPIC 213: GROUP BEHAVIOR

A group situation may counteract a particular bias or it may strengthen it.

Equally, the group situation could create new biases.

We typically use groups to make decisions in order to benefit from the range of knowledge and experience in a group.

A desire for social acceptance may encourage individuals with conflicting views to fall into line.

Or, those with opposing views may start to doubt their own convictions.

Evidence suggests that crowds – groups of unrelated individuals – are often able to identify correct answers to problems.

The research suggests the majority opinion of the audience is correct over 90% of the time.

TOPIC 214: MANAGING THE BIASES

We are unlikely to find a 'cure' for the biases, but if we are aware of them and their effect, we can possibly avoid the major pitfalls.

Audit trails

A clear understanding of why particular investment decisions have been taken can help mitigate the effects of behavioral biases.

As markets move and emotions take hold, this record can help prevent making snap judgments.

Framing

Framing is also a valuable adviser tool. Portfolio discussions should always be framed in terms of long-term goals and the client's total wealth picture.

Feedback

More generally, investors may be able to use feedback to mitigate behavioral biases.

Careful consideration of the outcomes of past decisions should help individuals learn to control and work around unhelpful decision-making biases.

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Lesson 41

THE EXPECTED UTILITY MODEL

TOPIC 215: THE EXPECTED UTILITY MODEL

- Expected utility model was developed by John von Neuman and Oskar Morgenstern to explain rational behaviour when people face uncertainty.
- It shows that when a consumer is faced with a choice of items or outcomes subject to various levels of chance, the optimal decision will be the one that maximizes the expected value of the utility (i.e., satisfaction) derived from the choice made.
- Expected value is the sum of the products of the various utilities and their associated probabilities.
- The consumer is expected to be able to rank the items or outcomes in terms of preference, but the expected value will be conditioned by their probability of occurrence.
- The von Neumann-Morgenstern utility function can be used to explain risk-averse, risk-neutral, and risk-loving behavior.

"Expected utility" is an economic term summarizing the utility that an entity or aggregate economy is expected to reach under any number of circumstances. The expected utility is calculated by taking the weighted average of all possible outcomes under certain circumstances. With the weights being assigned by the likelihood or probability, any particular event will occur.

The expected utility of an entity is derived from the expected utility hypothesis. This hypothesis states that under uncertainty, the weighted average of all possible levels of utility will best represent the utility at any given point in time. Expected utility model was developed by John von Neuman and Oskar Morgenstern to explain rational behavior when people face uncertainty.

Expected utility theory is used as a tool for analyzing situations in which individuals must make a decision without knowing the outcomes that may result from that decision, i.e., decision making under uncertainty. These individuals will choose the action that will result in the highest expected utility, which is the sum of the products of probability and utility over all possible outcomes. The decision made will also depend on the agent's risk aversion and the utility of other agents.

This theory also notes that the utility of money does not necessarily equate to the total value of money. This theory helps explain why people may take out insurance policies to cover themselves for various risks. The expected value from paying for insurance would be to lose out monetarily.

The possibility of large-scale losses could lead to a serious decline in utility because of the diminishing marginal utility of wealth.

Decisions involving expected utility are decisions involving uncertain outcomes. An individual calculates the probability of expected outcomes in such events and weighs them against the expected utility before making a decision.

For example, purchasing a lottery ticket represents two possible outcomes for the buyer. They could end up losing the amount they invested in buying the ticket, or they could end up making a smart profit by winning either a portion of the entire lottery. Assigning probability values to the costs involved (in this case, the nominal purchase price of a lottery ticket), it is not difficult to see that the expected utility to be gained from purchasing a lottery ticket is greater than not buying it. Expected utility is also used to evaluate situations without immediate payback, such as purchasing insurance. When one weighs the expected utility to be gained from making payments in an insurance product (possible tax breaks and guaranteed income at the end of a predetermined period) versus the expected utility of retaining the investment amount and spending it on other opportunities and products, insurance seems like a better option. Expected value is the sum of the products of the various utilities and their associated probabilities.

The consumer is expected to be able to rank the items or outcomes in terms of preference, but the expected value will be conditioned by their probability of occurrence.

The von Neumann-Morgenstern utility function can be used to explain risk-averse, risk-neutral, and risk-loving behavior.

TOPIC 216: THE NO-ARBITRAGE PRICING PRINCIPLE

- ✓ The idea behind a no-arbitrage condition is that if there is a mispriced security in the market, investors can always construct a portfolio with factor sensitivities similar to those of mispriced securities and exploit the arbitrage opportunity.
- ✓ As all investors would sell an overvalued and buy an undervalued portfolio, this would drive away any arbitrage profit.

TOPIC 217: ARBITRAGE PRICING MODEL

- Arbitrage pricing model is an alternative model to the capital asset pricing model
- It explains asset or portfolio returns with systematic factors and asset/portfolio sensitivities to such factors.
- It was developed by economist Stephen Ross in the 1970s.

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- The theory estimates the expected returns of well-diversified portfolios with the underlying assumption that portfolios are well-diversified and any discrepancy from the equilibrium price in the market would be instantaneously driven away by investors.
- APT is a multi-factor technical model based on the relationship between a financial asset's expected return and its risk. The model is designed to capture the sensitivity of the asset's returns to changes in certain macroeconomic variables. Investors and financial analysts can use these results to help price securities.

- These macroeconomic variables include:
  - Inflation
  - Industrial production
  - Risk premiums
  - The shape of the term structure of interest rates

How Options Work?

- An option is a contract giving the buyer the right—but not the obligation—to buy (in the case of a call) or sell (in the case of a put) the underlying asset at a specific price on or before a certain date.
- People use options for income, to speculate, and to hedge risk.
- An option to buy the specified item at a fixed price is a call
- An option to sell is a put.
- The fixed price specified in an option contract is called the option’s strike price or exercise price.
- The date after which an option can no longer be exercised is called its expiration date or maturity date.
- Exchange-traded options have standard terms defined by the options exchange.
- Options not traded on an exchange are called over-the-counter options.
- An option is identified by its strike price and its expiration date.

Index Options

- Index options are options contracts that utilize a benchmark index, or a futures contract based on that index, as its underlying instrument.
- Index options are typically European style and settle in cash for the value of the index at expiration.

- Like all options, index options will give the buyer the right, but not the obligation, to either go long (for a call) or short (for a put) the value of the index at a pre-specified strike price.

Example

- Suppose the S&P 500 is at 4,500 points. Jane decides to sell a call option to Tim with a strike price of 4,500. Tim pays a premium of $10. Just as a stock option typically covers 100 shares of a stock, index options typically use a multiplier of 100. So, in this case, Tim will pay $1,000 in total.
- $10 * 100 = $1,000
- If the S&P 500 gains value, Tim's call option will increase in price.
- If it falls, the call option will fall in price. For instance, if it rises to 4,525, Tim could exercise the option to get a payment of $2,500.
- (4,525 - 4,500) * 100 = $2,500

Arbitrage pricing theory (APT) is an alternative to the capital asset pricing model (CAPM) for explaining returns of assets or portfolios. It was developed by economist Stephen Ross in the 1970s. Over the years, arbitrage pricing theory has grown in popularity for its relatively simpler assumptions. However, arbitrage pricing theory is a lot more difficult to apply in practice because it requires a lot of data and complex statistical analysis.

APT is a multi-factor technical model based on the relationship between a financial asset's expected return and its risk. The model is designed to capture the sensitivity of the asset's returns to changes in certain macroeconomic variables. Investors and financial analysts can use these results to help price securities.

Inherent to the arbitrage pricing theory is the belief that mispriced securities can represent short-term, risk-free profit opportunities. APT differs from the more conventional CAPM, which uses only a single factor. Like CAPM, however, the APT assumes that a factor model can effectively describe the correlation between risk and return.

Underlying Assumptions of APT

Unlike the capital asset pricing model, arbitrage pricing theory does not assume that investors hold efficient portfolios.

The theory does, however, follow three underlying assumptions:

Asset returns are explained by systematic factors.

Investors can build a portfolio of assets where specific risk is eliminated through diversification.

No arbitrage opportunity exists among well-diversified portfolios. If any arbitrage opportunities do exist, they will be exploited away by investors. (This is how the theory got its name.)

However, according to the research of Stephen Ross and Richard Roll, the most important factors are the following:

1. Change in inflation
2. Change in the level of industrial production
3. Shifts in risk premiums
4. Change in the shape of the term structure of interest rates

TOPIC 218: FINANCIAL STRUCTURE OF A FIRM

- ✓ Financial structure refers to the mix of debt and equity that a company uses to finance its operations. It can also be known as capital structure.
- ✓ This composition directly affects the risk and value of the associated business. The financial managers of the business have the responsibility of deciding the best mixture of debt and equity for optimizing the financial structure.

In general, the financial structure of a company can also be referred to as the capital structure. In some cases, evaluating the financial structure may also include the decision between managing a private or public business and the capital opportunities that come with each.

Companies have several choices when it comes to setting up the business structure of their business. Companies can be either private or public. In each case, the framework for managing the capital structure is primarily the same but the financing options differ greatly.

Debt capital is received from credit investors and paid back over time with some form of interest. Equity capital is raised from shareholders giving them ownership in the business for their investment and a return on their equity that can come in the form of market value gains or distributions. Each business has a different mix of debt and equity depending on its needs, expenses, and investor demand.

Private versus Public

- ✓ Private and public companies have the same framework for developing their structure but several differences that distinguish the two.

Both types of companies can issue equity. Private equity is created and offered using the same concepts as public equity but private equity is only available to select investors rather than the public market on a stock exchange. As such the equity fundraising process is much different than

a formal initial public offering (IPO). Private companies can also go through multiple rounds of equity financing over time which affects their market valuation. Companies that mature and choose to issue shares in the public market do so through the support of an investment bank that helps them to pre-market the offering and value the initial shares. All shareholders are converted to public shareholders after an IPO and the market capitalization of the company is then valued based on shares outstanding times market price.

Debt capital follows similar processes in the credit market with private debt primarily only offered to select investors. In general, public companies are more closely followed by rating agencies with public ratings helping to classify debt investments for investors and the market at large. The debt obligations of a company take priority over equity for both private and public companies. Even though this helps debt to come with lower risks, private market companies can still usually expect to pay higher levels of interest because their businesses and cash flows are less established which increases risk.

Debt versus Equity

In building the financial structure of a company, financial managers can choose between either debt or equity. Investor demand for both classes of capital can heavily influence a company's financial structure. Ultimately, financial management seeks to finance the company at the lowest rate possible, reducing its capital obligations and allowing for greater capital investment in the business.

Overall, financial managers consider and evaluate the capital structure by seeking to optimize the weighted average cost of capital (WACC). WACC is a calculation that derives the average percentage of payout required by the company to its investors for all of its capital. A simplified determination of WACC is calculated by using a weighted average methodology that combines the payout rates of all of the company's debt and equity capital.

TOPIC 219: INTERNAL AND EXTERNAL SOURCES OF FINANCE

Internal Sources of Finance

- ✓ Internal sources of finance alludes to the sources of business finance that are generated within the business, from the existing assets or activities.
- ✓ Internal Sources of Finance include Sale of Stock, Sale of Fixed Assets, Retained Earnings and Debt Collection.

External Sources of Finance

- ✓ External sources of finance implies the arrangement of capital or funds from sources outside the business. External Sources of finance include Financial Institutions, Loan from banks, Preference Shares, Debenture, Public Deposits, Lease financing, Commercial paper, Trade Credit.
- ✓ Financial Institutions, Loan from banks, Preference Shares, Debenture, Public Deposits, Lease financing, Commercial paper, Trade Credit.

Lesson 42

EQUITY & DEBT FINANCING AND CAPITAL STRUCTURE

TOPIC 220: EQUITY FINANCING

- Equity financing is used when companies, often start-ups, have a short-term need for cash.
- It is typical for companies to use equity financing several times during the process of reaching maturity.
- There are two methods of equity financing: the private placement of stock with investors and public stock offerings.
- National and local governments keep a close watch on equity financing to ensure that everything done follows regulations.

Equity financing is the process of raising capital through the sale of shares. Companies raise money because they might have a short-term need to pay bills or need funds for a long-term project that promotes growth. By selling shares, a business effectively sells ownership in its company in return for cash.

Equity financing comes from a variety of sources. For example, an entrepreneur's friends and family, professional investors, or an initial public offering (IPO) may provide needed capital.

An IPO is a process that private companies undergo to offer shares of their business to the public in a new stock issuance. Public share issuance allows a company to raise capital from public investors. Industry giants, such as Google and Meta (formerly Facebook), raised billions in capital through IPOs.

While the term equity financing refers to the financing of public companies listed on an exchange, the term also applies to private company financing.

Equity financing involves the sale of common stock and the sale of other equity or quasi-equity instruments such as preferred stock, convertible preferred stock, and equity units that include common shares and warrants.

A startup that grows into a successful company will have several rounds of equity financing as it evolves. Since a startup typically attracts different types of investors at various stages of its evolution, it may use different equity instruments for its financing needs.

For example, angel investors and venture capitalists—generally the first investors in a startup—favor convertible preferred shares rather than common stock in exchange for funding new

companies because the former have more significant upside potential and some downside protection.

Once a company has grown large enough to consider going public, it may consider selling common stock to institutional and retail investors.

Later, if the company needs additional capital, it may choose secondary equity financing options, such as a rights offering or an offering of equity units that includes warrants as a sweetener.

TOPIC 221: DEBT FINANCING

- There are distinct advantages to both types of financing.
- Most companies use a combination of equity and debt financing.

Forms of Debt Financing

1. Corporate bond/Debt Securities

- A corporate bond is a type of debt security that is issued by a firm and sold to investors. The company gets the capital it needs and in return the investor is paid a pre-established number of interest payments at either a fixed or variable interest rate.

2. Secured Debt

Secured debt is debt backed or secured by collateral to reduce the risk associated with lending. If the borrower on a loan defaults on repayment, the bank seizes the collateral, sells it, and uses the proceeds to pay back the debt.

3. Long-term leases

A long-term lease is simply a lease in which the agreement term is ten years or longer.

4. Pension Liabilities

Pension Liabilities means any current or future obligations, liabilities, etc. in relation to pension plans, retiree medical, and any other long-term pension, welfare, or benefit plans.

Debt financing occurs when a firm raises money for working capital or capital expenditures by selling debt instruments to individuals and/or institutional investors. In return for lending the money, the individuals or institutions become creditors and receive a promise that the principal and interest on the debt will be repaid. The other way to raise capital in debt markets is to issue shares of stock in a public offering; this is called equity financing.

When a company needs money, there are three ways to obtain financing: sell equity, take on debt, or use some hybrid of the two. Equity represents an ownership stake in the company. It gives the shareholder a claim on future earnings, but it does not need to be paid back. If the company goes bankrupt, equity holders are the last in line to receive money.

A company can choose debt financing, which entails selling fixed income products, such as bonds, bills, or notes, to investors to obtain the capital needed to grow and expand its operations. When a company issues a bond, the investors that purchase the bond are lenders who are either retail or institutional investors that provide the company with debt financing. The amount of the investment loan—also known as the principal—must be paid back at some agreed date in the future. If the company goes bankrupt, lenders have a higher claim on any liquidated assets than shareholders.

TOPIC 222: CAPITAL STRUCTURE AND FACTORS CAUSING FRICTION

Factors that cause friction include:

1. Income Tax
2. Transaction Costs of Issuing debt or equity securities
3. Conflicts among the stakeholders

TOPIC 223: CREATING VALUE THROUGH FINANCING DECISIONS

By its choice of capital structure the firm can:

- Reduce its cost and circumvent regulations
- Reduce conflict of interest
- Provide stakeholders with financial assets, e.g. financial instruments

TOPIC 224: CAPITAL STRUCTURE AND REDUCING COSTS

- Taxes and Subsidies
- Costs of Financial Distress
- Financial distress is a condition in which a company cannot generate revenue or income because it is unable to meet or cannot pay its financial obligations.
- This is generally due to high fixed expenses (like overhead or salaries), illiquid assets, or revenues sensitive to economic downturns.
- Distress cost refers to the greater expense that a firm in financial distress incurs beyond the cost of doing business.

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- Distress costs can be tangible, such as having to pay higher interest rates or more money to suppliers upfront.
- Distress costs can also be intangible, such as a loss of employee morale and productivity.

TOPIC 225: TYPES OF CAPITALIZATION STRUCTURES

- A company's debt-to-equity ratio is a measure of risk for investors.
- Companies that use more debt than equity to finance their assets and fund operating activities have a high leverage ratio and an aggressive capital structure.
- A company that pays for assets with more equity than debt has a low leverage ratio and a conservative capital structure. That said, a high leverage ratio and an aggressive capital structure can also lead to higher growth rates, whereas a conservative capital structure can lead to lower growth rates.

Optimal Capital Structure

- It is the goal of company management to find the ideal mix of debt and equity, also referred to as the optimal capital structure, to finance operations.
- Analysts use the D/E ratio to compare capital structure. It is calculated by dividing total liabilities by total equity.
- Savvy companies have learned to incorporate both debt and equity into their corporate strategies. At times, however, companies may rely too heavily on external funding and debt in particular. Investors can monitor a firm's capital structure by tracking the D/E ratio and comparing it against the company's industry peers.

Lesson 43

FINANCING DECISIONS AND FINANCIAL METHODS

TOPIC 226: FINANCING DECISIONS

To explain how financial decisions are carried out, we take five examples and explain why and which financial decision is suggested for each of them.

1. Orr Oil Company needs $10 million to finance the test drilling of land it owns already.
2. Gormeh Foods operates as a chain of food stores. It is owned by 5 sisters. Each owns 1/5th of the outstanding stock.
3. Bombay Textile Company manufactures cotton clothes and exports about half of it to clothing companies in Singapore.
4. Holey wants to buy Burger Queen's franchise. He has $50,000 but is in need of another $50,000 to buy the franchise.
5. Lee production is a small movie production company. The 10 owners want to double the number of movies they produce in a year and want $10 million from outside investors.

Company	Financing Method	Reasons
Orr Oil Company	Debt with warrants	If oil is found, then the price of Orr stock will rise and new equity financing will be forthcoming to finance the additional investment needed.
Gormeh Foods, Inc.	Leasing arrangement	No more debt capacity. Leasing provides debt financing secured by the leased equipment.
Bombay Textile Company	Factoring	Rather than having to carry and finance receivables that might be costly to monitor and collect, it is more efficient to sell them to an institution that specializes in collecting receivables.
Holey's Burger Queen	Loans from friends or relatives	Cannot borrow from any institutional lender unless he has more capital of his own. Only those who know him and trust him would provide the capital to him.
Lee Productions	Common stock	Agency costs of debt financing would be very high.

TOPIC 227: FIVE FINANCING METHODS

1. Loans from Friends and Relatives

Appropriate when a firm is starting as a small enterprise. Future prospects are very uncertain.

2. Lease Arrangement

Lessor provides real assets and gets regular payments in return

3. Common Stock

A corporation goes public and issues common stock. Common stock represents shares of ownership in a corporation and the type of stock in which most people invest.

4. Debt with Warrants

Warrants are call options giving their owners the right to buy shares of the issuing company's stock at a fixed price. Warrants are attached to a debt issue when the company envisions a need for raising new equity sometime before the debt has to be repaid.

5. Factoring Receivables

When a company factors its receivables, it sells its accounts receivables (t a discount) to a factor, which is a firm that specializes in investing in receivables. Factoring is raising cash by selling a company asset rather than by borrowing or issuing new equity.

TOPIC 228: HOW TO EVALUATE LEVERED INVESTMENTS?

Levered Investments are evaluated using the following techniques:

1. Adjusted Present Value (APV)
2. Flows to Equity (FTE)
3. Weighted Average Cost of Capital (WACC)

Adjusted Present Value (APV)

- APV is the NPV of a project or company if financed solely by equity plus the present value of financing benefits.
- It is best used for leverage transactions, such as leveraged buyouts, but is more of an academic calculation.

The formula for APV is

Adjusted Present Value = Unlevered Firm Value + Net effect of debt

Example

Assume a multi-year projection calculation finds that the present value of Company ABC's free cash flow (FCF) plus terminal value is $100,000. The tax rate for the company is 30% and the interest rate is 7%. Its $50,000 debt load has an interest tax shield of $15,000, or ($50,000 * 30% * 7%) / 7%. Thus, the adjusted present value is $115,000, or $100,000 + $15,000.

TOPIC 229: FLOWS TO EQUITY (FTE)

The FTE approach requires that the cash flows from the project to the equity holders of the levered firm be discounted at the cost of equity capital. There are three steps to the FTE approach:

1.Calculating the levered cash flow
2.Calculating the discount rate for levered equity
3. NPV analysis

1. **Calculating the levered cash flow (LCF)**

   The levered cash flow is the cash flows of the project after cash costs, cash paid on interest, and cash paid in taxes.

$$LCF = UCF - (1 - T_C)r_B B$$

   rB = the interest rate

   B = the amount of the project financed by debt

   TC = the tax rate

   LCF = Levered cash flow

   UCF =Unlevered cash flow

2. **Calculating the discount rate for levered equity**

   The discount rate for levered equity is simply the rate of expected return that can be observed by the stock's price in the market.

   rs = the levered rate of return

   ro = the unlevered rate of return

   B/S = the debt to equity ratio

   Tc= Tax rate

$$r_s = r_0 + \frac{B}{S}(1 - T_C)(r_0 - r_B)$$

3. **NPV analysis**

   To find the present value of the cash flows to equity, we simply discount the levered cash flows by the expected return of the levered equity. To find net present value, the present value of the initial cash outflows are subtracted.

TOPIC 230: WEIGHTED AVERAGE COST OF CAPITAL (WACC)

- The weighted average cost of capital (WACC) represents a firm's average cost of capital from all sources, including common stock, preferred stock, bonds, and other forms of debt.
- The weighted average cost of capital is a common way to determine required rate of return.
- It expresses, in a single number, the return that both bondholders and shareholders demand in order to provide the company with capital. A firm's WACC is likely to be higher if its stock is relatively volatile or if its debt is seen as risky because investors will demand greater returns.

Weighted average cost of capital (WACC) represents a firm's average after-tax cost of capital from all sources, including common stock, preferred stock, bonds, and other forms of debt. WACC is the average rate that a company expects to pay to finance its assets.

WACC is a common way to determine required rate of return (RRR) because it expresses, in a single number, the return that both bondholders and shareholders demand to provide the company with capital. A firm's WACC is likely to be higher if its stock is relatively volatile or if its debt is seen as risky because investors will require greater returns.

WACC Formula and Calculation

WACC = $(\frac{E}{V} \mathbf{X}\ R_E)$ **+** $(\frac{D}{V}\ X\ R_d\ X\ (1\text{-}T_c)$

Where:

E=Market value of the firm's equity

D=Market value of the firm's debt

V=E+D

Re=Cost of equity

Rd=Cost of debt

Tc=Corporate tax rate

WACC = $(\frac{E}{V}X\ R_E)$ **+** $(\frac{D}{V}\ X\ R_d\ \ X\ (1\text{-}T_c)$

Where

$(\frac{E}{V}X\ R_E)$ = weighted value of equity capital

$(\frac{D}{V}\ X\ R_d\ \ X\ (1\text{-}T_c)$ = weighted value of debt capital

Example

- Suppose that a company obtained $1,000,000 in debt financing and $4,000,000 in equity financing by selling common shares. Proportion of equity based financing E/V would equal 0.8 ($4,000,000 ÷ $5,000,000 of total capital) and D/V would equal 0.2 ($1,000,000 ÷ $5,000,000 of total capital).

Lesson 44

HOW OPTIONS WORK?

TOPIC 231: HOW OPTIONS WORK?

- An option is a contract giving the buyer the right—but not the obligation—to buy (in the case of a call) or sell (in the case of a put) the underlying asset at a specific price on or before a certain date.
- People use options for income, to speculate, and to hedge risk. An option to buy the specified item at a fixed price is a call. An option to sell is a put.
- An option to buy the specified item at a fixed price is a call.
- An option to sell is a put.
- The fixed price specified in an option contract is called the option's strike price or exercise price.
- The date after which an option can no longer be exercised is called its expiration date or maturity date.
- Exchange-traded options have standard terms defined by the options exchange.
- Options not traded on an exchange are called over-the-counter options.
- An option is identified by its strike price and its expiration date.

The fixed price specified in an option contract is called the option's strike price or exercise price. The date after which an option can no longer be exercised is called its expiration date or maturity date. Exchange-traded options have standard terms defined by the options exchange.
Options not traded on an exchange are called over-the-counter options.
An option is identified by its strike price and its expiration date.

TOPIC 232: INDEX OPTIONS

- Index options are options contracts that utilize a benchmark index, or a futures contract based on that index, as its underlying instrument.
- Index options are typically European style and settle in cash for the value of the index at expiration.
- Like all options, index options will give the buyer the right, but not the obligation, to either go long (for a call) or short (for a put) the value of the index at a pre-specified strike price.

Example

- Suppose the S&P 500 is at 4,500 points. Jane decides to sell a call option to Tim with a strike price of 4,500. Tim pays a premium of $10. Just as a stock option typically covers 100 shares of a stock, index options typically use a multiplier of 100. So, in this case, Tim will pay $1,000 in total.
- If the S&P 500 gains value, Tim's call option will increase in price.

$10 * 100 = $1,000

If it falls, the call option will fall in price. For instance, if it rises to 4,525, Tim could exercise the option to get a payment of $2,500.

(4,525 - 4,500) * 100 = $2,500

- When Tim chooses to exercise his option, Jane and Tim will exchange money to settle the contract. However, if exercising the option would cause Tim to lose money, he'd likely simply let it expire, and no additional money would change hands. In this case, Jane profits from the premium Tim paid.

TOPIC 233: INVESTING WITH OPTIONS

Buying a Call

- Carla and Rick are bullish on GE and would like to buy the March call options.

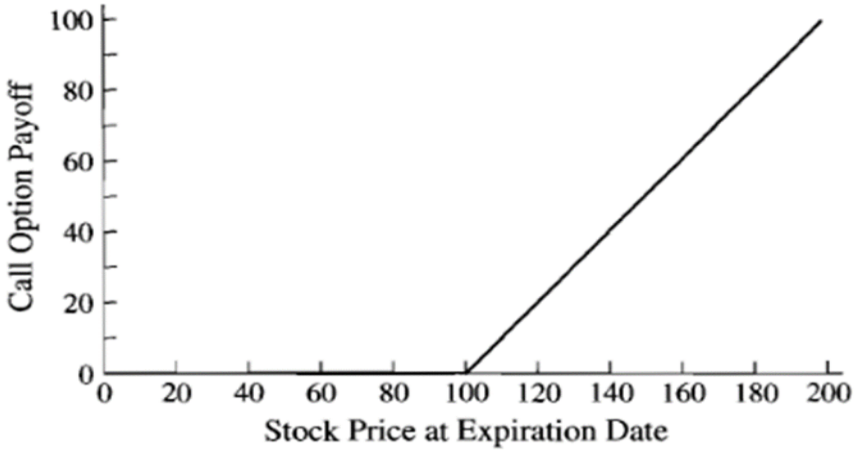
CALLS on GE (last price $27.20) - expiration March 2014					
Strike Price	Price	Implied Vol.	Delta	Intr. Value	Time Value
$24.00	$3.30	22.07	0.94	$3.20	$0.10
$25.00	$2.26	20.16	0.87	$2.20	$0.06
$26.00	$1.47	18.04	0.75	$1.20	$0.27
$27.00	$0.80	16.83	0.55	$0.20	$0.60
$28.00	$0.38	16.46	0.33	$0.00	$0.38
$29.00	$0.16	16.68	0.17	$0.00	$0.16
$30.00	$0.08	17.8	0.09	$0.00	$0.08

Investopedia

With GE trading at $27.20, Carla thinks it can trade up to $28 by March; in terms of downside risk, she thinks the stock could decline to $26. She, therefore, opts for the March $25 call (which is in-the-money) and pays $2.26 for it. The $2.26 is referred to as the premium or the cost of the option. As shown in Table 1, this call has an intrinsic value of $2.20 (i.e., the stock price of $27.20 less the strike price of $25) and the time value of $0.06 (i.e., the call price of $2.26 less intrinsic value of $2.20).

TOPIC 234: PUT OPTIONS PAYOFF DIAGRAM

- A put payoff diagram is a way of visualizing the value of a put option at expiration based on the value of the underlying stock.



TOPIC 235: PUT-CALL PARITY

- It shows the relationship that has to exist between European put and call options that have the same underlying asset, expiration, and strike prices.
- This concept says the price of a call option implies a certain fair price for the corresponding put option with the same strike price and expiration and vice versa.
- Put-call parity does not apply to American options because you can exercise them before the expiry date.
- If the put-call parity is violated, then arbitrage opportunities arise.
- You can determine the put-call party by using the formula C + PV(x) = P + S.
- Put-call party by using the formula C + PV(x) = P + S.
- C = price of the European call option
- PV(x) = the present value of the strike price (x), discounted from the value on the expiration date at the risk-free rate
- P = Price of the European put

- S = Spot price or the current market value of the underlying asset

TOPIC 236: SHORT SELLING

- Short selling occurs when an investor borrows a security and sells it on the open market, planning to buy it back later for less money.
- Short-sellers bet on, and profit from, a drop in a security's price. This can be contrasted with long investors who want the price to go up.
- Short selling has a high risk/reward ratio: It can offer big profits, but losses can mount quickly and infinitely due to margin calls.

Short selling is an investment or trading strategy that speculates on the decline in a stock or other security's price. It is an advanced strategy that should only be undertaken by experienced traders and investors.

Traders may use short selling as speculation, and investors or portfolio managers may use it as a hedge against the downside risk of a long position in the same security or a related one. Speculation carries the possibility of substantial risk and is an advanced trading method. Hedging is a more common transaction involving placing an offsetting position to reduce risk exposure.

In short selling, a position is opened by borrowing shares of a stock or other asset that the investor believes will decrease in value. The investor then sells these borrowed shares to buyers willing to pay the market price. Before the borrowed shares must be returned, the trader is betting that the price will continue to decline and they can purchase the shares at a lower cost. The risk of loss on a short sale is theoretically unlimited since the price of any asset can climb to infinity.

With short selling, a seller opens a short position by borrowing shares, usually from a broker-dealer, hoping to buy them back for a profit if the price declines. Shares must be borrowed because you cannot sell shares that do not exist. To close a short position, a trader buys the shares back on the market—hopefully at a price less than at which they borrowed the asset—and returns them to the lender or broker. Traders must account for any interest charged by the broker or commissions charged on trades.

To open a short position, a trader must have a margin account and will usually have to pay interest on the value of the borrowed shares while the position is open. Also, the Financial Industry Regulatory Authority (FINRA), which enforces the rules and regulations governing registered brokers and broker-dealer firms in the United States, the New York Stock Exchange (NYSE), and

the Federal Reserve have set minimum values for the amount that the margin account must maintain—known as the maintenance margin.

If an investor's account value falls below the maintenance margin, more funds are required, or the position might be sold by the broker.

The process of locating shares that can be borrowed and returning them at the end of the trade is handled behind the scenes by the broker. Opening and closing the trade can be made through the regular trading platforms with most brokers. However, each broker will have qualifications that the trading account must meet before they allow margin trading.

Example

- Imagine a trader who believes that XYZ stock—currently trading at $50—will decline in price in the next three months.
- They borrow 100 shares and sell them to another investor. The trader is now "short" 100 shares since they sold something that they did not own but had borrowed.
- The short sale was only made possible by borrowing the shares, which may not always be available if the stock is already heavily shorted by other traders.
- A week later, the company whose shares were shorted reports dismal financial results for the quarter, and the stock falls to $40.
- The trader decides to close the short position and buys 100 shares for $40 on the open market to replace the borrowed shares. The trader's profit on the short sale, excluding commissions and interest on the margin account, is $1,000: ($50 - $40 = $10 x 100 shares = $1,000).

TOPIC 237: VOLATILITY AND OPTION PRICES

- Option pricing, the amount per share at which an option is traded, is affected by a number of factors including volatility.
- Implied volatility is the real-time estimation of an asset's price as it trades.
- Implied volatility tends to increase when options markets experience a downtrend.
- Implied volatility falls when the options market shows an upward trend.
- Larger implied volatility means higher option prices
- Another form of volatility that affects options is historic volatility, also known as statistical volatility.
- This measures the speed at which underlying asset prices change over a given time period.

TOPIC 238: THE BLACK-SCHOLES MODEL

- The Black-Scholes model, aka the Black-Scholes-Merton (BSM) model, is a differential equation widely used to price options contracts.
- The Black-Scholes model requires five input variables: the strike price of an option, the current stock price, the time to expiration, the risk-free rate, and the volatility.
- The Black-Scholes model makes certain assumptions that can lead to prices that deviate from the real-world results.
- The standard BSM model is only used to price European options, as it does not take into account that American options could be exercised before the expiration date.

The Black-Scholes model, also known as the Black-Scholes-Merton (BSM) model, is one of the most important concepts in modern financial theory. This mathematical equation estimates the theoretical value of derivatives based on other investment instruments, taking into account the impact of time and other risk factors. Developed in 1973, it is still regarded as one of the best ways for pricing an options contract.

Developed in 1973 by Fischer Black, Robert Merton, and Myron Scholes, the Black-Scholes model was the first widely used mathematical method to calculate the theoretical value of an option contract, using current stock prices, expected dividends, the option's strike price, expected interest rates, time to expiration, and expected volatility.

The initial equation was introduced in Black and Scholes' 1973 paper, "The Pricing of Options and Corporate Liabilities," published in the Journal of Political Economy.

Robert C. Merton helped edit that paper. Later that year, he published his own article, "Theory of Rational Option Pricing," in The Bell Journal of Economics and Management Science, expanding the mathematical understanding and applications of the model, and coining the term "Black–Scholes theory of options pricing."

In 1997, Scholes and Merton were awarded the Nobel Memorial Prize in Economic Sciences for their work in finding "a new method to determine the value of derivatives." Black had passed away two years earlier, and so could not be a recipient, as Nobel Prizes are not given posthumously; however, the Nobel committee acknowledged his role in the Black-Scholes model.

Black-Scholes posits that instruments, such as stock shares or futures contracts, will have a lognormal distribution of prices following a random walk with constant drift and volatility. Using this assumption and factoring in other important variables, the equation derives the price of a European-style call option.

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The Black-Scholes equation requires five variables. These inputs are volatility, the price of the underlying asset, the strike price of the option, the time until expiration of the option, and the risk-free interest rate. With these variables, it is theoretically possible for options sellers to set rational prices for the options that they are selling.

Furthermore, the model predicts that the price of heavily traded assets follows a geometric Brownian motion with constant drift and volatility. When applied to a stock option, the model incorporates the constant price variation of the stock, the time value of money, the option's strike price, and the time to the option's expiry.

Black-Scholes Assumptions

The Black-Scholes model makes certain assumptions:

1. No dividends are paid out during the life of the option.
2. Markets are random (i.e., market movements cannot be predicted).
3. There are no transaction costs in buying the option.
4. The risk-free rate and volatility of the underlying asset are known and constant.
5. The returns of the underlying asset are normally distributed.
6. The option is European and can only be exercised at expiration.

While the original Black-Scholes model didn't consider the effects of dividends paid during the life of the option, the model is frequently adapted to account for dividends by determining the ex-dividend date value of the underlying stock. The model is also modified by many option-selling market makers to account for the effect of options that can be exercised before expiration.

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Formula

$$C = S_t N(d_1) - K e^{-rt} N(d_2)$$

where:

$$d_1 = \frac{ln\frac{S_t}{K} + (r + \frac{\sigma_v^2}{2})\, t}{\sigma_s \sqrt{t}}$$

and

$$d_2 = d_1 - \sigma_s \sqrt{t}$$

C = Call option price

S=Current stock price

K = Strike price

r = risk-free interest rate

t = time to maturity

N = normal distribution

The Black-Scholes call option formula is calculated by multiplying the stock price by the cumulative standard normal probability distribution function. Thereafter, the net present value (NPV) of the strike price multiplied by the cumulative standard normal distribution is subtracted from the resulting value of the previous calculation.

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